In [103…
```
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path
import os
from pathlib import Path
import numpy as np
```

In [104…
```
folder = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/GDP")
for f in folder.iterdir():
    print(f.name)
```

```
GDP per CAPITA - Maddison Project .xlsx
mpd2023_web.xlsx
```

In [105…
```
Project_Countries = [
    'Australia','Canada','Denmark','Finland','France','Germany',
    'Italy','Japan','Netherlands','Norway','Portugal','Spain',
    'Sweden','Switzerland','United Kingdom','United States'
]
```

DATA EXTRACTION MAKING THE VARIABLE ---> REAL GDP per CAPITA

In [106…
```
# In the Paper they use the MAddison Project Database, For Japan missing data (1882, 1884)
# they use the 1885 value of real GDP per capita and population figures for 1882–1884
# to back-calculate real GDP.
# Taking constant real GDP in 1885, we divide it by the annual population in 1882, 1883 and 1884
# to estimate real GDP per capita for those three years. This method provides a reasonable proxy
# in 2011 international dollars. At the end of the day, we end up with data for the full sample of 16 countries
# from 1882 to 2020.

# DATA file the Maddison Project Database found in the GDP folder in the DATA folder.

path_maddison_project = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/GDP/GDP per CAPITA - Maddison Project .xlsx")

df_maddison_project = pd.read_excel(path_maddison_project, sheet_name='Full data')

df_OECD_GDP = df_maddison_project[df_maddison_project['country'].isin(Project_Countries)]
df_OECD_GDP = df_OECD_GDP[['country', 'year', 'gdppc', 'pop']]


path_maddison_project_2022 = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/GDP")
df_maddison_project_2022 = pd.read_excel(path_maddison_project_2022 / "mpd2023_web.xlsx", sheet_name= 'GDPpc')
df_maddison_project_2022 = df_maddison_project_2022[['Australia','Canada','Denmark','Finland','France','Germany','Italy','Japan','Netherlands','Norway','Portugal','Spain','Sweden','Switzerland','Unit
df_maddison_project_2022 = df_maddison_project_2022.rename(columns={'GDP pc 2011 prices': 'year'})
df_maddison_project_2022['year'] = pd.to_numeric(df_maddison_project_2022['year'], errors='coerce')
df_maddison_project_2022 = df_maddison_project_2022.dropna(subset=['year']).astype({'year': 'int'})
df_maddison_project_2022_years = df_maddison_project_2022[df_maddison_project_2022["year"].isin([2018, 2019, 2020])]
df_maddison_project_2022_melted = df_maddison_project_2022_years.melt(id_vars=['year'], var_name='country', value_name='gdppc_2022')


years_to_add = [2019, 2020]

# long Maddison data for 2019–2020
new_rows = df_maddison_project_2022_melted[df_maddison_project_2022_melted['year'].isin(years_to_add)].copy()
new_rows['pop'] = np.nan
new_rows = new_rows.rename(columns={'gdppc_2022': 'gdppc'})
new_rows = new_rows[['country', 'year', 'gdppc', 'pop']]
df_OECD_GDP = pd.concat([df_OECD_GDP, new_rows], ignore_index=True)
df_OECD_GDP = df_OECD_GDP.sort_values(['country', 'year']).reset_index(drop=True)

df_OECD_GDP = pd.merge(df_OECD_GDP,df_maddison_project_2022_melted,on=['country', 'year'], how='left')

# overwrite old gdppc wherever you have a value from the 2022 Maddison data
mask = df_OECD_GDP['gdppc_2022'].notna()
df_OECD_GDP.loc[mask, 'gdppc'] = df_OECD_GDP.loc[mask, 'gdppc_2022']
df_OECD_GDP = df_OECD_GDP.drop(columns=['gdppc_2022'])

df_OECD_GDP = df_OECD_GDP[(df_OECD_GDP['year'] >= 1882) & (df_OECD_GDP['year'] <= 2020)]




# Sorting out the data for Japan for the years 1882, 1883, 1884
mask_japan = df_OECD_GDP['country'] == 'Japan'
jp = df_OECD_GDP[mask_japan]

rgdp_1885 = (jp.loc[jp['year'] == 1885, 'gdppc'].iloc[0]* jp.loc[jp['year'] == 1885, 'pop'].iloc[0])
years_to_fill = [1882, 1883, 1884]
mask_japan_early = mask_japan & df_OECD_GDP['year'].isin(years_to_fill)

df_OECD_GDP.loc[mask_japan_early, 'gdppc'] = (rgdp_1885 / df_OECD_GDP.loc[mask_japan_early, 'pop'])

#df_OECD_GDP.head()
```

Labour Productivity

In [107…
```
Project_Countries = [
    'Australia','Canada','Denmark','Finland','France','Germany',
    'Italy','Japan','Netherlands','Norway','Portugal','Spain',
    'Sweden','Switzerland','United Kingdom','United States'
]

path_labour_productivity = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/Labour Productivity/Long Term Labour Productivity.xlsx")
df_labour_productivity = pd.read_excel(path_labour_productivity, sheet_name='Labor Productivity')

df_labour_productivity = df_labour_productivity[
    ['Year', 'AUS', 'AUT', 'BEL', 'CAN', 'DNK', 'FIN', 'FRA', 'DEU',
     'GRC', 'IRL', 'ITA', 'JPN', 'NLD', 'NZL', 'NOR', 'PRT', 'ESP',
     'SWE', 'CHE', 'GBR', 'USA']
]

replace_name_dict = {
    'AUS': 'Australia',
    'AUT': 'Austria',
    'BEL': 'Belgium',
    'CAN': 'Canada',
    'DNK': 'Denmark',
    'FIN': 'Finland',
    'FRA': 'France',
    'DEU': 'Germany',
    'GRC': 'Greece',
    'IRL': 'Ireland',
    'ITA': 'Italy',
    'JPN': 'Japan',
    'NLD': 'Netherlands',
    'NZL': 'New Zealand',
    'NOR': 'Norway',
    'PRT': 'Portugal',
    'ESP': 'Spain',
    'SWE': 'Sweden',
    'CHE': 'Switzerland',
    'GBR': 'United Kingdom',
    'USA': 'United States'
}

df_labour_productivity = df_labour_productivity.rename(columns=replace_name_dict)

df_labour_productivity = df_labour_productivity[
    (df_labour_productivity['Year'] >= 1890) &
    (df_labour_productivity['Year'] <= 2020)
]

columns_keep_df_labour_productivity = ["Year"] + Project_Countries
df_labour_productivity = df_labour_productivity[columns_keep_df_labour_productivity]
df_labour_productivity.head()
```

Out[107…

	Year	Australia	Canada	Denmark	Finland	France	Germany	Italy	Japan	Netherlands	Norway	Portugal	Spain	Sweden	Switzerland	United Kingdom	United States
90	1890	11.275570	4.185372	3.550943	2.023169	2.792736	4.814139	2.018100	1.020471	5.493740	3.641319	1.925517	4.093740	2.689252	5.961463	5.974749	4.655917
91	1891	11.346384	3.856720	3.622624	2.013744	2.798662	4.832038	2.087370	0.970538	5.331186	3.699415	1.899728	4.170664	2.619799	5.536116	5.999465	4.703721
92	1892	9.399105	3.786976	3.705531	1.930385	2.841674	4.893941	2.119804	1.034053	5.364090	3.734071	1.892896	4.459632	2.826159	5.927149	5.962112	4.626473
93	1893	8.518302	3.718733	3.764063	1.999724	2.756078	5.231450	2.181473	1.031995	5.351097	3.797245	1.935931	4.282428	2.976881	6.074320	5.974519	5.022901
94	1894	8.448742	3.847008	3.809196	2.107358	2.992517	5.341485	2.184280	1.154029	5.415974	3.785117	1.918264	4.309023	2.973899	5.848441	6.268525	5.131884

Consumer Price Index

In [108…
```
# Missibg Data from Ireland --> paper claims full data set but data set missing 1882 to 1921, linear interpolation not possible since we dont have a stating points cannot
# draw a trend line.
path_CPI = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/CPI")
df_CPI_Jorda = pd.read_excel(path_CPI / " Jordá et al 2017  - .xlsx", sheet_name='Sheet1')
df_CPI_Jorda = df_CPI_Jorda[['country', 'year', 'cpi']]
df_CPI_Jorda = df_CPI_Jorda[(df_CPI_Jorda['year'] >= 1882) & (df_CPI_Jorda['year'] <= 2020)]

country_replace = {
    "AUS": "Australia",
    "AUT": "Austria",
    "BEL": "Belgium",
    "CAN": "Canada",
    "DNK": "Denmark",
    "FIN": "Finland",
    "FRA": "France",
    "DEU": "Germany",
    "GRC": "Greece",
    "IRL": "Ireland",
    "ITA": "Italy",
    "JPN": "Japan",
    "NLD": "Netherlands",
    "NZL": "New Zealand",
    "NOR": "Norway",
    "PRT": "Portugal",
    "ESP": "Spain",
    "SWE": "Sweden",
    "CHE": "Switzerland",
    "GBR": "United Kingdom",
    "UK": "United Kingdom",
    "USA": "United States",
    "US": "United States"
}

df_CPI_Jorda["country"] = df_CPI_Jorda["country"].replace(country_replace)

df_CPI_Jorda = df_CPI_Jorda[df_CPI_Jorda['country'].isin(Project_Countries)]
```

Trade Openness --> Exports - Imports / GDP

In [109…
```
# Trade Openness index --> Exports - Imports / GDP, keeping in line with the paper methodology
#Trade data are primarily sourced from the Jord`a-Schularick-Taylor Macrohistory Database (Jord`a et al., 2017)
#For specific country-year gaps, we first supplement with imports and exports data from Federico and Tena-Junguito (2016)
#Aus- tralia (1914), Switzerland (1882–1884), and Germany (1914–1923)
# In cases where the Jord`a-Schularick-Taylor Macrohistory Database provides data on imports and exports but lacks nominal GDP figures,
# we supplement with nominal GDP data from the Global Macro Database (Mu¨ller et al., 2025)
# we estimate trade openness by linearly interpolating the ratio of trade to GDP.
# full sample of 16 countries from 1882 to 2020.

path_TradeOpenness_Jorda_data_set = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/Trade Openness")
df_TradeOpenness_Jorda_data_set = pd.read_excel(path_CPI / " Jordá et al 2017  - .xlsx", sheet_name='Sheet1')
df_TradeOpenness_Jorda_data_set = df_TradeOpenness_Jorda_data_set[['country', 'year', 'exports', 'imports','gdp']]

country_replace = {
    "AUS": "Australia",
    "AUT": "Austria",
    "BEL": "Belgium",
    "CAN": "Canada",
    "DNK": "Denmark",
    "FIN": "Finland",
    "FRA": "France",
    "DEU": "Germany",
    "GRC": "Greece",
    "IRL": "Ireland",
    "ITA": "Italy",
    "JPN": "Japan",
    "NLD": "Netherlands",
    "NZL": "New Zealand",
    "NOR": "Norway",
    "PRT": "Portugal",
    "ESP": "Spain",
    "SWE": "Sweden",
    "CHE": "Switzerland",
    "GBR": "United Kingdom",
    "UK": "United Kingdom",
    "USA": "United States",
    "US": "United States"
}

df_TradeOpenness_Jorda_data_set["country"] = df_TradeOpenness_Jorda_data_set["country"].replace(country_replace)


df_TradeOpenness_Jorda_data_set = df_TradeOpenness_Jorda_data_set[df_TradeOpenness_Jorda_data_set["country"].isin(Project_Countries)]

df_TradeOpenness_GMD_nominal_GDP_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/Trade Openness")
df_TradeOpenness_GMD_nominal_GDP = pd.read_excel(df_TradeOpenness_GMD_nominal_GDP_path / "GMD.xlsx", sheet_name='data_final')
df_TradeOpenness_GMD_nominal_GDP = df_TradeOpenness_GMD_nominal_GDP[['countryname', 'year', 'nGDP', 'exports', 'imports']]
df_TradeOpenness_GMD_nominal_GDP = df_TradeOpenness_GMD_nominal_GDP.rename(columns={'countryname': 'country'})
df_TradeOpenness_GMD_nominal_GDP = df_TradeOpenness_GMD_nominal_GDP[df_TradeOpenness_GMD_nominal_GDP["country"].isin(Project_Countries)]

df_TradeOpenness = pd.merge(df_TradeOpenness_Jorda_data_set, df_TradeOpenness_GMD_nominal_GDP, on=['country', 'year'], how='left')

df_Spdf_Supplementary_STATS_Australia_1914_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/Trade Openness/Supplementary Trade STATS - AUSTRALIA (1914)")
df_Supplementary_STATS_Australia_1914 = pd.read_excel( df_Spdf_Supplementary_STATS_Australia_1914_path / "oceania_1826_1938_FTWTHD_201710_v01.xlsx", sheet_name='Current prices, 1913 borders')
```

Inputting the Missing Values for The exports and Imports and also the Nominal GDP

In [110…
```
# Australia (1914)
df_TradeOpenness.loc[
    (df_TradeOpenness["country"] == "Australia") &
    (df_TradeOpenness["year"] == 1914),
    "imports"
] = 343.33

df_TradeOpenness.loc[
    (df_TradeOpenness["country"] == "Australia") &
    (df_TradeOpenness["year"] == 1914),
    "exports"
] = 317.36


# Switzerland (1882, 1883, 1884)
swiss_years = [1882, 1883, 1884]

mask_ch = (
    (df_TradeOpenness["country"] == "Switzerland") &
    (df_TradeOpenness["year"].isin(swiss_years))
)

df_TradeOpenness.loc[mask_ch, "imports"] = [78.39, 87.47, 85.46]
df_TradeOpenness.loc[mask_ch, "exports"] = [134.97, 136.12, 136.68]


# Germany (1914–1923)
germany_years = [1914, 1915, 1916, 1917, 1918, 1919, 1920, 1921, 1922, 1923]

mask_de = (
    (df_TradeOpenness["country"] == "Germany") &
    (df_TradeOpenness["year"].isin(germany_years))
)

df_TradeOpenness.loc[mask_de, "imports"] = [
    2023.86, 1690.52, 2000.05, 1690.52, 1690.52, 1858.55, 2051.43, 2112.12, 2122.92, 1529.60,
]
df_TradeOpenness.loc[mask_de, "exports"] = [ 1761.95, 738.11, 904.78, 833.35, 1119.07, 1393.73, 1711.00, 1047.51, 1839.12, 1599.10 ]

df_TradeOpenness = df_TradeOpenness[(df_TradeOpenness['year'] >= 1882) & (df_TradeOpenness['year'] <= 2020)]
```

In [111…
```
def compute_trade_openness():
    global df_TradeOpenness   # <-- use your actual dataframe

    # 1. Primary measure: using exports_x and imports_x
    df_TradeOpenness['trade_openness_primary'] = (
        (df_TradeOpenness['exports_x'] + df_TradeOpenness['imports_x'])
        / df_TradeOpenness['nGDP']
    )

    # 2. Define conditions for when the value is "weird" or invalid
    weird = (
        df_TradeOpenness['trade_openness_primary'].isna() |
        (df_TradeOpenness['trade_openness_primary'] < 0) |
        (df_TradeOpenness['trade_openness_primary'] > 3)
    )

    # 3. Fallback using exports_y and imports_y
    df_TradeOpenness.loc[weird, 'trade_openness'] = (
        (df_TradeOpenness.loc[weird, 'exports_y'] + df_TradeOpenness.loc[weird, 'imports_y'])
        / df_TradeOpenness.loc[weird, 'nGDP']
    )

    # 4. Keep primary values where valid
    df_TradeOpenness.loc[~weird, 'trade_openness'] = (
        df_TradeOpenness.loc[~weird, 'trade_openness_primary']
    )

    # Cleanup
    df_TradeOpenness.drop(columns=['trade_openness_primary'], inplace=True)

    return df_TradeOpenness

df_TradeOpenness = compute_trade_openness()

def interpolate_openness_for_country_years(df, country, start_year, end_year):
    mask_country = df["country"] == country
    df_country = df.loc[mask_country].sort_values("year").copy()
    df_country["openness_interp"] = df_country["trade_openness"].interpolate(method="linear")
    mask_gap_years = (df_country["year"] >= start_year) & (df_country["year"] <= end_year)
    df.loc[mask_country & mask_gap_years, "trade_openness"] = df_country.loc[mask_gap_years, "openness_interp"].values
    return df
# Germany: 1944–1947
df_TradeOpenness = interpolate_openness_for_country_years(df_TradeOpenness, "Germany", 1944, 1947)
# Japan: 1944–1945
df_TradeOpenness = interpolate_openness_for_country_years(df_TradeOpenness, "Japan", 1944, 1945)
# Netherlands: 1944–1945
df_TradeOpenness = interpolate_openness_for_country_years(df_TradeOpenness, "Netherlands", 1944, 1945)
# Germany 1914–1920
df_TradeOpenness = interpolate_openness_for_country_years(df_TradeOpenness, "Germany", 1914, 1923)
# Switzerland 1882–1884
df_TradeOpenness = interpolate_openness_for_country_years(df_TradeOpenness, "Switzerland", 1882, 1884)
# Australia 1914
df_TradeOpenness = interpolate_openness_for_country_years(df_TradeOpenness, "Australia", 1914, 1914)



df_TradeOpenness.describe()
```

Out[111…

	year	exports_x	imports_x	gdp	nGDP	exports_y	imports_y	imports	exports	trade_openness
count	2224.000000	2.202000e+03	2.202000e+03	2.204000e+03	2.223000e+03	2.209000e+03	2.209000e+03	14.000000	14.000000	2.224000e+03
mean	1951.000000	7.722444e+05	7.984612e+05	2.723247e+06	1.019707e+07	1.380302e+06	1.298392e+06	1383.195714	976.632143	4.052065e-01
std	40.133829	5.240291e+06	5.242390e+06	1.760694e+07	6.605239e+07	8.896349e+06	8.533765e+06	831.755417	627.832361	6.466756e-01
min	1882.000000	2.856000e-12	2.873000e-12	1.707798e-11	9.669400e-09	1.823821e-09	2.362936e-09	78.390000	134.970000	1.282452e-08
25%	1916.000000	5.325250e+00	5.376575e+00	5.126680e+01	1.477916e+03	3.203133e+02	3.345548e+02	639.897500	422.547500	4.547730e-04
50%	1951.000000	4.129968e+02	4.537720e+02	1.794046e+03	1.960228e+04	3.419691e+03	3.201383e+03	1690.520000	976.145000	1.717040e-01
75%	1986.000000	7.359207e+03	9.192000e+03	4.556449e+04	4.431389e+05	1.128150e+05	1.126349e+05	2017.907500	1547.757500	4.825879e-01
max	2020.000000	7.237242e+07	6.630599e+07	2.070466e+08	5.579109e+08	1.019464e+08	1.038285e+08	2122.920000	1839.120000	2.996546e+00

Liquid Liabilities Data Set

In [112…
```
# Liquid Liabilities Data Set comes from Jorda-Schularick-Taylor Macrohistory Database (Jorda et al., 2017)
# computed as M2 or m3 / nominal Gdp
# e M2 estimates from Albert Ritschl's dataset on interwar Germany (Ritschl, 2014)
#For Spain (1936–1939), we use M1 estimates from Mart´ın-Acen˜a (2018) (Table 1, page 34)
# For France (1914–1919), we rely on Bordo and Hautcoeur (2003) estimates of money growth, applying their reported growth rates
# (30%, 70%, 27%, 28%, 40%, and 26%) to Jorda-Schularick-Taylor 1910 money stock level. Missing data for Japan (1945),
# the Netherlands (1942–1944), Spain (1940), Germany (1923–1924, 1945–1947) have been linearly interpolated.

path_Liquid_Liabilities = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/Liquid Liabilities")
df_Liquid_Liabilities = pd.read_excel(path_Liquid_Liabilities / "Jorda.xlsx", sheet_name="Sheet1")
df_Liquid_Liabilities = df_Liquid_Liabilities[["country", "year", "money", "gdp"]]
country_replace = {
    "AUS": "Australia",
    "AUT": "Austria",
    "BEL": "Belgium",
    "CAN": "Canada",
    "DNK": "Denmark",
    "FIN": "Finland",
    "FRA": "France",
    "DEU": "Germany",
    "GRC": "Greece",
    "IRL": "Ireland",
    "ITA": "Italy",
    "JPN": "Japan",
    "NLD": "Netherlands",
    "NZL": "New Zealand",
    "NOR": "Norway",
    "PRT": "Portugal",
    "ESP": "Spain",
    "SWE": "Sweden",
    "CHE": "Switzerland",
    "GBR": "United Kingdom",
    "UK": "United Kingdom",
    "USA": "United States",
    "US": "United States"
}

df_Liquid_Liabilities["country"] = df_Liquid_Liabilities["country"].replace(country_replace)
df_Liquid_Liabilities = df_Liquid_Liabilities[df_Liquid_Liabilities["country"].isin(Project_Countries)]

# Germany (1939–1945)

ger_money_ritschl = {
    1939:  6.3278,
    1940:  8.2882,
    1941:  9.9844,
    1942:  12.9153,
    1943:  16.4405,
    1944: 21.0798,
}

for year, value in ger_money_ritschl.items():
    df_Liquid_Liabilities.loc[
        (df_Liquid_Liabilities["country"] == "Germany") &
        (df_Liquid_Liabilities["year"] == year),
        "money"
    ] = value


# Spain (1936–1939)

spain_money_martin_acena = {
    1936:  11461,
    1937:  22219,
    1938:  35852,
    1939:  42900,
}
for year, value in spain_money_martin_acena.items():
    df_Liquid_Liabilities.loc[(df_Liquid_Liabilities["country"] == "Spain") & (df_Liquid_Liabilities["year"] == year),"money"] = value


# France (1914–1919)

French_Growth_Rates_Bordo_Hautcoeur = {
    1914: 0.30,
    1915: 0.70,
    1916: 0.27,
    1917: 0.28,
    1918: 0.40,
    1919: 0.26,
}
M_1910 = df_Liquid_Liabilities.loc[(df_Liquid_Liabilities["country"] == "France") & (df_Liquid_Liabilities["year"] == 1910),"money"].iloc[0]
current_level = M_1910
for year in sorted(French_Growth_Rates_Bordo_Hautcoeur.keys()):
    g = French_Growth_Rates_Bordo_Hautcoeur[year]
    current_level = current_level * (1 + g)
    df_Liquid_Liabilities.loc[
        (df_Liquid_Liabilities["country"] == "France") &
        (df_Liquid_Liabilities["year"] == year),
        "money"
    ] = current_level

df_Liquid_Liabilities["liquid_liabilities"] = (
    df_Liquid_Liabilities["money"] / df_Liquid_Liabilities["gdp"]
)

def interpolate_liq_for_country_years(df, country, start_year, end_year):
    mask_country = df["country"] == country
    df_country = df.loc[mask_country].sort_values("year").copy()
    df_country["liq_interp"] = df_country["liquid_liabilities"].interpolate(method="linear")
    mask_gap = (df_country["year"] >= start_year) & (df_country["year"] <= end_year)
    df.loc[mask_country & mask_gap, "liquid_liabilities"] = df_country.loc[mask_gap, "liq_interp"].values
    return df

# Japan (1945)
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Japan", 1945, 1945)

# Netherlands (1942–1944)
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Netherlands", 1942, 1944)

# Spain (1940)
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Spain", 1940, 1940)

# Germany (1923–1924)
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Germany", 1923, 1924)

# Germany (1945–1947)
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Germany", 1945, 1947)

# Wasn't mentioned in the paper but Norway has missing data for 1940 to 1945, applying same logic of linear interpolation
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Norway", 1940, 1945)

# Same for the Nethertheralns 1914 1921
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Netherlands", 1914, 1921)
df_Liquid_Liabilities = interpolate_liq_for_country_years(df_Liquid_Liabilities, "Netherlands", 1940, 1941)



df_Liquid_Liabilities = df_Liquid_Liabilities[(df_Liquid_Liabilities['year'] >= 1882) & (df_Liquid_Liabilities['year'] <= 2020)]

df_Liquid_Liabilities.describe()
```

Out[112…

	year	money	gdp	liquid_liabilities
count	2224.000000	2.215000e+03	2.204000e+03	2224.000000
mean	1951.000000	2.524930e+06	2.723247e+06	0.593804
std	40.133829	1.746533e+07	1.760694e+07	0.257660
min	1882.000000	6.204000e-12	1.707798e-11	0.057906
25%	1916.000000	1.992978e+01	5.126680e+01	0.431571
50%	1951.000000	1.041475e+03	1.794046e+03	0.548365
75%	1986.000000	2.518053e+04	4.556449e+04	0.707641
max	2020.000000	2.357367e+08	2.070466e+08	2.259868

Stock Market Capitalization

In [113…
```
df_stock_market_capitalization_Zimmermann_2022_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/Stock Market Capitalization/BBdatasetR1")
df_stock_market_capitalization_Zimmermann_2022 = pd.read_excel(df_stock_market_capitalization_Zimmermann_2022_path/ "Zimmermann 2022.xlsx", sheet_name="Data")
df_stock_market_capitalization_Zimmermann_2022 = df_stock_market_capitalization_Zimmermann_2022[["country", "year", "mcap_gdp"]]

country_replace = {
    "AUS": "Australia",
    "AUT": "Austria",
    "BEL": "Belgium",
    "CAN": "Canada",
    "DNK": "Denmark",
    "FIN": "Finland",
    "FRA": "France",
    "DEU": "Germany",
    "GRC": "Greece",
    "IRL": "Ireland",
    "ITA": "Italy",
    "JPN": "Japan",
    "NLD": "Netherlands",
    "NZL": "New Zealand",
    "NOR": "Norway",
    "PRT": "Portugal",
    "ESP": "Spain",
    "SWE": "Sweden",
    "CHE": "Switzerland",
    "GBR": "United Kingdom",
    "UK": "United Kingdom",
    "USA": "United States",
    "US": "United States"
}

df_stock_market_capitalization_Zimmermann_2022["country"] = df_stock_market_capitalization_Zimmermann_2022["country"].replace(country_replace)
excluded_countries = ["Finland", "Italy", "Japan", "Netherlands"]
df_stock_market_capitalization_Zimmermann_2022 = df_stock_market_capitalization_Zimmermann_2022[~df_stock_market_capitalization_Zimmermann_2022["country"].isin(excluded_countries)]

df_stock_market_capitalization_Zimmermann_2022 = df_stock_market_capitalization_Zimmermann_2022[df_stock_market_capitalization_Zimmermann_2022["country"].isin(Project_Countries)]

df_stock_market_capitalization_Zimmermann_2022["year"] = pd.to_datetime(
    df_stock_market_capitalization_Zimmermann_2022["year"]
)

df_stock_market_capitalization_Zimmermann_2022 = df_stock_market_capitalization_Zimmermann_2022[
    (df_stock_market_capitalization_Zimmermann_2022["year"].dt.year >= 1889) &
    (df_stock_market_capitalization_Zimmermann_2022["year"].dt.year <= 2016)
]

df_stock_market_capitalization_Zimmermann_2022["year"] = pd.to_datetime(
    df_stock_market_capitalization_Zimmermann_2022["year"], errors="coerce"
).dt.year

def interpolate_mcap_for_country_years(df, country, start_year, end_year):
    mask_country = df["country"] == country
    df_country = df.loc[mask_country].sort_values("year").copy()

    # interpolate the stock market capitalization ratio
    df_country["mcap_interp"] = df_country["mcap_gdp"].interpolate(method="linear")

    # now year is int, so this is fine
    mask_gap = (df_country["year"] >= start_year) & (df_country["year"] <= end_year)

    df.loc[mask_country & mask_gap, "mcap_gdp"] = df_country.loc[mask_gap, "mcap_interp"].values
    return df

# Canada (1927–1933)
df_stock_market_capitalization_Zimmermann_2022 = interpolate_mcap_for_country_years(
    df_stock_market_capitalization_Zimmermann_2022, "Canada", 1927, 1933
)

# Germany (1923)
df_stock_market_capitalization_Zimmermann_2022 = interpolate_mcap_for_country_years(
    df_stock_market_capitalization_Zimmermann_2022, "Germany", 1923, 1923
)

# Germany (1945)
df_stock_market_capitalization_Zimmermann_2022 = interpolate_mcap_for_country_years(
    df_stock_market_capitalization_Zimmermann_2022, "Germany", 1945, 1945
)

# France (1917–1919)
df_stock_market_capitalization_Zimmermann_2022 = interpolate_mcap_for_country_years(
    df_stock_market_capitalization_Zimmermann_2022, "France", 1917, 1919
)

# France (1939–1944)
df_stock_market_capitalization_Zimmermann_2022 = interpolate_mcap_for_country_years(
    df_stock_market_capitalization_Zimmermann_2022, "France", 1939, 1944
)

df_stock_market_capitalization_Zimmermann_2022 = interpolate_mcap_for_country_years(
    df_stock_market_capitalization_Zimmermann_2022, "Norway", 1940, 1945
)

print(df_stock_market_capitalization_Zimmermann_2022[df_stock_market_capitalization_Zimmermann_2022["mcap_gdp"].isna()])
```

```
            country  year  mcap_gdp
20        Australia  1889       NaN
21        Australia  1890       NaN
22        Australia  1891       NaN
23        Australia  1892       NaN
24        Australia  1893       NaN
25        Australia  1894       NaN
26        Australia  1895       NaN
27        Australia  1896       NaN
28        Australia  1897       NaN
314          Canada  1889       NaN
315          Canada  1890       NaN
316          Canada  1891       NaN
317          Canada  1892       NaN
318          Canada  1893       NaN
319          Canada  1894       NaN
320          Canada  1895       NaN
321          Canada  1896       NaN
322          Canada  1897       NaN
902           Spain  1889       NaN
903           Spain  1890       NaN
904           Spain  1891       NaN
905           Spain  1892       NaN
906           Spain  1893       NaN
907           Spain  1894       NaN
908           Spain  1895       NaN
909           Spain  1896       NaN
910           Spain  1897       NaN
2372  United States  1889       NaN
2373  United States  1890       NaN
2374  United States  1891       NaN
2375  United States  1892       NaN
2376  United States  1893       NaN
2377  United States  1894       NaN
2378  United States  1895       NaN
2379  United States  1896       NaN
2380  United States  1897       NaN
```

Bank Loans Data set --> Jorda, Muller and Verner, World Bank,

In [114…
```
df_Bank_Loan_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/CPI")
df_Bank_Loan_data = pd.read_excel(df_Bank_Loan_data_path / " Jordá et al 2017  - .xlsx", sheet_name='Sheet1')
df_Bank_Loan_GDP_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data/Trade Openness")
df_Bank_Loan_GDP_data = pd.read_excel(df_Bank_Loan_GDP_data_path / "GMD.xlsx", sheet_name='data_final')

country_replace = {
    "AUS": "Australia",
    "AUT": "Austria",
    "BEL": "Belgium",
    "CAN": "Canada",
    "DNK": "Denmark",
    "FIN": "Finland",
    "FRA": "France",
    "DEU": "Germany",
    "GRC": "Greece",
    "IRL": "Ireland",
    "ITA": "Italy",
    "JPN": "Japan",
    "NLD": "Netherlands",
    "NZL": "New Zealand",
    "NOR": "Norway",
    "PRT": "Portugal",
    "ESP": "Spain",
    "SWE": "Sweden",
    "CHE": "Switzerland",
    "GBR": "United Kingdom",
    "UK": "United Kingdom",
    "USA": "United States",
    "US": "United States"
}

df_Bank_Loan_data["country"] = df_Bank_Loan_data["country"].replace(country_replace)
df_Bank_Loan_data = df_Bank_Loan_data[['country', 'year', 'tloans', 'gdp']]
df_Bank_Loan_data = df_Bank_Loan_data[(df_Bank_Loan_data['year'] >= 1882) & (df_Bank_Loan_data['year'] <= 2020)]
df_Bank_Loan_data = df_Bank_Loan_data[df_Bank_Loan_data["country"].isin(Project_Countries)]
df_Bank_Loan_GDP_data = df_Bank_Loan_GDP_data[['countryname', 'year', 'nGDP']]




df_Bank_Loan_GDP_data = df_Bank_Loan_GDP_data[df_Bank_Loan_GDP_data["countryname"].isin(Project_Countries)]
df_Bank_Loan_GDP_data = df_Bank_Loan_GDP_data.rename(columns={'countryname': 'country'})
df_Bank_Loan_GDP_data = df_Bank_Loan_GDP_data[(df_Bank_Loan_GDP_data['year'] >= 1882) & (df_Bank_Loan_GDP_data['year'] <= 2020)]

df_Bank_Loan_data = pd.merge(df_Bank_Loan_data, df_Bank_Loan_GDP_data, on=['country', 'year'], how='left')
df_Bank_Loan_data['bank_loans_gdp'] = df_Bank_Loan_data.apply(
    lambda row: row['tloans'] / row['gdp'] if pd.notna(row['gdp'])
    else (row['tloans'] / row['nGDP'] if pd.notna(row['nGDP']) else None),
    axis=1
)

exclude_countries = ['Spain', 'France', 'Netherlands', 'Portugal']

df_Bank_Loan_data = df_Bank_Loan_data[~df_Bank_Loan_data["country"].isin(exclude_countries)]
df_Bank_Loan_data.head()

# Following the data cleaning steps illustrated in the paper
df_Bank_Loan_data = df_Bank_Loan_data.sort_values(['country', 'year'])
df_Bank_Loan_data['bank_loans_gdp_interp'] = (df_Bank_Loan_data.groupby('country')['bank_loans_gdp'].transform(lambda s: s.interpolate(method='linear')))
mask_germany = (
(df_Bank_Loan_data['country'] == 'Germany') &
(df_Bank_Loan_data['year'].between(1921, 1923) |df_Bank_Loan_data['year'].between(1941, 1945)))
mask_japan = ((df_Bank_Loan_data['country'] == 'Japan') &(df_Bank_Loan_data['year'] == 1945))
mask_Australia = ((df_Bank_Loan_data['country'] == 'Australia') & (df_Bank_Loan_data['year'].isin([1946, 1947])))
mask = mask_germany | mask_japan | mask_Australia
df_Bank_Loan_data.loc[mask, 'bank_loans_gdp'] = (df_Bank_Loan_data.loc[mask, 'bank_loans_gdp_interp'])
df_Bank_Loan_data = df_Bank_Loan_data.drop(columns='bank_loans_gdp_interp')

# getting missing data for Japan - World Bank

jp_fix = {
    2018: 147.01,
    2019: 147.83,
    2020: 154.29
}

for yr, val in jp_fix.items():
    mask = (df_Bank_Loan_data['country'] == 'Japan') & (df_Bank_Loan_data['year'] == yr)
    df_Bank_Loan_data.loc[mask, 'bank_loans_gdp'] = val / 100
```

Merging Everything Together

In [115…
```
df_gdp = df_OECD_GDP.copy()


df_lp_long = (
    df_labour_productivity
    .melt(id_vars='Year', var_name='country', value_name='labour_productivity')
    .rename(columns={'Year': 'year'})
)
df_cpi = df_CPI_Jorda[['country', 'year', 'cpi']].copy()
df_to = df_TradeOpenness[['country', 'year', 'trade_openness']].copy()
df_ll = df_Liquid_Liabilities[['country', 'year', 'liquid_liabilities']].copy()
df_mcap = df_stock_market_capitalization_Zimmermann_2022[['country', 'year', 'mcap_gdp']].copy()
df_loans = df_Bank_Loan_data[['country', 'year', 'bank_loans_gdp']].copy()

df_Final = (
    df_gdp
      .merge(df_lp_long, on=['country', 'year'], how='left')
      .merge(df_cpi,     on=['country', 'year'], how='left')
      .merge(df_to,      on=['country', 'year'], how='left')
      .merge(df_ll,      on=['country', 'year'], how='left')
      .merge(df_mcap,    on=['country', 'year'], how='left')
      .merge(df_loans,   on=['country', 'year'], how='left')
)

df_Final = df_Final[(df_Final['year'] >= 1882) & (df_Final['year'] <= 2020)]

output_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/final_panel_dataset.xlsx")

# Save df_all to Excel
df_Final.to_excel(output_path, index=False)

print("Saved to:", output_path)
```

```
Saved to: /Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/final_panel_dataset.xlsx
```

Question 14

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path
import os
from pathlib import Path
```

Descriptive Statistics Section

GDP

In [2]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'gdppc']]

periods = {
    '1882–1913': (1882, 1913),
    '1929–1971' : (1929, 1971),
    '1974–2020' : (1974, 2020)
}

results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['gdppc'].mean(),
        "Min": df_c['gdppc'].min(),
        "Max": df_c['gdppc'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'gdppc'].mean()

    results.append(row)

df_gdp_table = pd.DataFrame(results)

df_gdp_table = df_gdp_table.sort_values("Country").reset_index(drop=True)


df_gdp_table
```

Out[2]:

	Country	Avg	Min	Max	1882–1913	1929–1971	1974–2020
0	Australia	18181.714409	5451.0000	49814.669956	6878.343750	12220.511628	34459.942613
1	Canada	17379.594154	3304.0000	45284.213482	4652.812500	11967.860465	34199.352923
2	Denmark	17659.407083	3571.0000	48238.657869	4705.593750	11910.023256	35084.820948
3	Finland	13614.012432	1918.0000	40129.653300	2498.062500	7908.325581	29559.653788
4	France	15180.038225	3518.0000	39084.655420	4315.093750	9560.348837	30561.964111
5	Germany	15595.977800	3258.0000	46761.946056	4433.062500	9514.488372	31987.147111
6	Italy	13382.222730	2831.0000	36310.734900	3302.437500	7458.348837	28501.573606
7	Japan	13110.189602	1723.5686	38414.861800	2042.516029	5877.909449	30262.578890
8	Netherlands	17120.066007	4222.0000	48142.671760	5447.781250	10772.813953	33915.152658
9	Norway	21341.497371	2499.0000	85115.323086	3150.656250	9376.465116	50162.470949
10	Portugal	8766.608128	1581.0000	28056.056110	1862.593750	4145.441860	19730.585741
11	Spain	10539.900708	2330.0000	34957.465528	2661.812500	4826.906977	23397.110603
12	Sweden	16214.236925	2354.0000	45434.402948	3216.000000	11387.511628	32823.828353
13	Switzerland	20955.820699	3998.6616	62448.599363	6185.244103	13535.309081	41794.540964
14	United Kingdom	16398.416105	6132.0000	39113.011733	7148.531250	12140.465116	29177.358267
15	United States	22284.062755	6424.1432	56469.263659	7927.131712	15936.648963	41479.469879

Descripitve Stats Labour Productivity

In [3]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'labour_productivity']]

periods = {
    '1882–1913': (1882, 1913),
    '1929–1971' : (1929, 1971),
    '1974–2020' : (1974, 2020)
}
results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['labour_productivity'].mean(),
        "Min": df_c['labour_productivity'].min(),
        "Max": df_c['labour_productivity'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'labour_productivity'].mean()

    results.append(row)

df_lp_table = pd.DataFrame(results)
df_lp_table = df_lp_table.sort_values("Country").reset_index(drop=True)
df_lp_table
```

Out[3]:

	Country	Avg	Min	Max	1882–1913	1929–1971	1974–2020
0	Australia	24.412274	7.145286	57.693508	8.844059	17.531114	42.896392
1	Canada	23.736365	3.617792	59.533274	4.965576	16.858045	44.322236
2	Denmark	25.806922	3.550943	71.875021	4.742646	14.321432	52.951044
3	Finland	19.752029	1.930385	60.669460	2.515039	8.764999	43.751207
4	France	23.766845	2.756078	67.915392	3.276825	10.839148	52.094069
5	Germany	25.141762	4.585027	68.980046	6.014343	12.461173	52.173867
6	Italy	21.375320	2.018100	55.582627	2.378297	9.819482	47.023313
7	Japan	15.411422	0.970538	49.062035	1.324780	5.463195	35.819434
8	Netherlands	28.735044	5.239417	68.968044	5.925238	17.139801	56.715737
9	Norway	34.941293	3.641319	97.520531	4.286882	17.039040	75.829314
10	Portugal	14.053882	1.892896	38.907656	2.293712	6.964283	29.851783
11	Spain	20.032883	3.888716	53.113263	4.565802	8.862955	42.525098
12	Sweden	24.500651	2.619799	68.115160	3.722773	14.473205	49.860418
13	Switzerland	34.285447	5.536116	80.029640	7.020768	25.147592	63.801251
14	United Kingdom	23.198715	5.962112	59.286516	6.602303	13.397534	45.323436
15	United States	28.725957	4.626473	73.957562	5.621785	21.188733	53.373439

Descriptive Stats CPI

In [4]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'cpi']]

periods = {
    '1882–1913': (1882, 1913),
    '1929–1971' : (1929, 1971),
    '1974–2020' : (1974, 2020)
}

results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['cpi'].mean(),
        "Min": df_c['cpi'].min(),
        "Max": df_c['cpi'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'cpi'].mean()

    results.append(row)

df_cpi_table = pd.DataFrame(results)
df_cpi_table = df_cpi_table.sort_values("Country").reset_index(drop=True)
df_cpi_table
```

Out[4]:

	Country	Avg	Min	Max	1882–1913	1929–1971	1974–2020
0	Australia	43.544890	2.250000e+00	202.046014	2.682292e+00	9.146283	116.351147
1	Canada	45.877413	5.285346e+00	174.814103	6.240752e+00	15.904300	112.104647
2	Denmark	41.964554	2.109956e+00	161.815857	2.395390e+00	10.352823	110.231615
3	Finland	38.950207	4.620000e-02	162.232694	5.784375e-02	6.985444	107.648813
4	France	39.327044	6.050080e-02	155.324987	6.281631e-02	8.350713	107.488288
5	Germany	51.007171	1.150362e-11	168.789036	1.271163e-11	30.324740	118.420254
6	Italy	43.403157	2.208800e-02	200.084964	2.349424e-02	4.744669	123.435780
7	Japan	37.840419	1.604590e-02	111.607085	2.478576e-02	12.560825	98.654745
8	Netherlands	50.393610	5.380884e+00	181.424326	6.084630e+00	19.351761	121.941864
9	Norway	43.073465	2.293544e+00	187.737714	2.555589e+00	10.715319	112.724314
10	Portugal	45.323773	4.173523e-02	226.825215	5.211314e-02	2.170814	131.626799
11	Spain	44.773411	2.568748e-01	213.584964	3.040568e-01	3.726903	128.092626
12	Sweden	41.697069	2.407773e+00	161.660079	2.784503e+00	10.395410	109.000505
13	Switzerland	50.689569	9.257564e+00	132.437189	1.062807e+01	28.040517	108.051906
14	United Kingdom	43.987101	1.984892e+00	194.847983	2.110475e+00	8.816898	118.340113
15	United States	49.896845	6.736232e+00	192.622537	7.368184e+00	19.124733	119.524709

Descriptive Stats - Trade Openness

In [5]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'trade_openness']]

periods = {
    '1882–1913': (1882, 1913),
    '1929–1971' : (1929, 1971),
    '1974–2020' : (1974, 2020)
}
results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['trade_openness'].mean(),
        "Min": df_c['trade_openness'].min(),
        "Max": df_c['trade_openness'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'trade_openness'].mean()

    results.append(row)

df_to_table = pd.DataFrame(results)
df_to_table = df_to_table.sort_values("Country").reset_index(drop=True)
df_to_table
```

Out[5]:

	Country	Avg	Min	Max	1882–1913	1929–1971	1974–2020
0	Australia	2.417701e-01	1.422638e-01	4.240048e-01	2.292362e-01	2.196393e-01	2.877856e-01
1	Canada	4.049497e-04	2.345392e-04	6.922896e-04	3.493633e-04	3.198211e-04	5.070648e-04
2	Denmark	4.844653e-04	1.122112e-04	6.341054e-04	4.699919e-04	4.164150e-04	5.435144e-04
3	Finland	2.016736e+00	4.574830e-01	2.996546e+00	2.465905e+00	2.001051e+00	1.599314e+00
4	France	1.970264e-03	4.171811e-04	3.428153e-03	1.877088e-03	1.254822e-03	2.546570e-03
5	Germany	2.545903e-02	1.109059e-04	9.217843e-01	6.141566e-04	3.630010e-04	9.946653e-04
6	Italy	5.772409e-01	9.202175e-02	1.163041e+00	4.112590e-01	3.749590e-01	8.930931e-01
7	Japan	2.078527e-07	1.282452e-08	3.809018e-07	1.806363e-07	1.924190e-07	2.129923e-07
8	Netherlands	1.670221e+00	2.418701e-01	2.744003e+00	2.183510e+00	1.131527e+00	2.089003e+00
9	Norway	4.247826e-01	1.296565e-01	5.558980e-01	3.936148e-01	3.624605e-01	4.984302e-01
10	Portugal	3.620249e-01	8.010407e-02	8.677099e-01	9.682671e-02	2.982708e-01	6.349794e-01
11	Spain	2.502477e-01	3.128278e-02	6.700941e-01	1.517355e-01	1.211439e-01	4.716263e-01
12	Sweden	3.827747e-01	1.143868e-01	6.744272e-01	3.491635e-01	2.816653e-01	5.266224e-01
13	Switzerland	5.286803e-01	1.634748e-01	1.416275e+00	6.925174e-01	3.504960e-01	5.804088e-01
14	United Kingdom	3.659191e-04	1.011181e-04	5.634698e-04	4.673845e-04	2.725937e-04	3.636764e-04
15	United States	1.393867e-04	4.984706e-05	3.084248e-04	1.060479e-04	7.968947e-05	2.247321e-04

Descriptive Stats - liquid Liabiliteis

In [6]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'liquid_liabilities']].copy()

periods = {
    'Avg 1882–1913': (1882, 1913),
    'Avg 1929–1971': (1929, 1971),
    'Avg 1974–2020': (1974, 2020)
}

results = []
for country in sorted(df_Final['country'].unique()):
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['liquid_liabilities'].mean(),
        "Min": df_c['liquid_liabilities'].min(),
        "Max": df_c['liquid_liabilities'].max()
    }
    for label, (start, end) in periods.items():
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'liquid_liabilities'].mean()

    results.append(row)


df_liq_table = pd.DataFrame(results)
df_liq_table = df_liq_table.sort_values("Country").reset_index(drop=True)

df_liq_table
```

Out[6]:

	Country	Avg	Min	Max	Avg 1882–1913	Avg 1929–1971	Avg 1974–2020
0	Australia	0.588035	0.378306	1.219731	0.475271	0.564270	0.723271
1	Canada	0.438497	0.210120	0.971409	0.335605	0.412538	0.537484
2	Denmark	0.496879	0.324765	0.749691	0.577666	0.452793	0.451458
3	Finland	0.499915	0.242607	0.823183	0.497352	0.480582	0.510755
4	France	0.661069	0.287009	2.259868	0.651115	0.636227	0.626257
5	Germany	0.512251	0.057906	1.131326	0.601928	0.289691	0.644742
6	Italy	0.630216	0.304372	1.235554	0.444598	0.642336	0.761780
7	Japan	0.735814	0.209509	2.030276	0.287033	0.641858	1.206171
8	Netherlands	0.551982	0.199094	1.685238	0.255081	0.584656	0.762501
9	Norway	0.596664	0.388443	1.107734	0.578366	0.618113	0.524919
10	Portugal	0.511492	0.087546	1.168655	0.121681	0.510912	0.855755
11	Spain	0.608370	0.207639	1.262809	0.277856	0.603928	0.908790
12	Sweden	0.624740	0.385460	0.875163	0.593502	0.701271	0.548066
13	Switzerland	0.926073	0.481506	1.616489	0.567522	1.013753	1.142984
14	United Kingdom	0.589564	0.410412	1.183510	0.470901	0.571418	0.716867
15	United States	0.529296	0.291136	0.846065	0.399205	0.598438	0.559694

Descriptive Stats --> Stock Market Cap

In [7]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'mcap_gdp']]
periods = {
    '1882–1913': (1882, 1913),
    '1929–1971' : (1929, 1971),
    '1974–2020' : (1974, 2020)
}
results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['mcap_gdp'].mean(),
        "Min": df_c['mcap_gdp'].min(),
        "Max": df_c['mcap_gdp'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'mcap_gdp'].mean()

    results.append(row)
df_smc_table = pd.DataFrame(results)
df_smc_table = df_smc_table.sort_values("Country").reset_index(drop=True)
df_smc_table
```

Out[7]:

	Country	Avg	Min	Max	1882–1913	1929–1971	1974–2020
0	Australia	0.418503	0.132369	1.395296	0.208414	0.290691	0.703213
1	Canada	0.714749	0.268621	1.409225	0.469164	0.819846	0.800997
2	Denmark	0.289178	0.070842	1.220458	0.337489	0.142894	0.426838
3	Finland	NaN	NaN	NaN	NaN	NaN	NaN
4	France	0.396188	0.054870	0.967841	0.611027	0.274747	0.429348
5	Germany	0.239331	0.024887	0.714247	0.287374	0.147618	0.308788
6	Italy	NaN	NaN	NaN	NaN	NaN	NaN
7	Japan	NaN	NaN	NaN	NaN	NaN	NaN
8	Netherlands	NaN	NaN	NaN	NaN	NaN	NaN
9	Norway	0.209818	0.029932	0.832498	0.208851	0.109701	0.307934
10	Portugal	0.176810	0.004320	0.793364	0.078213	0.163738	0.253180
11	Spain	0.226914	0.055092	0.729985	0.080589	0.199170	0.348403
12	Sweden	0.382406	0.074692	1.409290	0.257910	0.211559	0.648783
13	Switzerland	0.679780	0.157315	2.797097	0.313177	0.400360	1.310204
14	United Kingdom	0.774818	0.198062	1.881063	0.706594	0.720806	0.951829
15	United States	0.674530	0.252113	1.534410	0.575888	0.561486	0.889553

Descriptive Stats --> Banks Loans

In [8]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'bank_loans_gdp']]
periods = {
    '1882–1913': (1882, 1913),
    '1929–1971' : (1929, 1971),
    '1974–2020' : (1974, 2020)
}

results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['bank_loans_gdp'].mean(),
        "Min": df_c['bank_loans_gdp'].min(),
        "Max": df_c['bank_loans_gdp'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'bank_loans_gdp'].mean()

    results.append(row)

df_bl_table = pd.DataFrame(results)
df_bl_table = df_bl_table.sort_values("Country").reset_index(drop=True)
df_bl_table
```

Out[8]:

	Country	Avg	Min	Max	1882–1913	1929–1971	1974–2020
0	Australia	0.456235	0.160158	1.606247	0.395751	0.225695	0.785653
1	Canada	0.469079	0.141676	1.124800	0.429858	0.263298	0.719027
2	Denmark	1.130445	0.547949	2.035726	1.028275	0.951024	1.347576
3	Finland	0.443136	0.086383	0.989226	0.295511	0.344873	0.642041
4	France	NaN	NaN	NaN	NaN	NaN	NaN
5	Germany	0.630132	0.050678	1.040243	0.743309	0.374665	0.844116
6	Italy	0.438776	0.128907	0.999568	0.207631	0.397922	0.679816
7	Japan	0.576814	0.107728	1.542900	0.257677	0.488570	0.869021
8	Netherlands	NaN	NaN	NaN	NaN	NaN	NaN
9	Norway	0.720617	0.229805	1.498531	0.651111	0.537671	0.891203
10	Portugal	NaN	NaN	NaN	NaN	NaN	NaN
11	Spain	NaN	NaN	NaN	NaN	NaN	NaN
12	Sweden	0.681006	0.376188	1.301502	0.644173	0.478102	0.910948
13	Switzerland	1.123329	0.647735	1.676905	0.952047	1.058768	1.308577
14	United Kingdom	0.421918	0.136823	1.182136	0.259581	0.250274	0.751273
15	United States	0.421697	0.115199	0.695372	0.329269	0.339959	0.555112

In [9]:

```
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'gdppc', 'labour_productivity', 'cpi', 'trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']]

df_Final.describe()
```

Out[9]:

	year	gdppc	labour_productivity	cpi	trade_openness	liquid_liabilities	mcap_gdp	bank_loans_gdp
count	2224.000000	2224.000000	2096.000000	2.224000e+03	2.224000e+03	2224.000000	1488.000000	1668.000000
mean	1951.000000	16107.735321	24.242301	4.448436e+01	4.052065e-01	0.593804	0.430062	0.626099
std	40.133829	14479.989948	21.727817	5.785014e+01	6.466756e-01	0.257660	0.372890	0.359545
min	1882.000000	1581.000000	0.970538	1.150362e-11	1.282452e-08	0.057906	0.004320	0.050678
25%	1916.000000	4997.250000	6.242636	2.666016e+00	4.547730e-04	0.431571	0.167840	0.341700
50%	1951.000000	9841.075950	14.581455	1.192234e+01	1.717040e-01	0.548365	0.321797	0.553202
75%	1986.000000	24807.509900	40.460965	8.554003e+01	4.825879e-01	0.707641	0.578042	0.838633
max	2020.000000	85115.323086	97.520531	2.268252e+02	2.996546e+00	2.259868	2.797097	2.035726

Looking for Discontinutiy

In [10]:

```
import pandas as pd

# Load your data (adjust path if needed)
df_Final_data_path = Path("/Users/lorenzozuberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")

# Make sure data are sorted by country and year
df = df.sort_values(["country", "year"])

def count_panel_stats(group):
    years = group["year"].tolist()
    n_obs = len(years)

    # Count gaps > 1 year
    discontinuities = sum(
        (years[i] - years[i-1] > 1)
        for i in range(1, n_obs)
    )

    # Count consecutive pairs (year_t - year_(t-1) == 1)
    consecutive_pairs = sum(
        (years[i] - years[i-1] == 1)
        for i in range(1, n_obs)
    )

    return pd.Series({
        "n_obs": n_obs,
        "discontinuities": discontinuities,
        "consecutive_pairs": consecutive_pairs
    })

# Apply function by country
panel_stats = df.groupby("country").apply(count_panel_stats)

# Display results
print(panel_stats)
```

```
                n_obs  discontinuities  consecutive_pairs
country
Australia         139                0                138
Canada            139                0                138
Denmark           139                0                138
Finland           139                0                138
France            139                0                138
Germany           139                0                138
Italy             139                0                138
Japan             139                0                138
Netherlands       139                0                138
Norway            139                0                138
Portugal          139                0                138
Spain             139                0                138
Sweden            139                0                138
Switzerland       139                0                138
United Kingdom    139                0                138
United States     139                0                138
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_62663/3757882027.py:33: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  panel_stats = df.groupby("country").apply(count_panel_stats)
```

Time Series Plot of the Explanatory Variable Part 1 Question 2

In [135…
```python
import pandas as pd
from pathlib import Path
import os
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde, skew, kurtosis, norm
from scipy.stats import pearsonr
import statsmodels.api as sm
import seaborn as sns
from statsmodels.nonparametric.smoothers_lowess import lowess
from difflib import get_close_matches
```

In [138…
```python
def find_column(df, name_candidates):
    cols = list(df.columns)

    for cand in name_candidates:
        if cand in cols:
            return cand

    lower_map = {c.lower(): c for c in cols}
    for cand in name_candidates:
        if cand.lower() in lower_map:
            return lower_map[cand.lower()]

    for cand in name_candidates:
        m = get_close_matches(cand, cols, n=1, cutoff=0.75)
        if m:
            return m[0]
    return None

def ensure_sorted_safe(df, id_col_hint='country', time_col_hint='year'):
    id_found = find_column(df, [id_col_hint, 'country', 'id', 'iso', 'country_name'])
    time_found = find_column(df, [time_col_hint, 'year', 'time', 'date', 'period'])
    if id_found is None or time_found is None:
        raise KeyError(f"Column not found. id_hint={id_col_hint} -> {id_found}, time_hint={time_col_hint} -> {time_found}. Available columns: {list(df.columns)}")
    # make it a numeric
    if pd.api.types.is_string_dtype(df[time_found]):
        sample = df[time_found].dropna().astype(str).head(10).tolist()
        if all(s.isdigit() and len(s) == 4 for s in sample):
            df[time_found] = pd.to_numeric(df[time_found], errors='coerce')
    sorted_df = df.sort_values([id_found, time_found], na_position='last').reset_index(drop=True)
    print(f"Sorted by detected columns: id='{id_found}', time='{time_found}'")
    return sorted_df

# Run diagnostic when cell executed
print('--- ensure_sorted_safe diagnostic ---')
if 'df' not in globals():
    print("No DataFrame named 'df' found in this kernel. Define 'df' and re-run this cell.")
else:
    print('DataFrame df found. Columns:')
    print(list(df.columns))
    print(Column dtypes:)
    print(df.dtypes)
    # pick hints from existing variables if present
    id_hint = globals().get('id_col', 'country')
    time_hint = globals().get('time_col', 'year')
    print(f'Using id hint: {id_hint}, time hint: {time_hint}')
    try:
        df0 = ensure_sorted_safe(df.copy(), id_hint, time_hint)
        print('
Sorted sample (first 8 rows):')
        display(df0.head(8))
    except Exception as e:
        print('
ERROR while sorting:')
        print(e)
        print('
Sample rows to help debugging:')
        display(df.head(10))
```

```
  Cell In[138], line 48
    print('
          ^
SyntaxError: unterminated string literal (detected at line 48)
```

Question 1 --> Within Transforms, Pooled etc

In [139…
```python
# Question 1 --> create the between-transformed and one-way within-transformed versions of each variable in the dataset.
# getting teh path to my data set

final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final = pd.read_excel(final_data_path / "final_panel_dataset.xlsx")

# In the study as one of teh explanatory varibles they create a 5 year lagged GDP per capita variable.
# Therefore, prior to plotting the distributions, we create this variable in the final data set.

df_Final = df_Final.sort_values(["country", "year"])
df_Final = df_Final[['country', 'year', 'gdppc', 'labour_productivity', 'cpi', 'trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']]
df_Final = df_Final.sort_values(["country", "year"])

# Logging GDP per capita first
df_Final['ln_gdppc'] = np.log(df_Final['gdppc'])
df_Final["ln_gdppc_lag5"] = df_Final.groupby("country")["ln_gdppc"].shift(5)
# Creating the 5 year lagged GDP per capita growth variable
df_Final["ln_gdp_growth_5y"] = df_Final["ln_gdppc"] - df_Final["ln_gdppc_lag5"]

# Doing of rthe alternative dependent variable: labour productivity growth
df_Final['ln_labour_productivity'] = np.log(df_Final['labour_productivity'])
df_Final["ln_labour_productivity_lag5"] = df_Final.groupby("country")["ln_labour_productivity"].shift(5)
df_Final["ln_labour_productivity_growth_5y"] = df_Final["ln_labour_productivity"] - df_Final["ln_labour_productivity_lag5"]

# taking teh logarithm of the other variables as well
df_Final['ln_cpi'] = np.log(df_Final['cpi'])
df_Final['ln_trade_openness'] = np.log(df_Final['trade_openness'])


variables = [ 'ln_gdppc','ln_labour_productivity', 'ln_cpi', 'ln_trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp', "ln_gdp_growth_5y", "ln_labour_productivity_growth_5y" ]

for var in variables:   # Between transformation
    df_Final[f'{var}_between'] = df_Final.groupby('country')[var].transform('mean')

    # One-way Within transformation
    df_Final[f'{var}_within'] = df_Final[var] - df_Final.groupby('country')[var].transform('mean')

rows = []

for var in variables:
    pooled_variance = df_Final[var].var(ddof=1)
    var_between = df_Final[f'{var}_between'].var(ddof=1)
    var_within = df_Final[f'{var}_within'].var(ddof=1)
    if pooled_variance != 0 and not np.isnan(pooled_variance):
        share_between = var_between / pooled_variance
        share_within = var_within / pooled_variance
    else:
        share_between = np.nan
        share_within = np.nan
    rows.append({"variable": var,"var_total_pooled": pooled_variance,"var_between": var_between,"var_within": var_within,"share_between": share_between,"share_within": share_within})

# vardecomp tabel
var_decomp = pd.DataFrame(rows).set_index("variable")


pd.set_option("display.float_format", "{:,.4f}".format)
print("Variance decomposition")
print(var_decomp)

df_Final.describe()
```

```
Variance decomposition
                                  var_total_pooled  var_between  var_within  \
variable
ln_gdppc                                    0.8785       0.0980      0.7805
ln_labour_productivity                      1.1241       0.1079      1.0162
ln_cpi                                     18.9989       3.4977     15.5012
ln_trade_openness                          20.0214      19.6485      0.3729
liquid_liabilities                          0.0664       0.0125      0.0539
mcap_gdp                                    0.1390       0.0447      0.0944
bank_loans_gdp                              0.1293       0.0602      0.0691
ln_gdp_growth_5y                            0.0172       0.0002      0.0170
ln_labour_productivity_growth_5y            0.0167       0.0004      0.0163

                                  share_between  share_within
variable
ln_gdppc                                 0.1115        0.8885
ln_labour_productivity                   0.0960        0.9040
ln_cpi                                   0.1841        0.8159
ln_trade_openness                        0.9814        0.0186
liquid_liabilities                       0.1877        0.8123
mcap_gdp                                 0.3212        0.6789
bank_loans_gdp                           0.4654        0.5346
ln_gdp_growth_5y                         0.0140        0.9860
ln_labour_productivity_growth_5y         0.0221        0.9779
```

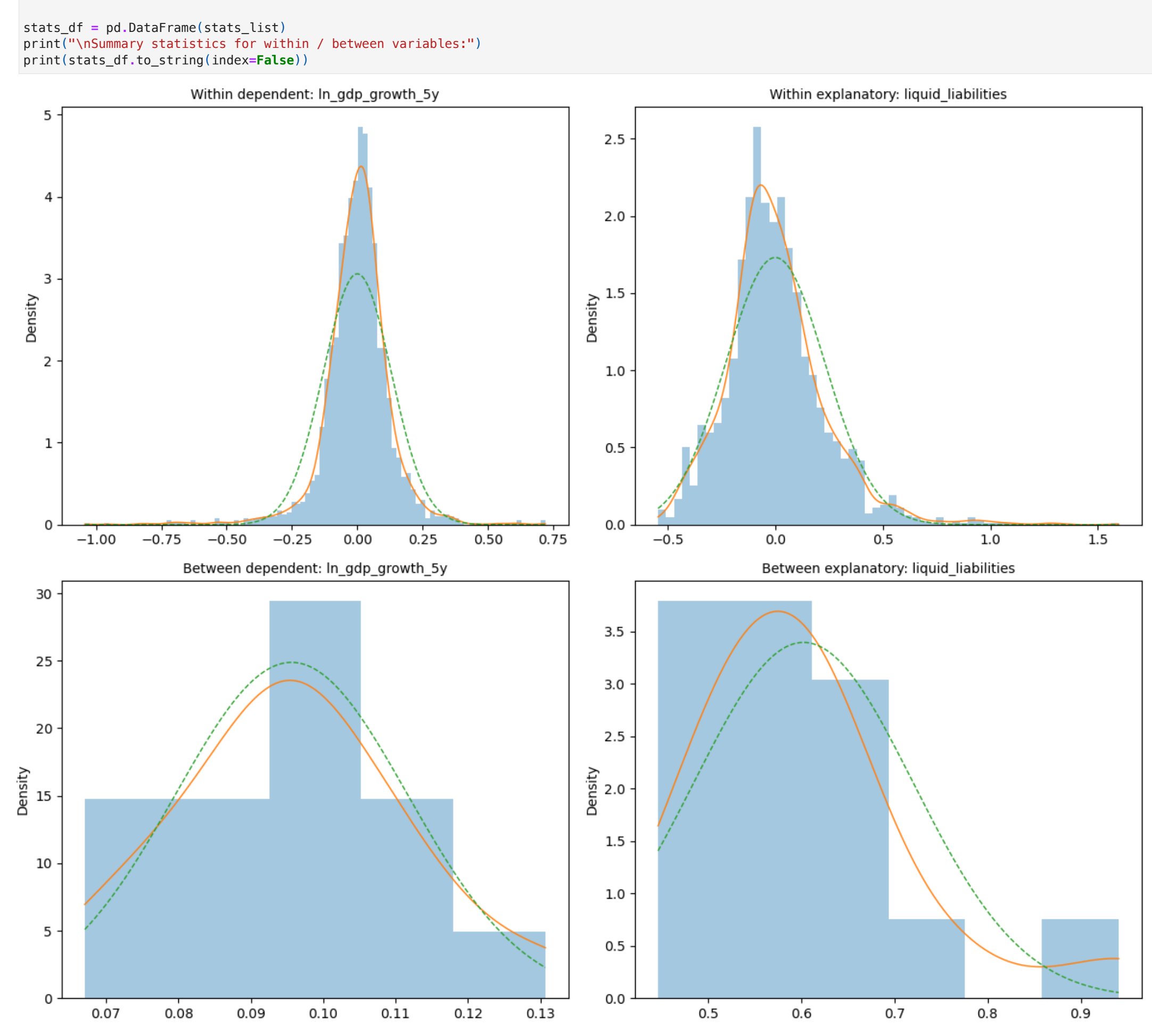
Out[139…

	year	gdppc	labour_productivity	cpi	trade_openness	liquid_liabilities	mcap_gdp	bank_loans_gdp	ln_gdppc	ln_gdppc_lag5	...	liquid_liabilities_between	liquid_liabilities_within	mcap_gdp_betw
count	2,224.0000	2,224.0000	2,096.0000	2,224.0000	2,224.0000	2,224.0000	1,488.0000	1,668.0000	2,224.0000	2,144.0000	...	2,224.0000	2,224.0000	1,668.0
mean	1,951.0000	16,107.7353	24.2423	44.4844	0.4052	0.5938	0.4301	0.6261	9.2763	9.2239	...	0.5938	-0.0000	0.4
std	40.1338	14,479.9899	21.7278	57.8501	0.6467	0.2577	0.3729	0.3595	0.9373	0.9123	...	0.1116	0.2322	0.2
min	1,882.0000	1,581.0000	0.9705	0.0000	0.0000	0.0579	0.0043	0.0507	7.3658	7.3658	...	0.4385	-0.5263	0.1
25%	1,916.0000	4,997.2500	6.2426	2.6660	0.0005	0.4316	0.1678	0.3417	8.5166	8.4918	...	0.5121	-0.1393	0.2
50%	1,951.0000	9,841.0760	14.5815	11.9223	0.1717	0.5484	0.3218	0.5532	9.1943	9.1301	...	0.5888	-0.0253	0.3
75%	1,986.0000	24,807.5099	40.4610	85.5400	0.4826	0.7076	0.5780	0.8386	10.1189	10.0575	...	0.6261	0.1109	0.6
max	2,020.0000	85,115.3231	97.5205	226.8252	2.9965	2.2599	2.7971	2.0357	11.3518	11.3231	...	0.9261	1.5988	0.7

8 rows × 34 columns

Question 2 --> Distributions Plots, etc --> Model is:

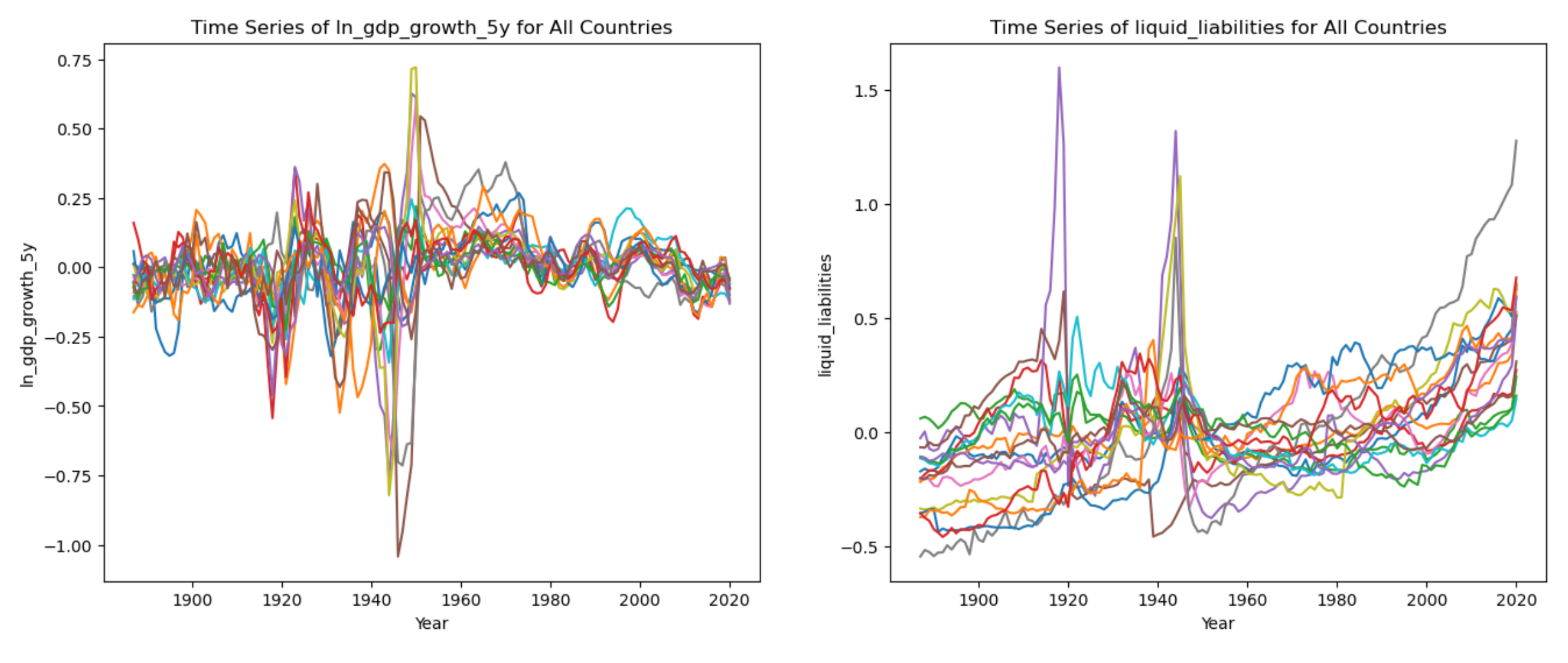
In [140…
```python
# Defining my Dependent and Explanatory Variables
Dependent_Variables_Growth = ['ln_gdp_growth_5y','ln_labour_productivity_growth_5y' ]
Explanatory_Variables = ['ln_gdppc_lag5', 'ln_labour_productivity_lag5', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']
Control_Variables = ['ln_cpi', 'ln_trade_openness']

# defining my individuals, time, dependent and explanatory variables

individuals = df_Final['country']
time = df_Final['year']
dep_var_col = 'ln_gdp_growth_5y'
exp_var_col = 'liquid_liabilities'
cols_to_keep = ["country", "year", dep_var_col, exp_var_col, "ln_cpi", "ln_trade_openness"]
df_Final_defined = df_Final[cols_to_keep].dropna()

df_Final_defined["dep_var_col_mean_i"] = df_Final_defined.groupby(individuals)[dep_var_col].transform("mean")
df_Final_defined["exp_var_col_mean_i"] = df_Final_defined.groupby(individuals)[exp_var_col].transform("mean")

df_Final_defined["dep_var_col_within"] = df_Final_defined[dep_var_col] - df_Final_defined['dep_var_col_mean_i']
df_Final_defined["exp_var_col_within"] = df_Final_defined[exp_var_col] - df_Final_defined['exp_var_col_mean_i']

between_df_Final_defined = (df_Final_defined.groupby("country").agg(dep_var_col_between=(dep_var_col, 'mean'),exp_var_col_between=(exp_var_col, 'mean')).reset_index())

def plot_dist_subplot(ax, values, title, bins='auto'):
    values = pd.to_numeric(values, errors='coerce')
    values = values[np.isfinite(values)]
    if len(values) == 0:
        ax.set_title(f"No valid data for {title}", fontsize=10)
        return
    mu = values.mean()
    sigma = values.std(ddof=1)
    # Histogram
    ax.hist(values, bins=bins, density=True, alpha=0.4)
    # KDE
    kde = gaussian_kde(values)
    xmin, xmax = values.min(), values.max()
    x_grid = np.linspace(xmin, xmax, 500)
    ax.plot(x_grid, kde(x_grid), linewidth=1)
    # Normal pdf
    if sigma > 0:
        normal_pdf = (
            1 / (np.sqrt(2 * np.pi) * sigma)
            * np.exp(-0.5 * ((x_grid - mu) / sigma) ** 2)
        )
        ax.plot(x_grid, normal_pdf, linestyle="--", linewidth=1)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel("")
    ax.set_ylabel("Density")
fig, axs = plt.subplots(2, 2, figsize=(12, 10))
plot_dist_subplot(axs[0, 0],df_Final_defined["dep_var_col_within"],f"Within dependent: {dep_var_col}")
plot_dist_subplot(axs[0, 1],df_Final_defined["exp_var_col_within"],f"Within explanatory: {exp_var_col}")
plot_dist_subplot(axs[1, 0],between_df_Final_defined["dep_var_col_between"],f"Between dependent: {dep_var_col}")
plot_dist_subplot(axs[1, 1],between_df_Final_defined["exp_var_col_between"],f"Between explanatory: {exp_var_col}")
plt.tight_layout()
plt.show()

def compute_stats(values, name):
    values = pd.to_numeric(values, errors='coerce')
    values = values[np.isfinite(values)]
    mu = values.mean()
    sd = values.std(ddof=1)
    sk = skew(values)
    kt = kurtosis(values, fisher=False)

    # cool high leverage figure calc make odds ones stand out
    if sd > 0:
        z = (values - mu) / sd
        high_lev = np.sum(np.abs(z) > 3)
    else:
        high_lev = 0

    return {
        "variable": name,
        "n": len(values),
        "mean": mu,
        "sd": sd,
        "skewness": sk,
        "kurtosis": kt,
        "high_leverage(|z|>3)": high_lev,
    }

stats_list = []
stats_list.append(
    compute_stats(df_Final_defined["dep_var_col_within"],
                  f"{dep_var_col}_within")
)
stats_list.append(
    compute_stats(df_Final_defined["exp_var_col_within"],
                  f"{exp_var_col}_within")
)
stats_list.append(
    compute_stats(between_df_Final_defined["dep_var_col_between"],
                  f"{dep_var_col}_between")
)
stats_list.append(
    compute_stats(between_df_Final_defined["exp_var_col_between"],
                  f"{exp_var_col}_between")
)

stats_df = pd.DataFrame(stats_list)
print("\nSummary statistics for within / between variables:")
print(stats_df.to_string(index=False))
```



```
Summary statistics for within / between variables:
               variable    n   mean     sd  skewness  kurtosis  high_leverage(|z|>3)
  ln_gdp_growth_5y_within 2144 0.0000 0.1304   -1.1084   12.7469                    37
liquid_liabilities_within 2144 0.0000 0.2305    1.1249    6.8723                    23
 ln_gdp_growth_5y_between   16 0.0956 0.0160    0.2492    2.9390                     0
liquid_liabilities_between   16 0.6026 0.1175    1.4965    5.4128                     0
```

In [141…
```python
df_timeseries_plot = df_Final_defined.sort_values(["country", "year"])
countries = sorted(df_timeseries_plot["country"].unique())
fig, axes = plt.subplots(1, 2, figsize=(16, 6), sharex=False)
for c in countries:
    df_c = df_timeseries_plot[df_timeseries_plot["country"] == c]
    axes[0].plot(df_c["year"], df_c['dep_var_col_within'], label=c)
axes[0].set_title(f"Time Series of {dep_var_col} for All Countries")
axes[0].set_xlabel("Year")
axes[0].set_ylabel(dep_var_col)
for c in countries:
    df_c = df_timeseries_plot[df_timeseries_plot["country"] == c]
    axes[1].plot(df_c["year"], df_c['exp_var_col_within'], label=c)

axes[1].set_title(f"Time Series of {exp_var_col} for All Countries")
axes[1].set_xlabel("Year")
axes[1].set_ylabel(exp_var_col)
```

Out[141…
```
Text(0, 0.5, 'liquid_liabilities')
```



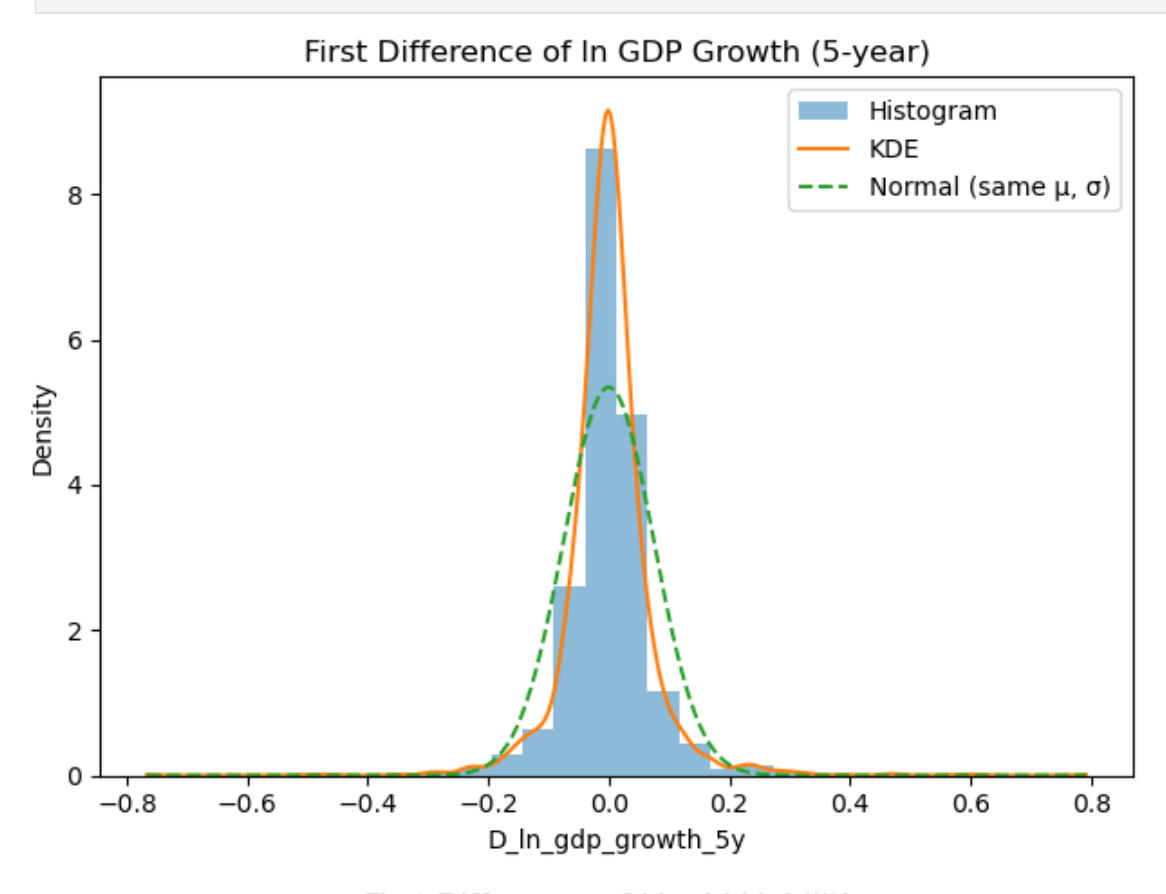
Question 3 --> 1st difference,

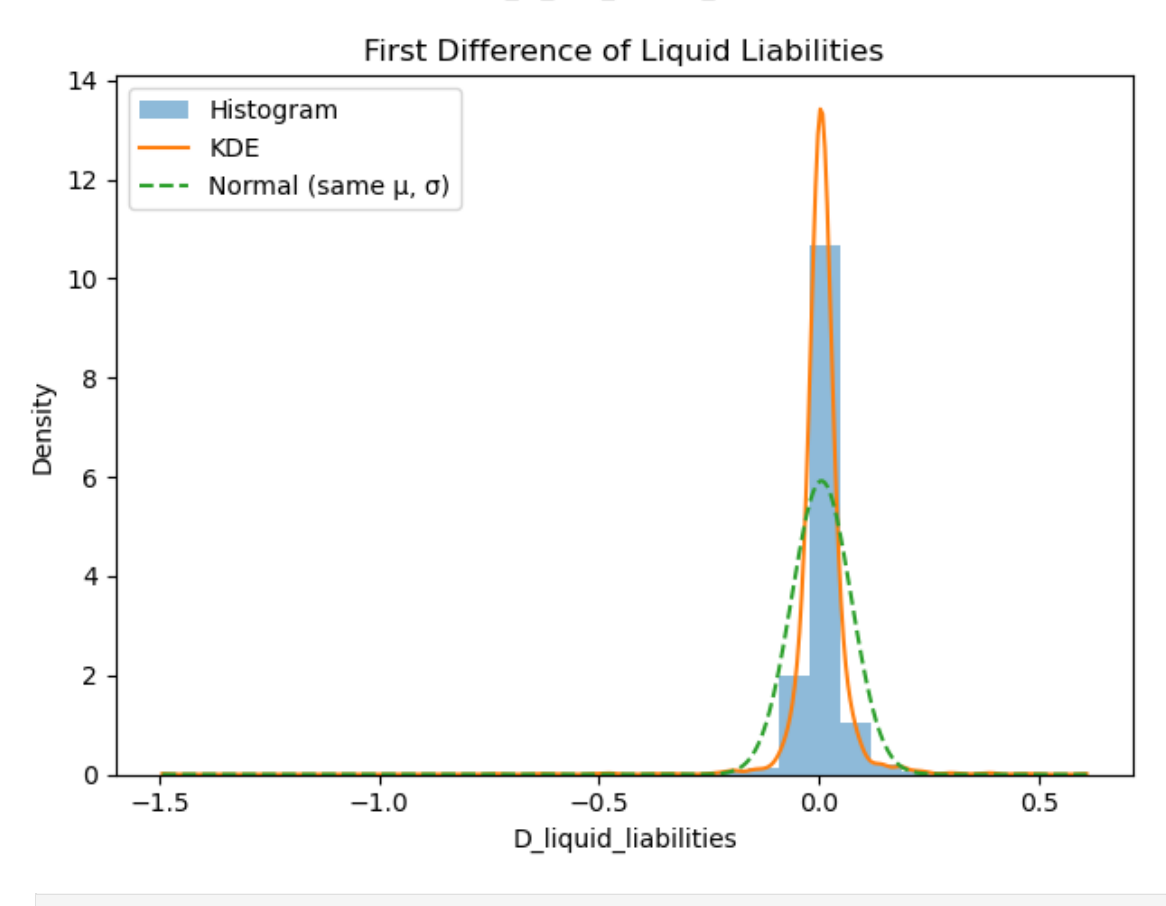
In [142…
```python
# Create first differences for the selected columns.
# Define which columns we want first differences for (use the dependent and the explanatory of interest)
x_col = ["ln_gdp_growth_5y", "liquid_liabilities"]

# Build df_Final_first_diff by grouping by country and taking first differences
df_Final_first_diff = df_Final.sort_values(["country", "year"]).copy()
for col in x_col:
    df_Final_first_diff[f"D_{col}"] = df_Final_first_diff.groupby("country")[col].diff()

# just tryna keep the first diff colums
first_diff = [f"D_{c}" for c in x_col]
fd_cols = first_diff

def check_first_diffs_boundaries(df, id_col, time_col, fd_cols):
    df = df.sort_values([id_col, time_col]).copy()
    first_id = df[id_col].iloc[0]
    T = (df[id_col] == first_id).sum()
    n = min(3 * T + 1, len(df))
    sub = df.iloc[:n].copy()
    changes = []
    ids = sub[id_col].to_numpy()
    for i in range(1, len(sub)):
        if ids[i] != ids[i - 1]:
            changes.append(i)
            if len(changes) == 3:
                break


    for idx in changes:
        row = sub.iloc[idx]
        print("\n change at stacked row", idx)
        print(row[[id_col, time_col]])


        for col in fd_cols:
            val = row.get(col, np.nan)
            if not pd.isna(val):

                print("\nProblem in column:", col)
                print(sub.iloc[max(idx-2, 0): idx+3][[id_col, time_col] + fd_cols])
                raise AssertionError(
                    f"First difference in column '{col}' at change of individual "
                    f"should be NaN, but is {val}."
                )


    print("\n✓ Check passed: at the first 3 changes of individuals,")
    print("   all specified first-difference columns are NaN at the boundary.")
    return True

check_first_diffs_boundaries(
    df_Final_first_diff,
    id_col="country",
    time_col="year",
    fd_cols=fd_cols
)
```

```
 change at stacked row 139
country    Canada
year         1882
Name: 139, dtype: object

 change at stacked row 278
country    Denmark
year          1882
Name: 278, dtype: object

 change at stacked row 417
country    Finland
year          1882
Name: 417, dtype: object

✓ Check passed: at the first 3 changes of individuals,
   all specified first-difference columns are NaN at the boundary.
```

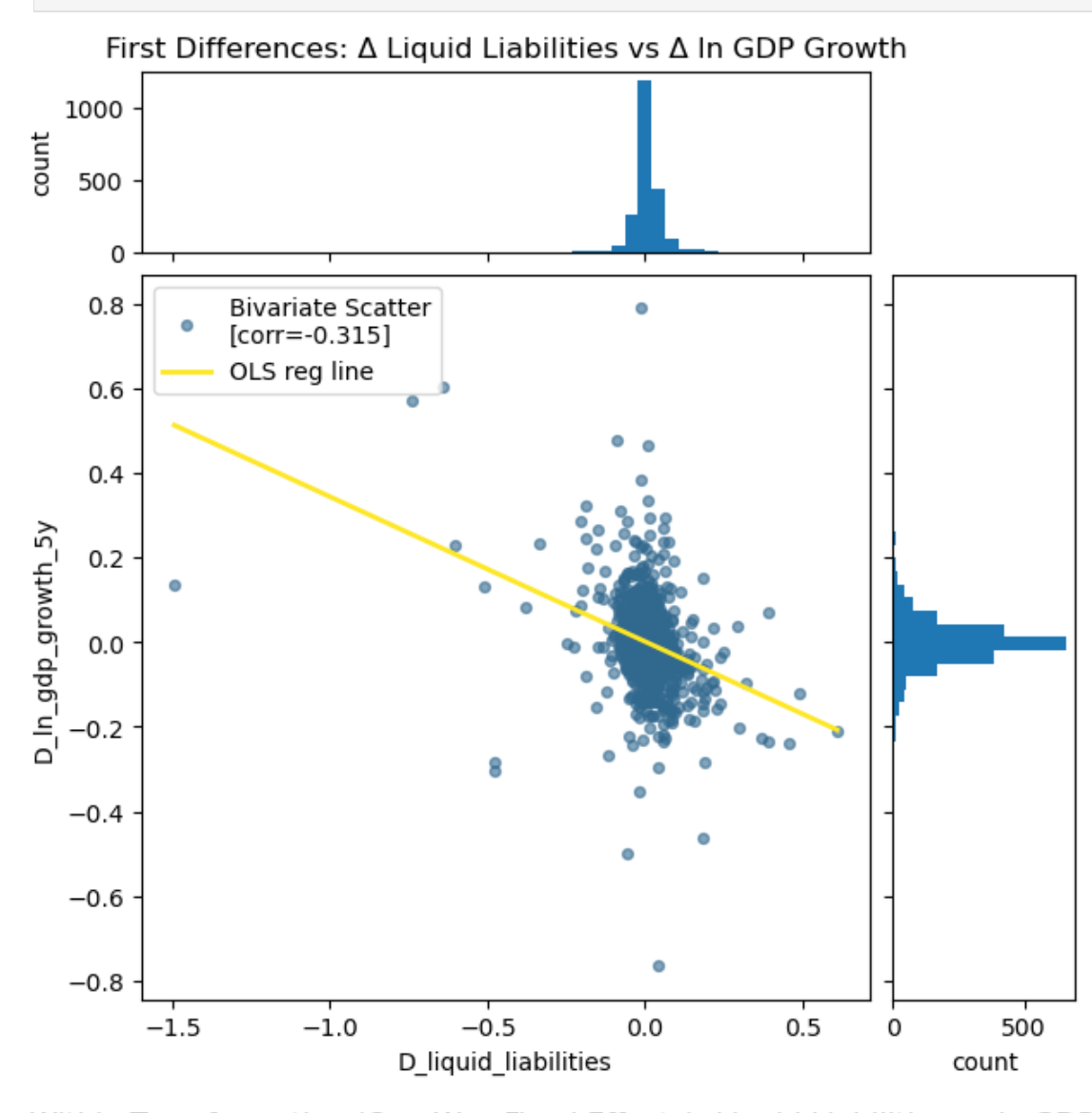
Out[142…
```
True
```

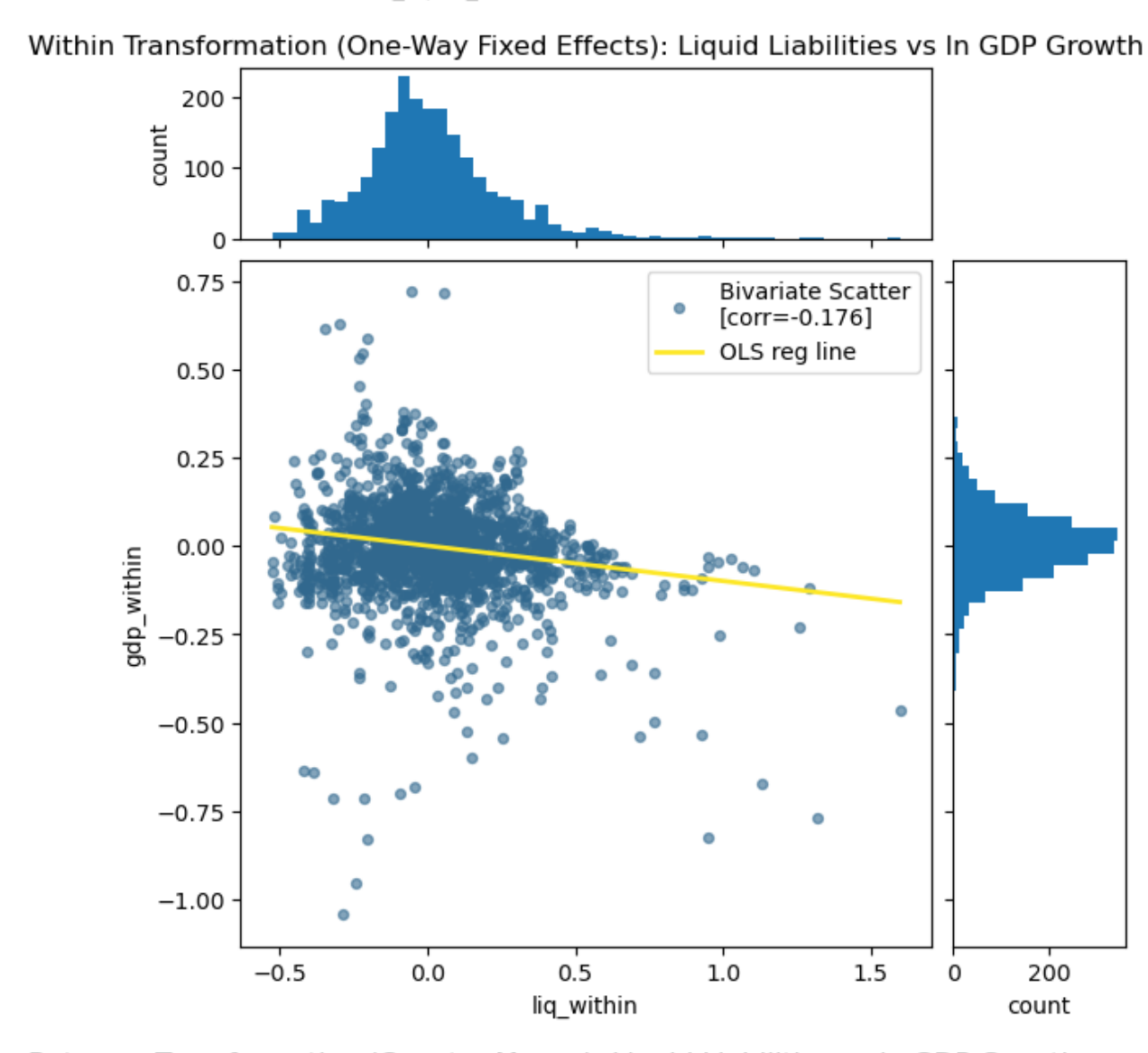
In [143…
```python
def plot_fd_distribution(df, col, title=None, bins=30):
    data = df[col].dropna().to_numpy()
    if data.size == 0:
        print(f"No non-missing data in column {col}.")
        return
    mu = data.mean()
    sigma = data.std(ddof=1)
    x_min, x_max = data.min(), data.max()
    x_vals = np.linspace(x_min, x_max, 400)
    kde = gaussian_kde(data)
    kde_vals = kde(x_vals)
    normal_vals = norm.pdf(x_vals, loc=mu, scale=sigma)
    plt.figure()
    plt.hist(data, bins=bins, density=True, alpha=0.5, label="Histogram")
    plt.plot(x_vals, kde_vals, label="KDE")
    plt.plot(x_vals, normal_vals, linestyle="--", label="Normal (same μ, σ)")
    plt.xlabel(col)
    if title is not None:
        plt.title(title)
    else:
        plt.title(f"Distribution of {col}")
    plt.ylabel("Density")
    plt.legend()
    plt.tight_layout()
    plt.show()

fd_gdp_col = "D_ln_gdp_growth_5y"
fd_liq_col = "D_liquid_liabilities"
plot_fd_distribution(df_Final_first_diff,col=fd_gdp_col,title="First Difference of ln GDP Growth (5-year)")
plot_fd_distribution(df_Final_first_diff,col=fd_liq_col,title="First Difference of Liquid Liabilities")
```

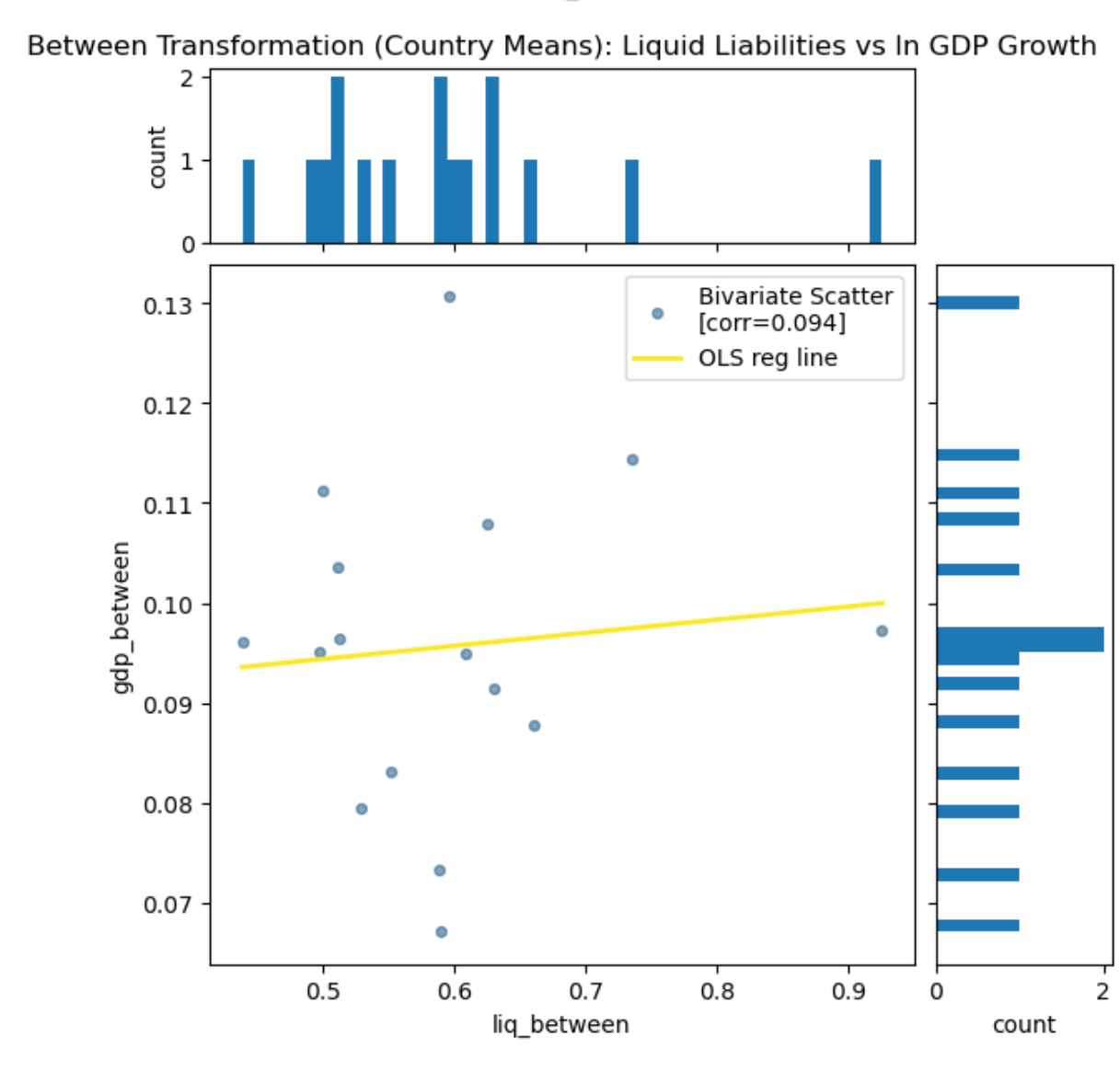




In [144…
```python
def scatter_with_marginals_cloud(df, x, y, title=None, bins=50):
    data = df[[x, y]].dropna()
    X = data[x].astype(float)
    Y = data[y].astype(float)
    corr = pearsonr(X, Y)[0]
    slope, intercept = np.polyfit(X, Y, 1)
    xx = np.linspace(X.min(), X.max(), 200)
    yy = slope * xx + intercept
    fig = plt.figure(figsize=(7, 7))
    gs = fig.add_gridspec(
        2, 2,
        height_ratios=(1, 4),
        width_ratios=(4, 1),
        hspace=0.05,
        wspace=0.05
    )
    ax_top   = fig.add_subplot(gs[0, 0])
    ax_main  = fig.add_subplot(gs[1, 0], sharex=ax_top)
    ax_right = fig.add_subplot(gs[1, 1], sharey=ax_main)
    ax_main.scatter(
        X, Y,
        s=18, alpha=0.6,
        color="#31688E",
        label=f"Bivariate Scatter\n[corr={corr:.3f}]"
    )
    ax_main.plot(xx, yy, linewidth=2, color="#FDE725", label="OLS reg line")
    ax_main.set_xlabel(x)
    ax_main.set_ylabel(y)
    ax_main.legend()
    ax_top.hist(X, bins=bins)
    ax_top.tick_params(axis="x", labelbottom=False)
    ax_top.set_ylabel("count")
    ax_right.hist(Y, bins=bins, orientation="horizontal")
    ax_right.tick_params(axis="y", labelleft=False)
    ax_right.set_xlabel("count")
    if title is None:
        title = f"{x} vs {y} (corr={corr:.3f})"
    ax_top.set_title(title)

    plt.show()

x_fd = "D_liquid_liabilities"
y_fd = "D_ln_gdp_growth_5y"

scatter_with_marginals_cloud( df_Final_first_diff,x=x_fd,y=y_fd,title="First Differences: Δ Liquid Liabilities vs Δ ln GDP Growth")


df_within = df_Final_first_diff.copy()
df_within["liq_within"] = df_within.groupby("country")["liquid_liabilities"] \
.transform(lambda v: v - v.mean())
df_within["gdp_within"] = df_within.groupby("country")["ln_gdp_growth_5y"] \
  .transform(lambda v: v - v.mean())

scatter_with_marginals_cloud(df_within,x="liq_within",y="gdp_within",title="Within Transformation (One-Way Fixed Effects): Liquid Liabilities vs ln GDP Growth")
df_between = df_Final_first_diff.groupby("country").agg({"liquid_liabilities": "mean","ln_gdp_growth_5y": "mean"}).reset_index()
df_between.rename(columns={ "liquid_liabilities": "liq_between","ln_gdp_growth_5y": "gdp_between"}, inplace=True)
scatter_with_marginals_cloud(df_between,x="liq_between",y="gdp_between",title="Between Transformation (Country Means): Liquid Liabilities vs ln GDP Growth")
```

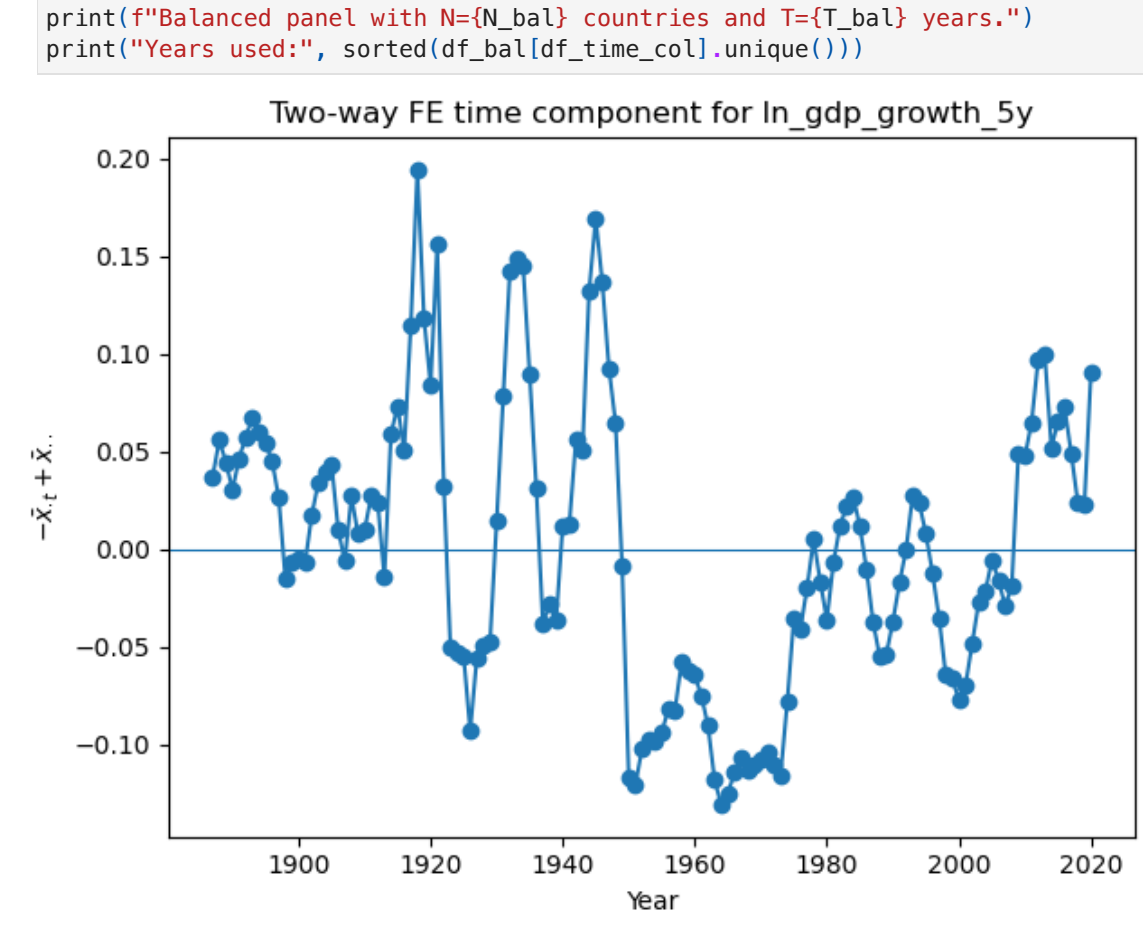






Question 9 --> TWFE

In [145…
```python
df_TWFE_Q = df_Final.copy()
df_id_col = "country"
df_time_col = "year"
T_per_country = df_TWFE_Q.groupby(df_id_col)[df_time_col].nunique()
T_max = T_per_country.max()
countries_longest = T_per_country[T_per_country == T_max].index
df_long = df_TWFE_Q[df_TWFE_Q[df_id_col].isin(countries_longest)].copy()
N = len(countries_longest)
year_counts = df_long.groupby(df_time_col)[df_id_col].nunique()
years_balanced = year_counts[year_counts == N].index
df_bal = df_long[df_long[df_time_col].isin(years_balanced)].copy()
df_bal = df_bal.sort_values([df_id_col, df_time_col])
N_bal = df_bal[df_id_col].nunique()
T_bal = df_bal[df_time_col].nunique()
x_var = "ln_gdp_growth_5y"
df_bal = df_bal.copy()
df_bal["x"] = df_bal[x_var]
x_overall = df_bal["x"].mean()
x_time_means = df_bal.groupby(df_time_col)["x"].mean()
twfe_time_component = x_overall - x_time_means
twfe_time_df = twfe_time_component.reset_index()
twfe_time_df.columns = [df_time_col, "-x(.t)+x(..)"]
plt.figure()
plt.plot(twfe_time_df[df_time_col], twfe_time_df["-x(.t)+x(..)"], marker="o")
plt.axhline(0, linewidth=0.8)
plt.xlabel("Year")
plt.ylabel(r"$- \bar{x}_{\cdot t} + \bar{x}_{\cdot \cdot}$")
plt.title(f"Two-way FE time component for {x_var}")
plt.tight_layout()
plt.show()
print(f"Balanced panel with N={N_bal} countries and T={T_bal} years.")
print("Years used:", sorted(df_bal[df_time_col].unique()))
```



```
Balanced panel with N=16 countries and T=139 years.
Years used: [np.int64(1882), np.int64(1883), np.int64(1884), np.int64(1885), np.int64(1886), np.int64(1887), np.int64(1888), np.int64(1889), np.int64(1890), np.int64(1891), np.int64(1892), np.int64(1
893), np.int64(1894), np.int64(1895), np.int64(1896), np.int64(1897), np.int64(1898), np.int64(1899), np.int64(1900), np.int64(1901), np.int64(1902), np.int64(1903), np.int64(1904), np.int64(1905), n
p.int64(1906), np.int64(1907), np.int64(1908), np.int64(1909), np.int64(1910), np.int64(1911), np.int64(1912), np.int64(1913), np.int64(1914), np.int64(1915), np.int64(1916), np.int64(1917), np.int6
4(1918), np.int64(1919), np.int64(1920), np.int64(1921), np.int64(1922), np.int64(1923), np.int64(1924), np.int64(1925), np.int64(1926), np.int64(1927), np.int64(1928), np.int64(1929), np.int64(193
0), np.int64(1931), np.int64(1932), np.int64(1933), np.int64(1934), np.int64(1935), np.int64(1936), np.int64(1937), np.int64(1938), np.int64(1939), np.int64(1940), np.int64(1941), np.int64(1942), np.
int64(1943), np.int64(1944), np.int64(1945), np.int64(1946), np.int64(1947), np.int64(1948), np.int64(1949), np.int64(1950), np.int64(1951), np.int64(1952), np.int64(1953), np.int64(1954), np.int64(1
955), np.int64(1956), np.int64(1957), np.int64(1958), np.int64(1959), np.int64(1960), np.int64(1961), np.int64(1962), np.int64(1963), np.int64(1964), np.int64(1965), np.int64(1966), np.int64(1967), n
p.int64(1968), np.int64(1969), np.int64(1970), np.int64(1971), np.int64(1972), np.int64(1973), np.int64(1974), np.int64(1975), np.int64(1976), np.int64(1977), np.int64(1978), np.int64(1979), np.int6
4(1980), np.int64(1981), np.int64(1982), np.int64(1983), np.int64(1984), np.int64(1985), np.int64(1986), np.int64(1987), np.int64(1988), np.int64(1989), np.int64(1990), np.int64(1991), np.int64(199
2), np.int64(1993), np.int64(1994), np.int64(1995), np.int64(1996), np.int64(1997), np.int64(1998), np.int64(1999), np.int64(2000), np.int64(2001), np.int64(2002), np.int64(2003), np.int64(2004), np.
int64(2005), np.int64(2006), np.int64(2007), np.int64(2008), np.int64(2009), np.int64(2010), np.int64(2011), np.int64(2012), np.int64(2013), np.int64(2014), np.int64(2015), np.int64(2016), np.int64(2
017), np.int64(2018), np.int64(2019), np.int64(2020)]
```

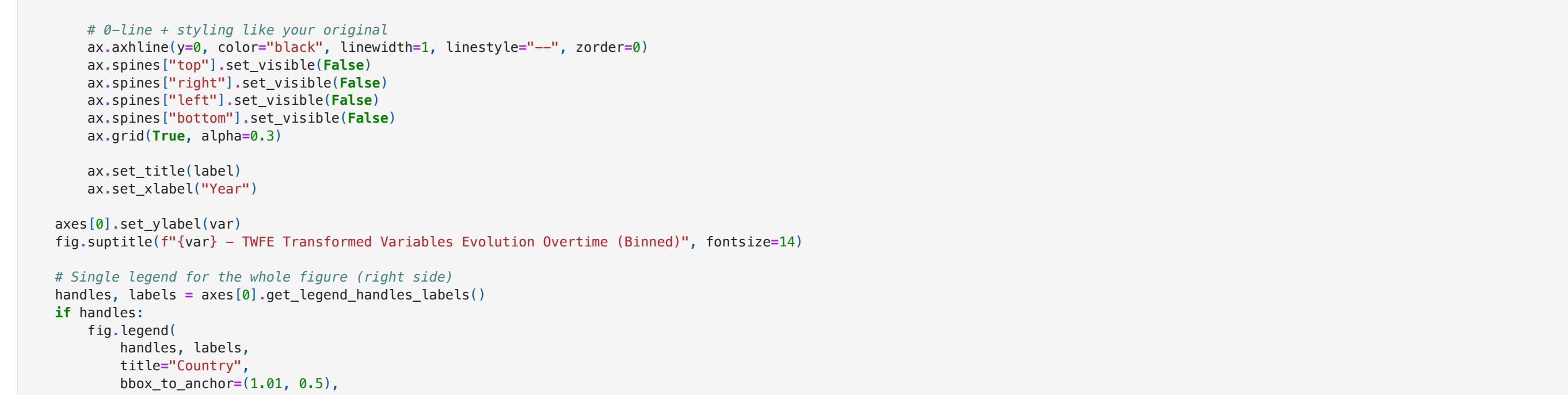
In [155…
```python
vars_for_twfe = ["ln_gdp_growth_5y", "liquid_liabilities", "ln_cpi", "ln_trade_openness", "mcap_gdp", "bank_loans_gdp", "ln_labour_productivity_growth_5y", 'ln_gdppc_lag5']

df_twfe = df_bal.copy()

for x_var in vars_for_twfe:
    x = df_twfe[x_var]
    x_i_dot = df_twfe.groupby(df_id_col)[x_var].transform("mean")      # x(i.)
    x_dot_t = df_twfe.groupby(df_time_col)[x_var].transform("mean")    # x(.t)
    x_dot_dot = df_twfe[x_var].mean()                                 # x(..)
    df_twfe[f"{x_var}_twfe"] = x - x_i_dot - x_dot_t + x_dot_dot
df_twfe[[df_id_col, df_time_col, "ln_gdp_growth_5y", "ln_gdp_growth_5y_twfe",
         "liquid_liabilities", "liquid_liabilities_twfe"]].head()
vars_for_twfe = ["ln_gdp_growth_5y", "liquid_liabilities"]
overall_means = df_bal[vars_for_twfe].mean()
time_means = df_bal.groupby(df_time_col)[vars_for_twfe].mean()
time_component = overall_means - time_means
table = time_component.reset_index()
table = table.rename(columns={df_time_col: "Time"})
new_cols = ["Time"] + [f"-{v}(.t)+{v}(..)" for v in vars_for_twfe]
table.columns = new_cols

print(table.to_string(index=False))
```

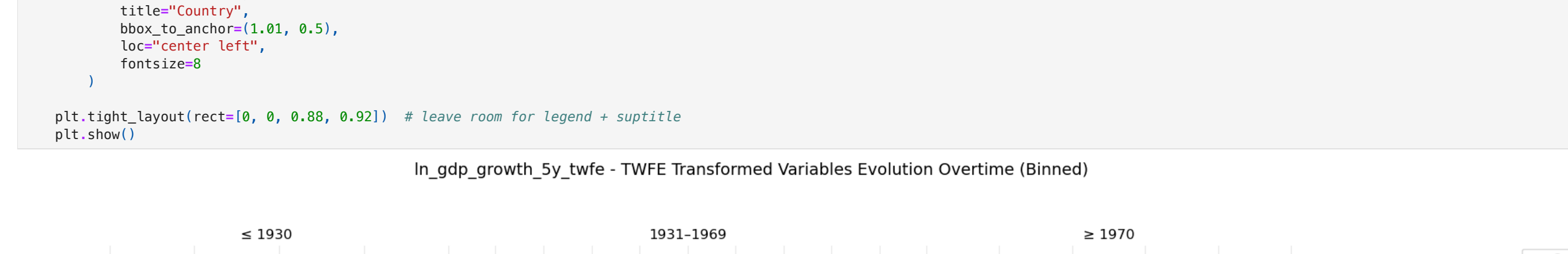
```
Time  -ln_gdp_growth_5y(.t)+ln_gdp_growth_5y(..)  -liquid_liabilities(.t)+liquid_liabilities(..)
1882                                         NaN                                           0.2593
1883                                         NaN                                           0.2499
1884                                         NaN                                           0.2346
1885                                         NaN                                           0.2225
1886                                         NaN                                           0.2125
1887                                      0.0372                                           0.1996
1888                                      0.0558                                           0.1947
```

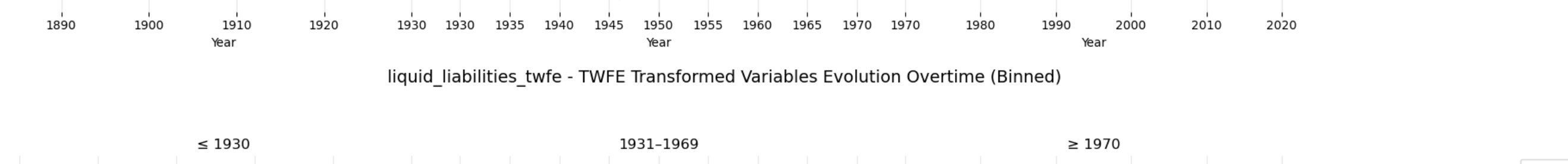
```
1889                              0.0446                             0.1992
1890                              0.0303                             0.2061
1891                              0.0464                             0.2124
1892                              0.0575                             0.1979
1893                              0.0670                             0.1837
1894                              0.0596                             0.1702
1895                              0.0542                             0.1588
1896                              0.0451                             0.1540
1897                              0.0263                             0.1496
1898                             -0.0147                             0.1525
1899                             -0.0071                             0.1408
1900                             -0.0050                             0.1409
1901                             -0.0068                             0.1266
1902                              0.0175                             0.1152
1903                              0.0341                             0.1167
1904                              0.0392                             0.0998
1905                              0.0435                             0.0864
1906                              0.0103                             0.0837
1907                             -0.0061                             0.0959
1908                              0.0272                             0.0651
1909                              0.0079                             0.0557
1910                              0.0099                             0.0517
1911                              0.0273                             0.0571
1912                              0.0236                             0.0682
1913                             -0.0143                             0.0711
1914                              0.0580                             0.0183
1915                              0.0733                             0.0040
1916                              0.0504                             0.0323
1917                              0.1147                            -0.0049
1918                              0.1941                            -0.0656
1919                              0.1181                            -0.0616
1920                              0.0840                             0.0968
1921                              0.1557                             0.0499
1922                              0.0321                             0.0394
1923                             -0.0504                             0.0609
1924                             -0.0530                             0.0899
1925                             -0.0545                             0.1025
1926                             -0.0928                             0.0850
1927                             -0.0554                             0.0662
1928                             -0.0492                             0.0660
1929                             -0.0470                             0.0638
1930                              0.0144                             0.0205
1931                              0.0786                            -0.0354
1932                              0.1422                            -0.0766
1933                              0.1490                            -0.0776
1934                              0.1453                            -0.0592
1935                              0.0895                            -0.0491
1936                              0.0315                            -0.0364
1937                             -0.0379                            -0.0155
1938                             -0.0283                            -0.0440
1939                             -0.0365                            -0.0338
1940                              0.0121                            -0.0429
1941                              0.0125                            -0.0726
1942                              0.0559                            -0.0946
1943                              0.0508                            -0.1500
1944                              0.1321                            -0.2553
1945                              0.1693                            -0.2183
1946                              0.1364                            -0.0824
1947                              0.0927                            -0.0121
1948                              0.0647                             0.0260
1949                             -0.0088                             0.0407
1950                             -0.1164                             0.0675
1951                             -0.1207                             0.1071
1952                             -0.1021                             0.1062
1953                             -0.0971                             0.0974
1954                             -0.0978                             0.0933
1955                             -0.0938                             0.0938
1956                             -0.0815                             0.1086
1957                             -0.0022                             0.1104
1958                             -0.0577                             0.0942
1959                             -0.0618                             0.0857
1960                             -0.0644                             0.0837
1961                             -0.0748                             0.0813
1962                             -0.0902                             0.0748
1963                             -0.1178                             0.0641
1964                             -0.1307                             0.0639
1965                             -0.1250                             0.0618
1966                             -0.1142                             0.0602
1967                             -0.1060                             0.0519
1968                             -0.1128                             0.0402
1969                             -0.1098                             0.0440
1970                             -0.1077                             0.0555
1971                             -0.1036                             0.0380
1972                             -0.1097                             0.0160
1973                             -0.1154                             0.0121
1974                             -0.0783                             0.0331
1975                             -0.0354                             0.0258
1976                             -0.0405                             0.0237
1977                             -0.0198                             0.0286
1978                              0.0051                             0.0037
1979                             -0.0163                            -0.0115
1980                             -0.0361                             0.0090
1981                             -0.0069                             0.0072
1982                              0.0122                            -0.0134
1983                              0.0219                            -0.0168
1984                              0.0270                            -0.0173
1985                              0.0117                            -0.0237
1986                             -0.0101                            -0.0292
1987                             -0.0372                            -0.0377
1988                             -0.0548                            -0.0425
1989                             -0.0539                            -0.0467
1990                             -0.0368                            -0.0483
1991                             -0.0172                            -0.0504
1992                             -0.0001                            -0.0549
1993                              0.0279                            -0.0674
1994                              0.0235                            -0.0580
1995                              0.0080                            -0.0587
1996                             -0.0121                            -0.0621
1997                             -0.0504                            -0.0584
1998                             -0.0643                            -0.0596
1999                             -0.0659                            -0.0704
2000                             -0.0773                            -0.0569
2001                             -0.0695                            -0.0762
2002                             -0.0483                            -0.0914
2003                             -0.0270                            -0.1107
2004                             -0.0216                            -0.1167
2005                             -0.0053                            -0.1378
2006                             -0.0157                            -0.1549
2007                             -0.0292                            -0.1864
2008                             -0.0189                            -0.2170
2009                              0.0486                            -0.2579
2010                              0.0476                            -0.2527
2011                              0.0649                            -0.2581
2012                              0.0970                            -0.2832
2013                              0.0996                            -0.2941
2014                              0.0520                            -0.3054
2015                              0.0653                            -0.3298
2016                              0.0725                            -0.3545
2017                              0.0491                            -0.3639
2018                              0.0239                            -0.3644
2019                              0.0333                            -0.3782
2020                              0.0909                            -0.5318
```

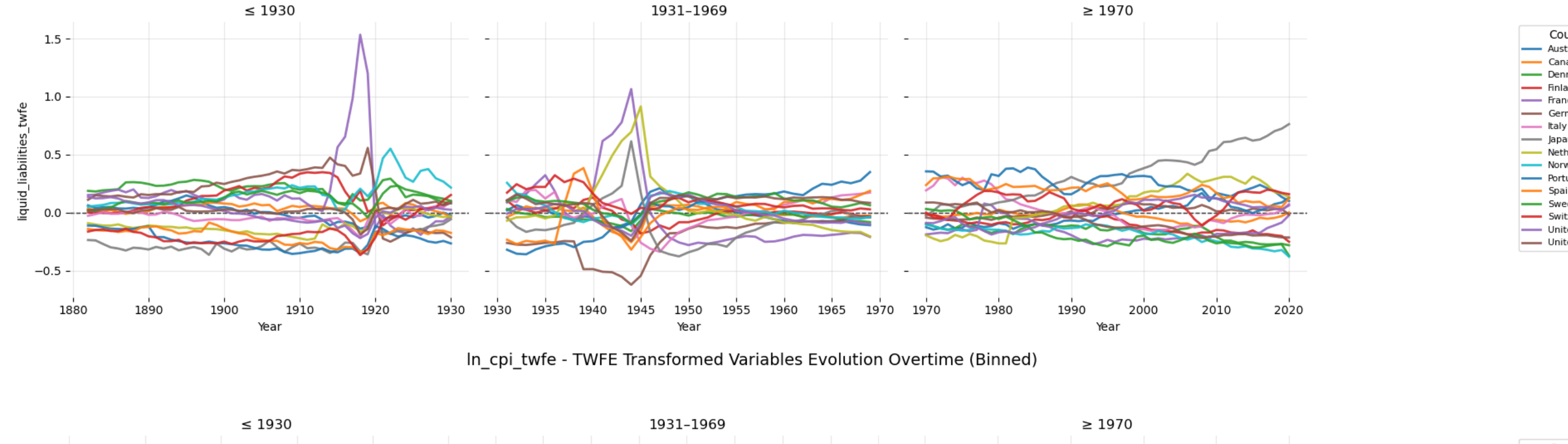
In [156…

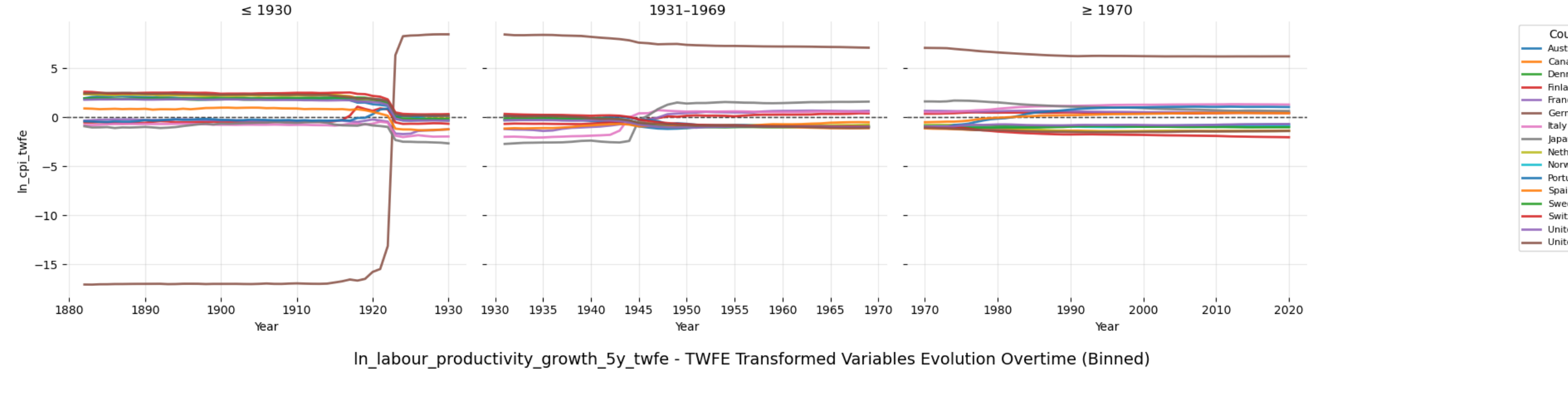
```python
id_col = "country"
time_col = "year"

# ALL countries (or replace with your subset list)
target_countries = sorted(df_twfe[id_col].unique())

vars_16 = [
    "ln_gdp_growth_5y",
    "liquid_liabilities",
    "ln_cpi",
    "ln_labour_productivity_growth_5y",
    "ln_gdppc_lag5",
    "bank_loans_gdp",
    "mcap_gdp",
    "ln_trade_openness"
]
target_variables = [v + "_twfe" for v in vars_16]

# Optional: custom colors (leave empty to let matplotlib choose automatically)
country_colors = {}  # e.g. {"Italy": "#576b6b", ...}

# Sort once
df_twfe_plot = df_twfe.sort_values([id_col, time_col]).copy()

# Time bins (label, min_year, max_year)
time_bins = [
    ("≤ 1930", None, 1930),
    ("1931–1969", 1931, 1969),
    ("≥ 1970", 1970, None),
]

def slice_by_period(d, min_year, max_year):
    if min_year is not None:
        d = d[d[time_col] >= min_year]
    if max_year is not None:
        d = d[d[time_col] <= max_year]
    return d

for var in target_variables:
    fig, axes = plt.subplots(1, 3, figsize=(18, 5), sharey=True)

    for ax, (label, min_y, max_y) in zip(axes, time_bins):
        for country in target_countries:
            d = df_twfe_plot[df_twfe_plot[id_col] == country][[time_col, var]].dropna()
            d = slice_by_period(d, min_y, max_y)

            if d.empty:
                continue

            ax.plot(
                d[time_col],
                d[var],
                linewidth=2,
                alpha=0.9,
                label=country,
                color=country_colors.get(country, None)  # None => default cycle
            )

        # 0-line + styling like your original
        ax.axhline(y=0, color="black", linewidth=1, linestyle="--", zorder=0)
        ax.spines["top"].set_visible(False)
        ax.spines["right"].set_visible(False)
        ax.spines["left"].set_visible(False)
        ax.spines["bottom"].set_visible(False)
        ax.grid(True, alpha=0.3)

        ax.set_title(label)
        ax.set_xlabel("Year")

    axes[0].set_ylabel(var)
    fig.suptitle(f"{var} - TWFE Transformed Variables Evolution Overtime (Binned)", fontsize=14)

    # Single legend for the whole figure (right side)
    handles, labels = axes[0].get_legend_handles_labels()
    if handles:
        fig.legend(
            handles, labels,
            title="Country",
            bbox_to_anchor=(1.01, 0.5),
            loc="center left",
            fontsize=8
        )

    plt.tight_layout(rect=[0, 0, 0.88, 0.92])  # leave room for legend + suptitle
    plt.show()
```

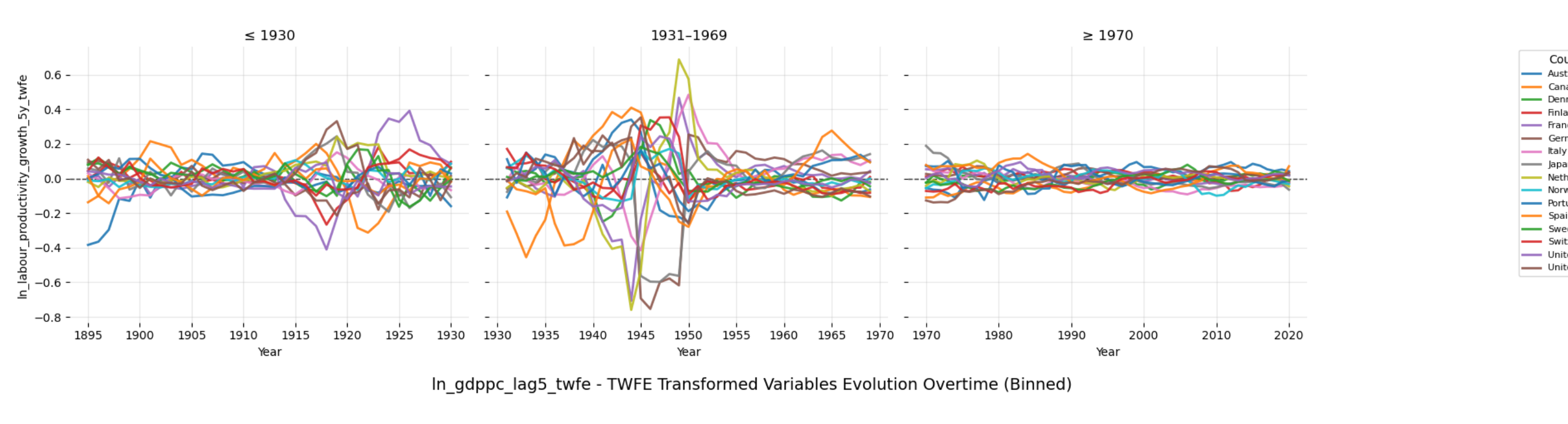


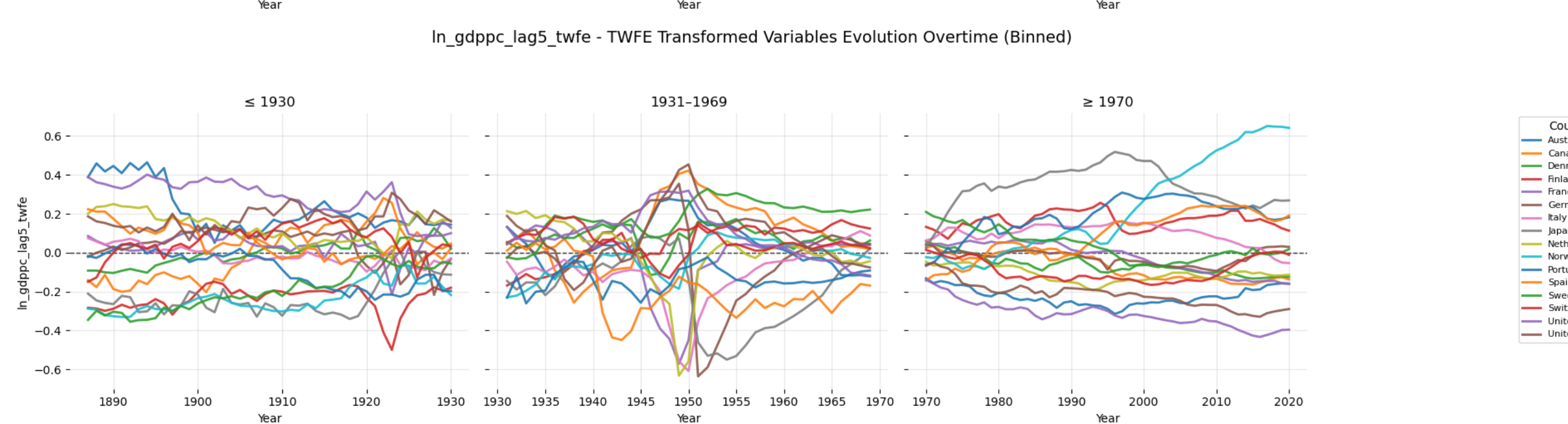


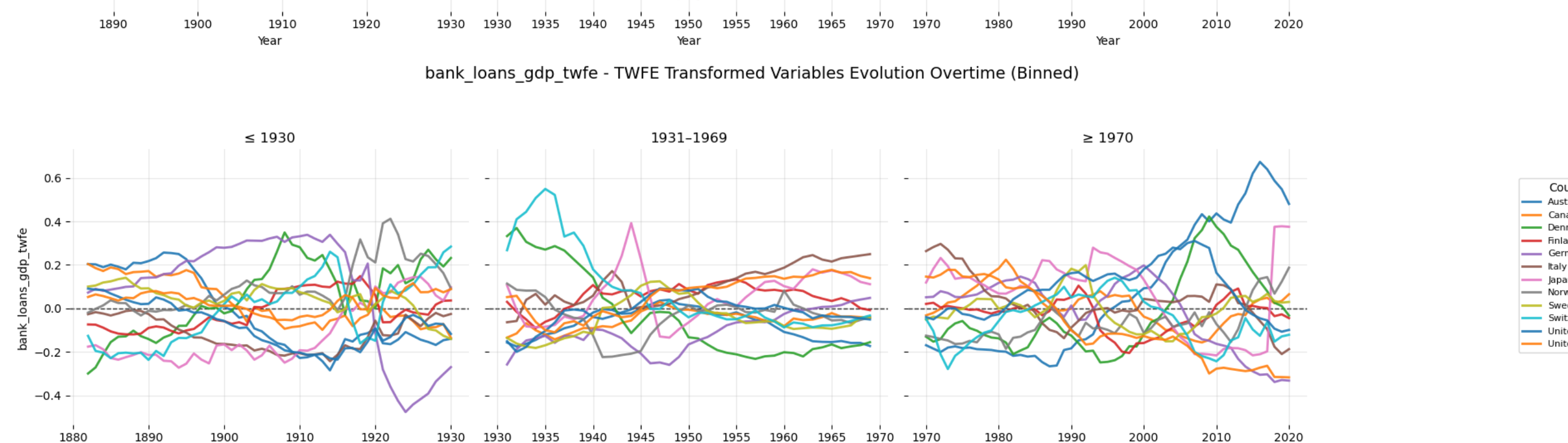












Box PLots --> Balanced Panel data for all countries

In [157…

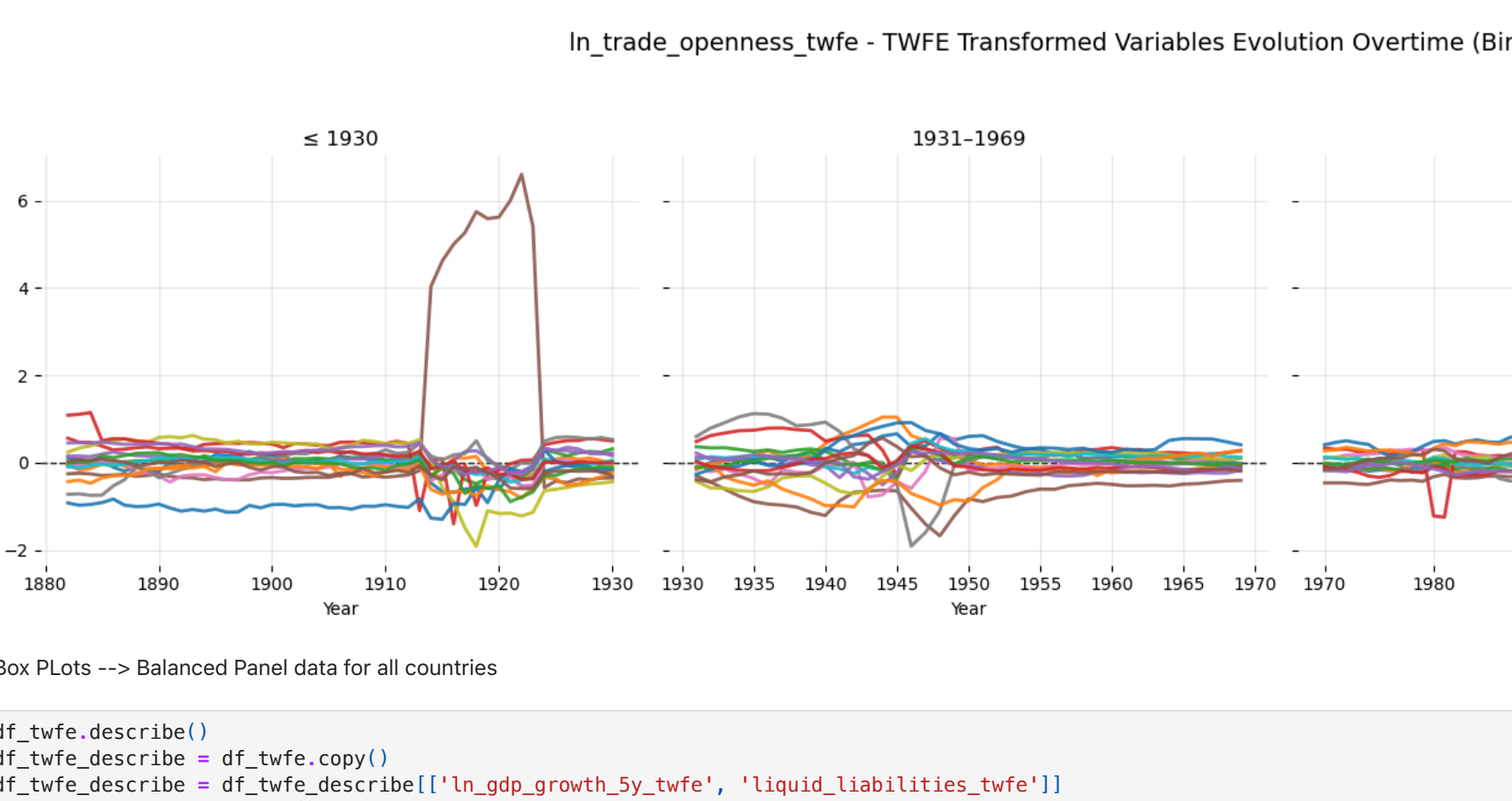
```python
df_twfe.describe()
df_twfe_describe = df_twfe.copy()
df_twfe_describe = df_twfe_describe[['ln_gdp_growth_5y_twfe', 'liquid_liabilities_twfe']]
df_twfe_describe.describe()
df_twfe.groupby("country")[f"{v}_twfe"].mean()
```

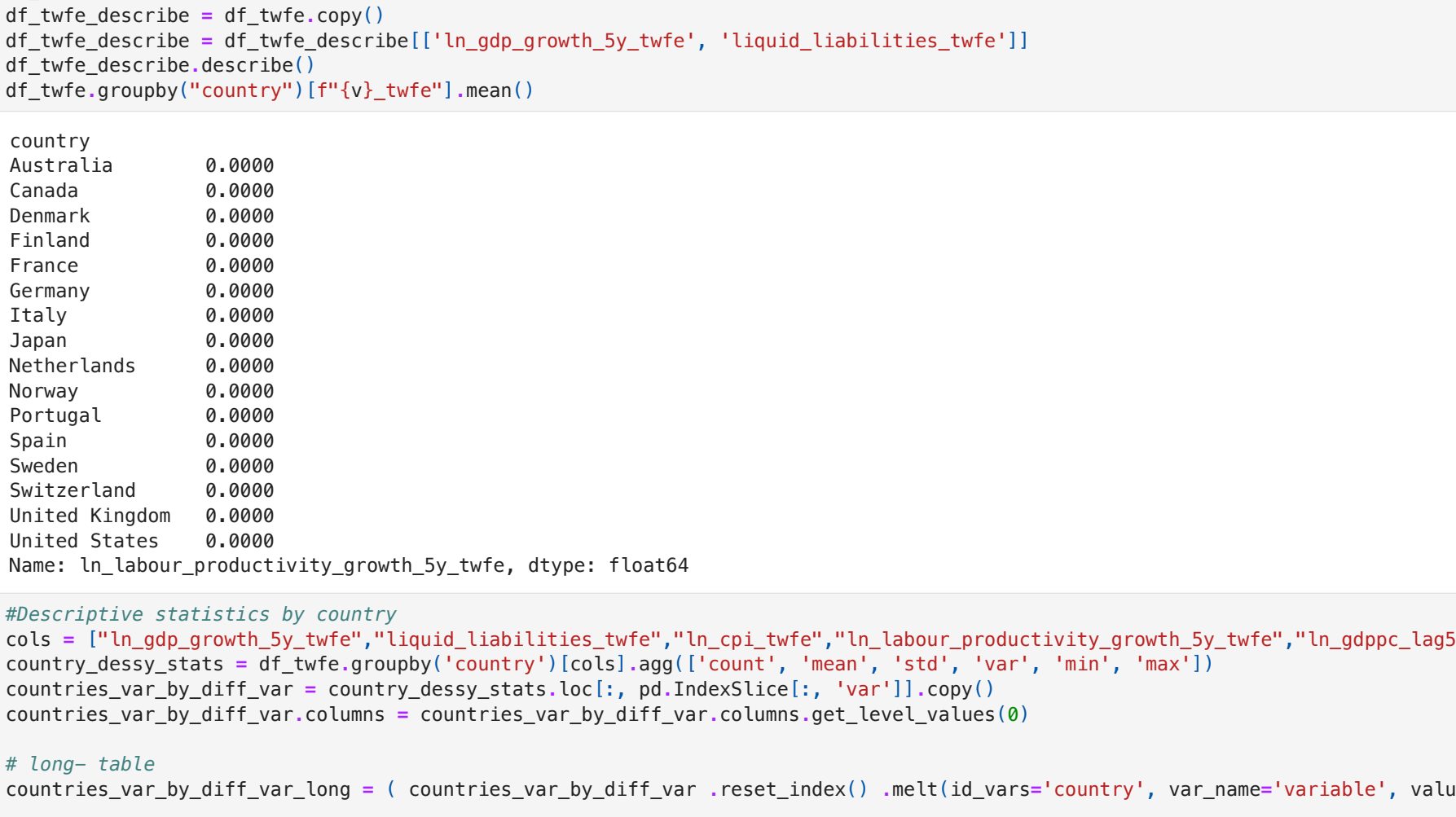
Out[157…

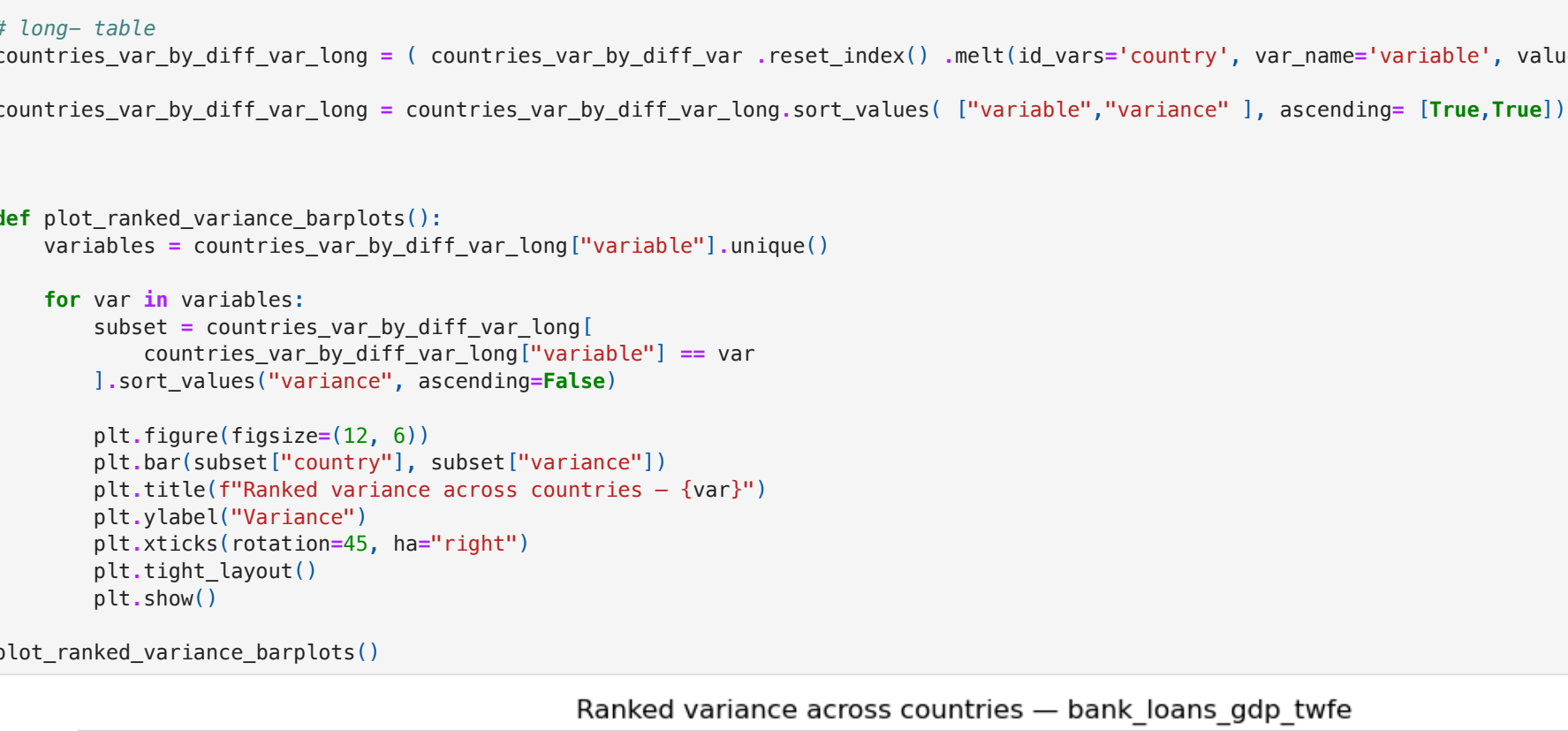
```
country
Australia         0.0000
Canada            0.0000
Denmark           0.0000
Finland           0.0000
France            0.0000
Germany           0.0000
Italy             0.0000
Japan             0.0000
Netherlands       0.0000
Norway            0.0000
Portugal          0.0000
Spain             0.0000
Sweden            0.0000
Switzerland       0.0000
United Kingdom    0.0000
United States     0.0000
Name: ln_labour_productivity_growth_5y_twfe, dtype: float64
```

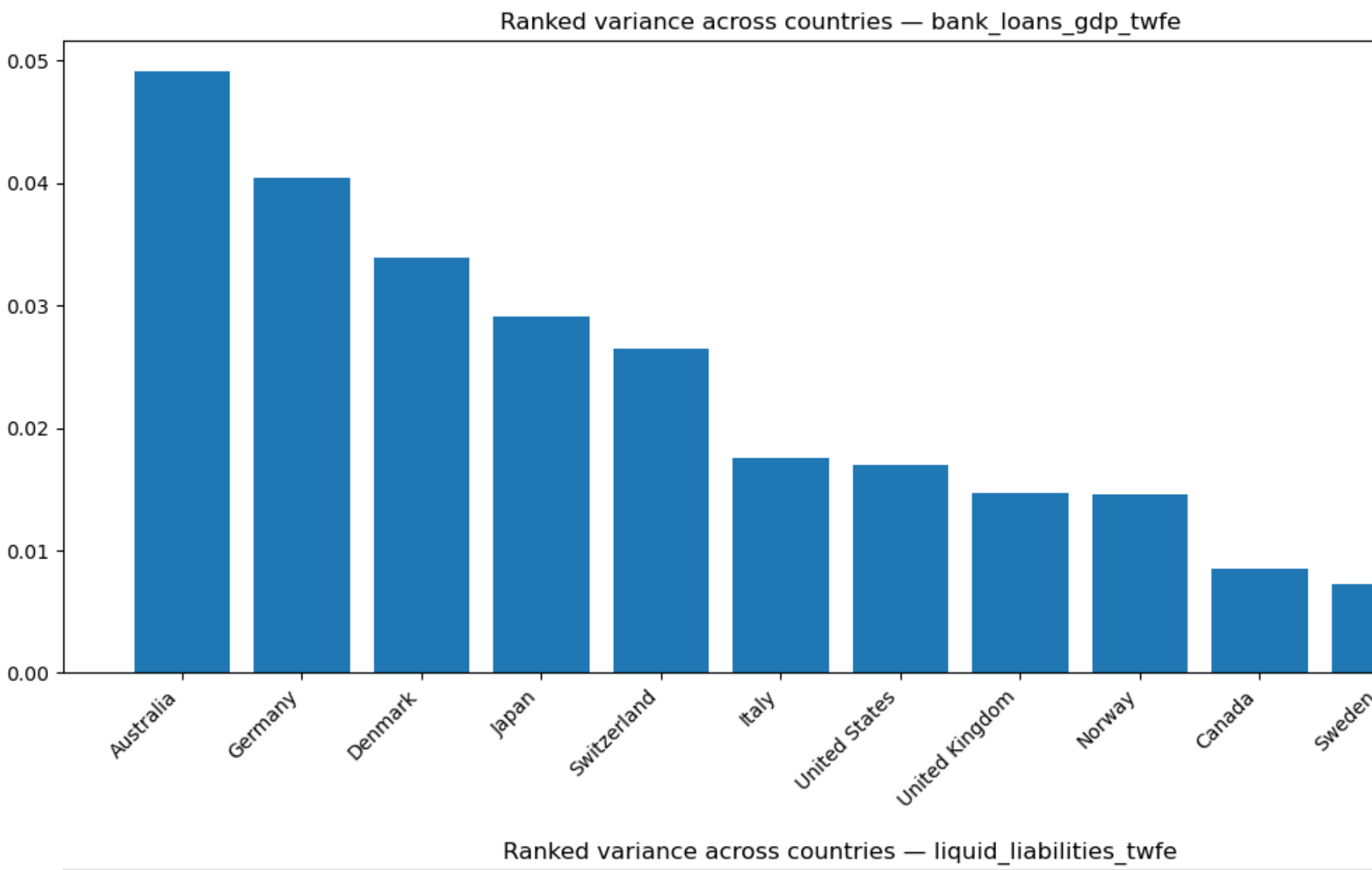
In [158…

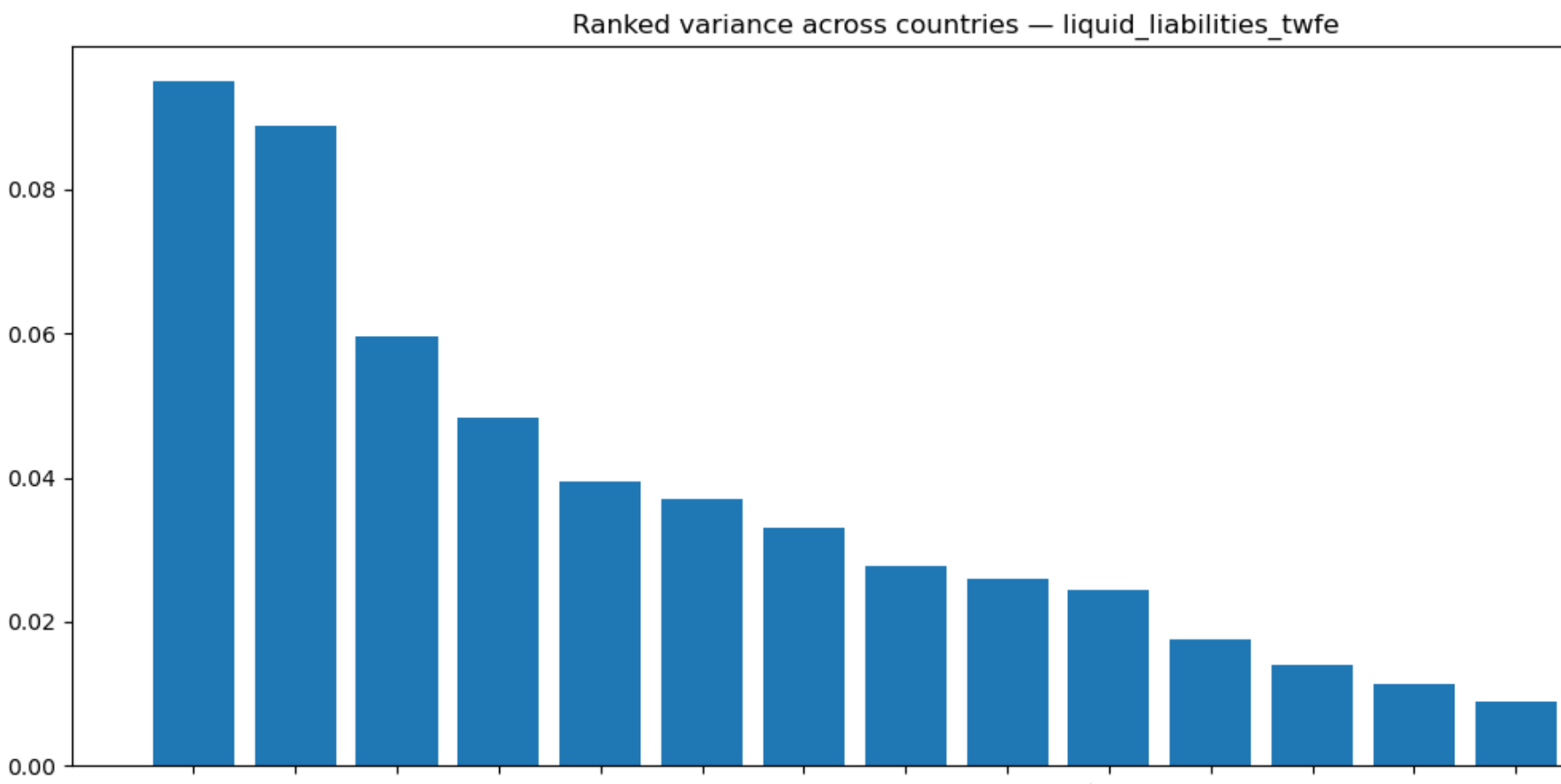
```python
#Descriptive statistics by country
cols = ["ln_gdp_growth_5y_twfe","liquid_liabilities_twfe","ln_cpi_twfe","ln_labour_productivity_growth_5y_twfe","ln_gdppc_lag5_twfe","bank_loans_gdp_twfe","mcap_gdp_twfe","ln_trade_openness_twfe",]
country_dessy_stats = df_twfe.groupby('country')[cols].agg(['count', 'mean', 'std', 'var', 'min', 'max'])
countries_var_by_diff_var = country_dessy_stats.loc[:, pd.IndexSlice[:, 'var']].copy()
countries_var_by_diff_var.columns = countries_var_by_diff_var.columns.get_level_values(0)

# long- table
countries_var_by_diff_var_long = ( countries_var_by_diff_var .reset_index() .melt(id_vars='country', var_name='variable', value_name='variance'))

countries_var_by_diff_var_long = countries_var_by_diff_var_long.sort_values( ["variable","variance" ], ascending= [True,True])


def plot_ranked_variance_barplots():
    variables = countries_var_by_diff_var_long["variable"].unique()

    for var in variables:
        subset = countries_var_by_diff_var_long[
            countries_var_by_diff_var_long["variable"] == var
        ].sort_values("variance", ascending=False)

        plt.figure(figsize=(12, 6))
        plt.bar(subset["country"], subset["variance"])
        plt.title(f"Ranked variance across countries — {var}")
        plt.ylabel("Variance")
        plt.xticks(rotation=45, ha="right")
        plt.tight_layout()
        plt.show()

plot_ranked_variance_barplots()
```

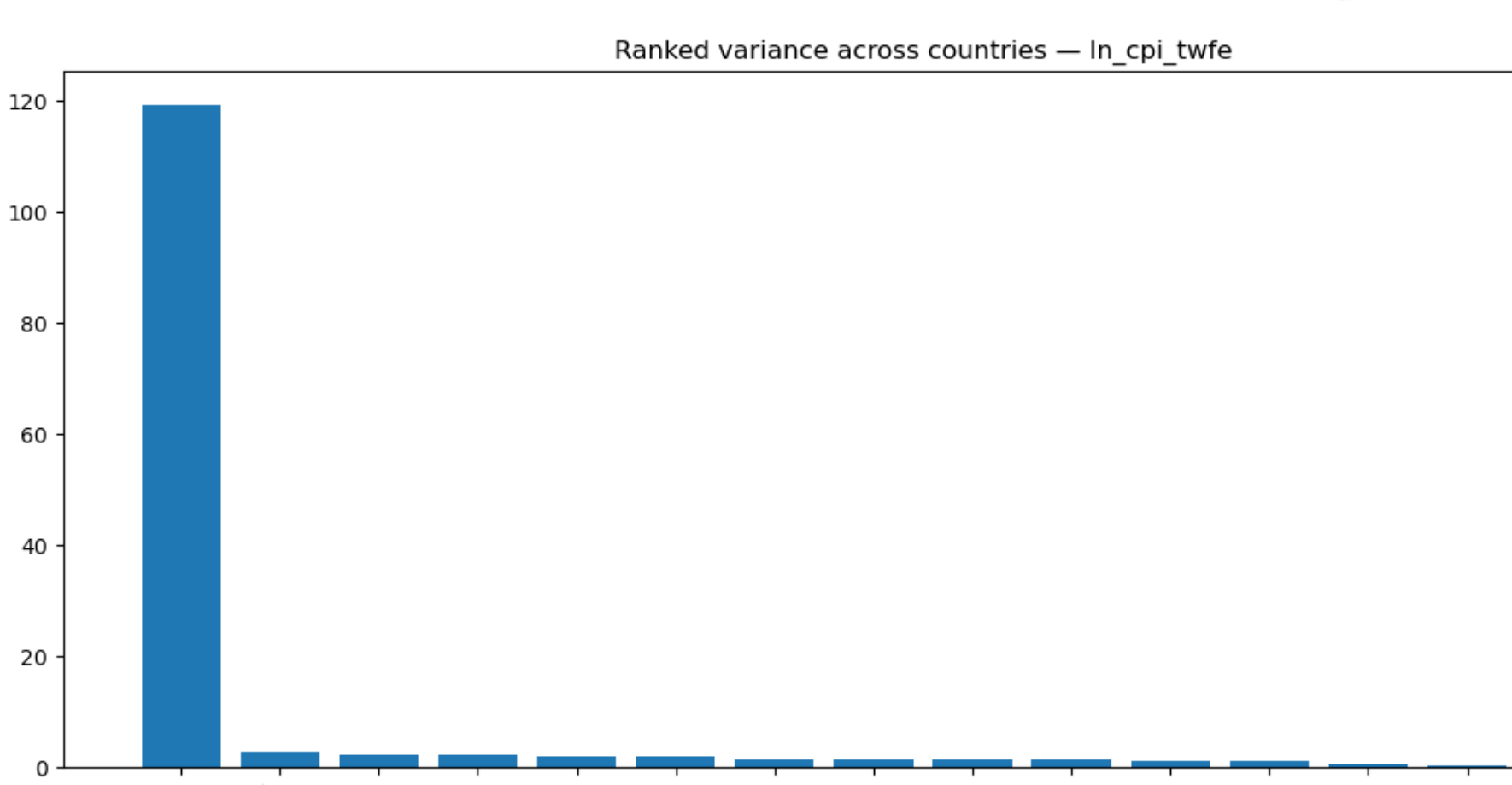


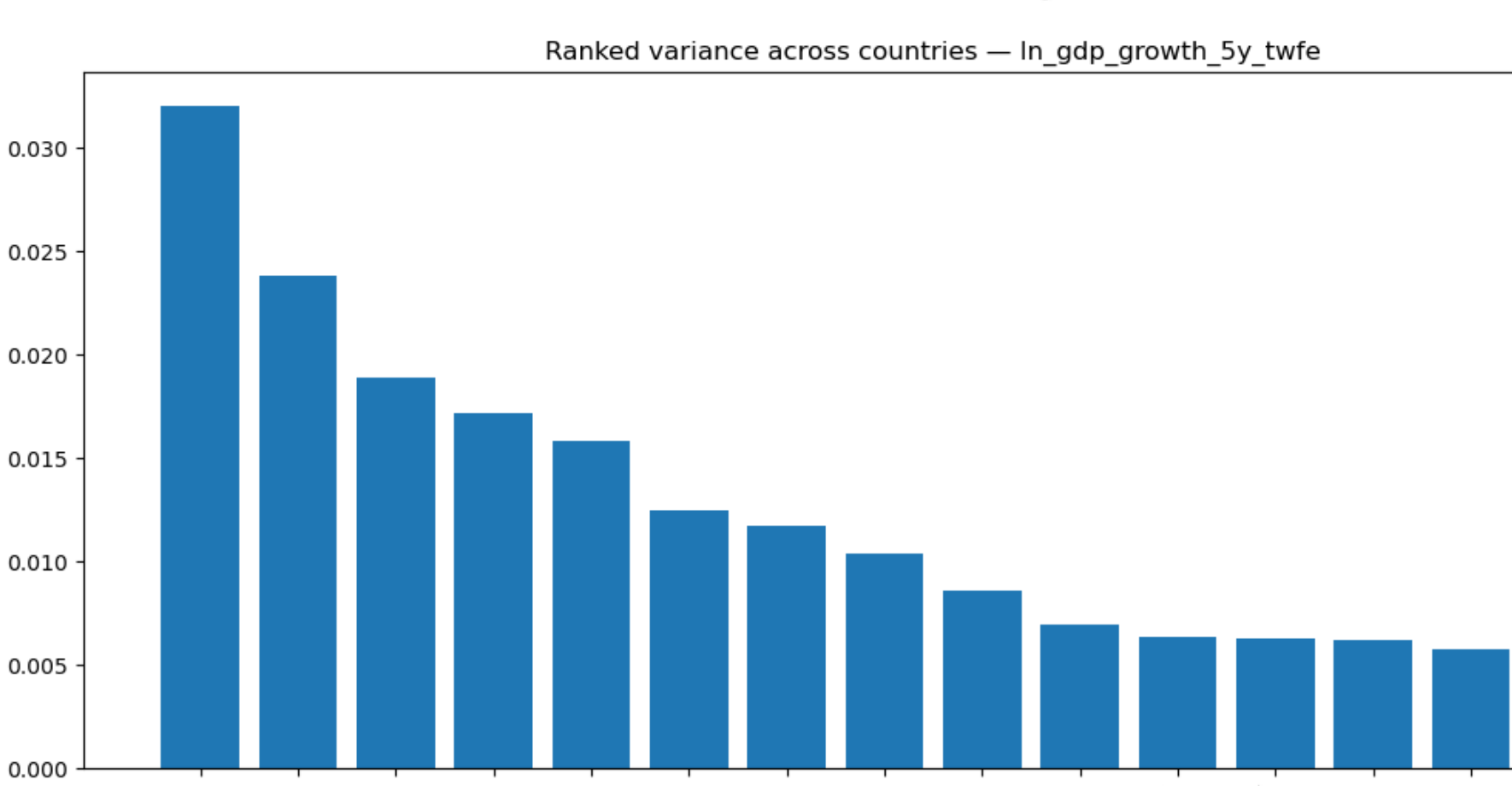


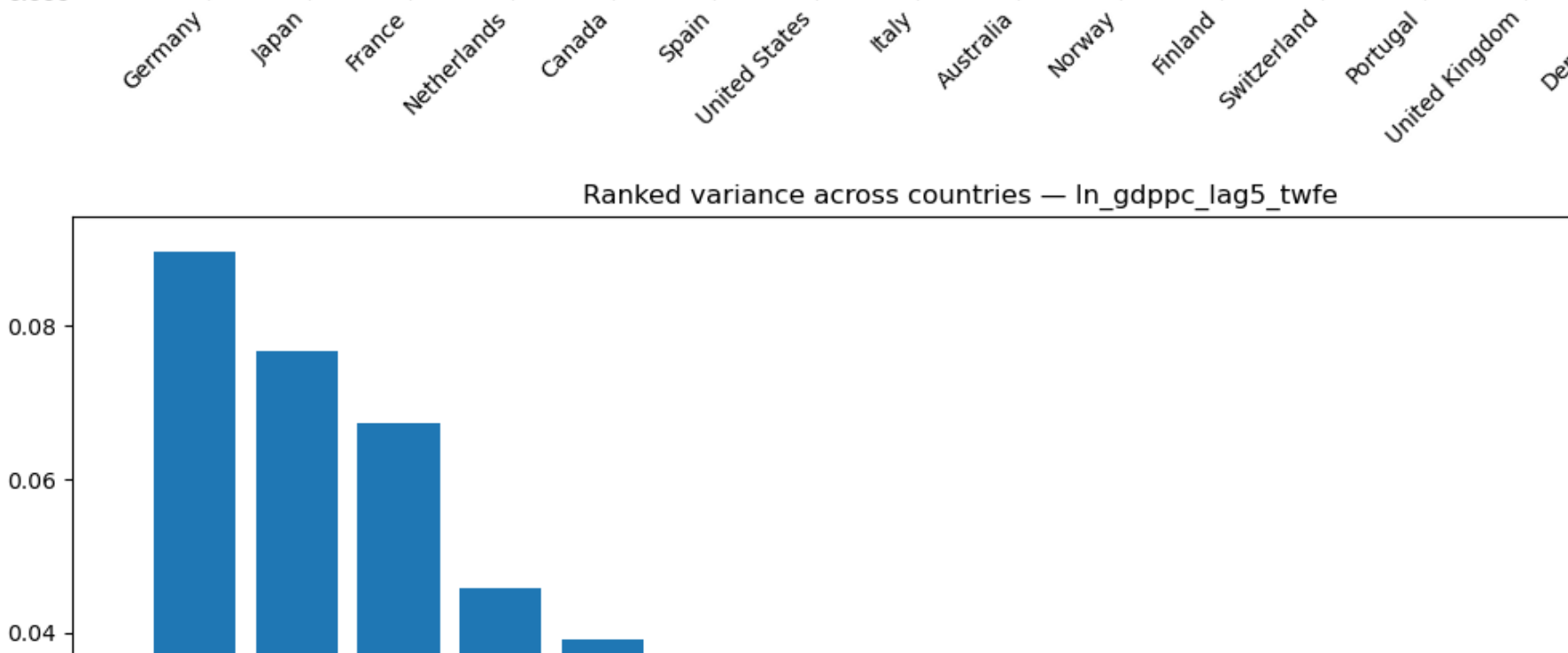












In [161…

```python
unique_countries = sorted(df_twfe['country'].unique())

# One colour per country
palette = sns.color_palette("tab20", n_colors=len(unique_countries))

fig, axes = plt.subplots(4, 2, figsize=(22, 28))
axes = axes.flatten()

for i, var in enumerate(cols):

    ax = axes[i]

    # variance ordering for this variable
    ordering = (
        countries_var_by_diff_var_long
        .query("variable == @var")
        .sort_values("variance")["country"]
        .tolist()
    )

    # dataset for plotting
    df_plot = df_twfe[["country", var]].copy()
    df_plot["country"] = pd.Categorical(df_plot["country"], categories=ordering, ordered=True)

    # country → colour mapping dictionary
    colour_map = {country: palette[j] for j, country in enumerate(unique_countries)}

    sns.boxplot(
        data=df_plot,
        x="country",
        y=var,
        order=ordering,
        ax=ax,
        palette=[colour_map[c] for c in ordering]  # colours aligned with ordering
    )

    ax.set_title(f"{var}", fontsize=14)
    ax.set_xlabel("")
    ax.set_ylabel("")
    ax.tick_params(axis='x', rotation=90)

plt.tight_layout()
plt.show()
```

```
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

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  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

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  sns.boxplot(
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/2342997286.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
```
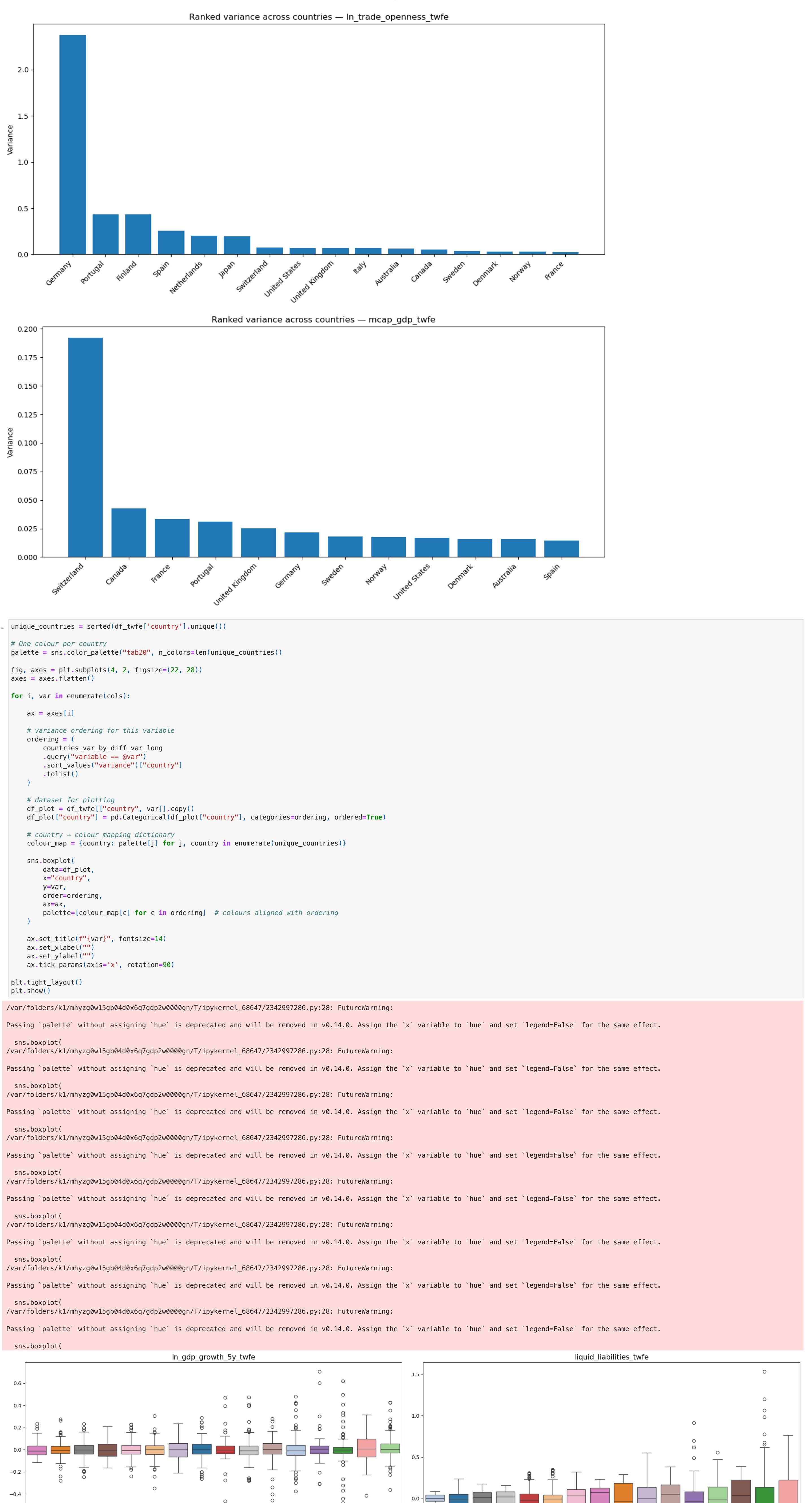
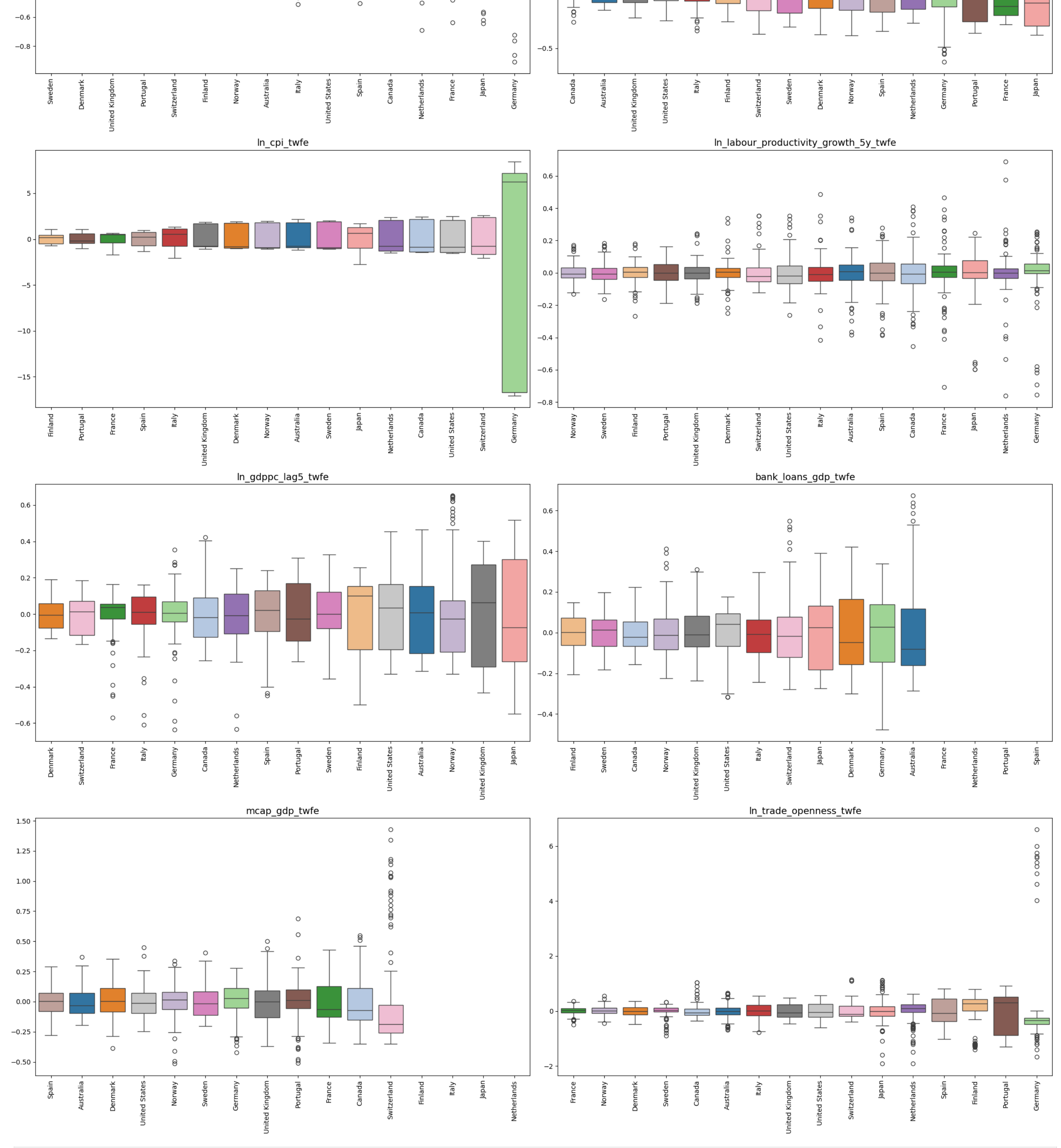





In [162…

```python
overall_desc = df_twfe[cols].describe().T
overall_desc['median'] = df_twfe[cols].median()
overall_desc = overall_desc[['count','mean','std','min','25%','median','75%','max']]
overall_desc
```

Out[162…

	count	mean	std	min	25%	median	75%	max
ln_gdp_growth_5y_twfe	2,144.0000	0.0000	0.1091	-0.9063	-0.0415	-0.0023	0.0418	0.7056
liquid_liabilities_twfe	2,224.0000	0.0000	0.1836	-0.6222	-0.1271	-0.0005	0.1041	1.5332
ln_cpi_twfe	2,224.0000	0.0000	2.9673	-17.0637	-0.9511	-0.2890	1.0618	8.4539
ln_labour_productivity_growth_5y_twfe	2,016.0000	0.0000	0.1052	-0.7599	-0.0408	-0.0015	0.0417	0.6881
ln_gdppc_lag5_twfe	2,144.0000	-0.0000	0.1841	-0.6361	-0.1243	0.0075	0.1186	0.6517
bank_loans_gdp_twfe	1,668.0000	-0.0000	0.1482	-0.4767	-0.1039	-0.0070	0.0868	0.6733
mcap_gdp_twfe	1,488.0000	0.0000	0.1928	-0.5146	-0.1173	-0.0123	0.0857	1.4266
ln_trade_openness_twfe	2,224.0000	-0.0000	0.5240	-1.9155	-0.1881	0.0001	0.1906	6.5972

In [163…

```python
country_stats = (
    df_twfe.groupby('country')[cols]
    .agg(['count','mean','std','var','min','max'])
)
country_stats
```

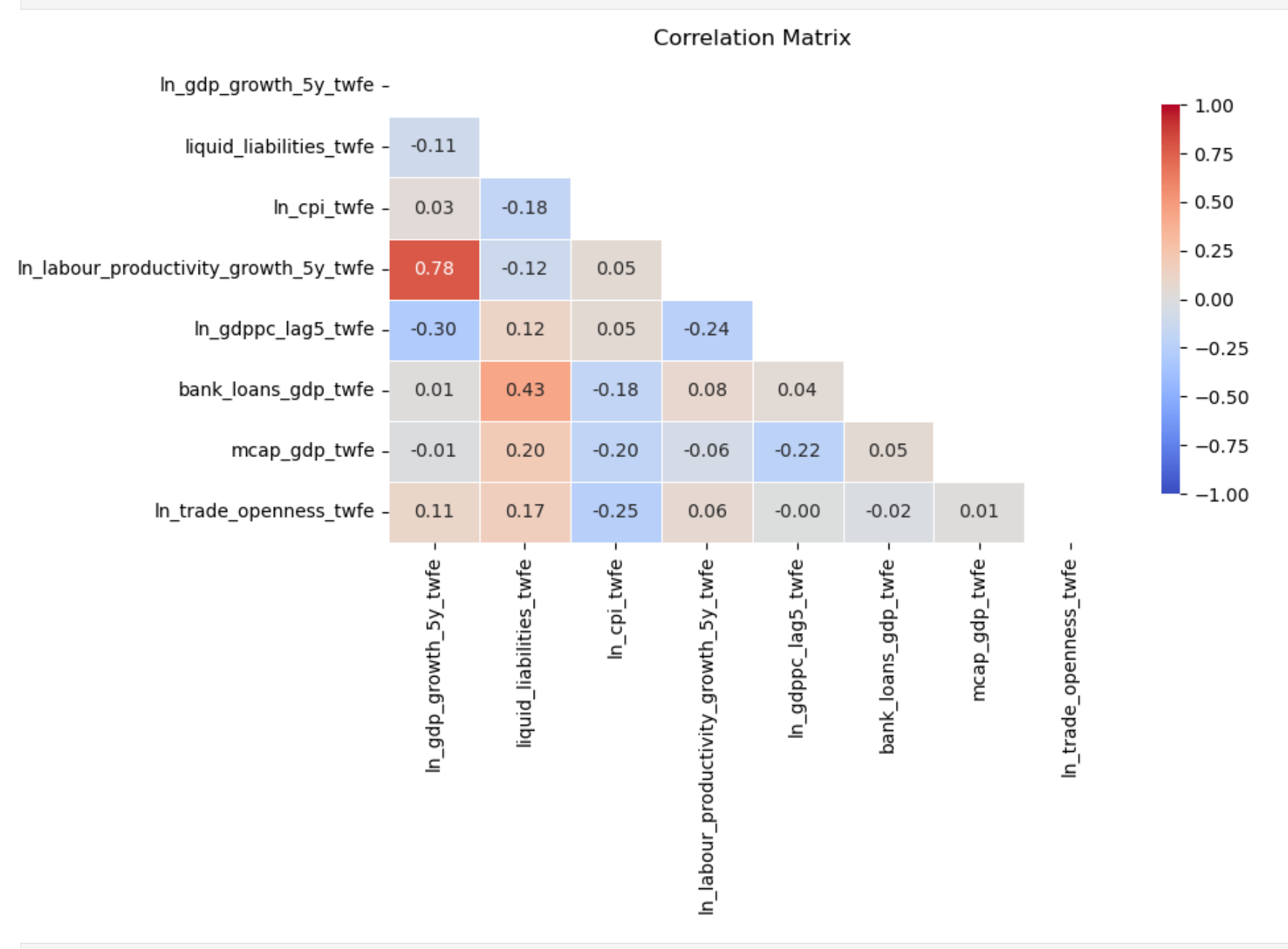
Out[163…

	ln_gdp_growth_5y_twfe						liquid_liabilities_twfe				…	mcap_gdp_twfe				ln_trade_openness_twfe					
	count	mean	std	var	min	max	count	mean	std	var	…	std	var	min	max	count	mean	std	var	min	max
country																					
Australia	134	0.0000	0.0929	0.0086	-0.2649	0.2893	139	0.0000	0.0934	0.0087	…	0.1252	0.0157	-0.1936	0.3696	139	-0.0000	0.2499	0.0624	-0.6827	0.6628
Canada	134	0.0000	0.1258	0.0158	-0.3753	0.4814	139	0.0000	0.0588	0.0035	…	0.2065	0.0426	-0.3517	0.5471	139	-0.0000	0.2338	0.0547	-0.3552	1.0431
Denmark	134	0.0000	0.0715	0.0051	-0.2804	0.2747	139	0.0000	0.1667	0.0278	…	0.1260	0.0159	-0.3848	0.3528	139	-0.0000	0.1782	0.0318	-0.4799	0.3642
Finland	134	-0.0000	0.0796	0.0063	-0.3494	0.3059	139	0.0000	0.1324	0.0175	…	NaN	NaN	NaN	NaN	139	-0.0000	0.6590	0.4343	-1.4003	0.7940
France	134	-0.0000	0.1375	0.0189	-0.6381	0.6174	139	0.0000	0.2980	0.0888	…	0.1829	0.0335	-0.3429	0.4288	139	-0.0000	0.1569	0.0246	-0.5060	0.3523
Germany	134	-0.0000	0.1790	0.0320	-0.9063	0.4267	139	0.0000	0.2196	0.0482	…	0.1465	0.0215	-0.4214	0.2782	139	-0.0000	1.5414	2.3760	-1.6752	6.5972
Italy	134	-0.0000	0.1018	0.0104	-0.5105	0.4711	139	0.0000	0.1181	0.0140	…	NaN	NaN	NaN	NaN	139	-0.0000	0.2592	0.0672	-0.7846	0.5501
Japan	134	-0.0000	0.1544	0.0238	-0.6420	0.3151	139	0.0000	0.3082	0.0950	…	NaN	NaN	NaN	NaN	139	-0.0000	0.4429	0.1961	-1.9085	1.1233
Netherlands	134	0.0000	0.1311	0.0172	-0.6897	0.7056	139	0.0000	0.1984	0.0394	…	NaN	NaN	NaN	NaN	139	-0.0000	0.4505	0.2029	-1.9155	0.6240
Norway	134	0.0000	0.0834	0.0070	-0.2120	0.2368	139	0.0000	0.1820	0.0331	…	0.1331	0.0177	-0.5146	0.3396	139	-0.0000	0.1668	0.0278	-0.4456	0.5469
Portugal	134	-0.0000	0.0790	0.0062	-0.1630	0.2096	139	0.0000	0.2442	0.0596	…	0.1756	0.0308	-0.5112	0.6898	139	-0.0000	0.6603	0.4360	-1.2967	0.9186
Spain	134	0.0000	0.1119	0.0125	-0.5064	0.2794	139	0.0000	0.1924	0.0370	…	0.1210	0.0146	-0.2784	0.2885	139	-0.0000	0.5059	0.2559	-1.0125	0.8168
Sweden	134	0.0000	0.0636	0.0040	-0.1138	0.2353	139	0.0000	0.1613	0.0260	…	0.1349	0.0182	-0.2019	0.4055	139	-0.0000	0.1953	0.0381	-0.8909	0.3304
Switzerland	134	0.0000	0.0796	0.0063	-0.2394	0.2273	139	0.0000	0.1560	0.0243	…	0.4384	0.1922	-0.3518	1.4266	139	-0.0000	0.2723	0.0741	-0.3853	1.1450
United Kingdom	134	0.0000	0.0761	0.0058	-0.2463	0.2326	139	0.0000	0.0941	0.0089	…	0.1595	0.0255	-0.3701	0.5012	139	-0.0000	0.2644	0.0699	-0.4668	0.4728
United States	134	0.0000	0.1085	0.0118	-0.2831	0.4725	139	0.0000	0.1066	0.0114	…	0.1289	0.0166	-0.2476	0.4476	139	0.0000	0.2672	0.0714	-0.5959	0.5633

16 rows × 48 columns

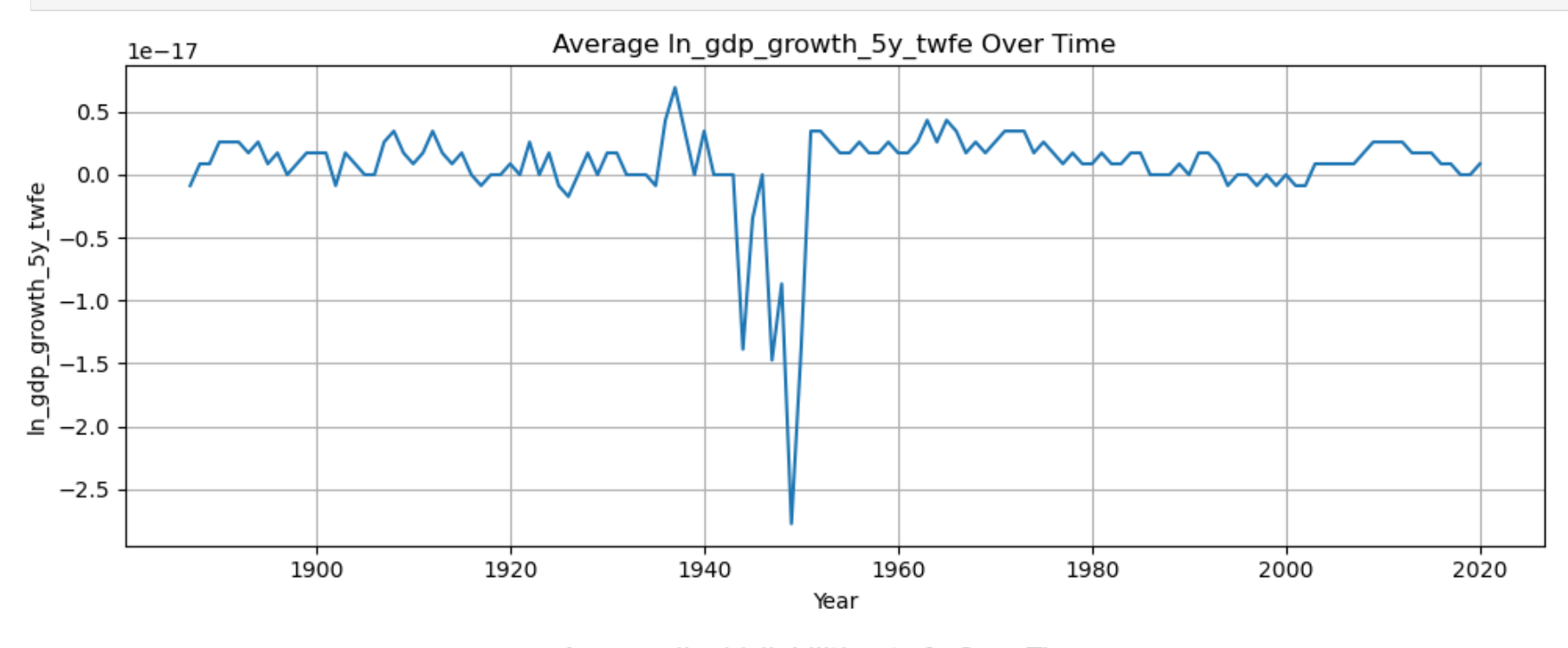
In [164…

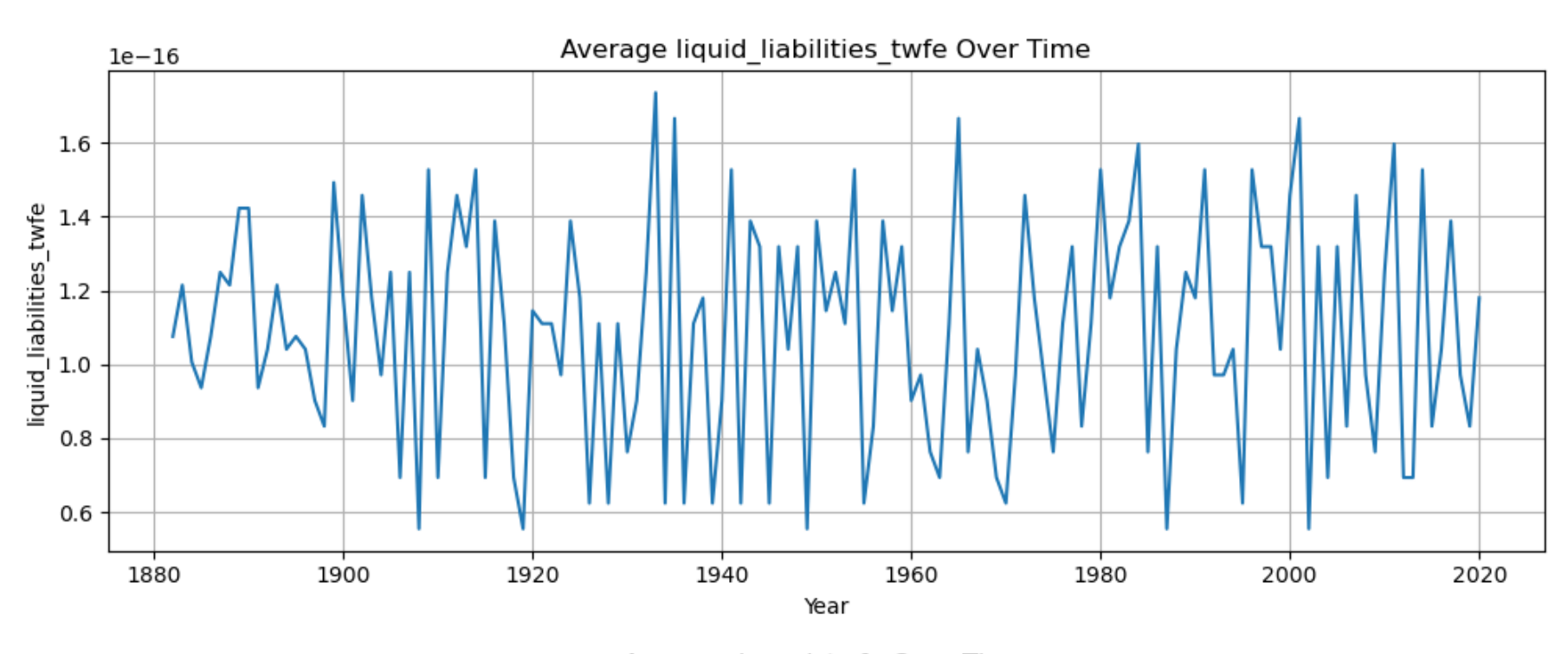
```python
corr = df_twfe[cols].corr()

# coverig up the uppie
mask = np.triu(np.ones_like(corr, dtype=bool))

plt.figure(figsize=(10, 7))
sns.heatmap(
        corr,
        mask=mask,
        annot=True,
        fmt=".2f",
        cmap="coolwarm",
        vmin=-1,
        vmax=1,
        linewidths=0.5,
        cbar_kws={"shrink": 0.8}
)

plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

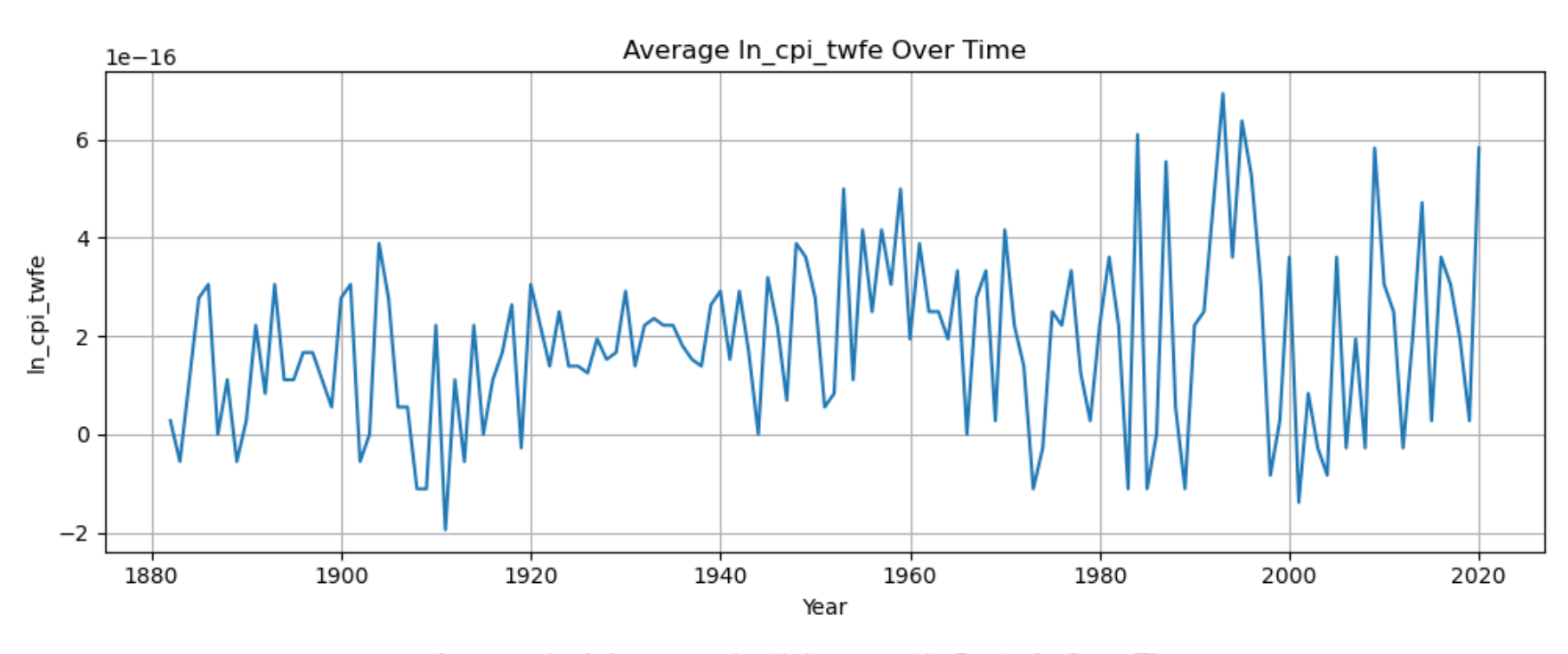

Correlation Matrix

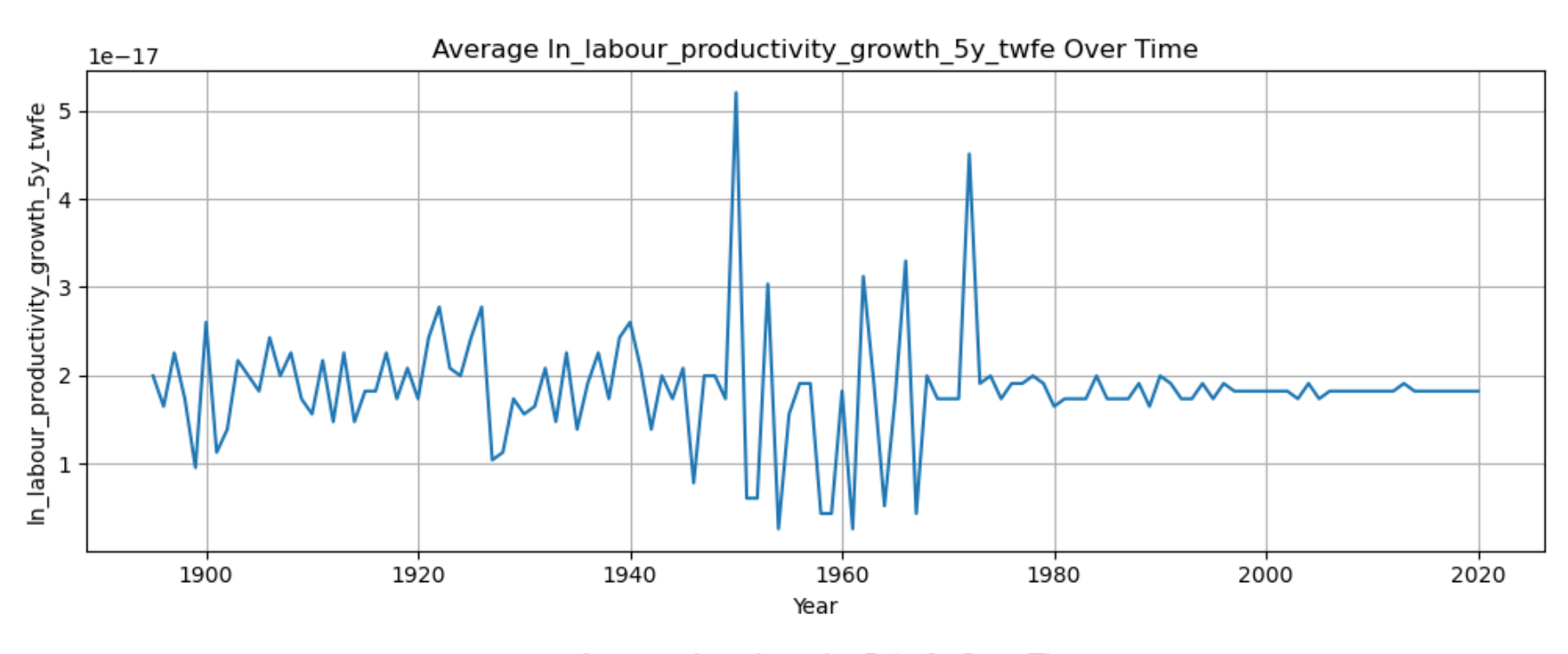
In [165…

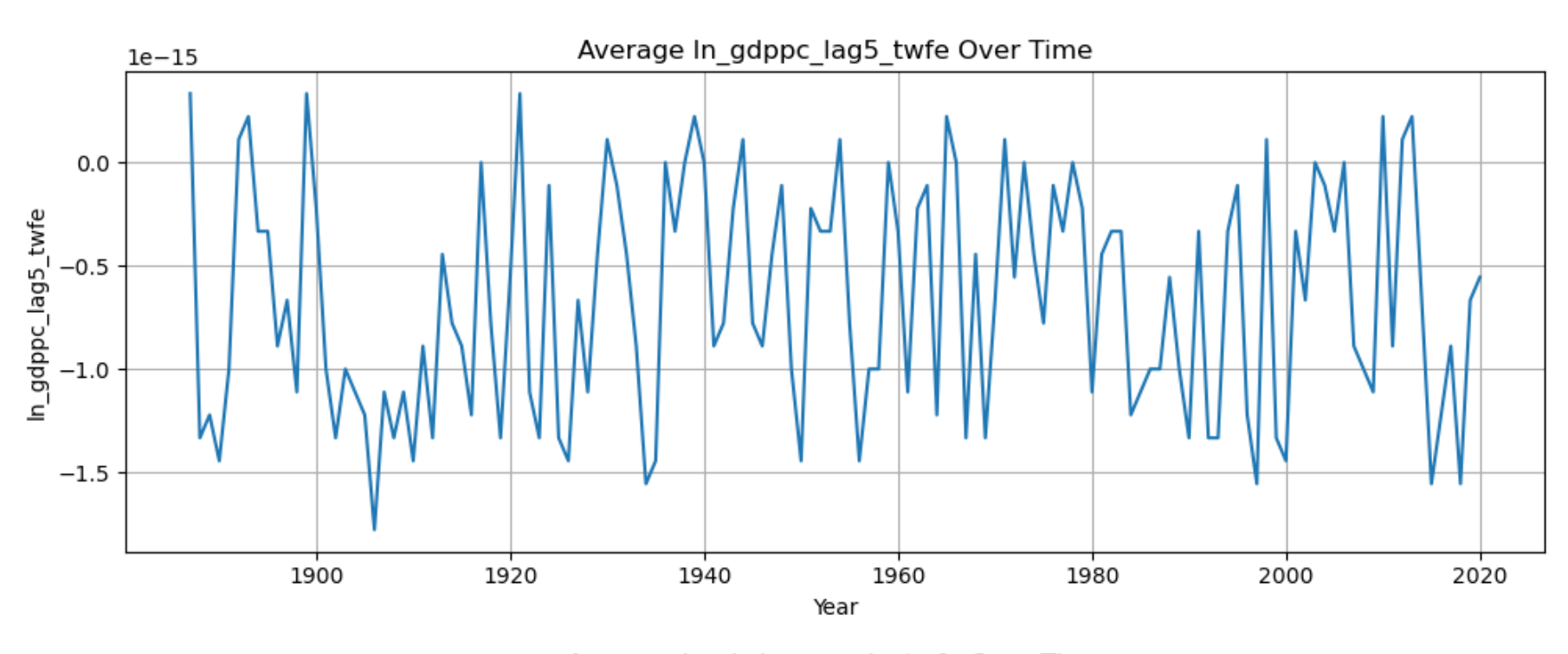
```python
for var in cols:
    df_twfe.groupby('year')[var].mean().plot(figsize=(10,4))
    plt.title(f"Average {var} Over Time")
    plt.xlabel("Year")
    plt.ylabel(var)
    plt.grid(True)
    plt.tight_layout()
    plt.show()
```

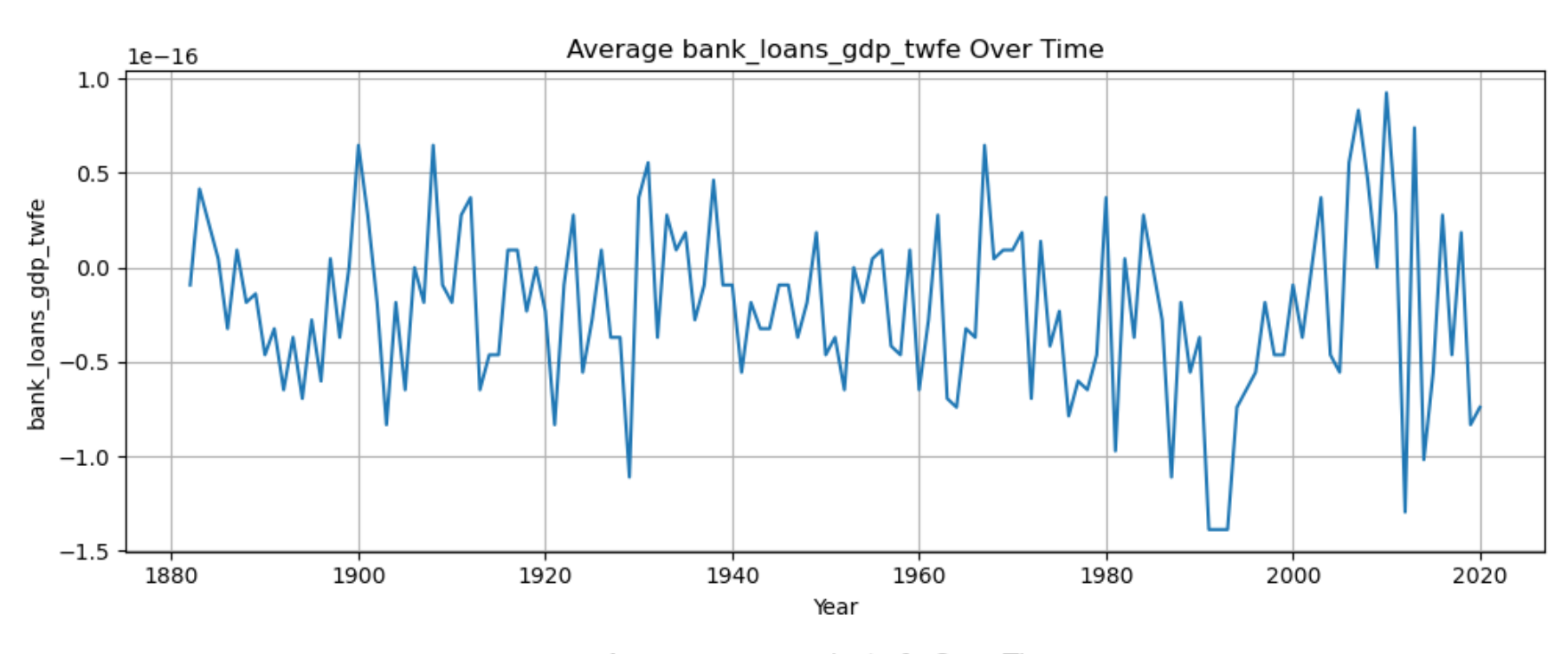

Average ln_gdp_growth_5y_twfe Over Time

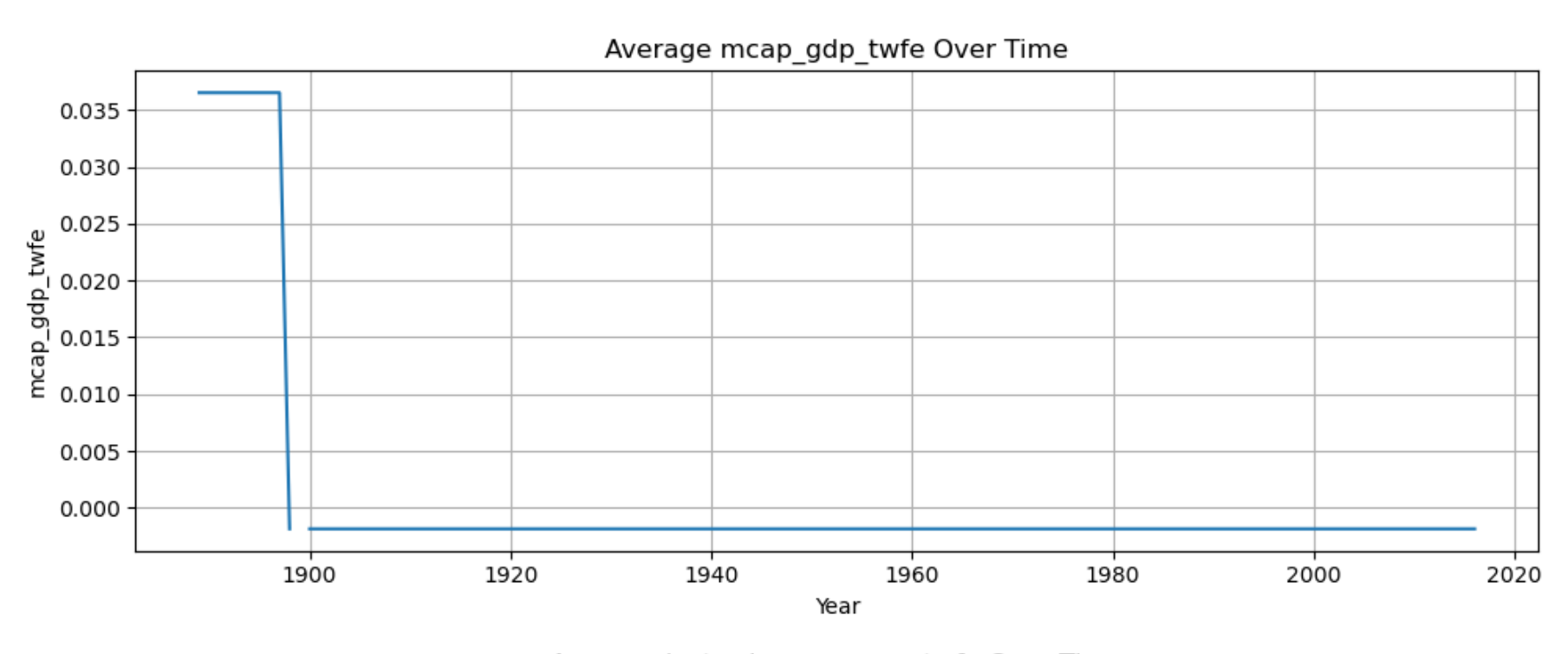

Average liquid_liabilities_twfe Over Time

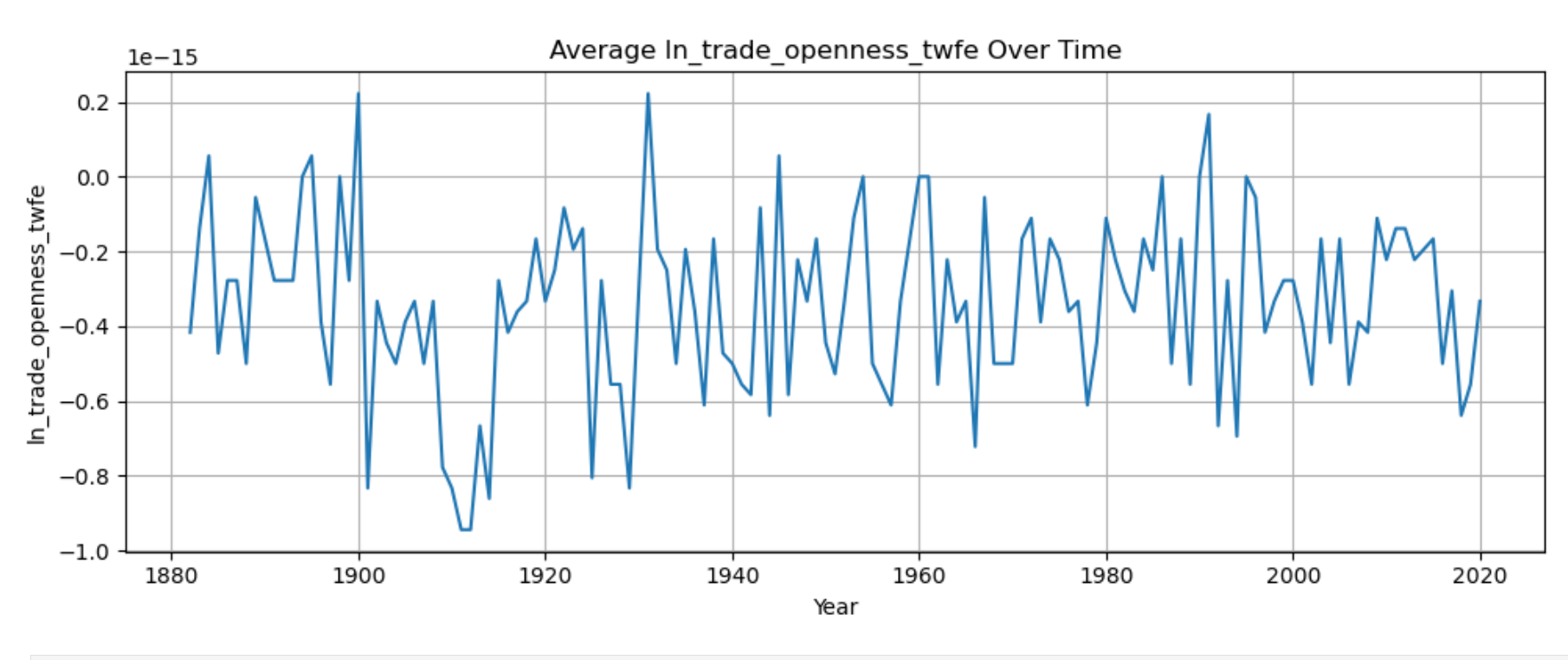

Average ln_cpi_twfe Over Time


Average ln_labour_productivity_growth_5y_twfe Over Time


Average ln_gdppc_lag5_twfe Over Time


Average bank_loans_gdp_twfe Over Time


Average mcap_gdp_twfe Over Time


Average ln_trade_openness_twfe Over Time

In [ ]:

```python
xtsum_table = {}

for var in cols:
    mean_overall = df_twfe[var].mean()

    mean_country = df_twfe.groupby('country')[var].mean()
    between_var = mean_country.var()

    df_twfe["wv"] = df_twfe[var] - df_twfe.groupby('country')[var].transform('mean')
    within_var = df_twfe["wv"].var()

    xtsum_table[var] = {
        "overall_mean": mean_overall,
        "overall_sd": df_twfe[var].std(),
        "between_sd": between_var**0.5,
        "within_sd": within_var**0.5
    }

xtsum_table
```

Out[ ]:

```
{'ln_gdp_growth_5y_twfe': {'overall_mean': np.float64(2.4791010869370043e-18),
  'overall_sd': 0.10912585709740846,
  'between_sd': 4.963734527954227e-18,
  'within_sd': 0.10912585709740846},
 'liquid_liabilities_twfe': {'overall_mean': np.float64(1.0837653716102585e-16),
  'overall_sd': 0.18363363000607902,
  'between_sd': 4.042370940802248e-17,
  'within_sd': 0.18363363000607902},
 'ln_cpi_twfe': {'overall_mean': np.float64(1.916931841079407e-16),
  'overall_sd': 2.9672821765078976,
  'between_sd': 2.1320899695677157e-16,
  'within_sd': 2.9672821765078976},
 'ln_labour_productivity_growth_5y_twfe': {'overall_mean': np.float64(1.9171447938870667e-17),
  'overall_sd': 0.1051628450342546,
  'between_sd': 8.817028499520694e-18,
  'within_sd': 0.1051628450342546},
 'ln_gdppc_lag5_twfe': {'overall_mean': np.float64(-6.744190612275205e-16),
  'overall_sd': 0.18407344415994994,
  'between_sd': 5.120898971657714e-16,
  'within_sd': 0.18407344415994992},
 'bank_loans_gdp_twfe': {'overall_mean': np.float64(-2.1432362945401702e-17),
  'overall_sd': 0.14815060196502217,
  'between_sd': 6.98769017106518e-17,
  'within_sd': 0.1481506019650222},
 'mcap_gdp_twfe': {'overall_mean': np.float64(1.0818705549102668e-17),
  'overall_sd': 0.19277883500406054,
  'between_sd': 0.004200566042937384,
  'within_sd': 0.1927379166514744},
 'ln_trade_openness_twfe': {'overall_mean': np.float64(-3.358624329891211e-16),
  'overall_sd': 0.5240223833901163,
  'between_sd': 6.492684364933169e-16,
  'within_sd': 0.5240223833901163}}
```

Question 12 --> Simple correlation coefficent

In [166…

```python
df_Twfe_r_coeffiecent_ordering = df_twfe

df_Twfe_r_coeffiecent_ordering = df_Twfe_r_coeffiecent_ordering[["country",  "ln_gdp_growth_5y_twfe", "liquid_liabilities_twfe",  "ln_labour_productivity_growth_5y_twfe", "ln_gdppc_lag5_twfe", "bank_

y = "ln_gdp_growth_5y_twfe"

X = ["liquid_liabilities_twfe","ln_gdppc_lag5_twfe","bank_loans_gdp_twfe", "mcap_gdp_twfe"]
corr_list = []
for x in X:
    tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
    tmp["variable"] = x
    corr_list.append(tmp)
corr_table = pd.concat(corr_list).reset_index(drop=True)
sd_table = ( df_Twfe_r_coeffiecent_ordering.groupby("country").agg(SD_Dependent = (y, "std"),SD_Explanatory  = (x, "std")).reset_index())
corr_table = corr_table.merge(sd_table, on="country", how="left")
corr_table["beta_simple"] = corr_table["corr"] * corr_table["SD_Dependent"] / corr_table["SD_Explanatory"]
corr_table = corr_table.sort_values("corr", ascending=False).reset_index(drop=True)

print(corr_table)
```

```
       country   corr                 variable  SD_Dependent  SD_Explanatory  \
0      Portugal 0.3518            mcap_gdp_twfe        0.0790          0.1756   
1      Portugal 0.3228  liquid_liabilities_twfe        0.0790          0.1756   
2       Denmark 0.2631  liquid_liabilities_twfe        0.0715          0.1260   
3        Sweden 0.2481  liquid_liabilities_twfe        0.0636          0.1349   
4         Japan 0.2173      bank_loans_gdp_twfe        0.1544             NaN   
..          ...    ...                      ...           ...             ...   
59        Spain    NaN      bank_loans_gdp_twfe        0.1119          0.1210   
60      Finland    NaN            mcap_gdp_twfe        0.0796             NaN   
61        Italy    NaN            mcap_gdp_twfe        0.1018             NaN   
62        Japan    NaN            mcap_gdp_twfe        0.1544             NaN   
63  Netherlands    NaN            mcap_gdp_twfe        0.1311             NaN   

    beta_simple  
0        0.1581  
1        0.1451  
2        0.1493  
3        0.1170  
4           NaN  
..          ...  
59          NaN  
60          NaN  
61          NaN  
62          NaN  
63          NaN  

[64 rows x 6 columns]
```

```
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_68647/4059540502.py:10: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  tmp = (df_Twfe_r_coeffiecent_ordering.groupby("country").apply(lambda z: z[[y, x]].corr().iloc[0, 1]).reset_index(name="corr"))
```

Question 12 --> TWFE transforms with the regression of the two Proxies of Economic Growth and then also for the explantory varibels.

In [167…

```python
counts = df_twfe.groupby("country").size()
df_twfe = df_twfe[df_twfe["country"].isin(counts[counts >= 2].index)].copy()

# makig my yr dummies

year_dummies = pd.get_dummies(df_twfe["year"], prefix ="year", drop_first=False)

# doing my regressions first for teh two dependent varibels;

dummy_reg_gdp = df_twfe["ln_gdp_growth_5y_within"].notna()
dummy_y_gdp_reg = df_twfe.loc[dummy_reg_gdp, "ln_gdp_growth_5y_within"]
dummy_x_gdp_reg = year_dummies.loc[dummy_reg_gdp]

ln_gdp_time_fe = sm.OLS(dummy_y_gdp_reg, dummy_x_gdp_reg).fit()

# second one

dummy_reg_gdp_labprod = df_twfe["ln_labour_productivity_growth_5y_within"].notna()
dummy_y_gdp_labprod = df_twfe.loc[dummy_reg_gdp_labprod, "ln_labour_productivity_growth_5y_within"]
dummy_x_gdo_labprod = year_dummies.loc[dummy_reg_gdp_labprod]
ln_labprod_time_fe = sm.OLS(dummy_y_gdp_labprod, dummy_x_gdo_labprod).fit()

# now explantory varibles

dummy_reg_liquid_liabilities = df_twfe["liquid_liabilities_within"].notna()
dummy_y_reg_liq = df_twfe.loc[dummy_reg_liquid_liabilities, "liquid_liabilities_within"]
dummy_x_reg_liq = year_dummies.loc[dummy_reg_liquid_liabilities]
liquid_liability_time_fe = sm.OLS(dummy_y_reg_liq, dummy_x_reg_liq).fit()


dummy_reg_mcap = df_twfe["mcap_gdp_within"].notna()
dummy_y_reg_mcap = df_twfe.loc[dummy_reg_mcap, "mcap_gdp_within"]
dummy_x_reg_mcap = year_dummies.loc[dummy_reg_mcap]

mcap_time_fe = sm.OLS(dummy_y_reg_mcap, dummy_x_reg_mcap).fit()


# bank_loans_gdp_within

dummy_reg_bank = df_twfe["bank_loans_gdp_within"].notna()
dummy_y_reg_bank = df_twfe.loc[dummy_reg_bank, "bank_loans_gdp_within"]
dummy_x_reg_bank = year_dummies.loc[dummy_reg_bank]

bank_loans_time_fe = sm.OLS(dummy_y_reg_bank, dummy_x_reg_bank).fit()

print("Time FE – ln GDP growth (5y)")
print(ln_gdp_time_fe.summary())

print("\nTime FE – ln labour productivity growth (5y)")
print(ln_labprod_time_fe.summary())

print("\nTime FE – liquid liabilities (within)")
print(liquid_liability_time_fe.summary())

print("\nTime FE – market cap / GDP (within)")
print(mcap_time_fe.summary())

print("\nTime FE – bank loans / GDP (within)")
print(bank_loans_time_fe.summary())
```

```
Time FE – ln GDP growth (5y)
                              OLS Regression Results                             
==================================================================================
Dep. Variable:     ln_gdp_growth_5y_within   R-squared:                       0.300
Model:                                 OLS   Adj. R-squared:                  0.253
Method:                      Least Squares   F-statistic:                     6.466
Date:                     Sat, 27 Dec 2025   Prob (F-statistic):           1.16e-84
Time:                             17:14:14   Log-Likelihood:                 1707.8
No. Observations:                     2144   AIC:                            -3148.
Df Residuals:                         2010   BIC:                            -2388.
Df Model:                              133                                         
Covariance Type:                 nonrobust                                         
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882  -9.301e-19   7.04e-19     -1.322      0.186   -2.31e-18     4.5e-19
year_1883           0          0        nan        nan           0           0
year_1884  -1.114e-18   7.04e-19     -1.583      0.114   -2.49e-18    2.66e-19
year_1885  -7.576e-19   7.04e-19     -1.077      0.282   -2.14e-18    6.22e-19
year_1886   -1.16e-18   7.04e-19     -1.649      0.099   -2.54e-18     2.2e-19
year_1887     -0.0372      0.028     -1.322      0.186      -0.092       0.018
year_1888     -0.0558      0.028     -1.981      0.048      -0.111      -0.001
year_1889     -0.0446      0.028     -1.583      0.114      -0.100       0.011
year_1890     -0.0303      0.028     -1.077      0.282      -0.086       0.025
year_1891     -0.0464      0.028     -1.649      0.099      -0.102       0.009
year_1892     -0.0575      0.028     -2.040      0.041      -0.113      -0.002
year_1893     -0.0670      0.028     -2.378      0.017      -0.122      -0.012
year_1894     -0.0596      0.028     -2.117      0.034      -0.115      -0.004
year_1895     -0.0542      0.028     -1.922      0.055      -0.109       0.001
year_1896     -0.0451      0.028     -1.601      0.110      -0.100       0.010
year_1897     -0.0263      0.028     -0.933      0.351      -0.082       0.029
year_1898      0.0147      0.028      0.522      0.601      -0.041       0.070
year_1899      0.0071      0.028      0.251      0.802      -0.048       0.062
year_1900      0.0050      0.028      0.179      0.858      -0.050       0.060
year_1901      0.0068      0.028      0.243      0.808      -0.048       0.062
year_1902     -0.0175      0.028     -0.620      0.535      -0.073       0.038
year_1903     -0.0341      0.028     -1.212      0.226      -0.089       0.021
year_1904     -0.0392      0.028     -1.393      0.164      -0.094       0.016
year_1905     -0.0435      0.028     -1.544      0.123      -0.099       0.012
year_1906     -0.0103      0.028     -0.367      0.714      -0.066       0.045
year_1907      0.0061      0.028      0.218      0.827      -0.049       0.061
year_1908     -0.0272      0.028     -0.964      0.335      -0.082       0.028
year_1909     -0.0079      0.028     -0.282      0.778      -0.063       0.047
year_1910     -0.0099      0.028     -0.350      0.726      -0.065       0.045
year_1911     -0.0273      0.028     -0.969      0.333      -0.083       0.028
year_1912     -0.0236      0.028     -0.838      0.402      -0.079       0.032
year_1913      0.0143      0.028      0.506      0.613      -0.041       0.070
year_1914     -0.0588      0.028     -2.088      0.037      -0.114      -0.004
year_1915     -0.0733      0.028     -2.602      0.009      -0.129      -0.018
year_1916     -0.0504      0.028     -1.789      0.074      -0.106       0.005
year_1917     -0.1147      0.028     -4.073      0.000      -0.170      -0.059
year_1918     -0.1941      0.028     -6.891      0.000      -0.249      -0.139
year_1919     -0.1181      0.028     -4.191      0.000      -0.173      -0.063
year_1920     -0.0840      0.028     -2.982      0.003      -0.139      -0.029
year_1921     -0.1557      0.028     -5.526      0.000      -0.211      -0.100
year_1922     -0.0321      0.028     -1.140      0.254      -0.087       0.023
year_1923      0.0504      0.028      1.789      0.074      -0.005       0.106
year_1924      0.0530      0.028      1.883      0.060      -0.002       0.108
year_1925      0.0545      0.028      1.935      0.053      -0.001       0.110
year_1926      0.0928      0.028      3.293      0.001       0.038       0.148
year_1927      0.0554      0.028      1.966      0.049       0.000       0.111
year_1928      0.0492      0.028      1.746      0.081      -0.006       0.104
year_1929      0.0470      0.028      1.668      0.095      -0.008       0.102
year_1930     -0.0144      0.028     -0.512      0.609      -0.070       0.041
year_1931     -0.0786      0.028     -2.789      0.005      -0.134      -0.023
year_1932     -0.1422      0.028     -5.049      0.000      -0.197      -0.087
year_1933     -0.1490      0.028     -5.289      0.000      -0.204      -0.094
year_1934     -0.1453      0.028     -5.158      0.000      -0.201      -0.090
year_1935     -0.0895      0.028     -3.177      0.002      -0.145      -0.034
year_1936     -0.0315      0.028     -1.118      0.264      -0.087       0.024
year_1937      0.0379      0.028      1.344      0.179      -0.017       0.093
year_1938      0.0283      0.028      1.004      0.316      -0.027       0.084
year_1939      0.0365      0.028      1.297      0.195      -0.019       0.092
year_1940     -0.0121      0.028     -0.428      0.669      -0.067       0.043
year_1941     -0.0125      0.028     -0.445      0.656      -0.068       0.043
year_1942     -0.0559      0.028     -1.984      0.047      -0.111      -0.001
year_1943     -0.0508      0.028     -1.802      0.072      -0.106       0.004
year_1944     -0.1321      0.028     -4.688      0.000      -0.187      -0.077
year_1945     -0.1693      0.028     -6.009      0.000      -0.225      -0.114
year_1946     -0.1364      0.028     -4.843      0.000      -0.192      -0.081
year_1947     -0.0927      0.028     -3.292      0.001      -0.148      -0.037
year_1948     -0.0647      0.028     -2.297      0.022      -0.120      -0.009
year_1949      0.0088      0.028      0.313      0.755      -0.046       0.064
year_1950      0.1164      0.028      4.134      0.000       0.061       0.172
year_1951      0.1207      0.028      4.285      0.000       0.065       0.176
year_1952      0.1021      0.028      3.625      0.000       0.047       0.157
year_1953      0.0971      0.028      3.446      0.001       0.042       0.152
year_1954      0.0978      0.028      3.474      0.001       0.043       0.153
year_1955      0.0938      0.028      3.328      0.001       0.039       0.149
year_1956      0.0815      0.028      2.892      0.004       0.026       0.137
year_1957      0.0822      0.028      2.919      0.004       0.027       0.137
year_1958      0.0577      0.028      2.048      0.041       0.002       0.113
year_1959      0.0618      0.028      2.195      0.028       0.007       0.117
year_1960      0.0644      0.028      2.286      0.022       0.009       0.120
year_1961      0.0748      0.028      2.656      0.008       0.020       0.130
year_1962      0.0902      0.028      3.202      0.001       0.035       0.145
year_1963      0.1178      0.028      4.183      0.000       0.063       0.173
year_1964      0.1307      0.028      4.638      0.000       0.075       0.186
year_1965      0.1250      0.028      4.437      0.000       0.070       0.180
year_1966      0.1142      0.028      4.054      0.000       0.059       0.169
year_1967      0.1060      0.028      3.765      0.000       0.051       0.161
year_1968      0.1128      0.028      4.003      0.000       0.058       0.168
year_1969      0.1098      0.028      3.896      0.000       0.055       0.165
year_1970      0.1077      0.028      3.824      0.000       0.052       0.163
year_1971      0.1036      0.028      3.679      0.000       0.048       0.159
year_1972      0.1097      0.028      3.895      0.000       0.054       0.165
year_1973      0.1154      0.028      4.095      0.000       0.060       0.171
year_1974      0.0783      0.028      2.778      0.006       0.023       0.133
year_1975      0.0354      0.028      1.257      0.209      -0.020       0.091
year_1976      0.0405      0.028      1.439      0.150      -0.015       0.096
year_1977      0.0198      0.028      0.702      0.483      -0.035       0.075
year_1978     -0.0051      0.028     -0.182      0.856      -0.060       0.050
year_1979      0.0163      0.028      0.580      0.562      -0.039       0.072
year_1980      0.0361      0.028      1.282      0.200      -0.019       0.091
year_1981      0.0069      0.028      0.244      0.807      -0.048       0.062
year_1982     -0.0122      0.028     -0.432      0.666      -0.067       0.043
year_1983     -0.0219      0.028     -0.779      0.436      -0.077       0.033
year_1984     -0.0270      0.028     -0.959      0.338      -0.082       0.028
year_1985     -0.0117      0.028     -0.414      0.679      -0.067       0.044
year_1986      0.0101      0.028      0.360      0.719      -0.045       0.065
year_1987      0.0372      0.028      1.319      0.187      -0.018       0.092
year_1988      0.0548      0.028      1.945      0.052      -0.000       0.110
year_1989      0.0539      0.028      1.912      0.056      -0.001       0.109
year_1990      0.0368      0.028      1.306      0.192      -0.018       0.092
year_1991      0.0172      0.028      0.610      0.542      -0.038       0.072
year_1992      0.0001      0.028      0.005      0.996      -0.055       0.055
year_1993     -0.0279      0.028     -0.989      0.323      -0.083       0.027
year_1994     -0.0235      0.028     -0.834      0.405      -0.079       0.032
year_1995     -0.0080      0.028     -0.283      0.778      -0.063       0.047
year_1996      0.0121      0.028      0.431      0.667      -0.043       0.067
year_1997      0.0352      0.028      1.249      0.212      -0.020       0.090
year_1998      0.0643      0.028      2.281      0.023       0.009       0.120
year_1999      0.0659      0.028      2.340      0.019       0.011       0.121
year_2000      0.0773      0.028      2.743      0.006       0.022       0.133
year_2001      0.0695      0.028      2.467      0.014       0.014       0.125
year_2002      0.0483      0.028      1.715      0.087      -0.007       0.104
year_2003      0.0270      0.028      0.959      0.338      -0.028       0.082
year_2004      0.0216      0.028      0.769      0.442      -0.034       0.077
year_2005      0.0053      0.028      0.187      0.852      -0.050       0.061
year_2006      0.0157      0.028      0.558      0.577      -0.040       0.071
year_2007      0.0292      0.028      1.035      0.301      -0.026       0.084
year_2008      0.0189      0.028      0.670      0.503      -0.036       0.074
year_2009     -0.0486      0.028     -1.726      0.085      -0.104       0.007
year_2010     -0.0476      0.028     -1.690      0.091      -0.103       0.008
year_2011     -0.0649      0.028     -2.304      0.021      -0.120      -0.010
year_2012     -0.0970      0.028     -3.445      0.001      -0.152      -0.042
year_2013     -0.0996      0.028     -3.535      0.000      -0.155      -0.044
year_2014     -0.0520      0.028     -1.845      0.065      -0.107       0.003
year_2015     -0.0653      0.028     -2.319      0.020      -0.121      -0.010
year_2016     -0.0725      0.028     -2.574      0.010      -0.128      -0.017
year_2017     -0.0491      0.028     -1.744      0.081      -0.104       0.006
year_2018     -0.0239      0.028     -0.848      0.396      -0.079       0.031
year_2019     -0.0233      0.028     -0.827      0.408      -0.079       0.032
year_2020     -0.0909      0.028     -3.226      0.001      -0.146      -0.036
==============================================================================
Omnibus:                      667.485   Durbin-Watson:                   0.375
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            14650.498
Skew:                          -0.934   Prob(JB):                         0.00
Kurtosis:                      15.669   Cond. No.                     1.03e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.5e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

Time FE – ln labour productivity growth (5y)
                                    OLS Regression Results                                    
==============================================================================================
Dep. Variable:     ln_labour_productivity_growth_5y_within   R-squared:                       0.322
Model:                                                 OLS   Adj. R-squared:                  0.277
Method:                                      Least Squares   F-statistic:                     7.173
Date:                                     Sat, 27 Dec 2025   Prob (F-statistic):           4.79e-91
Time:                                             17:14:14   Log-Likelihood:                 1680.4
No. Observations:                                     2016   AIC:                            -3109.
Df Residuals:                                         1890   BIC:                            -2402.
Df Model:                                              125                                         
Covariance Type:                                 nonrobust                                         
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882    -1.9e-18   6.78e-19     -2.803      0.005   -3.23e-18   -5.71e-19
year_1883           0          0        nan        nan           0           0
year_1884  -1.184e-18   6.78e-19     -1.746      0.081   -2.51e-18    1.46e-19
year_1885  -4.603e-19   6.78e-19     -0.679      0.497   -1.79e-18     8.7e-19
year_1886  -6.938e-19   6.78e-19     -1.023      0.306   -2.02e-18    6.36e-19
year_1887  -6.183e-19   6.78e-19     -0.912      0.362   -1.95e-18    7.12e-19
year_1888  -2.466e-19   6.78e-19     -0.364      0.716   -1.58e-18    1.08e-18
year_1889  -7.673e-19   6.78e-19     -1.132      0.258    -2.1e-18    5.63e-19
year_1890  -1.224e-18   6.78e-19     -1.806      0.071   -2.55e-18    1.06e-19
year_1891  -1.218e-18   6.78e-19     -1.796      0.073   -2.55e-18    1.12e-19
year_1892  -1.123e-18   6.78e-19     -1.656      0.098   -2.45e-18    2.07e-19
year_1893  -5.648e-19   6.78e-19     -0.833      0.405   -1.89e-18    7.65e-19
year_1894  -1.508e-19   6.78e-19     -0.222      0.824   -1.48e-18    1.18e-18
year_1895     -0.0761      0.027     -2.803      0.005      -0.129      -0.023
year_1896     -0.0691      0.027     -2.547      0.011      -0.122      -0.016
year_1897     -0.0474      0.027     -1.746      0.081      -0.101       0.006
year_1898     -0.0184      0.027     -0.679      0.497      -0.072       0.035
year_1899     -0.0278      0.027     -1.023      0.306      -0.081       0.025
year_1900     -0.0248      0.027     -0.912      0.362      -0.078       0.028
year_1901     -0.0099      0.027     -0.364      0.716      -0.063       0.043
year_1902     -0.0307      0.027     -1.132      0.258      -0.084       0.023
year_1903     -0.0490      0.027     -1.806      0.071      -0.102       0.004
year_1904     -0.0487      0.027     -1.796      0.073      -0.102       0.004
year_1905     -0.0449      0.027     -1.656      0.098      -0.098       0.008
year_1906     -0.0226      0.027     -0.833      0.405      -0.076       0.031
year_1907     -0.0060      0.027     -0.222      0.824      -0.059       0.047
year_1908     -0.0279      0.027     -1.028      0.304      -0.081       0.025
year_1909     -0.0134      0.027     -0.495      0.621      -0.067       0.040
year_1910     -0.0179      0.027     -0.658      0.510      -0.071       0.035
year_1911     -0.0213      0.027     -0.786      0.432      -0.075       0.032
year_1912     -0.0231      0.027     -0.852      0.394      -0.076       0.030
year_1913      0.0016      0.027      0.057      0.954      -0.052       0.055
year_1914     -0.0484      0.027     -1.783      0.075      -0.102       0.005
year_1915     -0.0649      0.027     -2.392      0.017      -0.118      -0.012
year_1916     -0.0413      0.027     -1.520      0.129      -0.095       0.012
year_1917     -0.0947      0.027     -3.489      0.000      -0.148      -0.041
year_1918     -0.1324      0.027     -4.879      0.000      -0.186      -0.079
year_1919     -0.0395      0.027     -1.453      0.146      -0.093       0.014
year_1920      0.0143      0.027      0.528      0.598      -0.039       0.068
year_1921     -0.0153      0.027     -0.564      0.573      -0.069       0.038
year_1922      0.0760      0.027      2.799      0.005       0.023       0.129
year_1923      0.1179      0.027      4.344      0.000       0.065       0.171
year_1924      0.0864      0.027      3.184      0.001       0.033       0.140
year_1925      0.0637      0.027      2.348      0.019       0.010       0.117
year_1926      0.0683      0.027      2.517      0.012       0.015       0.122
year_1927      0.0400      0.027      1.473      0.141      -0.013       0.093
year_1928      0.0346      0.027      1.276      0.202      -0.019       0.088
year_1929      0.0244      0.027      0.898      0.369      -0.029       0.078
year_1930     -0.0071      0.027     -0.263      0.792      -0.060       0.046
year_1931     -0.0417      0.027     -1.535      0.125      -0.095       0.012
year_1932     -0.0573      0.027     -2.112      0.035      -0.111      -0.004
year_1933     -0.0650      0.027     -2.395      0.017      -0.118      -0.012
year_1934     -0.0801      0.027     -2.950      0.003      -0.133      -0.027
year_1935     -0.0582      0.027     -2.144      0.032      -0.111      -0.005
year_1936     -0.0559      0.027     -2.059      0.040      -0.109      -0.003
year_1937     -0.0323      0.027     -1.189      0.235      -0.086       0.021
year_1938     -0.0265      0.027     -0.977      0.329      -0.080       0.027
year_1939     -0.0174      0.027     -0.642      0.521      -0.071       0.036
year_1940     -0.0760      0.027     -2.800      0.005      -0.129      -0.023
year_1941     -0.0512      0.027     -1.886      0.059      -0.104       0.002
year_1942     -0.0846      0.027     -3.116      0.002      -0.138      -0.031
year_1943     -0.0932      0.027     -3.432      0.001      -0.146      -0.040
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year_1944     -0.1488      0.027     -5.483      0.000      -0.202      -0.096
year_1945     -0.1919      0.027     -7.070      0.000      -0.245      -0.139
year_1946     -0.1046      0.027     -3.852      0.000      -0.158      -0.051
year_1947     -0.0382      0.027     -1.405      0.160      -0.091       0.015
year_1948     -0.0068      0.027     -0.250      0.802      -0.060       0.046
year_1949      0.0641      0.027      2.362      0.018       0.011       0.117
year_1950      0.1925      0.027      7.092      0.000       0.139       0.246
year_1951      0.1341      0.027      4.939      0.000       0.081       0.187
year_1952      0.0992      0.027      3.653      0.000       0.046       0.152
year_1953      0.0977      0.027      3.598      0.000       0.044       0.151
year_1954      0.1006      0.027      3.707      0.000       0.047       0.154
year_1955      0.1018      0.027      3.748      0.000       0.049       0.155
year_1956      0.0892      0.027      3.287      0.001       0.036       0.142
year_1957      0.0884      0.027      3.257      0.001       0.035       0.142
year_1958      0.0708      0.027      2.607      0.009       0.018       0.124
year_1959      0.0785      0.027      2.890      0.004       0.025       0.132
year_1960      0.0781      0.027      2.878      0.004       0.025       0.131
year_1961      0.0893      0.027      3.288      0.001       0.036       0.143
year_1962      0.1028      0.027      3.788      0.000       0.050       0.156
year_1963      0.1245      0.027      4.587      0.000       0.071       0.178
year_1964      0.1297      0.027      4.777      0.000       0.076       0.183
year_1965      0.1280      0.027      4.715      0.000       0.075       0.181
year_1966      0.1170      0.027      4.311      0.000       0.064       0.170
year_1967      0.1171      0.027      4.313      0.000       0.064       0.170
year_1968      0.1297      0.027      4.779      0.000       0.076       0.183
year_1969      0.1315      0.027      4.845      0.000       0.078       0.185
year_1970      0.1299      0.027      4.787      0.000       0.077       0.183
year_1971      0.1344      0.027      4.952      0.000       0.081       0.188
year_1972      0.1375      0.027      5.063      0.000       0.084       0.191
year_1973      0.1295      0.027      4.769      0.000       0.076       0.183
year_1974      0.0945      0.027      3.482      0.001       0.041       0.148
year_1975      0.0674      0.027      2.483      0.013       0.014       0.121
year_1976      0.0598      0.027      2.202      0.028       0.007       0.113
year_1977      0.0377      0.027      1.390      0.165      -0.015       0.091
year_1978      0.0201      0.027      0.739      0.460      -0.033       0.073
year_1979      0.0295      0.027      1.087      0.277      -0.024       0.083
year_1980      0.0306      0.027      1.127      0.260      -0.023       0.084
year_1981      0.0153      0.027      0.564      0.573      -0.038       0.069
year_1982      0.0024      0.027      0.089      0.929      -0.051       0.056
year_1983     -0.0053      0.027     -0.196      0.844      -0.059       0.048
year_1984     -0.0073      0.027     -0.269      0.788      -0.061       0.046
year_1985     -0.0003      0.027     -0.010      0.992      -0.054       0.053
year_1986     -0.0020      0.027     -0.075      0.941      -0.055       0.051
year_1987     -0.0023      0.027     -0.083      0.934      -0.055       0.051
year_1988     -0.0075      0.027     -0.276      0.783      -0.061       0.046
year_1989     -0.0117      0.027     -0.431      0.667      -0.065       0.042
year_1990     -0.0174      0.027     -0.641      0.522      -0.071       0.036
year_1991     -0.0181      0.027     -0.667      0.505      -0.071       0.035
year_1992     -0.0111      0.027     -0.408      0.683      -0.064       0.042
year_1993     -0.0121      0.027     -0.445      0.657      -0.065       0.041
year_1994     -0.0115      0.027     -0.425      0.671      -0.065       0.042
year_1995     -0.0133      0.027     -0.491      0.623      -0.067       0.040
year_1996     -0.0139      0.027     -0.514      0.608      -0.067       0.039
year_1997     -0.0168      0.027     -0.619      0.536      -0.070       0.036
year_1998     -0.0195      0.027     -0.720      0.472      -0.073       0.034
year_1999     -0.0283      0.027     -1.042      0.297      -0.082       0.025
year_2000     -0.0176      0.027     -0.649      0.517      -0.071       0.036
year_2001     -0.0185      0.027     -0.682      0.496      -0.072       0.035
year_2002     -0.0257      0.027     -0.946      0.344      -0.079       0.028
year_2003     -0.0273      0.027     -1.007      0.314      -0.081       0.026
year_2004     -0.0260      0.027     -0.957      0.339      -0.079       0.027
year_2005     -0.0386      0.027     -1.424      0.155      -0.092       0.015
year_2006     -0.0415      0.027     -1.527      0.127      -0.095       0.012
year_2007     -0.0475      0.027     -1.751      0.080      -0.101       0.006
year_2008     -0.0628      0.027     -2.312      0.021      -0.116      -0.010
year_2009     -0.0898      0.027     -3.306      0.001      -0.143      -0.037
year_2010     -0.0802      0.027     -2.953      0.003      -0.133      -0.027
year_2011     -0.0851      0.027     -3.135      0.002      -0.138      -0.032
year_2012     -0.0884      0.027     -3.257      0.001      -0.142      -0.035
year_2013     -0.0766      0.027     -2.823      0.005      -0.130      -0.023
year_2014     -0.0593      0.027     -2.184      0.029      -0.113      -0.006
year_2015     -0.0737      0.027     -2.716      0.007      -0.127      -0.020
year_2016     -0.0777      0.027     -2.862      0.004      -0.131      -0.024
year_2017     -0.0686      0.027     -2.528      0.012      -0.122      -0.015
year_2018     -0.0759      0.027     -2.797      0.005      -0.129      -0.023
year_2019     -0.0763      0.027     -2.810      0.005      -0.130      -0.023
year_2020     -0.0724      0.027     -2.667      0.008      -0.126      -0.019
==============================================================================
Omnibus:                      628.516   Durbin-Watson:                   0.432
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            11200.041
Skew:                          -0.996   Prob(JB):                         0.00
Kurtosis:                      14.374   Cond. No.                     1.03e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.5e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

Time FE — liquid liabilities (within)
                                OLS Regression Results                                
======================================================================================
Dep. Variable:     liquid_liabilities_within   R-squared:                       0.375
Model:                                   OLS   Adj. R-squared:                  0.333
Method:                        Least Squares   F-statistic:                     9.053
Date:                       Sat, 27 Dec 2025   Prob (F-statistic):          1.95e-132
Time:                               17:14:14   Log-Likelihood:                 614.04
No. Observations:                       2224   AIC:                            -950.1
Df Residuals:                           2085   BIC:                            -156.8
Df Model:                                138                                         
Covariance Type:                   nonrobust                                         
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882     -0.2593      0.047     -5.471      0.000      -0.352      -0.166
year_1883     -0.2499      0.047     -5.272      0.000      -0.343      -0.157
year_1884     -0.2346      0.047     -4.949      0.000      -0.328      -0.142
year_1885     -0.2225      0.047     -4.693      0.000      -0.315      -0.130
year_1886     -0.2125      0.047     -4.482      0.000      -0.305      -0.120
year_1887     -0.1996      0.047     -4.211      0.000      -0.293      -0.107
year_1888     -0.1947      0.047     -4.107      0.000      -0.288      -0.102
year_1889     -0.1992      0.047     -4.201      0.000      -0.292      -0.106
year_1890     -0.2061      0.047     -4.347      0.000      -0.299      -0.113
year_1891     -0.2124      0.047     -4.481      0.000      -0.305      -0.119
year_1892     -0.1979      0.047     -4.174      0.000      -0.291      -0.105
year_1893     -0.1837      0.047     -3.875      0.000      -0.277      -0.091
year_1894     -0.1702      0.047     -3.591      0.000      -0.263      -0.077
year_1895     -0.1588      0.047     -3.351      0.001      -0.252      -0.066
year_1896     -0.1540      0.047     -3.248      0.001      -0.247      -0.061
year_1897     -0.1496      0.047     -3.157      0.002      -0.243      -0.057
year_1898     -0.1525      0.047     -3.217      0.001      -0.245      -0.060
year_1899     -0.1408      0.047     -2.971      0.003      -0.234      -0.048
year_1900     -0.1409      0.047     -2.973      0.003      -0.234      -0.048
year_1901     -0.1266      0.047     -2.670      0.008      -0.220      -0.034
year_1902     -0.1152      0.047     -2.430      0.015      -0.208      -0.022
year_1903     -0.1167      0.047     -2.462      0.014      -0.210      -0.024
year_1904     -0.0998      0.047     -2.105      0.035      -0.193      -0.007
year_1905     -0.0864      0.047     -1.822      0.069      -0.179       0.007
year_1906     -0.0837      0.047     -1.765      0.078      -0.177       0.009
year_1907     -0.0959      0.047     -2.024      0.043      -0.189      -0.003
year_1908     -0.0651      0.047     -1.373      0.170      -0.158       0.028
year_1909     -0.0557      0.047     -1.176      0.240      -0.149       0.037
year_1910     -0.0517      0.047     -1.092      0.275      -0.145       0.041
year_1911     -0.0571      0.047     -1.205      0.228      -0.150       0.036
year_1912     -0.0682      0.047     -1.440      0.150      -0.161       0.025
year_1913     -0.0711      0.047     -1.499      0.134      -0.164       0.022
year_1914     -0.0183      0.047     -0.386      0.700      -0.111       0.075
year_1915     -0.0040      0.047     -0.085      0.932      -0.097       0.089
year_1916     -0.0323      0.047     -0.681      0.496      -0.125       0.061
year_1917      0.0049      0.047      0.104      0.917      -0.088       0.098
year_1918      0.0656      0.047      1.383      0.167      -0.027       0.159
year_1919      0.0616      0.047      1.300      0.194      -0.031       0.155
year_1920     -0.0968      0.047     -2.042      0.041      -0.190      -0.004
year_1921     -0.0499      0.047     -1.052      0.293      -0.143       0.043
year_1922     -0.0394      0.047     -0.831      0.406      -0.132       0.054
year_1923     -0.0609      0.047     -1.285      0.199      -0.154       0.032
year_1924     -0.0899      0.047     -1.897      0.058      -0.183       0.003
year_1925     -0.1025      0.047     -2.163      0.031      -0.195      -0.010
year_1926     -0.0850      0.047     -1.794      0.073      -0.178       0.008
year_1927     -0.0662      0.047     -1.396      0.163      -0.159       0.027
year_1928     -0.0660      0.047     -1.392      0.164      -0.159       0.027
year_1929     -0.0638      0.047     -1.347      0.178      -0.157       0.029
year_1930     -0.0205      0.047     -0.433      0.665      -0.113       0.072
year_1931      0.0354      0.047      0.747      0.455      -0.058       0.128
year_1932      0.0766      0.047      1.616      0.106      -0.016       0.170
year_1933      0.0776      0.047      1.637      0.102      -0.015       0.171
year_1934      0.0592      0.047      1.250      0.212      -0.034       0.152
year_1935      0.0491      0.047      1.036      0.300      -0.044       0.142
year_1936      0.0364      0.047      0.768      0.442      -0.057       0.129
year_1937      0.0155      0.047      0.327      0.744      -0.077       0.108
year_1938      0.0440      0.047      0.928      0.354      -0.049       0.137
year_1939      0.0338      0.047      0.714      0.475      -0.059       0.127
year_1940      0.0429      0.047      0.904      0.366      -0.050       0.136
year_1941      0.0726      0.047      1.532      0.126      -0.020       0.166
year_1942      0.0946      0.047      1.996      0.046       0.002       0.188
year_1943      0.1500      0.047      3.165      0.002       0.057       0.243
year_1944      0.2553      0.047      5.386      0.000       0.162       0.348
year_1945      0.2183      0.047      4.604      0.000       0.125       0.311
year_1946      0.0824      0.047      1.738      0.082      -0.011       0.175
year_1947      0.0121      0.047      0.255      0.799      -0.081       0.105
year_1948     -0.0260      0.047     -0.549      0.583      -0.119       0.067
year_1949     -0.0407      0.047     -0.858      0.391      -0.134       0.052
year_1950     -0.0675      0.047     -1.425      0.154      -0.160       0.025
year_1951     -0.1071      0.047     -2.258      0.024      -0.200      -0.014
year_1952     -0.1062      0.047     -2.240      0.025      -0.199      -0.013
year_1953     -0.0974      0.047     -2.055      0.040      -0.190      -0.004
year_1954     -0.0933      0.047     -1.967      0.049      -0.186      -0.000
year_1955     -0.0938      0.047     -1.979      0.048      -0.187      -0.001
year_1956     -0.1086      0.047     -2.291      0.022      -0.202      -0.016
year_1957     -0.1104      0.047     -2.329      0.020      -0.203      -0.017
year_1958     -0.0942      0.047     -1.987      0.047      -0.187      -0.001
year_1959     -0.0857      0.047     -1.808      0.071      -0.179       0.007
year_1960     -0.0837      0.047     -1.766      0.078      -0.177       0.009
year_1961     -0.0813      0.047     -1.715      0.087      -0.174       0.012
year_1962     -0.0748      0.047     -1.578      0.115      -0.168       0.018
year_1963     -0.0641      0.047     -1.353      0.176      -0.157       0.029
year_1964     -0.0639      0.047     -1.349      0.178      -0.157       0.029
year_1965     -0.0618      0.047     -1.304      0.192      -0.155       0.031
year_1966     -0.0602      0.047     -1.269      0.205      -0.153       0.033
year_1967     -0.0519      0.047     -1.095      0.273      -0.145       0.041
year_1968     -0.0402      0.047     -0.848      0.396      -0.133       0.053
year_1969     -0.0440      0.047     -0.928      0.354      -0.137       0.049
year_1970     -0.0555      0.047     -1.171      0.242      -0.148       0.037
year_1971     -0.0380      0.047     -0.801      0.423      -0.131       0.055
year_1972     -0.0160      0.047     -0.337      0.736      -0.109       0.077
year_1973     -0.0121      0.047     -0.255      0.799      -0.105       0.081
year_1974     -0.0331      0.047     -0.698      0.486      -0.126       0.060
year_1975     -0.0258      0.047     -0.543      0.587      -0.119       0.067
year_1976     -0.0237      0.047     -0.500      0.617      -0.117       0.069
year_1977     -0.0286      0.047     -0.603      0.546      -0.122       0.064
year_1978     -0.0037      0.047     -0.077      0.938      -0.097       0.089
year_1979      0.0115      0.047      0.243      0.808      -0.081       0.104
year_1980     -0.0090      0.047     -0.189      0.850      -0.102       0.084
year_1981     -0.0072      0.047     -0.151      0.880      -0.100       0.086
year_1982      0.0134      0.047      0.283      0.777      -0.080       0.106
year_1983      0.0168      0.047      0.353      0.724      -0.076       0.110
year_1984      0.0173      0.047      0.364      0.716      -0.076       0.110
year_1985      0.0237      0.047      0.499      0.618      -0.069       0.117
year_1986      0.0292      0.047      0.617      0.538      -0.064       0.122
year_1987      0.0377      0.047      0.795      0.427      -0.055       0.131
year_1988      0.0425      0.047      0.896      0.371      -0.051       0.135
year_1989      0.0467      0.047      0.984      0.325      -0.046       0.140
year_1990      0.0483      0.047      1.019      0.308      -0.045       0.141
year_1991      0.0504      0.047      1.063      0.288      -0.043       0.143
year_1992      0.0549      0.047      1.159      0.247      -0.038       0.148
year_1993      0.0674      0.047      1.422      0.155      -0.026       0.160
year_1994      0.0580      0.047      1.224      0.221      -0.035       0.151
year_1995      0.0587      0.047      1.239      0.215      -0.034       0.152
year_1996      0.0621      0.047      1.309      0.191      -0.031       0.155
year_1997      0.0584      0.047      1.231      0.218      -0.035       0.151
year_1998      0.0596      0.047      1.256      0.209      -0.033       0.153
year_1999      0.0704      0.047      1.486      0.137      -0.023       0.163
year_2000      0.0569      0.047      1.201      0.230      -0.036       0.150
year_2001      0.0762      0.047      1.606      0.108      -0.017       0.169
year_2002      0.0914      0.047      1.928      0.054      -0.002       0.184
year_2003      0.1107      0.047      2.336      0.020       0.018       0.204
year_2004      0.1167      0.047      2.463      0.014       0.024       0.210
year_2005      0.1378      0.047      2.908      0.004       0.045       0.231
year_2006      0.1549      0.047      3.267      0.001       0.062       0.248
year_2007      0.1864      0.047      3.933      0.000       0.093       0.279
year_2008      0.2170      0.047      4.579      0.000       0.124       0.310
year_2009      0.2579      0.047      5.441      0.000       0.165       0.351
year_2010      0.2527      0.047      5.331      0.000       0.160       0.346
year_2011      0.2581      0.047      5.444      0.000       0.165       0.351
year_2012      0.2832      0.047      5.974      0.000       0.190       0.376
year_2013      0.2941      0.047      6.204      0.000       0.201       0.387
year_2014      0.3054      0.047      6.442      0.000       0.212       0.398
year_2015      0.3298      0.047      6.958      0.000       0.237       0.423
year_2016      0.3545      0.047      7.479      0.000       0.262       0.447
year_2017      0.3639      0.047      7.676      0.000       0.271       0.457
year_2018      0.3644      0.047      7.686      0.000       0.271       0.457
year_2019      0.3782      0.047      7.979      0.000       0.285       0.471
year_2020      0.5318      0.047     11.219      0.000       0.439       0.625
==============================================================================
Omnibus:                      472.680   Durbin-Watson:                   0.125
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2197.817
Skew:                           0.940   Prob(JB):                         0.00
Kurtosis:                       7.492   Cond. No.                         1.00
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Time FE — market cap / GDP (within)
                            OLS Regression Results                            
==============================================================================
Dep. Variable:        mcap_gdp_within   R-squared:                       0.607
Model:                            OLS   Adj. R-squared:                  0.570
Method:                 Least Squares   F-statistic:                     16.67
Date:                Sat, 27 Dec 2025   Prob (F-statistic):          1.03e-198
Time:                        17:14:14   Log-Likelihood:                 339.75
No. Observations:                1488   AIC:                            -425.5
Df Residuals:                    1361   BIC:                             248.3
Df Model:                         126                                         
Covariance Type:            nonrobust                                         
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882  -3.387e-18   1.68e-18     -2.020      0.044   -6.68e-18    -9.8e-20
year_1883  -4.404e+12   1.53e+13     -0.288      0.774   -3.44e+13    2.56e+13
year_1884  -3.106e-18   1.68e-18     -1.852      0.064   -6.39e-18    1.83e-19
year_1885  -1.312e+13   4.23e+13     -0.310      0.757   -9.61e+13    6.99e+13
year_1886  -3.292e-18   1.68e-18     -1.963      0.050   -6.58e-18    -2.9e-21
year_1887   2.126e+12    6.8e+12      0.313      0.754   -1.12e+13    1.55e+13
year_1888  -1.509e-18   1.68e-18     -0.900      0.368    -4.8e-18    1.78e-18
year_1889     -0.0924      0.071     -1.297      0.195      -0.232       0.047
year_1890     -0.1089      0.071     -1.530      0.126      -0.249       0.031
year_1891     -0.0999      0.071     -1.403      0.161      -0.240       0.040
year_1892     -0.0927      0.071     -1.302      0.193      -0.232       0.047
year_1893     -0.0832      0.071     -1.168      0.243      -0.223       0.056
year_1894     -0.0712      0.071     -1.000      0.318      -0.211       0.068
year_1895     -0.0529      0.071     -0.744      0.457      -0.193       0.087
year_1896     -0.0539      0.071     -0.757      0.449      -0.194       0.086
year_1897     -0.0474      0.071     -0.666      0.506      -0.187       0.092
year_1898     -0.1174      0.058     -2.020      0.044      -0.231      -0.003
year_1899  -1.091e+12   1.57e+13     -0.069      0.945   -3.19e+13    2.97e+13
year_1900     -0.0784      0.213     -0.369      0.712      -0.495       0.338
year_1901     -0.1077      0.058     -1.852      0.064      -0.222       0.006
year_1902     -0.0843      0.069     -1.219      0.223      -0.220       0.051
year_1903     -0.1138      0.058     -1.957      0.051      -0.228       0.000
year_1904     -0.0769      0.058     -1.318      0.188      -0.191       0.038
year_1905     -0.0537      0.058     -0.921      0.357      -0.168       0.061
year_1906     -0.0521      0.058     -0.897      0.370      -0.166       0.062
year_1907     -0.0939      0.058     -1.615      0.107      -0.208       0.020
year_1908     -0.0517      0.058     -0.889      0.374      -0.166       0.062
year_1909     -0.0417      0.058     -0.717      0.474      -0.156       0.072
year_1910     -0.0324      0.058     -0.557      0.578      -0.146       0.082
year_1911     -0.0334      0.058     -0.575      0.565      -0.147       0.081
year_1912     -0.0478      0.058     -0.822      0.411      -0.162       0.066
year_1913     -0.0697      0.058     -1.198      0.231      -0.184       0.044
year_1914     -0.0835      0.058     -1.437      0.151      -0.198       0.030
year_1915     -0.0933      0.058     -1.604      0.109      -0.207       0.021
year_1916     -0.1234      0.069     -1.798      0.072      -0.258       0.011
year_1917     -0.1343      0.058     -2.311      0.021      -0.248      -0.020
year_1918     -0.1590      0.058     -2.735      0.006      -0.273      -0.045
year_1919     -0.1685      0.058     -2.899      0.004      -0.283      -0.054
year_1920     -0.2460      0.058     -4.232      0.000      -0.360      -0.132
year_1921     -0.2183      0.058     -3.756      0.000      -0.332      -0.104
year_1922     -0.1769      0.058     -3.043      0.002      -0.291      -0.063
year_1923     -0.1790      0.058     -3.079      0.002      -0.293      -0.065
year_1924     -0.1735      0.058     -2.986      0.003      -0.288      -0.060
year_1925     -0.1624      0.058     -2.794      0.005      -0.276      -0.048
year_1926     -0.1239      0.058     -2.131      0.033      -0.238      -0.010
year_1927     -0.0673      0.058     -1.157      0.247      -0.181       0.047
year_1928     -0.0007      0.058     -0.012      0.990      -0.115       0.113
year_1929     -0.0194      0.058     -0.333      0.739      -0.133       0.095
year_1930     -0.0595      0.058     -1.024      0.306      -0.174       0.055
year_1931     -0.1256      0.058     -2.161      0.031      -0.240      -0.012
year_1932     -0.0846      0.058     -1.455      0.146      -0.199       0.029
year_1933     -0.0230      0.058     -0.395      0.693      -0.137       0.091
year_1934     -0.0337      0.058     -0.579      0.562      -0.148       0.080
year_1935      0.0037      0.058      0.064      0.949      -0.110       0.118
year_1936      0.0668      0.058      1.149      0.251      -0.047       0.181
year_1937     -0.0471      0.058     -0.810      0.418      -0.161       0.067
year_1938     -0.0519      0.058     -0.893      0.372      -0.166       0.062
year_1939     -0.0897      0.058     -1.543      0.123      -0.204       0.024
year_1940     -0.1428      0.058     -2.457      0.014      -0.257      -0.029
year_1941     -0.1637      0.058     -2.816      0.005      -0.278      -0.050
year_1942     -0.1714      0.058     -2.948      0.003      -0.285      -0.057
year_1943     -0.1706      0.058     -2.935      0.003      -0.285      -0.057
year_1944     -0.1584      0.058     -2.725      0.007      -0.272      -0.044
year_1945     -0.1167      0.058     -2.007      0.045      -0.231      -0.003
year_1946     -0.1138      0.058     -1.958      0.050      -0.228       0.000
year_1947     -0.1397      0.058     -2.404      0.016      -0.254      -0.026
year_1948     -0.2036      0.058     -3.503      0.000      -0.318      -0.090
year_1949     -0.2150      0.058     -3.699      0.000      -0.329      -0.101
year_1950     -0.2061      0.058     -3.545      0.000      -0.320      -0.092
year_1951     -0.2070      0.058     -3.562      0.000      -0.321      -0.093
year_1952     -0.2076      0.058     -3.571      0.000      -0.322      -0.094
year_1953     -0.1948      0.058     -3.351      0.001      -0.309      -0.081
year_1954     -0.1273      0.058     -2.190      0.029      -0.241      -0.013
year_1955     -0.1018      0.058     -1.752      0.080      -0.216       0.012
year_1956     -0.1008      0.058     -1.734      0.083      -0.215       0.013
year_1957     -0.1345      0.058     -2.314      0.021      -0.249      -0.020
year_1958     -0.0879      0.058     -1.511      0.131      -0.202       0.026
year_1959     -0.0326      0.058     -0.561      0.575      -0.147       0.081
year_1960     -0.0424      0.058     -0.729      0.466      -0.156       0.072
year_1961      0.0063      0.058      0.108      0.914      -0.108       0.120
year_1962     -0.0494      0.058     -0.850      0.395      -0.163       0.065
year_1963     -0.0296      0.058     -0.510      0.610      -0.144       0.084
year_1964     -0.0415      0.058     -0.714      0.475      -0.156       0.073
year_1965     -0.0577      0.058     -0.992      0.321      -0.172       0.056
year_1966     -0.1137      0.058     -1.957      0.051      -0.228       0.000
year_1967     -0.0688      0.058     -1.184      0.237      -0.183       0.045
year_1968     -0.0199      0.058     -0.342      0.732      -0.134       0.094
year_1969     -0.0326      0.058     -0.561      0.575      -0.147       0.081
year_1970     -0.0900      0.058     -1.548      0.122      -0.204       0.024
year_1971     -0.0871      0.058     -1.498      0.134      -0.201       0.027
year_1972     -0.0384      0.058     -0.660      0.509      -0.152       0.076
year_1973     -0.0751      0.058     -1.293      0.196      -0.189       0.039
year_1974     -0.2085      0.058     -3.587      0.000      -0.323      -0.094
year_1975     -0.1870      0.058     -3.216      0.001      -0.301      -0.073
year_1976     -0.2178      0.058     -3.747      0.000      -0.332      -0.104
year_1977     -0.2443      0.058     -4.203      0.000      -0.358      -0.130
year_1978     -0.2464      0.058     -4.239      0.000      -0.360      -0.132
year_1979     -0.2384      0.058     -4.102      0.000      -0.352      -0.124
year_1980     -0.2094      0.058     -3.602      0.000      -0.323      -0.095
year_1981     -0.2314      0.058     -3.982      0.000      -0.345      -0.117
year_1982     -0.2305      0.058     -3.966      0.000      -0.345      -0.116
year_1983     -0.1719      0.058     -2.958      0.003      -0.286      -0.058
year_1984     -0.1800      0.058     -3.096      0.002      -0.294      -0.066
year_1985     -0.1083      0.058     -1.862      0.063      -0.222       0.006
year_1986     -0.0282      0.058     -0.486      0.627      -0.142       0.086
year_1987     -0.0422      0.058     -0.726      0.468      -0.156       0.072
year_1988     -0.0280      0.058     -0.482      0.630      -0.142       0.086
year_1989      0.0405      0.058      0.697      0.486      -0.073       0.155
year_1990     -0.0681      0.058     -1.172      0.241      -0.182       0.046
year_1991     -0.0310      0.058     -0.533      0.594      -0.145       0.083
year_1992     -0.0322      0.058     -0.554      0.580      -0.146       0.082
year_1993      0.0992      0.058      1.707      0.088      -0.015       0.213
year_1994      0.0653      0.058      1.124      0.261      -0.049       0.179
year_1995      0.1271      0.058      2.187      0.029       0.013       0.241
year_1996      0.2383      0.058      4.099      0.000       0.124       0.352
year_1997      0.4340      0.058      7.466      0.000       0.320       0.548
year_1998      0.5024      0.058      8.642      0.000       0.388       0.616
year_1999      0.6879      0.058     11.835      0.000       0.574       0.802
year_2000      0.6889      0.058     11.851      0.000       0.575       0.803
year_2001      0.4963      0.058      8.537      0.000       0.382       0.610
year_2002      0.2506      0.058      4.312      0.000       0.137       0.365
year_2003      0.3757      0.058      6.463      0.000       0.262       0.490
year_2004      0.4389      0.058      7.550      0.000       0.325       0.553
year_2005      0.5846      0.058     10.057      0.000       0.471       0.699
year_2006      0.7156      0.058     12.311      0.000       0.602       0.830
year_2007      0.6849      0.058     11.782      0.000       0.571       0.799
year_2008      0.2069      0.058      3.560      0.000       0.093       0.321
year_2009      0.4241      0.058      7.297      0.000       0.310       0.538
year_2010      0.4866      0.058      8.372      0.000       0.373       0.601
year_2011      0.3345      0.058      5.755      0.000       0.221       0.449
year_2012      0.3973      0.058      6.834      0.000       0.283       0.511
year_2013      0.5462      0.058      9.397      0.000       0.432       0.660
year_2014      0.5658      0.058      9.733      0.000       0.452       0.680
year_2015      0.5766      0.058      9.920      0.000       0.463       0.691
year_2016      0.6045      0.058     10.399      0.000       0.490       0.719
year_2017           0          0        nan        nan           0           0
year_2018           0          0        nan        nan           0           0
year_2019           0          0        nan        nan           0           0
year_2020           0          0        nan        nan           0           0
==============================================================================
Omnibus:                      681.683   Durbin-Watson:                   0.160
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             6313.782
Skew:                           1.904   Prob(JB):                         0.00
Kurtosis:                      12.345   Cond. No.                     1.05e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 1.1e-31. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.

Time FE — bank loans / GDP (within)
                                OLS Regression Results                                
=================================================================================
Dep. Variable:     bank_loans_gdp_within   R-squared:                       0.682
Model:                               OLS   Adj. R-squared:                  0.654
Method:                    Least Squares   F-statistic:                     23.81
Date:                   Sat, 27 Dec 2025   Prob (F-statistic):          7.71e-292
Time:                           17:14:14   Log-Likelihood:                 818.80
No. Observations:                   1668   AIC:                            -1360.
Df Residuals:                       1529   BIC:                            -606.3
Df Model:                            138                                         
Covariance Type:               nonrobust                                         
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
year_1882     -0.2828      0.045     -6.333      0.000      -0.370      -0.195
year_1883     -0.2797      0.045     -6.264      0.000      -0.367      -0.192
year_1884     -0.2580      0.045     -5.778      0.000      -0.346      -0.170
year_1885     -0.2407      0.045     -5.390      0.000      -0.328      -0.153
year_1886     -0.2255      0.045     -5.049      0.000      -0.313      -0.138
year_1887     -0.2148      0.045     -4.810      0.000      -0.302      -0.127
year_1888     -0.2069      0.045     -4.634      0.000      -0.295      -0.119
year_1889     -0.2015      0.045     -4.511      0.000      -0.289      -0.114
year_1890     -0.2038      0.045     -4.564      0.000      -0.291      -0.116
year_1891     -0.1979      0.045     -4.432      0.000      -0.285      -0.110
year_1892     -0.1849      0.045     -4.140      0.000      -0.272      -0.097
year_1893     -0.1650      0.045     -3.695      0.000      -0.253      -0.077
year_1894     -0.1492      0.045     -3.341      0.001      -0.237      -0.062
year_1895     -0.1430      0.045     -3.202      0.001      -0.231      -0.055
year_1896     -0.1231      0.045     -2.756      0.006      -0.211      -0.035
year_1897     -0.1237      0.045     -2.769      0.006      -0.211      -0.036
year_1898     -0.1159      0.045     -2.596      0.010      -0.204      -0.028
year_1899     -0.0995      0.045     -2.228      0.026      -0.187      -0.012
year_1900     -0.0959      0.045     -2.147      0.032      -0.183      -0.008
year_1901     -0.0844      0.045     -1.891      0.059      -0.172       0.003
year_1902     -0.0722      0.045     -1.616      0.106      -0.160       0.015
year_1903     -0.0638      0.045     -1.428      0.153      -0.151       0.024
year_1904     -0.0258      0.045     -0.578      0.563      -0.113       0.062
year_1905     -0.0169      0.045     -0.378      0.706      -0.104       0.071
year_1906     -0.0051      0.045     -0.114      0.909      -0.093       0.083
year_1907      0.0038      0.045      0.086      0.931      -0.084       0.091
year_1908      0.0321      0.045      0.719      0.472      -0.055       0.120
year_1909      0.0376      0.045      0.842      0.400      -0.050       0.125
year_1910      0.0488      0.045      1.093      0.274      -0.039       0.136
year_1911      0.0497      0.045      1.112      0.266      -0.038       0.137
year_1912      0.0476      0.045      1.065      0.287      -0.040       0.135
year_1913      0.0429      0.045      0.961      0.336      -0.045       0.131
year_1914      0.0718      0.045      1.608      0.108      -0.016       0.159
year_1915      0.0168      0.045      0.377      0.706      -0.071       0.104
year_1916     -0.0672      0.045     -1.505      0.132      -0.155       0.020
year_1917     -0.0793      0.045     -1.775      0.076      -0.167       0.008
year_1918     -0.0835      0.045     -1.869      0.062      -0.171       0.004
year_1919     -0.0647      0.045     -1.449      0.148      -0.152       0.023
year_1920     -0.1321      0.045     -2.957      0.003      -0.220      -0.044
year_1921     -0.0478      0.045     -1.071      0.284      -0.135       0.040
year_1922     -0.0523      0.045     -1.171      0.242      -0.140       0.035
year_1923     -0.0710      0.045     -1.590      0.112      -0.159       0.017
year_1924     -0.1028      0.045     -2.302      0.021      -0.190      -0.015
year_1925     -0.1060      0.045     -2.374      0.018      -0.194      -0.018
year_1926     -0.0616      0.045     -1.379      0.168      -0.149       0.026
year_1927     -0.0349      0.045     -0.781      0.435      -0.122       0.053
year_1928     -0.0305      0.045     -0.683      0.495      -0.118       0.057
year_1929     -0.0198      0.045     -0.443      0.658      -0.107       0.068
year_1930      0.0190      0.045      0.426      0.670      -0.069       0.107
year_1931      0.0774      0.045      1.732      0.083      -0.010       0.165
year_1932      0.1000      0.045      2.239      0.025       0.012       0.188
year_1933      0.0699      0.045      1.565      0.118      -0.018       0.157
year_1934      0.0286      0.045      0.641      0.522      -0.059       0.116
year_1935      0.0045      0.045      0.101      0.919      -0.083       0.092
year_1936     -0.0237      0.045     -0.532      0.595      -0.111       0.064
year_1937     -0.0702      0.045     -1.573      0.116      -0.158       0.017
year_1938     -0.0761      0.045     -1.704      0.089      -0.164       0.011
year_1939     -0.0883      0.045     -1.977      0.048      -0.176      -0.001
year_1940     -0.1357      0.045     -3.038      0.002      -0.223      -0.048
year_1941     -0.1864      0.045     -4.174      0.000      -0.274      -0.099
year_1942     -0.2231      0.045     -4.995      0.000      -0.311      -0.135
year_1943     -0.2566      0.045     -5.746      0.000      -0.344      -0.169
year_1944     -0.2716      0.045     -6.082      0.000      -0.359      -0.184
year_1945     -0.2944      0.045     -6.592      0.000      -0.382      -0.207
year_1946     -0.3033      0.045     -6.791      0.000      -0.391      -0.216
year_1947     -0.3110      0.045     -6.965      0.000      -0.399      -0.223
year_1948     -0.3048      0.045     -6.826      0.000      -0.392      -0.217
year_1949     -0.2878      0.045     -6.445      0.000      -0.375      -0.200
year_1950     -0.2748      0.045     -6.153      0.000      -0.362      -0.187
year_1951     -0.2822      0.045     -6.320      0.000      -0.370      -0.195
year_1952     -0.2721      0.045     -6.093      0.000      -0.360      -0.185
year_1953     -0.2598      0.045     -5.817      0.000      -0.347      -0.172
year_1954     -0.2464      0.045     -5.518      0.000      -0.334      -0.159
year_1955     -0.2393      0.045     -5.359      0.000      -0.327      -0.152
year_1956     -0.2370      0.045     -5.307      0.000      -0.325      -0.149
year_1957     -0.2345      0.045     -5.251      0.000      -0.322      -0.147
year_1958     -0.2216      0.045     -4.962      0.000      -0.309      -0.134
year_1959     -0.2054      0.045     -4.601      0.000      -0.293      -0.118
year_1960     -0.1785      0.045     -3.997      0.000      -0.266      -0.091
year_1961     -0.1712      0.045     -3.834      0.000      -0.259      -0.084
year_1962     -0.1525      0.045     -3.415      0.001      -0.240      -0.065
year_1963     -0.1318      0.045     -2.950      0.003      -0.219      -0.044
year_1964     -0.1282      0.045     -2.871      0.004      -0.216      -0.041
year_1965     -0.1174      0.045     -2.628      0.009      -0.205      -0.030
year_1966     -0.1130      0.045     -2.531      0.011      -0.201      -0.025
year_1967     -0.1021      0.045     -2.287      0.022      -0.190      -0.015
year_1968     -0.0912      0.045     -2.043      0.041      -0.179      -0.004
year_1969     -0.0807      0.045     -1.806      0.071      -0.168       0.007
year_1970     -0.0904      0.045     -2.025      0.043      -0.178      -0.003
year_1971     -0.0750      0.045     -1.679      0.093      -0.163       0.013
year_1972     -0.0588      0.045     -1.317      0.188      -0.146       0.029
year_1973     -0.0525      0.045     -1.175      0.240      -0.140       0.035
year_1974     -0.0447      0.045     -1.000      0.317      -0.132       0.043
year_1975     -0.0416      0.045     -0.932      0.351      -0.129       0.046
year_1976     -0.0438      0.045     -0.981      0.327      -0.131       0.044
year_1977     -0.0353      0.045     -0.790      0.430      -0.123       0.052
year_1978     -0.0281      0.045     -0.629      0.529      -0.116       0.060
year_1979     -0.0200      0.045     -0.448      0.654      -0.108       0.068
year_1980     -0.0198      0.045     -0.444      0.657      -0.107       0.068
year_1981     -0.0120      0.045     -0.269      0.788      -0.100       0.076
year_1982     -0.0071      0.045     -0.158      0.875      -0.095       0.081
year_1983     -0.0019      0.045     -0.042      0.967      -0.089       0.086
year_1984      0.0202      0.045      0.451      0.652      -0.067       0.108
year_1985      0.0568      0.045      1.272      0.203      -0.031       0.144
year_1986      0.1037      0.045      2.323      0.020       0.016       0.191
year_1987      0.1421      0.045      3.183      0.001       0.055       0.230
year_1988      0.1776      0.045      3.977      0.000       0.090       0.265
year_1989      0.2099      0.045      4.701      0.000       0.122       0.298
year_1990      0.2251      0.045      5.040      0.000       0.137       0.313
year_1991      0.2140      0.045      4.793      0.000       0.126       0.302
year_1992      0.2167      0.045      4.852      0.000       0.129       0.304
year_1993      0.2064      0.045      4.621      0.000       0.119       0.294
year_1994      0.1743      0.045      3.903      0.000       0.087       0.262
year_1995      0.1691      0.045      3.787      0.000       0.082       0.257
year_1996      0.1674      0.045      3.748      0.000       0.080       0.255
year_1997      0.1728      0.045      3.870      0.000       0.085       0.260
year_1998      0.1931      0.045      4.323      0.000       0.105       0.281
year_1999      0.2093      0.045      4.688      0.000       0.122       0.297
year_2000      0.2096      0.045      4.693      0.000       0.122       0.297
year_2001      0.2318      0.045      5.191      0.000       0.144       0.319
year_2002      0.2529      0.045      5.663      0.000       0.165       0.340
year_2003      0.2743      0.045      6.143      0.000       0.187       0.362
year_2004      0.2877      0.045      6.442      0.000       0.200       0.375
year_2005      0.3245      0.045      7.267      0.000       0.237       0.412
year_2006      0.3564      0.045      7.982      0.000       0.269       0.444
year_2007      0.4041      0.045      9.050      0.000       0.317       0.492
year_2008      0.4376      0.045      9.800      0.000       0.350       0.525
year_2009      0.4828      0.045     10.812      0.000       0.395       0.570
year_2010      0.4506      0.045     10.091      0.000       0.363       0.538
year_2011      0.4381      0.045      9.810      0.000       0.350       0.526
year_2012      0.4381      0.045      9.810      0.000       0.350       0.526
year_2013      0.4398      0.045      9.849      0.000       0.352       0.527
year_2014      0.4522      0.045     10.125      0.000       0.365       0.540
year_2015      0.4656      0.045     10.427      0.000       0.378       0.553
year_2016      0.4767      0.045     10.674      0.000       0.389       0.564
year_2017      0.4719      0.045     10.568      0.000       0.384       0.560
year_2018      0.5182      0.045     11.605      0.000       0.431       0.606
year_2019      0.5240      0.045     11.735      0.000       0.436       0.612
year_2020      0.5910      0.045     13.235      0.000       0.503       0.679
==============================================================================
Omnibus:                       96.861   Durbin-Watson:                   0.085
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              135.646
Skew:                           0.509   Prob(JB):                     3.51e-30
Kurtosis:                       3.956   Cond. No.                         1.00
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```
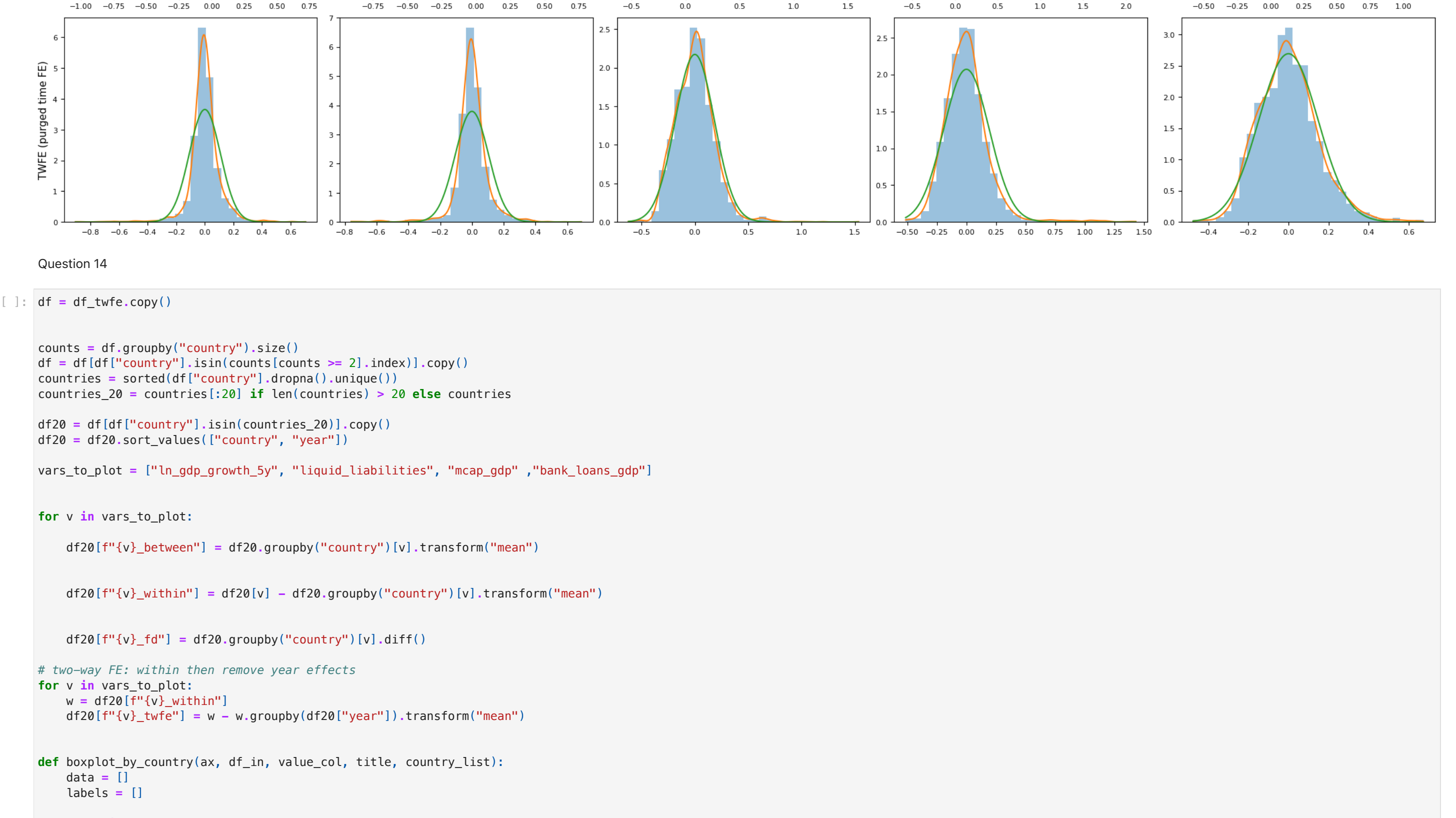
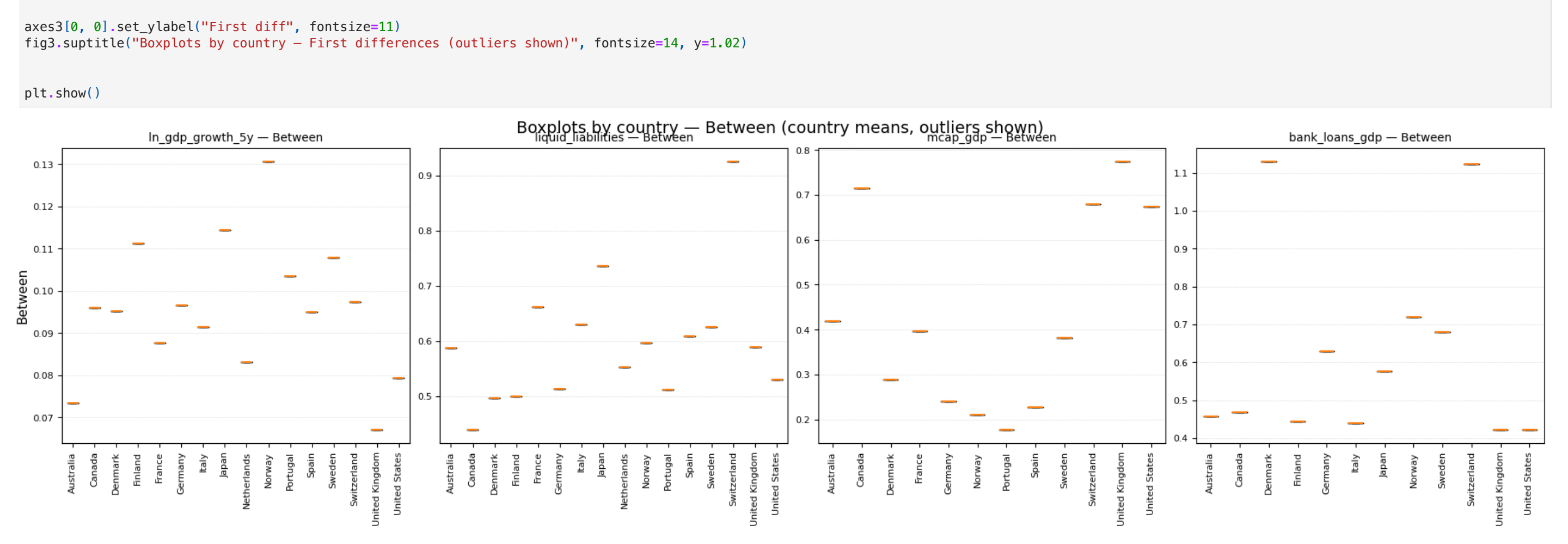

Question 13

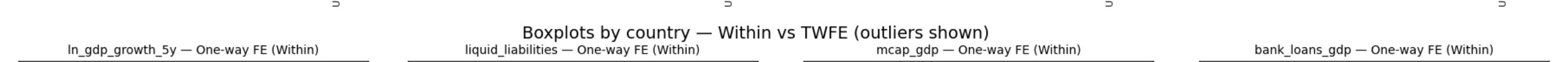
In [ ]:
```python
df_twfe["ln_gdp_growth_5y_twfe"] = np.nan
df_twfe.loc[dummy_reg_gdp, "ln_gdp_growth_5y_twfe"] = ln_gdp_time_fe.resid

df_twfe["ln_labour_productivity_growth_5y_twfe"] = np.nan
df_twfe.loc[dummy_reg_gdp_labprod, "ln_labour_productivity_growth_5y_twfe"] = ln_labprod_time_fe.resid

df_twfe["liquid_liabilities_twfe"] = np.nan
df_twfe.loc[dummy_reg_liquid_liabilities, "liquid_liabilities_twfe"] = liquid_liability_time_fe.resid

df_twfe["mcap_gdp_twfe"] = np.nan
df_twfe.loc[dummy_reg_mcap, "mcap_gdp_twfe"] = mcap_time_fe.resid

df_twfe["bank_loans_gdp_twfe"] = np.nan
df_twfe.loc[dummy_reg_bank, "bank_loans_gdp_twfe"] = bank_loans_time_fe.resid

df_twfe["ln_gdp_growth_5y_between"] = df_twfe.groupby("country")["ln_gdp_growth_5y"].transform("mean")
df_twfe["ln_labour_productivity_growth_5y_between"] = df_twfe.groupby("country")["ln_labour_productivity_growth_5y"].transform("mean")
df_twfe["liquid_liabilities_between"] = df_twfe.groupby("country")["liquid_liabilities"].transform("mean")
df_twfe["mcap_gdp_between"] = df_twfe.groupby("country")["mcap_gdp"].transform("mean")
df_twfe["bank_loans_gdp_between"] = df_twfe.groupby("country")["bank_loans_gdp"].transform("mean")

vars_info = [
    ("ln_gdp_growth_5y", "ln_gdp_growth_5y_between", "ln_gdp_growth_5y_within", "ln_gdp_growth_5y_twfe", "GDP growth (5y, log diff)"),
    ("ln_labour_productivity_growth_5y", "ln_labour_productivity_growth_5y_between", "ln_labour_productivity_growth_5y_within", "ln_labour_productivity_growth_5y_twfe", "Labour prod growth (5y)"),
    ("liquid_liabilities", "liquid_liabilities_between", "liquid_liabilities_within", "liquid_liabilities_twfe", "Liquid liabilities"),
    ("mcap_gdp", "mcap_gdp_between", "mcap_gdp_within", "mcap_gdp_twfe", "Market cap / GDP"),
    ("bank_loans_gdp", "bank_loans_gdp_between", "bank_loans_gdp_within", "bank_loans_gdp_twfe", "Bank loans / GDP"),
]

row_names = ["Between", "Within (1-way FE)", "TWFE (purged time FE)"]

nvars = len(vars_info)
fig, axes = plt.subplots(3, nvars, figsize=(4.2*nvars, 10.5), constrained_layout=True)

def plot_dist(ax, s, title=None):
    s = pd.Series(s).dropna().astype(float)

    if s.empty or s.var(ddof=1) == 0:
        ax.text(0.5, 0.5, "No variation / empty", ha="center", va="center")
        ax.set_axis_off()
        return


    ax.hist(s, bins=30, density=True, alpha=0.45)


    kde = gaussian_kde(s.values)
    x = np.linspace(s.min(), s.max(), 400)
    ax.plot(x, kde(x), linewidth=1.5)


    mu = s.mean()
    sigma = s.std(ddof=1)
    ax.plot(x, norm.pdf(x, loc=mu, scale=sigma), linewidth=1.5)

    if title is not None:
        ax.set_title(title, fontsize=10)

    ax.tick_params(axis="both", labelsize=8)

for j, (base, between_col, within_col, twfe_col, nice) in enumerate(vars_info):
    plot_dist(axes[0, j], df_twfe[between_col], title=nice)   # Between
    plot_dist(axes[1, j], df_twfe[within_col])                 # Within
    plot_dist(axes[2, j], df_twfe[twfe_col])                   # TWFE

for i in range(3):
    axes[i, 0].set_ylabel(row_names[i], fontsize=11)

plt.suptitle("Distributions: Between vs Within vs TWFE (Hist + KDE + Normal fit)", fontsize=14)
plt.show()
```

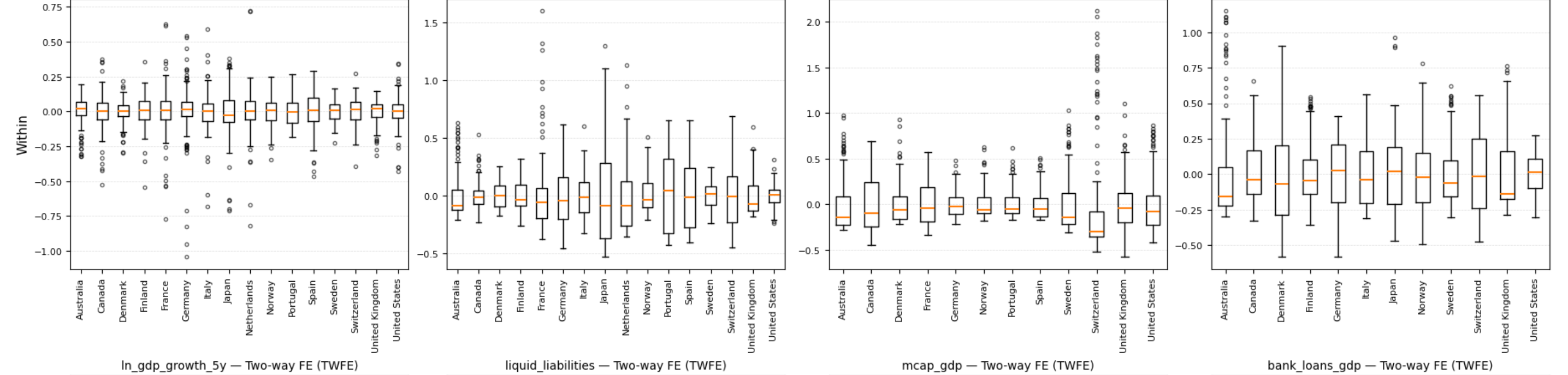


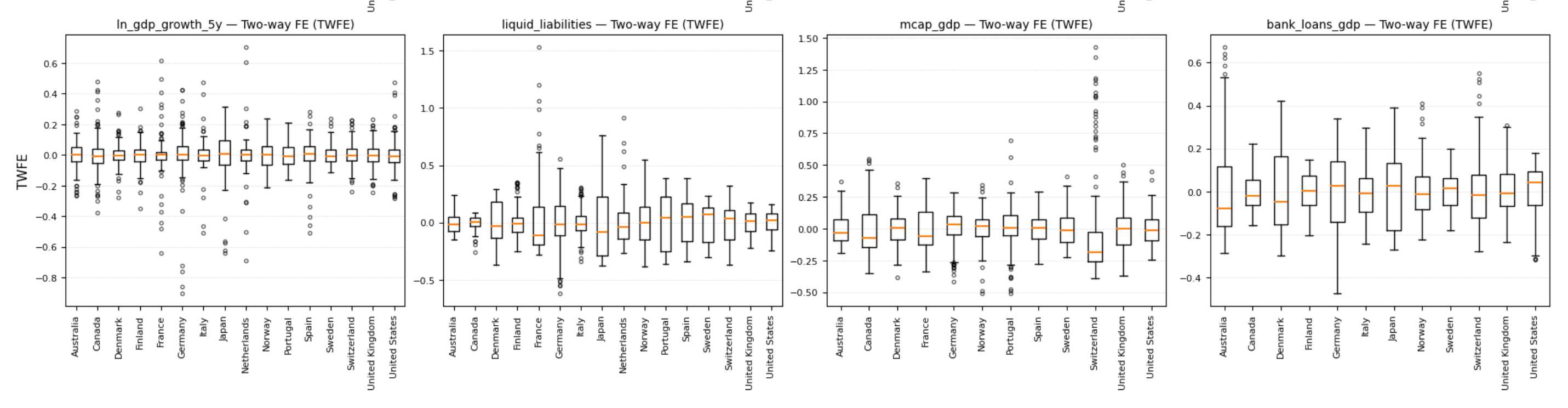
Question 14

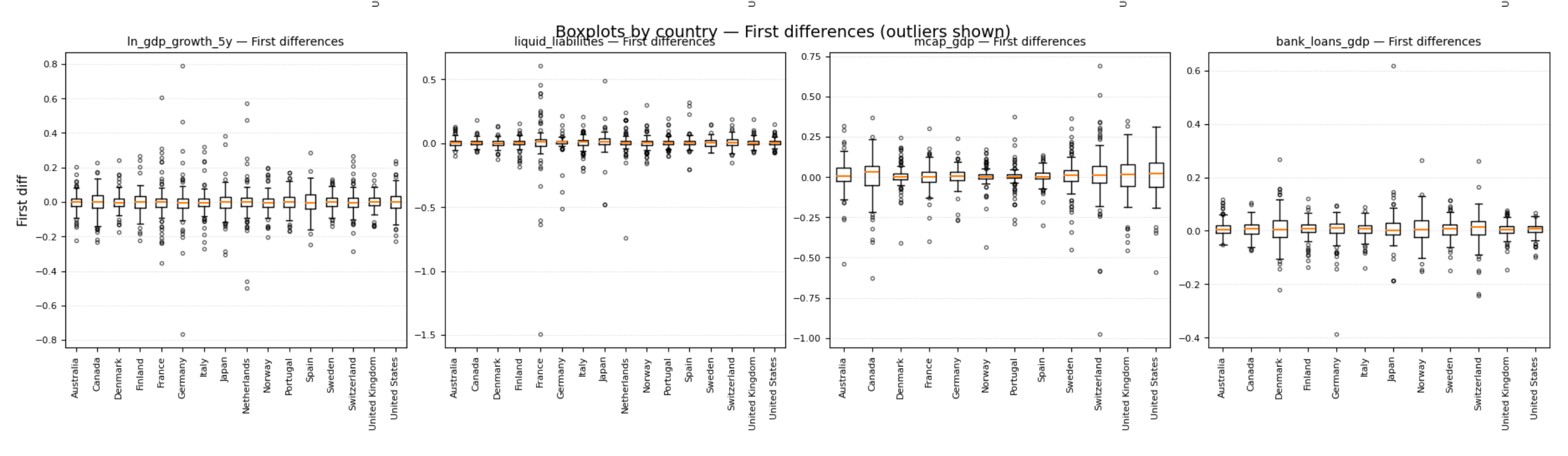
In [ ]:
```python
df = df_twfe.copy()


counts = df.groupby("country").size()
df = df[df["country"].isin(counts[counts >= 2].index)].copy()
countries = sorted(df["country"].dropna().unique())
countries_20 = countries[:20] if len(countries) > 20 else countries

df20 = df[df["country"].isin(countries_20)].copy()
df20 = df20.sort_values(["country", "year"])

vars_to_plot = ["ln_gdp_growth_5y", "liquid_liabilities", "mcap_gdp" ,"bank_loans_gdp"]


for v in vars_to_plot:

    df20[f"{v}_between"] = df20.groupby("country")[v].transform("mean")


    df20[f"{v}_within"] = df20[v] - df20.groupby("country")[v].transform("mean")


    df20[f"{v}_fd"] = df20.groupby("country")[v].diff()

# two-way FE: within then remove year effects
for v in vars_to_plot:
    w = df20[f"{v}_within"]
    df20[f"{v}_twfe"] = w - w.groupby(df20["year"]).transform("mean")


def boxplot_by_country(ax, df_in, value_col, title, country_list):
    data = []
    labels = []

    for c in country_list:
        s = df_in.loc[df_in["country"] == c, value_col].dropna()
        if len(s) > 0:
            data.append(s.to_numpy())
            labels.append(str(c))

    if len(data) == 0:
        ax.set_title(title, fontsize=10)
        ax.text(0.5, 0.5, "No data", ha="center", va="center", transform=ax.transAxes)
        ax.set_xticks([])
        return

    ax.boxplot(
        data,
        showfliers=True,
        flierprops=dict(marker="o", markersize=3, alpha=0.6),
        medianprops=dict(linewidth=1.5),
        whiskerprops=dict(linewidth=1.0),
        capprops=dict(linewidth=1.0),
        boxprops=dict(linewidth=1.0),
    )
    ax.set_title(title, fontsize=10)
    ax.set_xticklabels(labels, rotation=90, fontsize=8)
    ax.tick_params(axis="y", labelsize=8)
    ax.yaxis.grid(True, linestyle="--", linewidth=0.5, alpha=0.4)


def _axes_2d(axes, nrows, ncols):

    axes = np.array(axes)
    if nrows == 1 and ncols == 1:
        return axes.reshape(1, 1)
    if nrows == 1:
        return axes.reshape(1, -1)
    if ncols == 1:
        return axes.reshape(-1, 1)
    return axes

# between
cols = len(vars_to_plot)
fig1, axes1 = plt.subplots(1, cols, figsize=(4.6 * cols, 4.8), constrained_layout=True)
axes1 = _axes_2d(axes1, 1, cols)

for j, v in enumerate(vars_to_plot):
    boxplot_by_country(axes1[0, j], df20, f"{v}_between", f"{v} — Between", countries_20)

axes1[0, 0].set_ylabel("Between", fontsize=11)
fig1.suptitle("Boxplots by country — Between (country means, outliers shown)", fontsize=14, y=1.02)

plt.show()

# within
fig2, axes2 = plt.subplots(2, cols, figsize=(4.6 * cols, 9.0), constrained_layout=True)
axes2 = _axes_2d(axes2, 2, cols)

for j, v in enumerate(vars_to_plot):
    boxplot_by_country(axes2[0, j], df20, f"{v}_within", f"{v} — One-way FE (Within)", countries_20)
    boxplot_by_country(axes2[1, j], df20, f"{v}_twfe",   f"{v} — Two-way FE (TWFE)",   countries_20)

axes2[0, 0].set_ylabel("Within", fontsize=11)
axes2[1, 0].set_ylabel("TWFE", fontsize=11)
fig2.suptitle("Boxplots by country — Within vs TWFE (outliers shown)", fontsize=14, y=1.02)

plt.show()

#firsyt diff
fig3, axes3 = plt.subplots(1, cols, figsize=(4.6 * cols, 4.8), constrained_layout=True)
axes3 = _axes_2d(axes3, 1, cols)

for j, v in enumerate(vars_to_plot):
    boxplot_by_country(axes3[0, j], df20, f"{v}_fd", f"{v} — First differences", countries_20)

axes3[0, 0].set_ylabel("First diff", fontsize=11)
fig3.suptitle("Boxplots by country — First differences (outliers shown)", fontsize=14, y=1.02)

plt.show()
```











Question 14 --> Dessy Stats

In [ ]:
```
def descriptive_table(df, col):
    """
    Return a Series with descriptive stats for df[col].
    Handles empty / all-NaN columns safely.
    """
    # select series and drop missing values
    s = df[col].dropna().astype(float)

    n = len(s)
    if n == 0:
        return pd.Series({
            "Mean": np.nan,
            "SE": np.nan,
            "Q1": np.nan,
            "Median": np.nan,
            "Q3": np.nan,
            "Std_MAX": np.nan,
            "Std_MIN": np.nan
        })

    mean = s.mean()
    se = s.std(ddof=1) / np.sqrt(n) if n > 1 else np.nan

    #
    q1 = s.quantile(0.25)
    median = s.median()
    q3 = s.quantile(0.75)

    std_max = (s.max() - mean) / se if se and not np.isclose(se, 0) else np.nan
    std_min = (s.min() - mean) / se if se and not np.isclose(se, 0) else np.nan

    return pd.Series({
        "Mean": mean,
        "SE": se,
        "Q1": q1,
        "Median": median,
        "Q3": q3,
        "Std_MAX": std_max,
        "Std_MIN": std_min
    })


tables = {}

for v in vars_to_plot:
    tables[v] = pd.DataFrame({
        "Between": descriptive_table(df20, f"{v}_between"),
        "Within": descriptive_table(df20, f"{v}_within"),
        "TWFE": descriptive_table(df20, f"{v}_twfe"),
        "FirstDiff": descriptive_table(df20, f"{v}_fd"),
    })

for v in ["ln_gdp_growth_5y", "liquid_liabilities", "mcap_gdp", "bank_loans_gdp"]:
    print(f"\n=== {v} ===")
    print(tables[v])
```

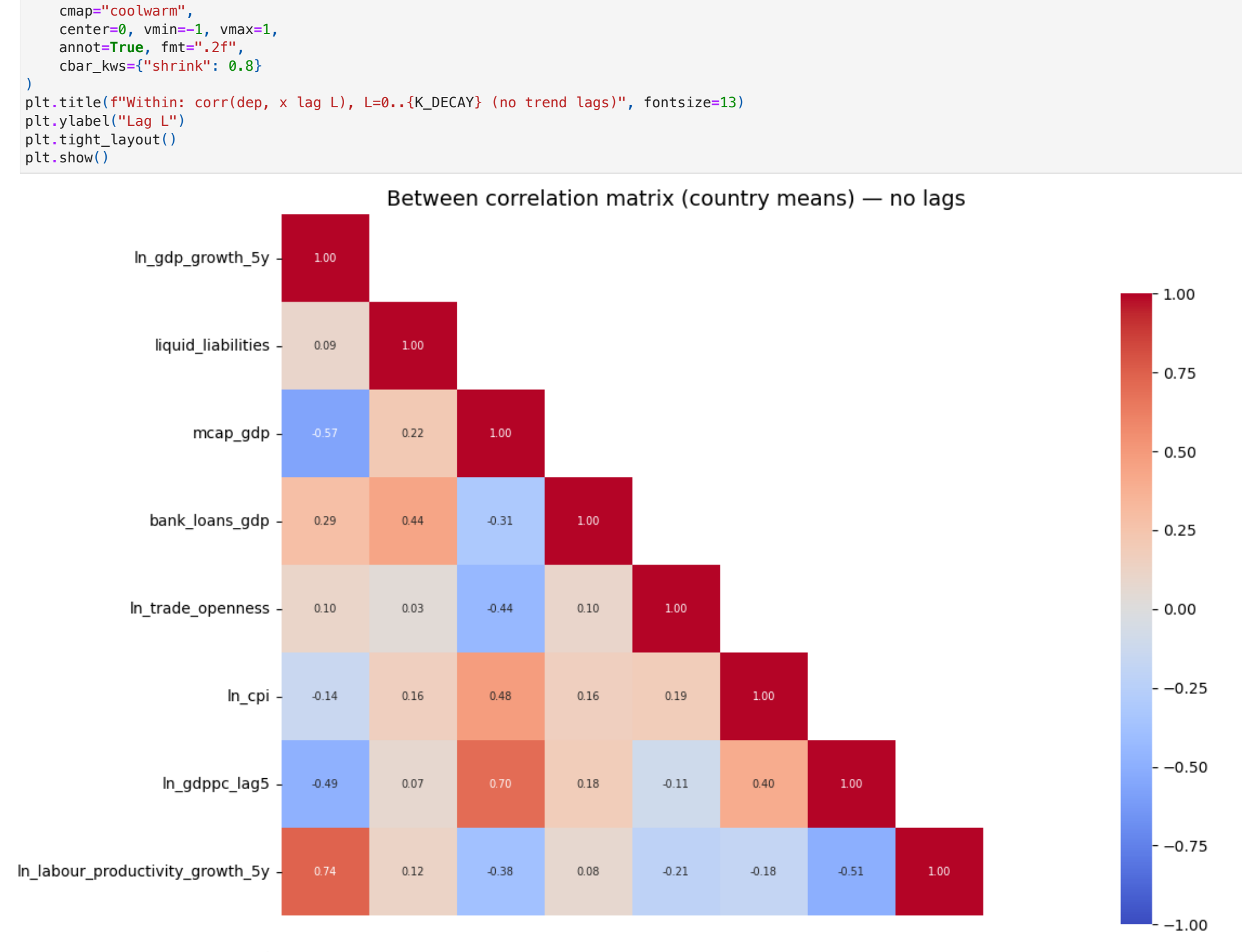
```
=== ln_gdp_growth_5y ===
         Between    Within      TWFE  FirstDiff
Mean      0.0956    0.0000    0.0000    -0.0004
SE        0.0003    0.0028    0.0024     0.0016
Q1        0.0866   -0.0574   -0.0415    -0.0289
Median    0.0956    0.0064   -0.0023    -0.0017
Q3        0.1047    0.0617    0.0418     0.0267
Std_MAX 106.5262  255.6753  299.3752   489.1175
Std_MIN -86.7047 -370.2491 -384.5361  -473.1952

=== liquid_liabilities ===
         Between    Within      TWFE   FirstDiff
Mean      0.5938   -0.0000    0.0000      0.0057
SE        0.0024    0.0049    0.0039      0.0014
Q1        0.5121   -0.1393   -0.1271     -0.0107
Median    0.5888   -0.0253   -0.0005      0.0056
Q3        0.6261    0.1109    0.1041      0.0224
Std_MAX 140.3621  324.6846  393.7554    420.9124
Std_MIN -65.6070 -106.8822 -159.7807 -1,046.0670

=== mcap_gdp ===
         Between   Within      TWFE  FirstDiff
Mean      0.4319  -0.0000   -0.0000     0.0061
SE        0.0052   0.0080    0.0050     0.0028
Q1        0.2362  -0.1747   -0.1155    -0.0237
Median    0.3893  -0.0742   -0.0116     0.0041
Q3        0.6758   0.0952    0.0836     0.0445
Std_MAX  66.2633 265.8348  286.0876   249.4248
Std_MIN -49.2983 -72.4132 -102.6897  -356.5606

=== bank_loans_gdp ===
         Between   Within      TWFE  FirstDiff
Mean      0.6261   0.0000   -0.0000     0.0063
SE        0.0060   0.0064    0.0036     0.0011
Q1        0.4420  -0.1897   -0.1039    -0.0112
Median    0.5229  -0.0508   -0.0070     0.0072
Q3        0.6909   0.1562    0.0868     0.0249
Std_MAX  83.9762 178.6632  185.6218   541.8276
Std_MIN -34.0340 -90.4951 -131.4073  -348.0149
```

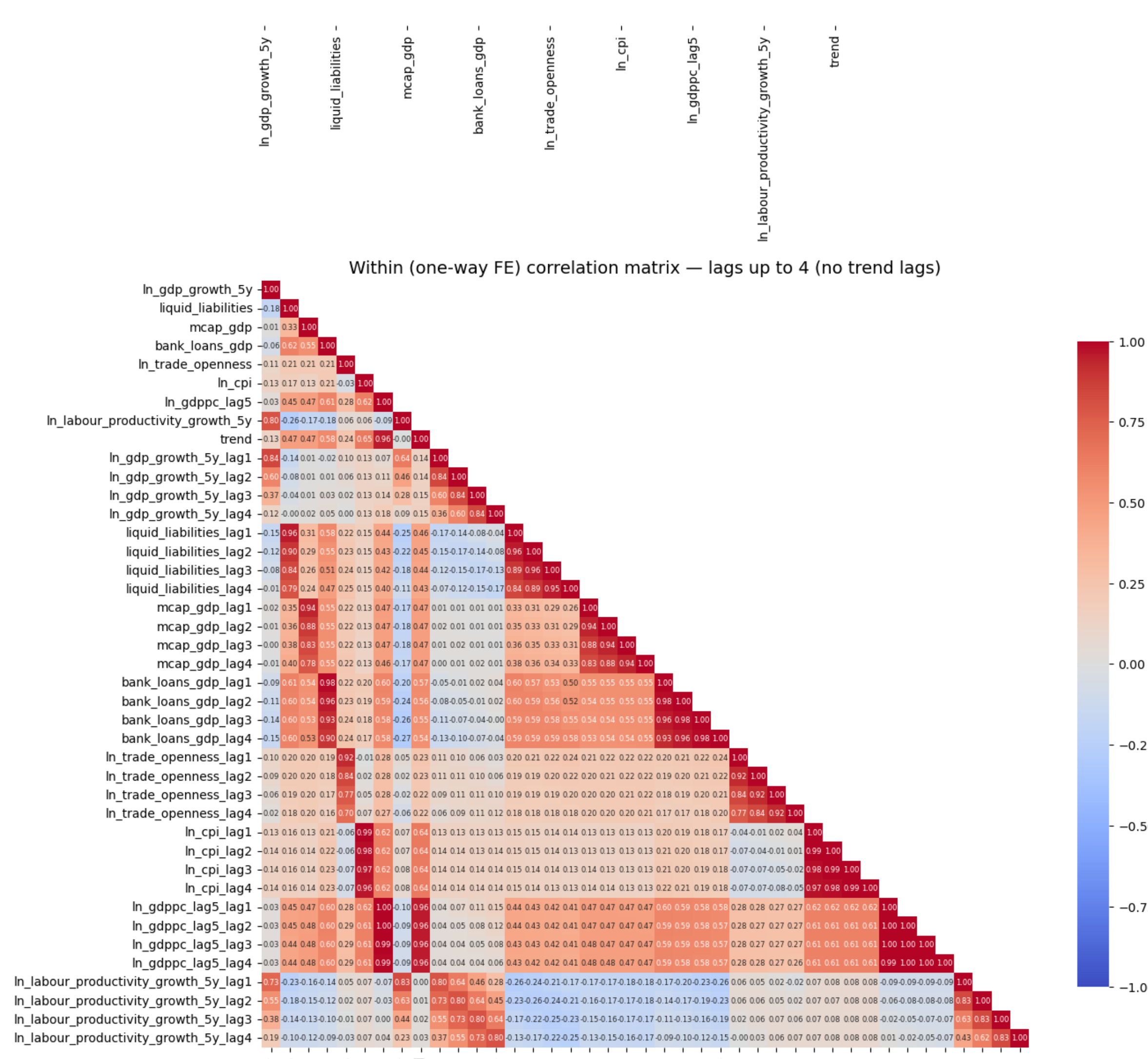
Question 16

In [171…
```
df_Q16 = df_twfe.copy()
df_Q16["trend"] = df_Q16.groupby("country").cumcount() + 1
dep_var = "ln_gdp_growth_5y"
x_vars = [
    "liquid_liabilities",
    "mcap_gdp",
    "bank_loans_gdp",
    "ln_trade_openness",
    "ln_cpi",
    "ln_gdppc_lag5",
    "ln_labour_productivity_growth_5y"
]
all_vars = [dep_var] + x_vars + ["trend"]


for v in [dep_var] + x_vars:
    df_Q16[f"{v}_lag1"] = df_Q16.groupby("country")[v].shift(1)

lag_vars = [f"{v}_lag1" for v in [dep_var] + x_vars]


df_between = (
    df_Q16
   .groupby("country")[all_vars + lag_vars].mean())

corr_between = df_between.corr()

corr_dep_between = (corr_between.loc[dep_var].sort_values())


df_within = df_Q16.copy()

# demean by country
for v in all_vars + lag_vars:
    df_within[f"{v}_within"] = (
        df_within[v] - df_within.groupby("country")[v].transform("mean")
    )

within_vars = [f"{v}_within" for v in all_vars + lag_vars]

corr_within = df_within[within_vars].corr()

# Correlations with dependent variable
corr_dep_within = (
    corr_within.loc[f"{dep_var}_within"]
    .sort_values()
)


corr_between, corr_within, corr_dep_between, corr_dep_within
```

Out[171…
```
(                                       ln_gdp_growth_5y  liquid_liabilities  \
 ln_gdp_growth_5y                                1.0000              0.0942
 liquid_liabilities                              0.0942              1.0000
 mcap_gdp                                       -0.5680              0.2152
 bank_loans_gdp                                  0.2851              0.4389
 ln_trade_openness                               0.1020              0.0317
 ln_cpi                                         -0.1434              0.1583
 ln_gdppc_lag5                                  -0.4902              0.0733
 ln_labour_productivity_growth_5y                0.7401              0.1193
 trend                                              NaN                 NaN
 ln_gdp_growth_5y_lag1                           0.9999              0.0973
 liquid_liabilities_lag1                         0.0968              0.9999
 mcap_gdp_lag1                                  -0.5680              0.2152
 bank_loans_gdp_lag1                             0.2834              0.4377
 ln_trade_openness_lag1                          0.1020              0.0317
 ln_cpi_lag1                                    -0.1431              0.1587
 ln_gdppc_lag5_lag1                             -0.4937              0.0723
 ln_labour_productivity_growth_5y_lag1           0.7398              0.1210

                                        mcap_gdp  bank_loans_gdp  \
 ln_gdp_growth_5y                        -0.5680          0.2851
 liquid_liabilities                       0.2152          0.4389
 mcap_gdp                                 1.0000         -0.3147
 bank_loans_gdp                          -0.3147          1.0000
 ln_trade_openness                       -0.4438          0.0965
 ln_cpi                                   0.4840          0.1569
 ln_gdppc_lag5                            0.7031          0.1813
 ln_labour_productivity_growth_5y        -0.3787          0.0751
 trend                                       NaN             NaN
 ln_gdp_growth_5y_lag1                   -0.5655          0.2800
 liquid_liabilities_lag1                  0.2134          0.4458
 mcap_gdp_lag1                            1.0000         -0.3147
 bank_loans_gdp_lag1                     -0.3131          1.0000
 ln_trade_openness_lag1                  -0.4437          0.0966
 ln_cpi_lag1                              0.4842          0.1572
 ln_gdppc_lag5_lag1                       0.7050          0.1789
 ln_labour_productivity_growth_5y_lag1   -0.3849          0.0734

                                        ln_trade_openness  ln_cpi  \
 ln_gdp_growth_5y                                  0.1020 -0.1434
 liquid_liabilities                                0.0317  0.1583
 mcap_gdp                                         -0.4438  0.4840
 bank_loans_gdp                                    0.0965  0.1569
 ln_trade_openness                                 1.0000  0.1861
 ln_cpi                                            0.1861  1.0000
 ln_gdppc_lag5                                    -0.1071  0.4035
 ln_labour_productivity_growth_5y                 -0.2136 -0.1751
 trend                                                NaN     NaN
 ln_gdp_growth_5y_lag1                             0.0993 -0.1442
 liquid_liabilities_lag1                           0.0387  0.1627
 mcap_gdp_lag1                                    -0.4438  0.4840
 bank_loans_gdp_lag1                               0.0969  0.1547
 ln_trade_openness_lag1                            1.0000  0.1862
 ln_cpi_lag1                                       0.1858  1.0000
 ln_gdppc_lag5_lag1                               -0.1079  0.4033
 ln_labour_productivity_growth_5y_lag1            -0.2103 -0.1758

                                        ln_gdppc_lag5  \
 ln_gdp_growth_5y                             -0.4902
 liquid_liabilities                            0.0733
 mcap_gdp                                      0.7031
 bank_loans_gdp                                0.1813
 ln_trade_openness                            -0.1071
 ln_cpi                                        0.4035
 ln_gdppc_lag5                                 1.0000
 ln_labour_productivity_growth_5y             -0.5062
 trend                                            NaN
 ln_gdp_growth_5y_lag1                        -0.4931
 liquid_liabilities_lag1                       0.0800
 mcap_gdp_lag1                                 0.7031
 bank_loans_gdp_lag1                           0.1830
 ln_trade_openness_lag1                       -0.1068
 ln_cpi_lag1                                   0.4036
 ln_gdppc_lag5_lag1                            1.0000
 ln_labour_productivity_growth_5y_lag1        -0.5097

                                        ln_labour_productivity_growth_5y  \
 ln_gdp_growth_5y                                                 0.7401
 liquid_liabilities                                               0.1193
 mcap_gdp                                                        -0.3787
 bank_loans_gdp                                                   0.0751
 ln_trade_openness                                               -0.2136
 ln_cpi                                                          -0.1751
 ln_gdppc_lag5                                                   -0.5062
 ln_labour_productivity_growth_5y                                 1.0000
 trend                                                               NaN
 ln_gdp_growth_5y_lag1                                            0.7405
 liquid_liabilities_lag1                                          0.1190
 mcap_gdp_lag1                                                   -0.3787
 bank_loans_gdp_lag1                                              0.0748
 ln_trade_openness_lag1                                          -0.2136
 ln_cpi_lag1                                                     -0.1747
 ln_gdppc_lag5_lag1                                              -0.5076
 ln_labour_productivity_growth_5y_lag1                            0.9999

                                        trend  ln_gdp_growth_5y_lag1  \
 ln_gdp_growth_5y                         NaN                 0.9999
 liquid_liabilities                       NaN                 0.0973
 mcap_gdp                                 NaN                -0.5655
 bank_loans_gdp                           NaN                 0.2800
 ln_trade_openness                        NaN                 0.0993
 ln_cpi                                   NaN                -0.1442
 ln_gdppc_lag5                            NaN                -0.4931
 ln_labour_productivity_growth_5y         NaN                 0.7405
 trend                                    NaN                    NaN
 ln_gdp_growth_5y_lag1                    NaN                 1.0000
 liquid_liabilities_lag1                  NaN                 0.0999
 mcap_gdp_lag1                            NaN                -0.5655
 bank_loans_gdp_lag1                      NaN                 0.2783
 ln_trade_openness_lag1                   NaN                 0.0992
 ln_cpi_lag1                              NaN                -0.1440
 ln_gdppc_lag5_lag1                       NaN                -0.4966
 ln_labour_productivity_growth_5y_lag1    NaN                 0.7403

                                        liquid_liabilities_lag1  mcap_gdp_lag1  \
 ln_gdp_growth_5y                                        0.0968        -0.5680
 liquid_liabilities                                      0.9999         0.2152
 mcap_gdp                                                0.2134         1.0000
 bank_loans_gdp                                          0.4458        -0.3147
 ln_trade_openness                                       0.0387        -0.4438
 ln_cpi                                                  0.1627         0.4840
 ln_gdppc_lag5                                           0.0800         0.7031
 ln_labour_productivity_growth_5y                        0.1190        -0.3787
 trend                                                      NaN            NaN
 ln_gdp_growth_5y_lag1                                   0.0999        -0.5655
 liquid_liabilities_lag1                                 1.0000         0.2134
 mcap_gdp_lag1                                           0.2134         1.0000
 bank_loans_gdp_lag1                                     0.4447        -0.3131
 ln_trade_openness_lag1                                  0.0387        -0.4437
 ln_cpi_lag1                                             0.1631         0.4842
 ln_gdppc_lag5_lag1                                      0.0789         0.7050
 ln_labour_productivity_growth_5y_lag1                   0.1206        -0.3849

                                        bank_loans_gdp_lag1  \
 ln_gdp_growth_5y                                    0.2834
 liquid_liabilities                                  0.4377
 mcap_gdp                                           -0.3131
 bank_loans_gdp                                      1.0000
 ln_trade_openness                                   0.0969
 ln_cpi                                              0.1547
 ln_gdppc_lag5                                       0.1830
 ln_labour_productivity_growth_5y                    0.0748
 trend                                                  NaN
 ln_gdp_growth_5y_lag1                               0.2783
 liquid_liabilities_lag1                             0.4447
 mcap_gdp_lag1                                      -0.3131
 bank_loans_gdp_lag1                                 1.0000
 ln_trade_openness_lag1                              0.0969
 ln_cpi_lag1                                         0.1550
 ln_gdppc_lag5_lag1                                  0.1807
 ln_labour_productivity_growth_5y_lag1               0.0732

                                        ln_trade_openness_lag1  ln_cpi_lag1  \
 ln_gdp_growth_5y                                       0.1020      -0.1431
 liquid_liabilities                                     0.0317       0.1587
 mcap_gdp                                              -0.4437       0.4842
 bank_loans_gdp                                         0.0966       0.1572
 ln_trade_openness                                      1.0000       0.1858
 ln_cpi                                                 0.1862       1.0000
 ln_gdppc_lag5                                         -0.1068       0.4036
 ln_labour_productivity_growth_5y                      -0.2136      -0.1747
 trend                                                     NaN          NaN
 ln_gdp_growth_5y_lag1                                  0.0992      -0.1440
 liquid_liabilities_lag1                                0.0387       0.1631
 mcap_gdp_lag1                                         -0.4437       0.4842
 bank_loans_gdp_lag1                                    0.0969       0.1550
 ln_trade_openness_lag1                                 1.0000       0.1859
 ln_cpi_lag1                                            0.1859       1.0000
 ln_gdppc_lag5_lag1                                    -0.1076       0.4034
 ln_labour_productivity_growth_5y_lag1                 -0.2103      -0.1754

                                        ln_gdppc_lag5_lag1  \
 ln_gdp_growth_5y                                  -0.4937
 liquid_liabilities                                 0.0723
 mcap_gdp                                           0.7050
 bank_loans_gdp                                     0.1789
 ln_trade_openness                                 -0.1079
 ln_cpi                                             0.4033
 ln_gdppc_lag5                                      1.0000
 ln_labour_productivity_growth_5y                  -0.5076
 trend                                                 NaN
 ln_gdp_growth_5y_lag1                             -0.4966
 liquid_liabilities_lag1                            0.0789
 mcap_gdp_lag1                                      0.7050
 bank_loans_gdp_lag1                                0.1807
 ln_trade_openness_lag1                            -0.1076
 ln_cpi_lag1                                        0.4034
 ln_gdppc_lag5_lag1                                 1.0000
 ln_labour_productivity_growth_5y_lag1             -0.5111

                                        ln_labour_productivity_growth_5y_lag1
 ln_gdp_growth_5y                                                     0.7398
 liquid_liabilities                                                   0.1210
 mcap_gdp                                                            -0.3849
 bank_loans_gdp                                                       0.0734
 ln_trade_openness                                                   -0.2103
 ln_cpi                                                              -0.1758
 ln_gdppc_lag5                                                       -0.5097
 ln_labour_productivity_growth_5y                                     0.9999
 trend                                                                   NaN
 ln_gdp_growth_5y_lag1                                                0.7403
 liquid_liabilities_lag1                                              0.1206
 mcap_gdp_lag1                                                       -0.3849
 bank_loans_gdp_lag1                                                  0.0732
 ln_trade_openness_lag1                                              -0.2103
 ln_cpi_lag1                                                         -0.1754
 ln_gdppc_lag5_lag1                                                  -0.5111
 ln_labour_productivity_growth_5y_lag1                                1.0000  ,
                                               ln_gdp_growth_5y_within  \
 ln_gdp_growth_5y_within                                        1.0000
 liquid_liabilities_within                                     -0.1762
 mcap_gdp_within                                                0.0108
 bank_loans_gdp_within                                         -0.0636
 ln_trade_openness_within                                       0.1072
 ln_cpi_within                                                  0.1283
 ln_gdppc_lag5_within                                           0.0332
 ln_labour_productivity_growth_5y_within                        0.7965
 trend_within                                                   0.1263
 ln_gdp_growth_5y_lag1_within                                   0.8367
 liquid_liabilities_lag1_within                                -0.1541
 mcap_gdp_lag1_within                                           0.0226
 bank_loans_gdp_lag1_within                                    -0.0865
 ln_trade_openness_lag1_within                                  0.1024
 ln_cpi_lag1_within                                             0.1349
 ln_gdppc_lag5_lag1_within                                      0.0278
 ln_labour_productivity_growth_5y_lag1_within                   0.7261

                                               liquid_liabilities_within  \
 ln_gdp_growth_5y_within                                         -0.1762
 liquid_liabilities_within                                        1.0000
 mcap_gdp_within                                                  0.3312
 bank_loans_gdp_within                                            0.6160
 ln_trade_openness_within                                         0.2073
 ln_cpi_within                                                    0.1650
 ln_gdppc_lag5_within                                             0.4506
 ln_labour_productivity_growth_5y_within                         -0.2637
 trend_within                                                     0.4698
 ln_gdp_growth_5y_lag1_within                                    -0.1367
 liquid_liabilities_lag1_within                                   0.9572
 mcap_gdp_lag1_within                                             0.3451
 bank_loans_gdp_lag1_within                                       0.6119
 ln_trade_openness_lag1_within                                    0.2010
 ln_cpi_lag1_within                                               0.1620
 ln_gdppc_lag5_lag1_within                                        0.4496
 ln_labour_productivity_growth_5y_lag1_within                    -0.2326

                                               mcap_gdp_within  \
 ln_gdp_growth_5y_within                                0.0108
 liquid_liabilities_within                              0.3312
 mcap_gdp_within                                        1.0000
 bank_loans_gdp_within                                  0.5476
 ln_trade_openness_within                               0.2097
 ln_cpi_within                                          0.1328
 ln_gdppc_lag5_within                                   0.4716
 ln_labour_productivity_growth_5y_within               -0.1708
 trend_within                                           0.4714
 ln_gdp_growth_5y_lag1_within                           0.0076
 liquid_liabilities_lag1_within                         0.3085
 mcap_gdp_lag1_within                                   0.9408
 bank_loans_gdp_lag1_within                             0.5435
 ln_trade_openness_lag1_within                          0.2033
 ln_cpi_lag1_within                                     0.1341
 ln_gdppc_lag5_lag1_within                              0.4740
 ln_labour_productivity_growth_5y_lag1_within          -0.1629

                                               bank_loans_gdp_within  \
 ln_gdp_growth_5y_within                                     -0.0636
 liquid_liabilities_within                                    0.6160
 mcap_gdp_within                                              0.5476
 bank_loans_gdp_within                                        1.0000
 ln_trade_openness_within                                     0.2056
 ln_cpi_within                                                0.2094
 ln_gdppc_lag5_within                                         0.6057
 ln_labour_productivity_growth_5y_within                     -0.1776
 trend_within                                                 0.5772
 ln_gdp_growth_5y_lag1_within                                -0.0187
 liquid_liabilities_lag1_within                               0.5819
 mcap_gdp_lag1_within                                         0.5492
 bank_loans_gdp_lag1_within                                   0.9845
 ln_trade_openness_lag1_within                                0.1910
 ln_cpi_lag1_within                                           0.2136
 ln_gdppc_lag5_lag1_within                                    0.6040
 ln_labour_productivity_growth_5y_lag1_within                -0.1434

                                               ln_trade_openness_within  \
 ln_gdp_growth_5y_within                                          0.1072
 liquid_liabilities_within                                        0.2073
 mcap_gdp_within                                                  0.2097
 bank_loans_gdp_within                                            0.2056
 ln_trade_openness_within                                         1.0000
 ln_cpi_within                                                   -0.0325
 ln_gdppc_lag5_within                                             0.2848
 ln_labour_productivity_growth_5y_within                          0.0575
 trend_within                                                     0.2404
 ln_gdp_growth_5y_lag1_within                                     0.0967
 liquid_liabilities_lag1_within                                   0.2162
 mcap_gdp_lag1_within                                             0.2179
 bank_loans_gdp_lag1_within                                       0.2170
 ln_trade_openness_lag1_within                                    0.9235
 ln_cpi_lag1_within                                              -0.0607
 ln_gdppc_lag5_lag1_within                                        0.2846
 ln_labour_productivity_growth_5y_lag1_within                     0.0521

                                               ln_cpi_within  \
 ln_gdp_growth_5y_within                              0.1283
 liquid_liabilities_within                            0.1650
 mcap_gdp_within                                      0.1328
 bank_loans_gdp_within                                0.2094
 ln_trade_openness_within                            -0.0325
 ln_cpi_within                                        1.0000
 ln_gdppc_lag5_within                                 0.6181
 ln_labour_productivity_growth_5y_within              0.0648
 trend_within                                         0.6451
 ln_gdp_growth_5y_lag1_within                         0.1316
 liquid_liabilities_lag1_within                       0.1549
 mcap_gdp_lag1_within                                 0.1332
 bank_loans_gdp_lag1_within                           0.1970
 ln_trade_openness_lag1_within                       -0.0052
 ln_cpi_lag1_within                                   0.9933
 ln_gdppc_lag5_lag1_within                            0.6163
 ln_labour_productivity_growth_5y_lag1_within         0.0666

                                               ln_gdppc_lag5_within  \
 ln_gdp_growth_5y_within                                     0.0332
 liquid_liabilities_within                                   0.4506
 mcap_gdp_within                                             0.4716
 bank_loans_gdp_within                                       0.6057
 ln_trade_openness_within                                    0.2848
 ln_cpi_within                                               0.6181
 ln_gdppc_lag5_within                                        1.0000
 ln_labour_productivity_growth_5y_within                    -0.0943
 trend_within                                                0.9584
 ln_gdp_growth_5y_lag1_within                                0.0701
 liquid_liabilities_lag1_within                              0.4371
 mcap_gdp_lag1_within                                        0.4693
 bank_loans_gdp_lag1_within                                  0.5979
 ln_trade_openness_lag1_within                               0.2807
 ln_cpi_lag1_within                                          0.6178
 ln_gdppc_lag5_lag1_within                                   0.9983
 ln_labour_productivity_growth_5y_lag1_within               -0.0676

                                               ln_labour_productivity_growth_5y_within  \
 ln_gdp_growth_5y_within                                                        0.7965
 liquid_liabilities_within                                                     -0.2637
 mcap_gdp_within                                                               -0.1708
 bank_loans_gdp_within                                                         -0.1776
 ln_trade_openness_within                                                       0.0575
 ln_cpi_within                                                                  0.0648
 ln_gdppc_lag5_within                                                          -0.0943
 ln_labour_productivity_growth_5y_within                                        1.0000
 trend_within                                                                  -0.0038
 ln_gdp_growth_5y_lag1_within                                                   0.6368
 liquid_liabilities_lag1_within                                                -0.2455
 mcap_gdp_lag1_within                                                          -0.1740
 bank_loans_gdp_lag1_within                                                    -0.2045
 ln_trade_openness_lag1_within                                                  0.0469
 ln_cpi_lag1_within                                                             0.0711
 ln_gdppc_lag5_lag1_within                                                     -0.0961
 ln_labour_productivity_growth_5y_lag1_within                                   0.8297

                                               trend_within  \
 ln_gdp_growth_5y_within                             0.1263
 liquid_liabilities_within                           0.4698
 mcap_gdp_within                                     0.4714
 bank_loans_gdp_within                               0.5772
 ln_trade_openness_within                            0.2404
 ln_cpi_within                                       0.6451
 ln_gdppc_lag5_within                                0.9584
 ln_labour_productivity_growth_5y_within            -0.0038
 trend_within                                        1.0000
 ln_gdp_growth_5y_lag1_within                        0.1371
 liquid_liabilities_lag1_within                      0.4571
 mcap_gdp_lag1_within                                0.4714
 bank_loans_gdp_lag1_within                          0.5674
 ln_trade_openness_lag1_within                       0.2347
 ln_cpi_lag1_within                                  0.6435
 ln_gdppc_lag5_lag1_within                           0.9577
 ln_labour_productivity_growth_5y_lag1_within        0.0041

                                               ln_gdp_growth_5y_lag1_within  \
 ln_gdp_growth_5y_within                                             0.8367
 liquid_liabilities_within                                          -0.1367
 mcap_gdp_within                                                     0.0076
 bank_loans_gdp_within                                              -0.0187
 ln_trade_openness_within                                            0.0967
 ln_cpi_within                                                       0.1316
 ln_gdppc_lag5_within                                                0.0701
 ln_labour_productivity_growth_5y_within                             0.6368
 trend_within                                                        0.1371
 ln_gdp_growth_5y_lag1_within                                        1.0000
 liquid_liabilities_lag1_within                                     -0.1681
 mcap_gdp_lag1_within                                                0.0108
 bank_loans_gdp_lag1_within                                         -0.0523
 ln_trade_openness_lag1_within                                       0.1113
 ln_cpi_lag1_within                                                  0.1336
 ln_gdppc_lag5_lag1_within                                           0.0426
 ln_labour_productivity_growth_5y_lag1_within                        0.7962

                                               liquid_liabilities_lag1_within  \
 ln_gdp_growth_5y_within                                              -0.1541
 liquid_liabilities_within                                             0.9572
 mcap_gdp_within                                                       0.3085
 bank_loans_gdp_within                                                 0.5819
 ln_trade_openness_within                                              0.2162
 ln_cpi_within                                                         0.1549
 ln_gdppc_lag5_within                                                  0.4371
 ln_labour_productivity_growth_5y_within                              -0.2455
 trend_within                                                          0.4571
 ln_gdp_growth_5y_lag1_within                                         -0.1681
 liquid_liabilities_lag1_within                                        1.0000
 mcap_gdp_lag1_within                                                  0.3312
 bank_loans_gdp_lag1_within                                            0.6032
 ln_trade_openness_lag1_within                                         0.2001
 ln_cpi_lag1_within                                                    0.1543
 ln_gdppc_lag5_lag1_within                                             0.4374
 ln_labour_productivity_growth_5y_lag1_within                         -0.2599

                                               mcap_gdp_lag1_within  \
 ln_gdp_growth_5y_within                                     0.0226
 liquid_liabilities_within                                   0.3451
 mcap_gdp_within                                             0.9408
 bank_loans_gdp_within                                       0.5492
 ln_trade_openness_within                                    0.2179
 ln_cpi_within                                               0.1332
 ln_gdppc_lag5_within                                        0.4693
 ln_labour_productivity_growth_5y_within                    -0.1740
 trend_within                                                0.4714
 ln_gdp_growth_5y_lag1_within                                0.0108
 liquid_liabilities_lag1_within                              0.3312
 mcap_gdp_lag1_within                                        1.0000
 bank_loans_gdp_lag1_within                                  0.5478
 ln_trade_openness_lag1_within                               0.2097
 ln_cpi_lag1_within                                          0.1328
 ln_gdppc_lag5_lag1_within                                   0.4717
 ln_labour_productivity_growth_5y_lag1_within               -0.1708

                                               bank_loans_gdp_lag1_within  \
 ln_gdp_growth_5y_within                                          -0.0865
 liquid_liabilities_within                                         0.6119
 mcap_gdp_within                                                   0.5435
 bank_loans_gdp_within                                             0.9845
 ln_trade_openness_within                                          0.2170
 ln_cpi_within                                                     0.1970
 ln_gdppc_lag5_within                                              0.5979
 ln_labour_productivity_growth_5y_within                          -0.2045
 trend_within                                                      0.5674
 ln_gdp_growth_5y_lag1_within                                     -0.0523
 liquid_liabilities_lag1_within                                    0.6032
 mcap_gdp_lag1_within                                              0.5478
 bank_loans_gdp_lag1_within                                        1.0000
 ln_trade_openness_lag1_within                                     0.2023
 ln_cpi_lag1_within                                                0.2027
 ln_gdppc_lag5_lag1_within                                         0.5965
 ln_labour_productivity_growth_5y_lag1_within                     -0.1715

                                               ln_trade_openness_lag1_within  \
 ln_gdp_growth_5y_within                                              0.1024
 liquid_liabilities_within                                            0.2010
 mcap_gdp_within                                                      0.2033
 bank_loans_gdp_within                                                0.1910
 ln_trade_openness_within                                             0.9235
 ln_cpi_within                                                       -0.0052
 ln_gdppc_lag5_within                                                 0.2807
 ln_labour_productivity_growth_5y_within                              0.0469
 trend_within                                                         0.2347
 ln_gdp_growth_5y_lag1_within                                         0.1113
 liquid_liabilities_lag1_within                                       0.2001
 mcap_gdp_lag1_within                                                 0.2097
 bank_loans_gdp_lag1_within                                           0.2023
 ln_trade_openness_lag1_within                                        1.0000
 ln_cpi_lag1_within                                                  -0.0375
 ln_gdppc_lag5_lag1_within                                            0.2796
 ln_labour_productivity_growth_5y_lag1_within                         0.0610

                                               ln_cpi_lag1_within  \
 ln_gdp_growth_5y_within                                   0.1349
 liquid_liabilities_within                                 0.1620
 mcap_gdp_within                                           0.1341
 bank_loans_gdp_within                                     0.2136
 ln_trade_openness_within                                 -0.0607
 ln_cpi_within                                             0.9933
 ln_gdppc_lag5_within                                      0.6178
 ln_labour_productivity_growth_5y_within                   0.0711
 trend_within                                              0.6435
 ln_gdp_growth_5y_lag1_within                              0.1336
 liquid_liabilities_lag1_within                            0.1543
 mcap_gdp_lag1_within                                      0.1328
 bank_loans_gdp_lag1_within                                0.2027
 ln_trade_openness_lag1_within                            -0.0375
 ln_cpi_lag1_within                                        1.0000
 ln_gdppc_lag5_lag1_within                                 0.6160
 ln_labour_productivity_growth_5y_lag1_within              0.0690

                                               ln_gdppc_lag5_lag1_within  \
 ln_gdp_growth_5y_within                                          0.0278
 liquid_liabilities_within                                        0.4496
 mcap_gdp_within                                                  0.4740
 bank_loans_gdp_within                                            0.6040
 ln_trade_openness_within                                         0.2846
 ln_cpi_within                                                    0.6163
 ln_gdppc_lag5_within                                             0.9983
 ln_labour_productivity_growth_5y_within                         -0.0961
 trend_within                                                     0.9577
 ln_gdp_growth_5y_lag1_within                                     0.0426
 liquid_liabilities_lag1_within                                   0.4374
 mcap_gdp_lag1_within                                             0.4717
 bank_loans_gdp_lag1_within                                       0.5965
 ln_trade_openness_lag1_within                                    0.2796
 ln_cpi_lag1_within                                               0.6160
 ln_gdppc_lag5_lag1_within                                        1.0000
 ln_labour_productivity_growth_5y_lag1_within                    -0.0876

                                               ln_labour_productivity_growth_5y_lag1_within
 ln_gdp_growth_5y_within                                                             0.7261
 liquid_liabilities_within                                                          -0.2326
 mcap_gdp_within                                                                    -0.1629
 bank_loans_gdp_within                                                              -0.1434
 ln_trade_openness_within                                                            0.0521
 ln_cpi_within                                                                       0.0666
 ln_gdppc_lag5_within                                                               -0.0676
 ln_labour_productivity_growth_5y_within                                             0.8297
 trend_within                                                                        0.0041
 ln_gdp_growth_5y_lag1_within                                                        0.7962
 liquid_liabilities_lag1_within                                                     -0.2599
 mcap_gdp_lag1_within                                                               -0.1708
 bank_loans_gdp_lag1_within                                                         -0.1715
 ln_trade_openness_lag1_within                                                       0.0610
 ln_cpi_lag1_within                                                                  0.0690
 ln_gdppc_lag5_lag1_within                                                          -0.0876
 ln_labour_productivity_growth_5y_lag1_within                                        1.0000  ,
 mcap_gdp                                -0.5680
 mcap_gdp_lag1                           -0.5680
 ln_gdppc_lag5_lag1                      -0.4937
 ln_gdppc_lag5                           -0.4902
 ln_cpi                                  -0.1434
 ln_cpi_lag1                             -0.1431
 liquid_liabilities                       0.0942
 liquid_liabilities_lag1                  0.0968
 ln_trade_openness_lag1                   0.1020
 ln_trade_openness                        0.1020
 bank_loans_gdp_lag1                      0.2834
 bank_loans_gdp                           0.2851
 ln_labour_productivity_growth_5y_lag1    0.7398
 ln_labour_productivity_growth_5y         0.7401
 ln_gdp_growth_5y_lag1                    0.9999
 ln_gdp_growth_5y                         1.0000
 trend                                       NaN
 Name: ln_gdp_growth_5y, dtype: float64,
 liquid_liabilities_within                       -0.1762
 liquid_liabilities_lag1_within                  -0.1541
 bank_loans_gdp_lag1_within                      -0.0865
 bank_loans_gdp_within                           -0.0636
 mcap_gdp_within                                  0.0108
 mcap_gdp_lag1_within                             0.0226
 ln_gdppc_lag5_lag1_within                        0.0278
 ln_gdppc_lag5_within                             0.0332
 ln_trade_openness_lag1_within                    0.1024
 ln_trade_openness_within                         0.1072
 trend_within                                     0.1263
 ln_cpi_within                                    0.1283
 ln_cpi_lag1_within                               0.1349
 ln_labour_productivity_growth_5y_lag1_within     0.7261
 ln_labour_productivity_growth_5y_within          0.7965
 ln_gdp_growth_5y_lag1_within                     0.8367
 ln_gdp_growth_5y_within                          1.0000
 Name: ln_gdp_growth_5y_within, dtype: float64)
```

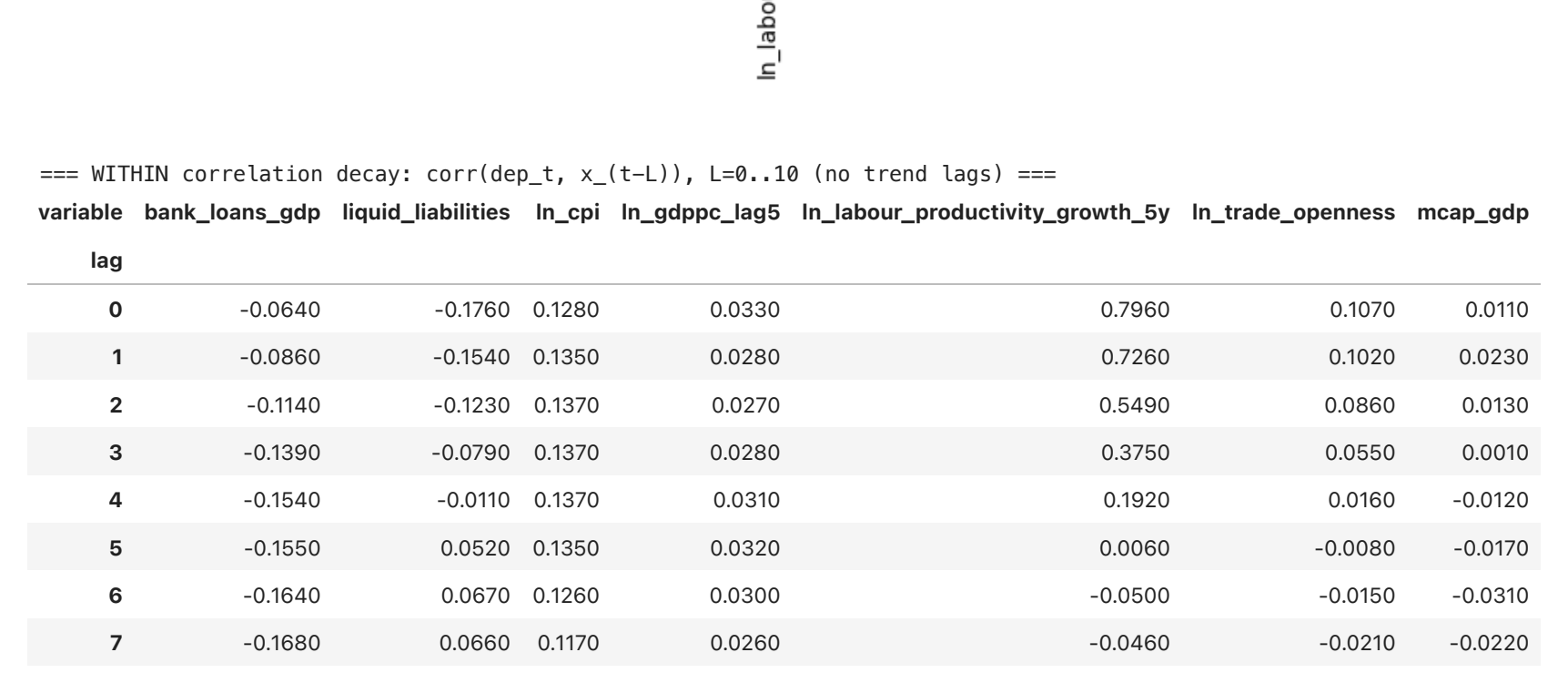
In [172…
```
df_Q16 = df_twfe.copy()


df_Q16["trend"] = df_Q16.groupby("country").cumcount() + 1

dep_var = "ln_gdp_growth_5y"
x_vars = [
    "liquid_liabilities",
    "mcap_gdp",
    "bank_loans_gdp",
    "ln_trade_openness",
    "ln_cpi",
    "ln_gdppc_lag5",
    "ln_labour_productivity_growth_5y"
]
all_vars = [dep_var] + x_vars + ["trend"]

K_HEAT  = 4
K_DECAY = 10


for v in [dep_var] + x_vars:
    for L in range(1, K_DECAY + 1):
        df_Q16[f"{v}_lag{L}"] = df_Q16.groupby("country")[v].shift(L)


df_between = df_Q16.groupby("country")[all_vars].mean()
corr_between = df_between.corr()


lag_vars_heat = [f"{v}_lag{L}" for v in ([dep_var] + x_vars) for L in range(1, K_HEAT + 1)]
within_cols_heat = ([dep_var] + x_vars) + ["trend"] + lag_vars_heat

df_within_heat = df_Q16.copy()
for c in within_cols_heat:
    df_within_heat[c] = df_within_heat[c] - df_within_heat.groupby("country")[c].transform("mean")

corr_within = df_within_heat[within_cols_heat].corr()


mask_between = np.triu(np.ones_like(corr_between, dtype=bool), k=1)
mask_within  = np.triu(np.ones_like(corr_within,  dtype=bool), k=1)

plt.figure(figsize=(12, 10))
sns.heatmap(
    corr_between,
    mask=mask_between,
    cmap="coolwarm",
    center=0, vmin=-1, vmax=1,
    square=True,
    annot=True, fmt=".2f",
    annot_kws={"size": 7},
    cbar_kws={"shrink": 0.8}
)
plt.title("Between correlation matrix (country means) — no lags", fontsize=14)
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()
plt.show()

plt.figure(figsize=(14, 12))
sns.heatmap(
    corr_within,
    mask=mask_within,
    cmap="coolwarm",
    center=0, vmin=-1, vmax=1,
    square=True,
    annot=True, fmt=".2f",
    annot_kws={"size": 6},
    cbar_kws={"shrink": 0.8}
)
plt.title(f"Within (one-way FE) correlation matrix — lags up to {K_HEAT} (no trend lags)", fontsize=14)
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()


lag_vars_decay = [f"{v}_lag{L}" for v in ([dep_var] + x_vars) for L in range(1, K_DECAY + 1)]
within_cols_decay = ([dep_var] + x_vars) + ["trend"] + lag_vars_decay

df_within_decay = df_Q16.copy()
for c in within_cols_decay:
    df_within_decay[c] = df_within_decay[c] - df_within_decay.groupby("country")[c].transform("mean")

corr_within_decay = df_within_decay[within_cols_decay].corr()

def corr_decay_table(corr_mat, dep_var, x_vars, max_lag=10):
    rows = []
    for x in x_vars:
        rows.append({"variable": x, "lag": 0, "corr": corr_mat.loc[dep_var, x]})
        for L in range(1, max_lag + 1):
            col = f"{x}_lag{L}"
            if col in corr_mat.columns:
                rows.append({"variable": x, "lag": L, "corr": corr_mat.loc[dep_var, col]})
    return pd.DataFrame(rows).pivot(index="lag", columns="variable", values="corr")

decay_within = corr_decay_table(corr_within_decay, dep_var, x_vars, max_lag=K_DECAY)

print(f"\n=== WITHIN correlation decay: corr(dep_t, x_(t-L)), L=0..{K_DECAY} (no trend lags) ===")
display(decay_within.round(3))

plt.figure(figsize=(12, 4))
sns.heatmap(
    decay_within,
    cmap="coolwarm",
    center=0, vmin=-1, vmax=1,
    annot=True, fmt=".2f",
    cbar_kws={"shrink": 0.8}
)
plt.title(f"Within: corr(dep, x lag L), L=0..{K_DECAY} (no trend lags)", fontsize=13)
plt.ylabel("Lag L")
plt.tight_layout()
plt.show()
```





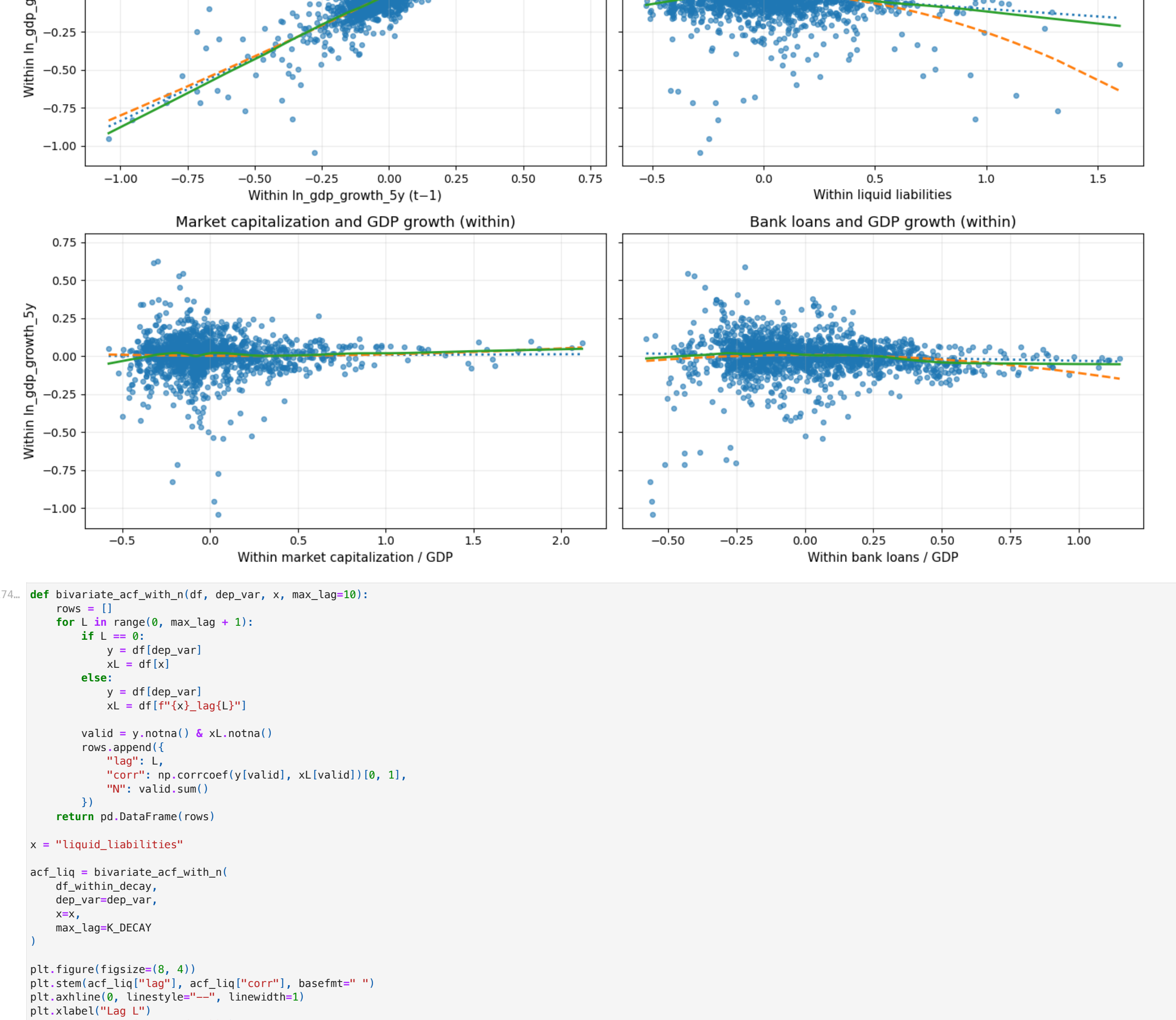
```
=== WITHIN correlation decay: corr(dep_t, x_(t-L)), L=0..10 (no trend lags) ===
```

variable	bank_loans_gdp	liquid_liabilities	ln_cpi	ln_gdppc_lag5	ln_labour_productivity_growth_5y	ln_trade_openness	mcap_gdp
lag							
0	-0.0640	-0.1760	0.1280	0.0330	0.7960	0.1070	0.0110
1	-0.0860	-0.1540	0.1350	0.0280	0.7260	0.1020	0.0230
2	-0.1140	-0.1230	0.1370	0.0270	0.5490	0.0860	0.0130
3	-0.1390	-0.0790	0.1370	0.0280	0.3750	0.0550	0.0010
4	-0.1540	-0.0110	0.1370	0.0310	0.1920	0.0160	-0.0120
5	-0.1550	0.0520	0.1350	0.0320	0.0060	-0.0080	-0.0170
6	-0.1640	0.0670	0.1260	0.0300	-0.0500	-0.0150	-0.0310
7	-0.1680	0.0660	0.1170	0.0260	-0.0460	-0.0210	-0.0220
8	-0.1650	0.0570	0.1100	0.0230	-0.0380	-0.0280	-0.0190
9	-0.1600	0.0400	0.1070	0.0180	-0.0220	-0.0400	-0.0290
10	-0.1600	0.0210	0.1060	0.0140	0.0020	-0.0470	-0.0430



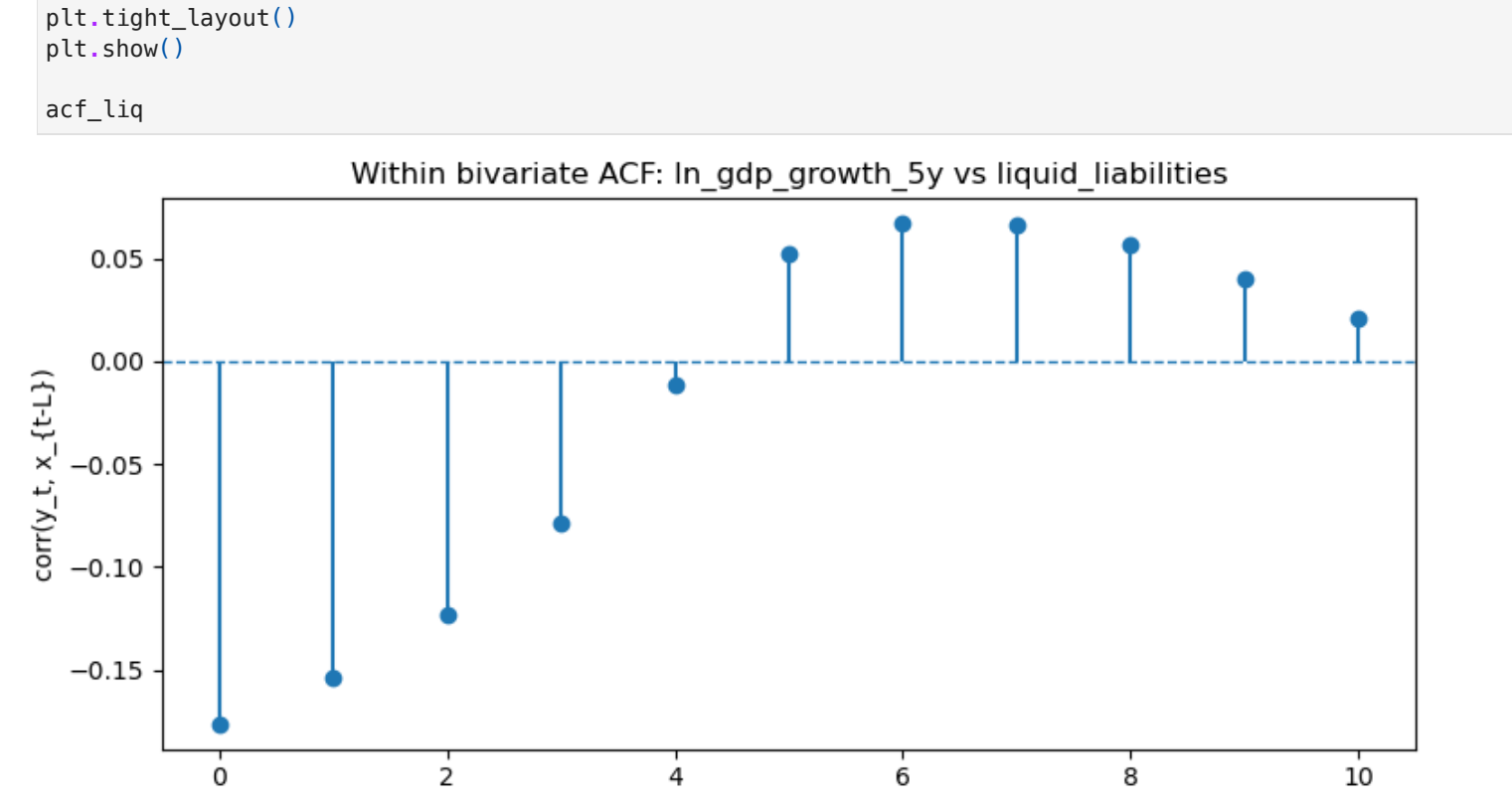
In [ ]:

```
df_plot = df_twfe.copy()
df_plot = df_plot.sort_values(["country", "year"])

y = "ln_gdp_growth_5y"
x_list = ["liquid_liabilities", "mcap_gdp", "bank_loans_gdp"]

# lag of dependent variable
df_plot[f"{y}_lag1"] = df_plot.groupby("country")[y].shift(1)

def within_transform(df, var, group="country"):
    return df[var] - df.groupby(group)[var].transform("mean")

# dependent
df_plot[f"{y}_within"] = within_transform(df_plot, y)
df_plot[f"{y}_lag1_within"] = within_transform(df_plot, f"{y}_lag1")

# explanatory variables
for x in x_list:
    df_plot[f"{x}_within"] = within_transform(df_plot, x)
# ----------------------------
# Helper to plot one panel
# ----------------------------
def bivar_within_panel(ax, df, xcol, ycol, title, xlabel, ylabel, lowess_frac=0.35):
    d = df[[xcol, ycol]].dropna()
    X = d[xcol].to_numpy()
    Y = d[ycol].to_numpy()

    order = np.argsort(X)
    Xs, Ys = X[order], Y[order]

    # Scatter
    ax.scatter(X, Y, s=18, alpha=0.6, label="Actual data")

    # Linear fit
    b1 = np.polyfit(X, Y, 1)
    ax.plot(Xs, np.polyval(b1, Xs), linestyle=":", linewidth=2, label="Linear fit")

    # Quadratic fit
    b2 = np.polyfit(X, Y, 2)
    ax.plot(Xs, np.polyval(b2, Xs), linestyle="--", linewidth=2, label="Quadratic fit")

    # LOWESS
    lo = lowess(Y, X, frac=lowess_frac, return_sorted=True)
    ax.plot(lo[:, 0], lo[:, 1], linewidth=2, label="LOWESS")

    ax.set_title(title, fontsize=13)
    ax.set_xlabel(xlabel, fontsize=11)
    ax.set_ylabel(ylabel, fontsize=11)
    ax.grid(alpha=0.25)

fig, axes = plt.subplots(2, 2, figsize=(14, 10), sharey=True)


bivar_within_panel(
    axes[0, 0],
    df_plot,
    xcol=f"{y}_lag1_within",
    ycol=f"{y}_within",
    title="GDP growth vs lagged GDP growth (within)",
    xlabel="Within ln_gdp_growth_5y (t-1)",
    ylabel="Within ln_gdp_growth_5y"
)


bivar_within_panel(
    axes[0, 1],
    df_plot,
    xcol="liquid_liabilities_within",
    ycol=f"{y}_within",
    title="Liquid liabilities and GDP growth (within)",
    xlabel="Within liquid liabilities",
    ylabel=""
)

bivar_within_panel(
    axes[1, 0],
    df_plot,
    xcol="mcap_gdp_within",
    ycol=f"{y}_within",
    title="Market capitalization and GDP growth (within)",
    xlabel="Within market capitalization / GDP",
    ylabel="Within ln_gdp_growth_5y"
)


bivar_within_panel(
    axes[1, 1],
    df_plot,
    xcol="bank_loans_gdp_within",
    ycol=f"{y}_within",
    title="Bank loans and GDP growth (within)",
    xlabel="Within bank loans / GDP",
    ylabel=""
)
handles, labels = axes[0, 0].get_legend_handles_labels()
fig.legend(handles, labels, loc="upper center", ncol=4, frameon=True)

plt.suptitle(
    "Bivariate relationships (within-transformed): GDP growth and financial variables",
    fontsize=18,
    y=0.98
)

plt.tight_layout(rect=[0, 0, 1, 0.94])
plt.show()
```



In [174…

```
def bivariate_acf_with_n(df, dep_var, x, max_lag=10):
    rows = []
    for L in range(0, max_lag + 1):
        if L == 0:
            y = df[dep_var]
            xL = df[x]
        else:
            y = df[dep_var]
            xL = df[f"{x}_lag{L}"]

        valid = y.notna() & xL.notna()
        rows.append({
            "lag": L,
            "corr": np.corrcoef(y[valid], xL[valid])[0, 1],
            "N": valid.sum()
        })
    return pd.DataFrame(rows)

x = "liquid_liabilities"

acf_liq = bivariate_acf_with_n(
    df_within_decay,
    dep_var=dep_var,
    x=x,
    max_lag=K_DECAY
)

plt.figure(figsize=(8, 4))
plt.stem(acf_liq["lag"], acf_liq["corr"], basefmt=" ")
plt.axhline(0, linestyle="--", linewidth=1)
plt.xlabel("Lag L")
plt.ylabel("corr(y_t, x_{t-L})")
plt.title(f"Within bivariate ACF: {dep_var} vs {x}")
plt.tight_layout()
plt.show()

acf_liq
```



Out[174…

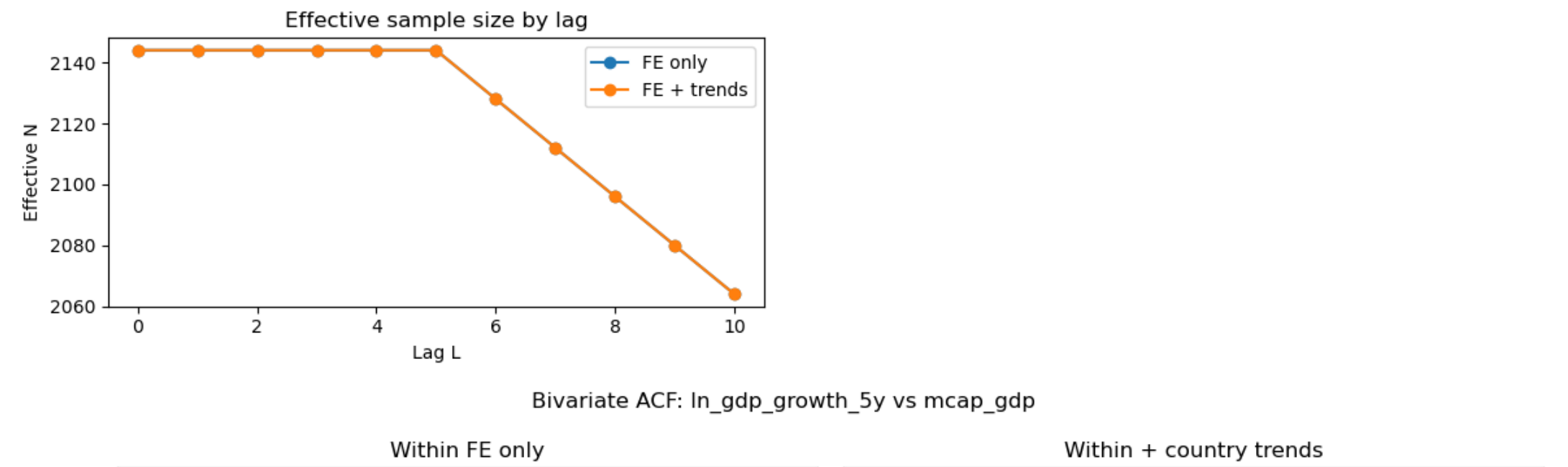
	lag	corr	N
0	0	-0.1762	2144
1	1	-0.1541	2144
2	2	-0.1228	2144
3	3	-0.0787	2144
4	4	-0.0113	2144
5	5	0.0524	2144
6	6	0.0669	2128
7	7	0.0659	2112
8	8	0.0566	2096
9	9	0.0401	2080
10	10	0.0212	2064

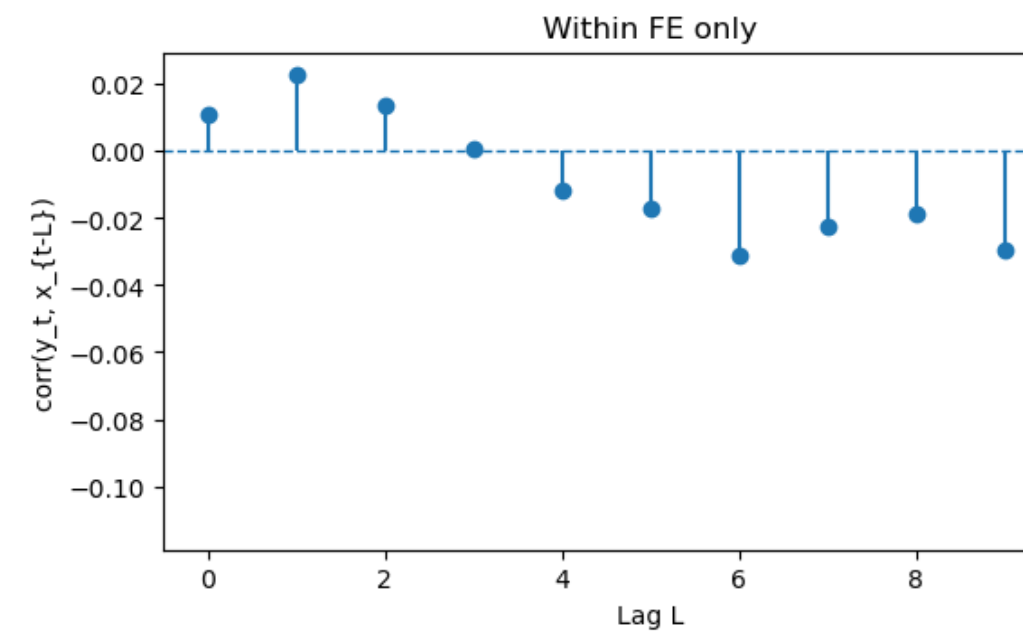
In [175…

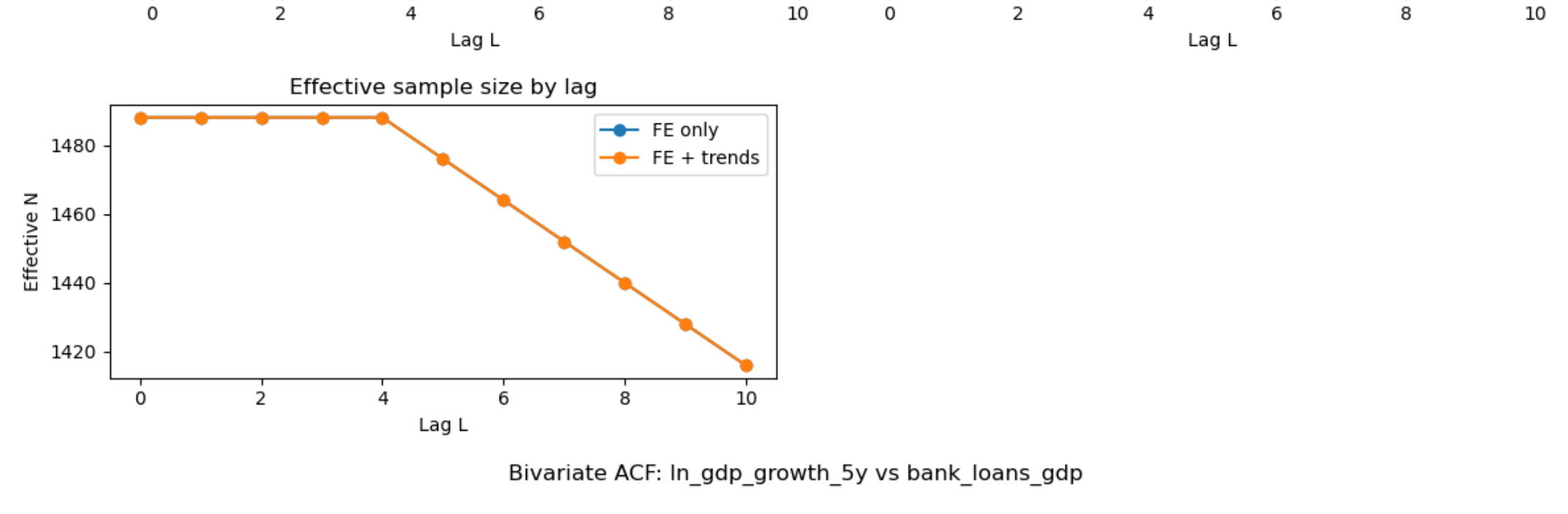
```
def detrend_within_safe(df, var, trend_col="trend", group_col="country", min_n=3):
    out = pd.Series(np.nan, index=df.index, dtype=float)
    for c, g in df.groupby(group_col):
        valid = g[var].notna() & g[trend_col].notna()
        gv = g.loc[valid, [var, trend_col]]
        if gv.shape[0] < min_n:
            continue
        y = gv[var].astype(float)
        X = add_constant(gv[trend_col].astype(float))
        res = OLS(y, X).fit().resid
        out.loc[gv.index] = res
    return out
df_trended = df_Q16.copy()

df_trended[dep_var] = detrend_within_safe(df_Q16, dep_var)

for x in x_vars:
    df_trended[x] = detrend_within_safe(df_Q16, x)
    for L in range(1, K_DECAY + 1):
        df_trended[f"{x}_lag{L}"] = df_trended.groupby("country")[x].shift(L)
```

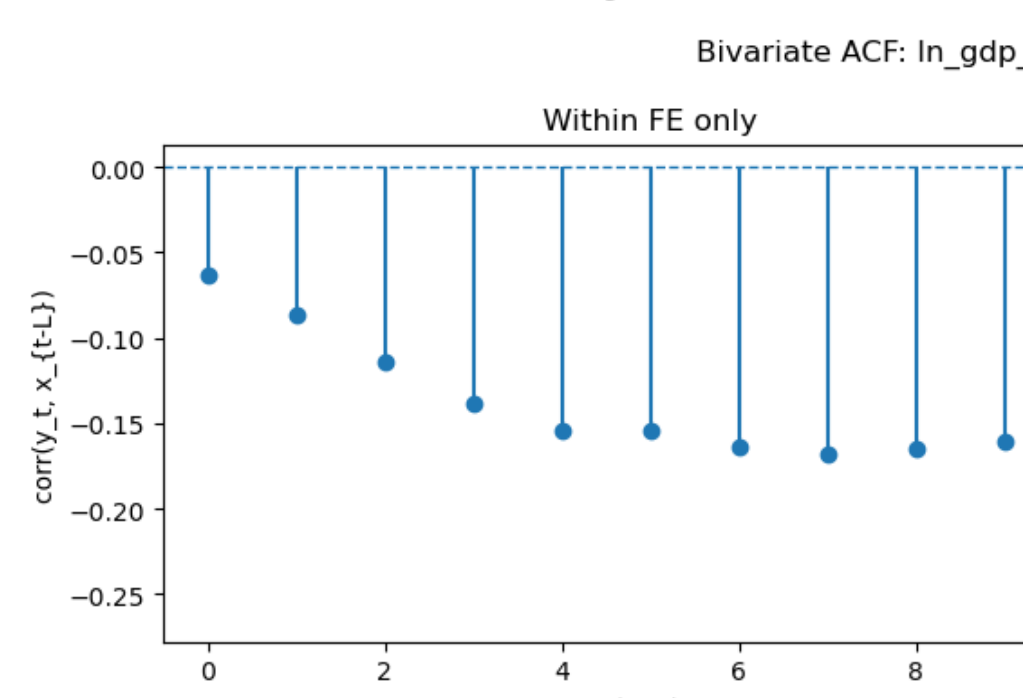
In [177…

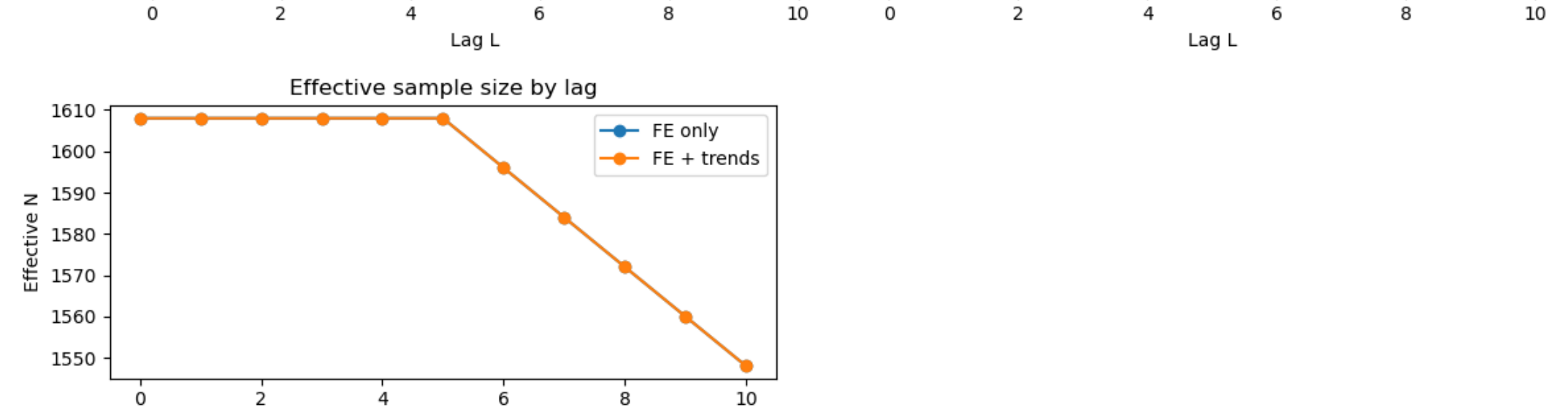
```
def bivariate_acf(df, dep_var, x, max_lag=10):
    rows = []
    for L in range(0, max_lag + 1):
        if L == 0:
            y = df[dep_var]
            xL = df[x]
        else:
            y = df[dep_var]
            xL = df[f"{x}_lag{L}"]

        valid = y.notna() & xL.notna()
        rows.append({
            "lag": L,
            "corr": np.corrcoef(y[valid], xL[valid])[0, 1] if valid.sum() > 2 else np.nan,
            "N": valid.sum()
        })
    return pd.DataFrame(rows)

acf_raw = bivariate_acf(df_within_decay, dep_var, "liquid_liabilities", K_DECAY)

for L in range(1, K_DECAY + 1):
    df_trended[f"liquid_liabilities_lag{L}"] = (
        df_trended.groupby("country")["liquid_liabilities"].shift(L)
    )

acf_trended = bivariate_acf(df_trended, dep_var, "liquid_liabilities", K_DECAY)

fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharey=True)

# BEFORE detrending
axes[0].stem(acf_raw["lag"], acf_raw["corr"], basefmt=" ")
axes[0].axhline(0, linestyle="--", linewidth=1)
axes[0].set_title("Within FE only")
axes[0].set_xlabel("Lag L")
axes[0].set_ylabel("corr(y_t, x_{t-L})")

# AFTER detrending
axes[1].stem(acf_trended["lag"], acf_trended["corr"], basefmt=" ")
axes[1].axhline(0, linestyle="--", linewidth=1)
axes[1].set_title("Within + country trends")
axes[1].set_xlabel("Lag L")

plt.suptitle("Bivariate ACF: ln_gdp_growth_5y vs liquid_liabilities")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 3))
plt.plot(acf_raw["lag"], acf_raw["N"], marker="o", label="FE only")
plt.plot(acf_trended["lag"], acf_trended["N"], marker="o", label="FE + trends")
plt.xlabel("Lag L")
plt.ylabel("Effective N")
plt.title("Effective sample size by lag")
plt.legend()
plt.tight_layout()
plt.show()



acf_raw = bivariate_acf(df_within_decay, dep_var, "mcap_gdp", K_DECAY)

for L in range(1, K_DECAY + 1):
    df_trended[f"mcap_gdp_lag{L}"] = (
        df_trended.groupby("country")["mcap_gdp"].shift(L)
    )

acf_trended = bivariate_acf(df_trended, dep_var, "mcap_gdp", K_DECAY)

fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharey=True)

# BEFORE detrending
axes[0].stem(acf_raw["lag"], acf_raw["corr"], basefmt=" ")
axes[0].axhline(0, linestyle="--", linewidth=1)
axes[0].set_title("Within FE only")
axes[0].set_xlabel("Lag L")
axes[0].set_ylabel("corr(y_t, x_{t-L})")

# AFTER detrending
axes[1].stem(acf_trended["lag"], acf_trended["corr"], basefmt=" ")
axes[1].axhline(0, linestyle="--", linewidth=1)
axes[1].set_title("Within + country trends")
axes[1].set_xlabel("Lag L")

plt.suptitle("Bivariate ACF: ln_gdp_growth_5y vs mcap_gdp")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 3))
plt.plot(acf_raw["lag"], acf_raw["N"], marker="o", label="FE only")
plt.plot(acf_trended["lag"], acf_trended["N"], marker="o", label="FE + trends")
plt.xlabel("Lag L")
plt.ylabel("Effective N")
plt.title("Effective sample size by lag")
plt.legend()
plt.tight_layout()
plt.show()


acf_raw = bivariate_acf(df_within_decay, dep_var, "bank_loans_gdp", K_DECAY)

for L in range(1, K_DECAY + 1):
    df_trended[f"bank_loans_gdp_lag{L}"] = (
        df_trended.groupby("country")["bank_loans_gdp"].shift(L)
    )

acf_trended = bivariate_acf(df_trended, dep_var, "bank_loans_gdp", K_DECAY)

fig, axes = plt.subplots(1, 2, figsize=(12, 4), sharey=True)

# BEFORE detrending
axes[0].stem(acf_raw["lag"], acf_raw["corr"], basefmt=" ")
axes[0].axhline(0, linestyle="--", linewidth=1)
axes[0].set_title("Within FE only")
axes[0].set_xlabel("Lag L")
axes[0].set_ylabel("corr(y_t, x_{t-L})")

# AFTER detrending
axes[1].stem(acf_trended["lag"], acf_trended["corr"], basefmt=" ")
axes[1].axhline(0, linestyle="--", linewidth=1)
axes[1].set_title("Within + country trends")
axes[1].set_xlabel("Lag L")

plt.suptitle("Bivariate ACF: ln_gdp_growth_5y vs bank_loans_gdp")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 3))
plt.plot(acf_raw["lag"], acf_raw["N"], marker="o", label="FE only")
plt.plot(acf_trended["lag"], acf_trended["N"], marker="o", label="FE + trends")
plt.xlabel("Lag L")
plt.ylabel("Effective N")
plt.title("Effective sample size by lag")
plt.legend()
plt.tight_layout()
plt.show()
```

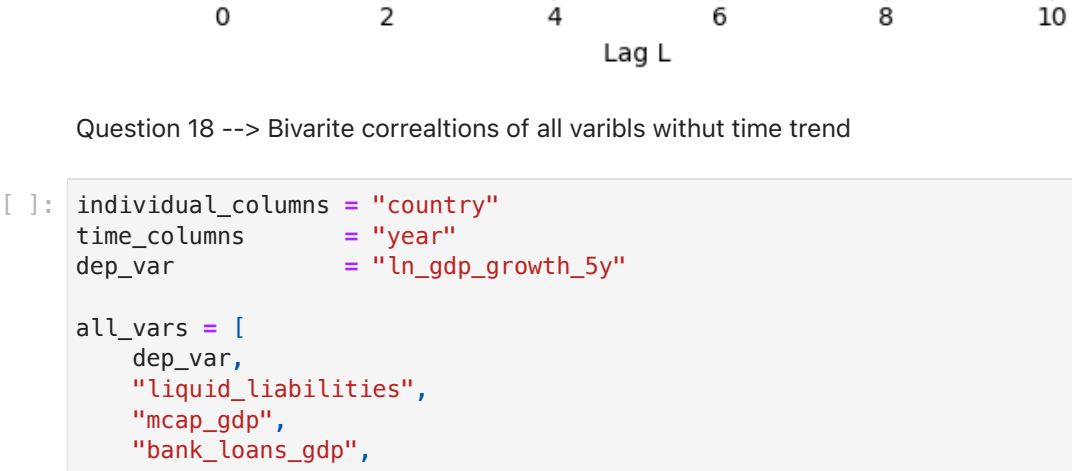












Question 18 --> Bivarite correaltions of all varibls withut time trend

In [ ]:

```
individual_columns = "country"
time_columns       = "year"
dep_var            = "ln_gdp_growth_5y"

all_vars = [
    dep_var,
    "liquid_liabilities",
    "mcap_gdp",
    "bank_loans_gdp",
    "ln_cpi",
    "trade_openness"
]

LOW_Threshold  = 0.10
HIGH_Threshold = 0.80

def ensure_sorted(df, id_col, time_col):
    return df.sort_values([id_col, time_col]).reset_index(drop=True)

def twoway_within_transform(df, cols, id_col, time_col):
    out = pd.DataFrame(index=df.index)
    overall = df[cols].mean(skipna=True)
    mean_i = df.groupby(id_col)[cols].transform("mean")
    mean_t = df.groupby(time_col)[cols].transform("mean")
    out[cols] = df[cols] - mean_i - mean_t + overall
    return out

def first_difference(df, cols, id_col, time_col):
    df = ensure_sorted(df, id_col, time_col)
    d = df.groupby(id_col)[cols].diff()
    d.columns = [f"d_{c}" for c in cols]
    return d

def lag_within_id(df, cols, id_col, L=1):
    lagged = df.groupby(id_col)[cols].shift(L)
    lagged.columns = [f"{c}_lag{L}" for c in cols]
    return lagged

def corr_matrix(df, cols, method="pearson", min_periods=3):
    return df[cols].corr(method=method, min_periods=min_periods)

def flag_low_corr_with_y(corr, y_name, low_thr=0.1):
    s = corr.loc[:, y_name].drop(y_name, errors="ignore")
    return s[abs(s) < low_thr].sort_values(key=lambda x: abs(x))

def flag_high_corr_among_x(corr, y_name, high_thr=0.8):
    xs = [c for c in corr.columns if c != y_name]
    C = corr.loc[xs, xs].copy()
    pairs = []
    for i, a in enumerate(xs):
        for b in xs[i+1:]:
            val = C.loc[a, b]
            if pd.notna(val) and abs(val) > high_thr:
                pairs.append((a, b, val))
    return pd.DataFrame(pairs, columns=["var1", "var2", "corr"]).sort_values(
        by="corr", key=lambda x: abs(x), ascending=False
    )

def check_boundary_missing_dots(df, id_col, cols_to_check):
    boundary = df[id_col].ne(df[id_col].shift(1))
    boundary_rows = df.loc[boundary, [id_col] + cols_to_check].copy()
    miss = boundary_rows[cols_to_check].isna()
    miss.columns = [f"{c}_isna_at_boundary" for c in miss.columns]
    return pd.concat([boundary_rows[[id_col]], miss], axis=1)

id_col, time_col = individual_columns, time_columns

required = [id_col, time_col] + all_vars
missing = [c for c in required if c not in df.columns]
if missing:
    raise KeyError(
        "Missing columns in df:\n"
        + "\n".join(missing)
        + "\n\nTip: print(df.columns) and fix typos (e.g., liquid_liabilities)."
    )

df0 = ensure_sorted(df.copy(), id_col, time_col)
df_tw = twoway_within_transform(df0, all_vars, id_col, time_col)
df_tw[id_col] = df0[id_col].values
df_tw[time_col] = df0[time_col].values
df_d = first_difference(df0, all_vars, id_col, time_col)
df_d[id_col] = df0[id_col].values
df_d[time_col] = df0[time_col].values
```

```python
d_cols = [f"d_{c}" for c in all_vars]
df_dlag = lag_within_id(df_d, d_cols, id_col, L=1)
df_fd = pd.concat([df_d, df_dlag], axis=1)
corr_tw = corr_matrix(df_tw, all_vars)
fd_cols_all = d_cols + list(df_dlag.columns)
corr_fd = corr_matrix(df_fd, fd_cols_all)
low_tw  = flag_low_corr_with_y(corr_tw, dep_var, low_thr=LOW_Threshold)
high_tw = flag_high_corr_among_x(corr_tw, dep_var, high_thr=HIGH_Threshold)

d_dep = f"d_{dep_var}"
low_fd  = flag_low_corr_with_y(corr_fd, d_dep, low_thr=LOW_Threshold)
high_fd = flag_high_corr_among_x(corr_fd, d_dep, high_thr=HIGH_Threshold)

print("\n=== TWO-WAY-WITHIN correlation matrix (rounded) ===")
print(corr_tw.round(3))

print("\n=== TWO-WAY-WITHIN: low |corr(y,x)| < 0.1 ===")
print(low_tw)

print("\n=== TWO-WAY-WITHIN: high |corr(x1,x2)| > 0.8 (among Xs) ===")
print(high_tw)

print("\n=== FIRST-DIFF correlation matrix (rounded) ===")
print(corr_fd.round(3))

print("\n=== FIRST-DIFF: low |corr(dy, dX or lag(dX))| < 0.1 ===")
print(low_fd)

print("\n=== FIRST-DIFF: high |corr| > 0.8 (among dX and lag(dX), excluding dy) ===")
print(high_fd)
cols_show = [id_col, time_col] + d_cols + list(df_dlag.columns)
print("\n=== First 30 observations: first differences + lag(first differences) ===")
print(df_fd[cols_show].head(30))
boundary_check = check_boundary_missing_dots(df_fd, id_col, d_cols + list(df_dlag.columns))
print("\n=== Boundary missing-dot check (should be True everywhere) ===")
print(boundary_check.head(30))
all_missing = boundary_check.drop(columns=[id_col]).all().all()
print("\nAll Δ and lag(Δ) missing at boundaries:", all_missing)
```

```
=== TWO-WAY-WITHIN correlation matrix (rounded) ===
                    ln_gdp_growth_5y  liquid_liabilities  mcap_gdp  \
ln_gdp_growth_5y              1.0000             -0.1070   -0.0130
liquid_liabilities           -0.1070              1.0000    0.1970
mcap_gdp                     -0.0130              0.1970    1.0000
bank_loans_gdp                0.0100              0.4330    0.0520
ln_cpi                        0.0320             -0.1840   -0.2020
trade_openness                0.0740              0.0710    0.0490

                    bank_loans_gdp  ln_cpi  trade_openness
ln_gdp_growth_5y            0.0100  0.0320          0.0740
liquid_liabilities          0.4330 -0.1840          0.0710
mcap_gdp                    0.0520 -0.2020          0.0490
bank_loans_gdp              1.0000 -0.1810         -0.0120
ln_cpi                     -0.1810  1.0000         -0.0080
trade_openness             -0.0120 -0.0080          1.0000

=== TWO-WAY-WITHIN: low |corr(y,x)| < 0.1 ===
bank_loans_gdp    0.0103
mcap_gdp         -0.0132
ln_cpi            0.0318
trade_openness    0.0741
Name: ln_gdp_growth_5y, dtype: float64

=== TWO-WAY-WITHIN: high |corr(x1,x2)| > 0.8 (among Xs) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== FIRST-DIFF correlation matrix (rounded) ===
                          d_ln_gdp_growth_5y  d_liquid_liabilities  \
d_ln_gdp_growth_5y                    1.0000               -0.3150
d_liquid_liabilities                 -0.3150                1.0000
d_mcap_gdp                           -0.0310                0.0790
d_bank_loans_gdp                     -0.1100                0.5430
d_ln_cpi                             -0.0670               -0.0610
d_trade_openness                      0.0360               -0.0380
d_ln_gdp_growth_5y_lag1               0.2330               -0.1370
d_liquid_liabilities_lag1            -0.0620                0.2230
d_mcap_gdp_lag1                       0.1260               -0.0410
d_bank_loans_gdp_lag1                 0.0280                0.1290
d_ln_cpi_lag1                         0.0660               -0.0460
d_trade_openness_lag1                 0.0090               -0.0020

                          d_mcap_gdp  d_bank_loans_gdp  d_ln_cpi  \
d_ln_gdp_growth_5y           -0.0310           -0.1100   -0.0670
d_liquid_liabilities          0.0790            0.5430   -0.0610
d_mcap_gdp                    1.0000            0.0520   -0.0390
d_bank_loans_gdp              0.0520            1.0000   -0.1140
d_ln_cpi                     -0.0390           -0.1140    1.0000
d_trade_openness              0.0140            0.0170   -0.0750
d_ln_gdp_growth_5y_lag1      -0.0450            0.0770    0.0060
d_liquid_liabilities_lag1     0.0130            0.0970   -0.0250
d_mcap_gdp_lag1               0.0360           -0.0530    0.0120
d_bank_loans_gdp_lag1         0.0130            0.2900   -0.0860
d_ln_cpi_lag1                -0.0350           -0.0800    0.2280
d_trade_openness_lag1         0.0080           -0.0150    0.0530

                          d_trade_openness  d_ln_gdp_growth_5y_lag1  \
d_ln_gdp_growth_5y                  0.0360                   0.2330
d_liquid_liabilities               -0.0380                  -0.1370
d_mcap_gdp                          0.0140                  -0.0450
d_bank_loans_gdp                    0.0170                   0.0770
d_ln_cpi                           -0.0750                   0.0060
d_trade_openness                    1.0000                   0.0510
d_ln_gdp_growth_5y_lag1             0.0510                   1.0000
d_liquid_liabilities_lag1          -0.0170                  -0.3050
d_mcap_gdp_lag1                     0.1130                  -0.0310
d_bank_loans_gdp_lag1               0.0050                  -0.1030
d_ln_cpi_lag1                      -0.0310                  -0.0680
d_trade_openness_lag1              -0.3240                   0.0380

                          d_liquid_liabilities_lag1  d_mcap_gdp_lag1  \
d_ln_gdp_growth_5y                          -0.0620           0.1260
d_liquid_liabilities                         0.2230          -0.0410
d_mcap_gdp                                   0.0130           0.0360
d_bank_loans_gdp                             0.0970          -0.0530
d_ln_cpi                                    -0.0250           0.0120
d_trade_openness                            -0.0170           0.1130
d_ln_gdp_growth_5y_lag1                     -0.3050          -0.0310
d_liquid_liabilities_lag1                    1.0000           0.0790
d_mcap_gdp_lag1                              0.0790           1.0000
d_bank_loans_gdp_lag1                        0.5360           0.0520
d_ln_cpi_lag1                               -0.0600          -0.0390
d_trade_openness_lag1                       -0.0450           0.0140

                          d_bank_loans_gdp_lag1  d_ln_cpi_lag1  \
d_ln_gdp_growth_5y                       0.0280         0.0660
d_liquid_liabilities                     0.1290        -0.0460
d_mcap_gdp                               0.0130        -0.0350
d_bank_loans_gdp                         0.2900        -0.0800
d_ln_cpi                                -0.0860         0.2280
d_trade_openness                         0.0050        -0.0310
d_ln_gdp_growth_5y_lag1                 -0.1030        -0.0680
d_liquid_liabilities_lag1                0.5360        -0.0600
d_mcap_gdp_lag1                          0.0520        -0.0390
d_bank_loans_gdp_lag1                    1.0000        -0.1150
d_ln_cpi_lag1                           -0.1150         1.0000
d_trade_openness_lag1                    0.0130        -0.0770

                          d_trade_openness_lag1
d_ln_gdp_growth_5y                       0.0090
d_liquid_liabilities                    -0.0020
d_mcap_gdp                               0.0080
d_bank_loans_gdp                        -0.0150
d_ln_cpi                                 0.0530
d_trade_openness                        -0.3240
d_ln_gdp_growth_5y_lag1                  0.0380
d_liquid_liabilities_lag1               -0.0450
d_mcap_gdp_lag1                          0.0140
d_bank_loans_gdp_lag1                    0.0130
d_ln_cpi_lag1                           -0.0770
d_trade_openness_lag1                    1.0000

=== FIRST-DIFF: low |corr(dy, dX or lag(dX))| < 0.1 ===
d_trade_openness_lag1        0.0088
d_bank_loans_gdp_lag1        0.0277
d_mcap_gdp                  -0.0315
d_trade_openness             0.0362
d_liquid_liabilities_lag1   -0.0618
d_ln_cpi_lag1                0.0662
d_ln_cpi                    -0.0670
Name: d_ln_gdp_growth_5y, dtype: float64

=== FIRST-DIFF: high |corr| > 0.8 (among dX and lag(dX), excluding dy) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== First 30 observations: first differences + lag(first differences) ===
      country  year  d_ln_gdp_growth_5y  d_liquid_liabilities  d_mcap_gdp  \
0   Australia  1882                 NaN                   NaN         NaN
1   Australia  1883                 NaN               -0.0122         NaN
2   Australia  1884                 NaN                0.0210         NaN
3   Australia  1885                 NaN                0.0148         NaN
4   Australia  1886                 NaN               -0.0028         NaN
5   Australia  1887                 NaN                0.0113         NaN
6   Australia  1888             -0.1246                0.0119         NaN
7   Australia  1889              0.0915               -0.0099         NaN
8   Australia  1890             -0.0899                0.0032         NaN
9   Australia  1891              0.0670                0.0139         NaN
10  Australia  1892             -0.2230                0.0436         NaN
11  Australia  1893             -0.0466                0.0198         NaN
12  Australia  1894             -0.0370                0.0347         NaN
13  Australia  1895             -0.0142                0.0366         NaN
14  Australia  1896              0.0096               -0.0116         NaN
15  Australia  1897              0.0806               -0.0323         NaN
16  Australia  1898              0.2037               -0.0346         NaN
17  Australia  1899             -0.0288               -0.0096         NaN
18  Australia  1900              0.1211                0.0056         NaN
19  Australia  1901             -0.0997                0.0036     -0.0216
20  Australia  1902              0.0706               -0.0166     -0.0157
21  Australia  1903             -0.0650                0.0162      0.0148
22  Australia  1904              0.0655               -0.0196     -0.0131
23  Australia  1905             -0.0472                0.0290      0.0390
24  Australia  1906              0.0957                0.0011      0.0164
25  Australia  1907              0.0280               -0.0242     -0.0598
26  Australia  1908             -0.0475                0.0188      0.0129
27  Australia  1909              0.0072               -0.0130      0.0130
28  Australia  1910              0.0480                0.0012     -0.0063
29  Australia  1911             -0.0718                0.0081     -0.0163

    d_bank_loans_gdp  d_ln_cpi  d_trade_openness  d_ln_gdp_growth_5y_lag1  \
0                NaN       NaN               NaN                      NaN
1             0.0015    0.1038           -0.0089                      NaN
2             0.0089    0.0140           -0.0185                      NaN
3             0.0292    0.0000            0.0011                      NaN
4             0.0055    0.0000           -0.0423                      NaN
5            -0.0015   -0.0282           -0.0119                      NaN
6             0.0382   -0.0438            0.0261                      NaN
7             0.0034    0.0000           -0.0116                  -0.1246
8             0.0075    0.0000           -0.0107                   0.0915
9             0.0194   -0.0150            0.0388                  -0.0899
10            0.0388   -0.0153           -0.0180                   0.0670
11            0.0177   -0.0313            0.0131                  -0.2230
12            0.0111   -0.0656            0.0060                  -0.0466
13           -0.0176   -0.0702            0.0404                  -0.0370
14           -0.0260    0.0000            0.0051                  -0.0142
15           -0.0427    0.0000           -0.0016                   0.0096
16           -0.0534   -0.0183            0.0294                   0.0806
17           -0.0422    0.0541           -0.0184                   0.2037
18           -0.0064    0.0174            0.0068                  -0.0288
19           -0.0057    0.0171           -0.0122                   0.1211
20           -0.0215    0.0656           -0.0375                  -0.0997
21            0.0165    0.0157            0.0124                   0.0706
22           -0.0198   -0.0480            0.0098                  -0.0650
23           -0.0045    0.0000            0.0022                   0.0655
24           -0.0180    0.0163            0.0274                  -0.0472
25           -0.0149    0.0000           -0.0063                   0.0957
26            0.0200    0.0317           -0.0204                   0.0280
27           -0.0259    0.0155           -0.0127                  -0.0475
28           -0.0176    0.0153            0.0152                   0.0072
29            0.0042    0.0730           -0.0027                   0.0480

    d_liquid_liabilities_lag1  d_mcap_gdp_lag1  d_bank_loans_gdp_lag1  \
0                         NaN              NaN                    NaN
1                         NaN              NaN                    NaN
2                     -0.0122              NaN                 0.0015
3                      0.0210              NaN                 0.0089
4                      0.0148              NaN                 0.0292
5                     -0.0028              NaN                 0.0055
6                      0.0113              NaN                -0.0015
7                      0.0119              NaN                 0.0382
8                     -0.0099              NaN                 0.0034
9                      0.0032              NaN                 0.0075
10                     0.0139              NaN                 0.0194
11                     0.0436              NaN                 0.0388
12                     0.0198              NaN                 0.0177
13                     0.0347              NaN                 0.0111
14                     0.0366              NaN                -0.0176
15                    -0.0116              NaN                -0.0260
16                    -0.0323              NaN                -0.0427
17                    -0.0346              NaN                -0.0534
18                    -0.0096              NaN                -0.0422
19                     0.0056              NaN                -0.0064
20                     0.0036          -0.0216                -0.0057
21                    -0.0166          -0.0157                -0.0215
22                     0.0162           0.0148                 0.0165
23                    -0.0196          -0.0131                -0.0198
24                     0.0290           0.0390                -0.0045
25                     0.0011           0.0164                -0.0180
26                    -0.0242          -0.0598                -0.0149
27                     0.0188           0.0129                 0.0200
28                    -0.0130           0.0130                -0.0259
29                     0.0012          -0.0063                -0.0176

    d_ln_cpi_lag1  d_trade_openness_lag1
0             NaN                    NaN
1             NaN                    NaN
2          0.1038                -0.0089
3          0.0140                -0.0185
4          0.0000                 0.0011
5          0.0000                -0.0423
6         -0.0282                -0.0119
7         -0.0438                 0.0261
8          0.0000                -0.0116
9          0.0000                -0.0107
10        -0.0150                 0.0388
11        -0.0153                -0.0180
12        -0.0313                 0.0131
13        -0.0656                 0.0060
14        -0.0702                 0.0404
15         0.0000                 0.0051
16         0.0000                -0.0016
17        -0.0183                 0.0294
18         0.0541                -0.0184
19         0.0174                 0.0068
20         0.0171                -0.0122
21         0.0656                -0.0375
22         0.0157                 0.0124
23        -0.0480                 0.0098
24         0.0000                 0.0022
25         0.0163                 0.0274
26         0.0000                -0.0063
27         0.0317                -0.0204
28         0.0155                -0.0127
29         0.0153                 0.0152

=== Boundary missing-dot check (should be True everywhere) ===
             country  d_ln_gdp_growth_5y_isna_at_boundary  \
0          Australia                                 True
139           Canada                                 True
278          Denmark                                 True
417          Finland                                 True
556           France                                 True
695          Germany                                 True
834            Italy                                 True
973            Japan                                 True
1112     Netherlands                                 True
1251          Norway                                 True
1390        Portugal                                 True
1529           Spain                                 True
1668          Sweden                                 True
1807     Switzerland                                 True
1946  United Kingdom                                 True
2085   United States                                 True

      d_liquid_liabilities_isna_at_boundary  d_mcap_gdp_isna_at_boundary  \
0                                      True                         True
139                                    True                         True
278                                    True                         True
417                                    True                         True
556                                    True                         True
695                                    True                         True
834                                    True                         True
973                                    True                         True
1112                                   True                         True
1251                                   True                         True
1390                                   True                         True
1529                                   True                         True
1668                                   True                         True
1807                                   True                         True
1946                                   True                         True
2085                                   True                         True

      d_bank_loans_gdp_isna_at_boundary  d_ln_cpi_isna_at_boundary  \
0                                  True                       True
139                                True                       True
278                                True                       True
417                                True                       True
556                                True                       True
695                                True                       True
834                                True                       True
973                                True                       True
1112                               True                       True
1251                               True                       True
1390                               True                       True
1529                               True                       True
1668                               True                       True
1807                               True                       True
1946                               True                       True
2085                               True                       True

      d_trade_openness_isna_at_boundary  \
0                                  True
139                                True
278                                True
417                                True
556                                True
695                                True
834                                True
973                                True
1112                               True
1251                               True
1390                               True
1529                               True
1668                               True
1807                               True
1946                               True
2085                               True

      d_ln_gdp_growth_5y_lag1_isna_at_boundary  \
0                                         True
139                                       True
278                                       True
417                                       True
556                                       True
695                                       True
834                                       True
973                                       True
1112                                      True
1251                                      True
1390                                      True
1529                                      True
1668                                      True
1807                                      True
1946                                      True
2085                                      True

      d_liquid_liabilities_lag1_isna_at_boundary  \
0                                           True
139                                         True
278                                         True
417                                         True
556                                         True
695                                         True
834                                         True
973                                         True
1112                                        True
1251                                        True
1390                                        True
1529                                        True
1668                                        True
1807                                        True
1946                                        True
2085                                        True

      d_mcap_gdp_lag1_isna_at_boundary  \
0                                 True
139                               True
278                               True
417                               True
556                               True
695                               True
834                               True
973                               True
1112                              True
1251                              True
1390                              True
1529                              True
1668                              True
1807                              True
1946                              True
2085                              True

      d_bank_loans_gdp_lag1_isna_at_boundary  d_ln_cpi_lag1_isna_at_boundary  \
0                                       True                            True
139                                     True                            True
278                                     True                            True
417                                     True                            True
556                                     True                            True
695                                     True                            True
834                                     True                            True
973                                     True                            True
1112                                    True                            True
1251                                    True                            True
1390                                    True                            True
1529                                    True                            True
1668                                    True                            True
1807                                    True                            True
1946                                    True                            True
2085                                    True                            True

      d_trade_openness_lag1_isna_at_boundary
0                                       True
139                                     True
278                                     True
417                                     True
556                                     True
695                                     True
834                                     True
973                                     True
1112                                    True
1251                                    True
1390                                    True
1529                                    True
1668                                    True
1807                                    True
1946                                    True
2085                                    True

All Δ and lag(Δ) missing at boundaries: True
```

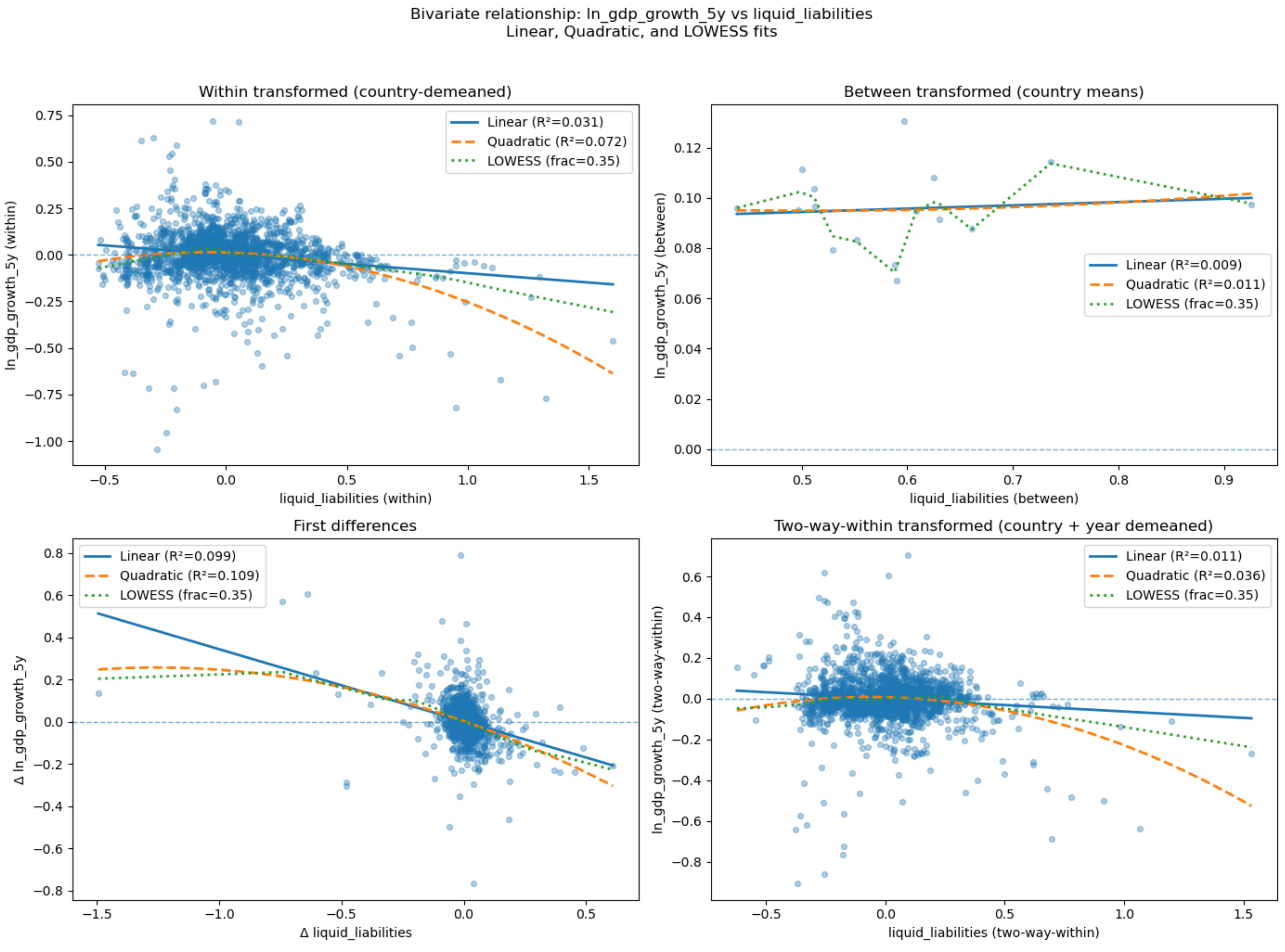
```
In [ ]: z
```

```
=== TWO-WAY-WITHIN correlation matrix (rounded) ===
                    ln_gdp_growth_5y  liquid_liabilities  mcap_gdp  \
ln_gdp_growth_5y              1.0000             -0.1070   -0.0130
liquid_liabilities           -0.1070              1.0000    0.1970
mcap_gdp                     -0.0130              0.1970    1.0000
bank_loans_gdp                0.0100              0.4330    0.0520
ln_cpi                        0.0320             -0.1840   -0.2020
trade_openness                0.0740              0.0710    0.0490

                    bank_loans_gdp  ln_cpi  trade_openness
ln_gdp_growth_5y            0.0100  0.0320          0.0740
liquid_liabilities          0.4330 -0.1840          0.0710
mcap_gdp                    0.0520 -0.2020          0.0490
bank_loans_gdp              1.0000 -0.1810         -0.0120
ln_cpi                     -0.1810  1.0000         -0.0080
trade_openness             -0.0120 -0.0080          1.0000

=== TWO-WAY-WITHIN: low |corr(y,x)| < 0.1 ===
bank_loans_gdp    0.0103
mcap_gdp         -0.0132
ln_cpi            0.0318
trade_openness    0.0741
Name: ln_gdp_growth_5y, dtype: float64

=== TWO-WAY-WITHIN: high |corr(x1,x2)| > 0.8 (among Xs) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== FIRST-DIFF correlation matrix (rounded) ===
                          d_ln_gdp_growth_5y  d_liquid_liabilities  \
d_ln_gdp_growth_5y                    1.0000               -0.3150
d_liquid_liabilities                 -0.3150                1.0000
d_mcap_gdp                           -0.0310                0.0790
d_bank_loans_gdp                     -0.1100                0.5430
d_ln_cpi                             -0.0670               -0.0610
d_trade_openness                      0.0360               -0.0380
d_ln_gdp_growth_5y_lag1               0.2330               -0.1370
d_liquid_liabilities_lag1            -0.0620                0.2230
d_mcap_gdp_lag1                       0.1260               -0.0410
d_bank_loans_gdp_lag1                 0.0280                0.1290
d_ln_cpi_lag1                         0.0660               -0.0460
d_trade_openness_lag1                 0.0090               -0.0020

                          d_mcap_gdp  d_bank_loans_gdp  d_ln_cpi  \
d_ln_gdp_growth_5y           -0.0310           -0.1100   -0.0670
d_liquid_liabilities          0.0790            0.5430   -0.0610
d_mcap_gdp                    1.0000            0.0520   -0.0390
d_bank_loans_gdp              0.0520            1.0000   -0.1140
d_ln_cpi                     -0.0390           -0.1140    1.0000
d_trade_openness              0.0140            0.0170   -0.0750
d_ln_gdp_growth_5y_lag1      -0.0450            0.0770    0.0060
d_liquid_liabilities_lag1     0.0130            0.0970   -0.0250
d_mcap_gdp_lag1               0.0360           -0.0530    0.0120
d_bank_loans_gdp_lag1         0.0130            0.2900   -0.0860
d_ln_cpi_lag1                -0.0350           -0.0800    0.2280
d_trade_openness_lag1         0.0080           -0.0150    0.0530

                          d_trade_openness  d_ln_gdp_growth_5y_lag1  \
d_ln_gdp_growth_5y                  0.0360                   0.2330
d_liquid_liabilities               -0.0380                  -0.1370
d_mcap_gdp                          0.0140                  -0.0450
d_bank_loans_gdp                    0.0170                   0.0770
d_ln_cpi                           -0.0750                   0.0060
d_trade_openness                    1.0000                   0.0510
d_ln_gdp_growth_5y_lag1             0.0510                   1.0000
d_liquid_liabilities_lag1          -0.0170                  -0.3050
d_mcap_gdp_lag1                     0.1130                  -0.0310
d_bank_loans_gdp_lag1               0.0050                  -0.1030
d_ln_cpi_lag1                      -0.0310                  -0.0680
d_trade_openness_lag1              -0.3240                   0.0380

                          d_liquid_liabilities_lag1  d_mcap_gdp_lag1  \
d_ln_gdp_growth_5y                          -0.0620           0.1260
d_liquid_liabilities                         0.2230          -0.0410
d_mcap_gdp                                   0.0130           0.0360
d_bank_loans_gdp                             0.0970          -0.0530
d_ln_cpi                                    -0.0250           0.0120
d_trade_openness                            -0.0170           0.1130
d_ln_gdp_growth_5y_lag1                     -0.3050          -0.0310
d_liquid_liabilities_lag1                    1.0000           0.0790
d_mcap_gdp_lag1                              0.0790           1.0000
d_bank_loans_gdp_lag1                        0.5360           0.0520
d_ln_cpi_lag1                               -0.0600          -0.0390
d_trade_openness_lag1                       -0.0450           0.0140

                          d_bank_loans_gdp_lag1  d_ln_cpi_lag1  \
d_ln_gdp_growth_5y                       0.0280         0.0660
d_liquid_liabilities                     0.1290        -0.0460
d_mcap_gdp                               0.0130        -0.0350
d_bank_loans_gdp                         0.2900        -0.0800
d_ln_cpi                                -0.0860         0.2280
d_trade_openness                         0.0050        -0.0310
d_ln_gdp_growth_5y_lag1                 -0.1030        -0.0680
d_liquid_liabilities_lag1                0.5360        -0.0600
d_mcap_gdp_lag1                          0.0520        -0.0390
d_bank_loans_gdp_lag1                    1.0000        -0.1150
d_ln_cpi_lag1                           -0.1150         1.0000
d_trade_openness_lag1                    0.0130        -0.0770

                          d_trade_openness_lag1
d_ln_gdp_growth_5y                       0.0090
d_liquid_liabilities                    -0.0020
d_mcap_gdp                               0.0080
d_bank_loans_gdp                        -0.0150
d_ln_cpi                                 0.0530
d_trade_openness                        -0.3240
d_ln_gdp_growth_5y_lag1                  0.0380
d_liquid_liabilities_lag1               -0.0450
d_mcap_gdp_lag1                          0.0140
d_bank_loans_gdp_lag1                    0.0130
d_ln_cpi_lag1                           -0.0770
d_trade_openness_lag1                    1.0000

=== FIRST-DIFF: low |corr(dy, dX or lag(dX))| < 0.1 ===
d_trade_openness_lag1        0.0088
d_bank_loans_gdp_lag1        0.0277
d_mcap_gdp                  -0.0315
d_trade_openness             0.0362
d_liquid_liabilities_lag1   -0.0618
d_ln_cpi_lag1                0.0662
d_ln_cpi                    -0.0670
Name: d_ln_gdp_growth_5y, dtype: float64

=== FIRST-DIFF: high |corr| > 0.8 (among dX and lag(dX), excluding dy) ===
Empty DataFrame
Columns: [var1, var2, corr]
Index: []

=== First 30 observations: first differences + lag(first differences) ===
      country  year  d_ln_gdp_growth_5y  d_liquid_liabilities  d_mcap_gdp  \
0   Australia  1882                 NaN                   NaN         NaN
1   Australia  1883                 NaN               -0.0122         NaN
2   Australia  1884                 NaN                0.0210         NaN
3   Australia  1885                 NaN                0.0148         NaN
4   Australia  1886                 NaN               -0.0028         NaN
5   Australia  1887                 NaN                0.0113         NaN
6   Australia  1888             -0.1246                0.0119         NaN
7   Australia  1889              0.0915               -0.0099         NaN
8   Australia  1890             -0.0899                0.0032         NaN
9   Australia  1891              0.0670                0.0139         NaN
10  Australia  1892             -0.2230                0.0436         NaN
11  Australia  1893             -0.0466                0.0198         NaN
12  Australia  1894             -0.0370                0.0347         NaN
13  Australia  1895             -0.0142                0.0366         NaN
14  Australia  1896              0.0096               -0.0116         NaN
15  Australia  1897              0.0806               -0.0323         NaN
16  Australia  1898              0.2037               -0.0346         NaN
17  Australia  1899             -0.0288               -0.0096         NaN
18  Australia  1900              0.1211                0.0056         NaN
19  Australia  1901             -0.0997                0.0036     -0.0216
20  Australia  1902              0.0706               -0.0166     -0.0157
21  Australia  1903             -0.0650                0.0162      0.0148
22  Australia  1904              0.0655               -0.0196     -0.0131
23  Australia  1905             -0.0472                0.0290      0.0390
24  Australia  1906              0.0957                0.0011      0.0164
25  Australia  1907              0.0280               -0.0242     -0.0598
26  Australia  1908             -0.0475                0.0188      0.0129
27  Australia  1909              0.0072               -0.0130      0.0130
28  Australia  1910              0.0480                0.0012     -0.0063
29  Australia  1911             -0.0718                0.0081     -0.0163

    d_bank_loans_gdp  d_ln_cpi  d_trade_openness  d_ln_gdp_growth_5y_lag1  \
0                NaN       NaN               NaN                      NaN
1             0.0015    0.1038           -0.0089                      NaN
2             0.0089    0.0140           -0.0185                      NaN
3             0.0292    0.0000            0.0011                      NaN
4             0.0055    0.0000           -0.0423                      NaN
5            -0.0015   -0.0282           -0.0119                      NaN
6             0.0382   -0.0438            0.0261                      NaN
7             0.0034    0.0000           -0.0116                  -0.1246
8             0.0075    0.0000           -0.0107                   0.0915
9             0.0194   -0.0150            0.0388                  -0.0899
10            0.0388   -0.0153           -0.0180                   0.0670
11            0.0177   -0.0313            0.0131                  -0.2230
12            0.0111   -0.0656            0.0060                  -0.0466
13           -0.0176   -0.0702            0.0404                  -0.0370
14           -0.0260    0.0000            0.0051                  -0.0142
15           -0.0427    0.0000           -0.0016                   0.0096
16           -0.0534   -0.0183            0.0294                   0.0806
17           -0.0422    0.0541           -0.0184                   0.2037
18           -0.0064    0.0174            0.0068                  -0.0288
19           -0.0057    0.0171           -0.0122                   0.1211
20           -0.0215    0.0656           -0.0375                  -0.0997
21            0.0165    0.0157            0.0124                   0.0706
22           -0.0198   -0.0480            0.0098                  -0.0650
23           -0.0045    0.0000            0.0022                   0.0655
24           -0.0180    0.0163            0.0274                  -0.0472
25           -0.0149    0.0000           -0.0063                   0.0957
26            0.0200    0.0317           -0.0204                   0.0280
27           -0.0259    0.0155           -0.0127                  -0.0475
28           -0.0176    0.0153            0.0152                   0.0072
29            0.0042    0.0730           -0.0027                   0.0480

    d_liquid_liabilities_lag1  d_mcap_gdp_lag1  d_bank_loans_gdp_lag1  \
0                         NaN              NaN                    NaN
1                         NaN              NaN                    NaN
2                     -0.0122              NaN                 0.0015
3                      0.0210              NaN                 0.0089
4                      0.0148              NaN                 0.0292
5                     -0.0028              NaN                 0.0055
6                      0.0113              NaN                -0.0015
7                      0.0119              NaN                 0.0382
8                     -0.0099              NaN                 0.0034
9                      0.0032              NaN                 0.0075
10                     0.0139              NaN                 0.0194
11                     0.0436              NaN                 0.0388
12                     0.0198              NaN                 0.0177
13                     0.0347              NaN                 0.0111
14                     0.0366              NaN                -0.0176
15                    -0.0116              NaN                -0.0260
16                    -0.0323              NaN                -0.0427
17                    -0.0346              NaN                -0.0534
18                    -0.0096              NaN                -0.0422
19                     0.0056              NaN                -0.0064
20                     0.0036          -0.0216                -0.0057
21                    -0.0166          -0.0157                -0.0215
22                     0.0162           0.0148                 0.0165
23                    -0.0196          -0.0131                -0.0198
24                     0.0290           0.0390                -0.0045
25                     0.0011           0.0164                -0.0180
26                    -0.0242          -0.0598                -0.0149
27                     0.0188           0.0129                 0.0200
28                    -0.0130           0.0130                -0.0259
29                     0.0012          -0.0063                -0.0176

    d_ln_cpi_lag1  d_trade_openness_lag1
0             NaN                    NaN
1             NaN                    NaN
2          0.1038                -0.0089
3          0.0140                -0.0185
4          0.0000                 0.0011
5          0.0000                -0.0423
6         -0.0282                -0.0119
7         -0.0438                 0.0261
8          0.0000                -0.0116
9          0.0000                -0.0107
10        -0.0150                 0.0388
11        -0.0153                -0.0180
12        -0.0313                 0.0131
13        -0.0656                 0.0060
14        -0.0702                 0.0404
15         0.0000                 0.0051
16         0.0000                -0.0016
17        -0.0183                 0.0294
18         0.0541                -0.0184
19         0.0174                 0.0068
20         0.0171                -0.0122
21         0.0656                -0.0375
22         0.0157                 0.0124
23        -0.0480                 0.0098
24         0.0000                 0.0022
25         0.0163                 0.0274
26         0.0000                -0.0063
27         0.0317                -0.0204
28         0.0155                -0.0127
29         0.0153                 0.0152

=== Boundary missing-dot check (should be True everywhere) ===
             country  d_ln_gdp_growth_5y_isna_at_boundary  \
0          Australia                                 True
139           Canada                                 True
278          Denmark                                 True
417          Finland                                 True
556           France                                 True
695          Germany                                 True
834            Italy                                 True
973            Japan                                 True
1112     Netherlands                                 True
1251          Norway                                 True
1390        Portugal                                 True
1529           Spain                                 True
1668          Sweden                                 True
1807     Switzerland                                 True
1946  United Kingdom                                 True
2085   United States                                 True

      d_liquid_liabilities_isna_at_boundary  d_mcap_gdp_isna_at_boundary  \
0                                      True                         True
139                                    True                         True
278                                    True                         True
417                                    True                         True
556                                    True                         True
695                                    True                         True
834                                    True                         True
973                                    True                         True
1112                                   True                         True
1251                                   True                         True
1390                                   True                         True
1529                                   True                         True
1668                                   True                         True
1807                                   True                         True
1946                                   True                         True
2085                                   True                         True

      d_bank_loans_gdp_isna_at_boundary  d_ln_cpi_isna_at_boundary  \
0                                  True                       True
139                                True                       True
278                                True                       True
417                                True                       True
556                                True                       True
695                                True                       True
834                                True                       True
973                                True                       True
1112                               True                       True
1251                               True                       True
1390                               True                       True
1529                               True                       True
1668                               True                       True
1807                               True                       True
1946                               True                       True
2085                               True                       True

      d_trade_openness_isna_at_boundary  \
0                                  True
139                                True
278                                True
417                                True
556                                True
695                                True
834                                True
973                                True
1112                               True
1251                               True
1390                               True
1529                               True
1668                               True
1807                               True
1946                               True
2085                               True

      d_ln_gdp_growth_5y_lag1_isna_at_boundary  \
0                                         True
139                                       True
278                                       True
417                                       True
556                                       True
695                                       True
834                                       True
973                                       True
1112                                      True
1251                                      True
1390                                      True
1529                                      True
1668                                      True
1807                                      True
1946                                      True
2085                                      True

      d_liquid_liabilities_lag1_isna_at_boundary  \
0                                           True
139                                         True
278                                         True
417                                         True
556                                         True
695                                         True
834                                         True
973                                         True
1112                                        True
1251                                        True
1390                                        True
1529                                        True
1668                                        True
1807                                        True
1946                                        True
2085                                        True

      d_mcap_gdp_lag1_isna_at_boundary  \
0                                 True
139                               True
278                               True
417                               True
556                               True
695                               True
834                               True
973                               True
1112                              True
1251                              True
1390                              True
1529                              True
1668                              True
1807                              True
1946                              True
```

```
      d_bank_loans_gdp_lag1_isna_at_boundary  d_ln_cpi_lag1_isna_at_boundary  \
0                                       True                            True
139                                     True                            True
278                                     True                            True
417                                     True                            True
556                                     True                            True
695                                     True                            True
834                                     True                            True
973                                     True                            True
1112                                    True                            True
1251                                    True                            True
1390                                    True                            True
1529                                    True                            True
1668                                    True                            True
1807                                    True                            True
1946                                    True                            True
2085                                    True                            True

      d_trade_openness_lag1_isna_at_boundary
0                                       True
139                                     True
278                                     True
417                                     True
556                                     True
695                                     True
834                                     True
973                                     True
1112                                    True
1251                                    True
1390                                    True
1529                                    True
1668                                    True
1807                                    True
1946                                    True
2085                                    True

All Δ and lag(Δ) missing at boundaries: True
```

In [180…

```python
id_col   = "country"
time_col = "year"

y_col = "ln_gdp_growth_5y"
x_col = "liquid_liabilities"

LOWESS_FRAC = 0.35
LOWESS_IT   = 1

# ----------------------------

def ensure_sorted(df, id_col, time_col):
    return df.sort_values([id_col, time_col]).reset_index(drop=True)

def within_transform(df, cols, id_col):
    return df[cols] - df.groupby(id_col)[cols].transform("mean")

def between_transform(df, cols, id_col):
    return df.groupby(id_col)[cols].mean().reset_index()

def twoway_within_transform(df, cols, id_col, time_col):
    overall = df[cols].mean(skipna=True)
    mean_i = df.groupby(id_col)[cols].transform("mean")
    mean_t = df.groupby(time_col)[cols].transform("mean")
    return df[cols] - mean_i - mean_t + overall

def first_difference(df, cols, id_col, time_col):
    df = ensure_sorted(df, id_col, time_col)
    d = df.groupby(id_col)[cols].diff()
    d.columns = [f"d_{c}" for c in cols]
    return d



def _fit_linear(x, y):
    X = sm.add_constant(x)
    return sm.OLS(y, X, missing="drop").fit()

def _fit_quadratic(x, y):
    X = np.column_stack([x, x**2])
    X = sm.add_constant(X)
    return sm.OLS(y, X, missing="drop").fit()

def _predict_on_grid(x, model_type="linear", params=None, xgrid=None):
    if xgrid is None:
        xgrid = np.linspace(np.nanmin(x), np.nanmax(x), 200)

    params = np.asarray(params, dtype=float)

    if model_type == "linear":
        b0, b1 = params
        yhat = b0 + b1 * xgrid

    elif model_type == "quadratic":
        b0, b1, b2 = params
        yhat = b0 + b1 * xgrid + b2 * xgrid**2

    else:
        raise ValueError("model_type must be 'linear' or 'quadratic'")

    return xgrid, yhat

def plot_bivariate_with_fits(ax, x, y, title, xlabel, ylabel):
    mask = np.isfinite(x) & np.isfinite(y)
    x = np.asarray(x[mask], dtype=float)
    y = np.asarray(y[mask], dtype=float)

    ax.scatter(x, y, alpha=0.35, s=18)

    if x.size < 10:
        ax.set_title(f"{title}\n(too few obs for smooth fits)")
        ax.set_xlabel(xlabel)
        ax.set_ylabel(ylabel)
        return


    lin = _fit_linear(x, y)
    xg, y_lin = _predict_on_grid(x, "linear", lin.params)   # <-- FIXED

    # Quad
    quad = _fit_quadratic(x, y)
    xg2, y_quad = _predict_on_grid(x, "quadratic", quad.params, xgrid=xg)  # <-- FIXED

    # LOWESS
    lo = lowess(y, x, frac=LOWESS_FRAC, it=LOWESS_IT, return_sorted=True)
    x_lo, y_lo = lo[:, 0], lo[:, 1]

    ax.plot(xg, y_lin, linewidth=2, label=f"Linear (R²={lin.rsquared:.3f})")
    ax.plot(xg2, y_quad, linewidth=2, linestyle="--", label=f"Quadratic (R²={quad.rsquared:.3f})")
    ax.plot(x_lo, y_lo, linewidth=2, linestyle=":", label=f"LOWESS (frac={LOWESS_FRAC})")

    ax.axhline(0, linewidth=1, linestyle="--", alpha=0.6)
    ax.set_title(title)
    ax.set_xlabel(xlabel)
    ax.set_ylabel(ylabel)
    ax.legend()


df0 = ensure_sorted(df.copy(), id_col, time_col)
needed = [id_col, time_col, y_col, x_col]
missing = [c for c in needed if c not in df0.columns]
if missing:
    raise KeyError(f"Missing columns in df: {missing}")

w = within_transform(df0, [y_col, x_col], id_col)
df_within = pd.concat([df0[[id_col, time_col]], w], axis=1)
df_between = between_transform(df0, [y_col, x_col], id_col)
d = first_difference(df0, [y_col, x_col], id_col, time_col)
df_fd = pd.concat([df0[[id_col, time_col]], d], axis=1)
tw = twoway_within_transform(df0, [y_col, x_col], id_col, time_col)
df_twoway = pd.concat([df0[[id_col, time_col]], tw], axis=1)
fig, axes = plt.subplots(2, 2, figsize=(14, 10), sharey=False)

plot_bivariate_with_fits(
    axes[0, 0], df_within[x_col], df_within[y_col],
    title="Within transformed (country-demeaned)",
    xlabel="liquid_liabilities (within)",
    ylabel="ln_gdp_growth_5y (within)"
)

plot_bivariate_with_fits(
    axes[0, 1], df_between[x_col], df_between[y_col],
    title="Between transformed (country means)",
    xlabel="liquid_liabilities (between)",
    ylabel="ln_gdp_growth_5y (between)"
)

plot_bivariate_with_fits(
    axes[1, 0], df_fd[f"d_{x_col}"], df_fd[f"d_{y_col}"],
    title="First differences",
    xlabel="Δ liquid_liabilities",
    ylabel="Δ ln_gdp_growth_5y"
)

plot_bivariate_with_fits(
    axes[1, 1], df_twoway[x_col], df_twoway[y_col],
    title="Two-way-within transformed (country + year demeaned)",
    xlabel="liquid_liabilities (two-way-within)",
    ylabel="ln_gdp_growth_5y (two-way-within)"
)

plt.suptitle(
    "Bivariate relationship: ln_gdp_growth_5y vs liquid_liabilities\n"
    "Linear, Quadratic, and LOWESS fits",
    y=1.02
)
plt.tight_layout()
plt.show()
```



In [9]:

```python
import pandas as pd
from pathlib import Path
import os
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde, skew, kurtosis, norm
from scipy.stats import pearsonr
import statsmodels.api as sm
import seaborn as sns
from statsmodels.nonparametric.smoothers_lowess import lowess
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
from scipy import stats
from linearmodels.panel import RandomEffects
from linearmodels.panel import PanelOLS
import statsmodels.api as sm
from linearmodels.iv import IV2SLS
import matplotlib.pyplot as plt

from statsmodels.tsa.stattools import adfuller
from scipy.stats import chi2
```

In [4]:

```python
# loading in my data frame classic first steps same as the other part in part 1

final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final = pd.read_excel(final_data_path / "final_panel_dataset.xlsx")

# In the study as one of teh explanatory varibles they create a 5 year lagged GDP per capita variable.
# Therefore, prior to plotting the distributions, we create this variable in the final data set.

df_Final = df_Final.sort_values(["country", "year"])
df_Final = df_Final[['country', 'year', 'gdppc', 'labour_productivity', 'cpi', 'trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']]
df_Final = df_Final.sort_values(["country", "year"])

# Logging GDP per capita first
df_Final['ln_gdppc'] = np.log(df_Final['gdppc'])
df_Final["ln_gdppc_lag5"] = df_Final.groupby("country")["ln_gdppc"].shift(5)
# Creating the 5 year lagged GDP per capita growth variable
df_Final["ln_gdp_growth_5y"] = df_Final["ln_gdppc"] - df_Final["ln_gdppc_lag5"]

# Doing of rthe alternative dependent variable: labour productivity growth
df_Final['ln_labour_productivity'] = np.log(df_Final['labour_productivity'])
df_Final["ln_labour_productivity_lag5"] = df_Final.groupby("country")["ln_labour_productivity"].shift(5)
df_Final["ln_labour_productivity_growth_5y"] = df_Final["ln_labour_productivity"] - df_Final["ln_labour_productivity_lag5"]

# taking teh logarithm of the other variables as well
df_Final['ln_cpi'] = np.log(df_Final['cpi'])
df_Final['ln_trade_openness'] = np.log(df_Final['trade_openness'])


variables = [ 'ln_gdppc','ln_labour_productivity', 'ln_cpi', 'ln_trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp', "ln_gdp_growth_5y", "ln_labour_productivity_growth_5y" ]

for var in variables:   # Between transformation
    df_Final[f'{var}_between'] = df_Final.groupby('country')[var].transform('mean')

    # One-way Within transformation
    df_Final[f'{var}_within'] = df_Final[var] - df_Final.groupby('country')[var].transform('mean')

rows = []

# Defining my Dependent and Explanatory Variables
Dependent_Variables_Growth = ['ln_gdp_growth_5y','ln_labour_productivity_growth_5y' ]
Explanatory_Variables = ['ln_gdppc_lag5', 'ln_labour_productivity_lag5', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']
Control_Variables = ['ln_cpi', 'ln_trade_openness']

individuals = df_Final['country']
time = df_Final['year']
dep_var_col = 'ln_gdp_growth_5y'
exp_var_col = 'liquid_liabilities'
cols_to_keep = ["country", "year", dep_var_col, exp_var_col, "ln_cpi", "ln_trade_openness"]
df_Final_defined = df_Final[cols_to_keep].dropna()

df_Final_defined["dep_var_col_mean_i"] = df_Final_defined.groupby(individuals)[dep_var_col].transform("mean")
df_Final_defined["exp_var_col_mean_i"] = df_Final_defined.groupby(individuals)[exp_var_col].transform("mean")

df_Final_defined["dep_var_col_within"] = df_Final_defined[dep_var_col] - df_Final_defined['dep_var_col_mean_i']
df_Final_defined["exp_var_col_within"] = df_Final_defined[exp_var_col] - df_Final_defined['exp_var_col_mean_i']

between_df_Final_defined = (
    df_Final_defined
    .groupby("country")
    .agg(
        dep_var_col_between=(dep_var_col, 'mean'),
        exp_var_col_between=(exp_var_col, 'mean')
    )
    .reset_index()
)

# TWFE transformation
df_twfe = df_Final.copy()


id_col = "country"
time_col = "year"
cols_twfe = ["liquid_liabilities","mcap_gdp","bank_loans_gdp","ln_gdp_growth_5y","ln_labour_productivity_growth_5y", "ln_cpi", "ln_trade_openness"]
for col in cols_twfe:
    mi = df_twfe.groupby(id_col)[col].transform("mean")
    mt = df_twfe.groupby(time_col)[col].transform("mean")
    mall = df_twfe[col].mean()
    df_twfe[col + "_twfe"] = df_twfe[col] - mi - mt + mall

# all the the data witgh the respecive data frame is on df_TWFE
```

Question 20

In [5]:

```python
id_col = "country"
time_col = "year"

dep_var = "ln_gdp_growth_5y"
x_vars = ["ln_gdppc_lag5", "liquid_liabilities", "mcap_gdp", "bank_loans_gdp"]
control_vars = ["ln_cpi", "ln_trade_openness"]

VIF_cutoff = 10.0
cluster = True

# variance inflation factor calculation
def calculate_vif(df, features):
    X = df[features]
    X = sm.add_constant(X, has_constant="add")
    vals = [variance_inflation_factor(X.values, i) for i in range(X.shape[1])]
    cols = list(X.columns)
    vif_series = pd.Series(vals, index=cols, name="VIF")
    if "const" in vif_series.index:
        vif_series = vif_series.drop("const")
    else:
        vif_series = vif_series.iloc[1:] if len(vif_series) > 0 else vif_series
    return vif_series

def drop_high_vif(X: pd.DataFrame, vif_cut: float, keep: set = None, verbose: bool = True):
    if keep is None:
        keep = set()

    Xw = X.copy()

    while True:
        v = calculate_vif(Xw, list(Xw.columns))
        droppable = [c for c in Xw.columns if c not in keep]
        if not droppable:
            break
        v_droppable = v.reindex(droppable).astype(float)
        worst = v_droppable.idxmax()
        worst_val = v_droppable.loc[worst]
        if np.isfinite(worst_val) and worst_val > vif_cut:
            if verbose:
                print(f"    [VIF] dropping '{worst}' (VIF={worst_val:.2f} > {vif_cut})")
            Xw = Xw.drop(columns=[worst])
        else:
            break
    return Xw, calculate_vif(Xw, list(Xw.columns))


def fit_ols(y: pd.Series, X: pd.DataFrame, clusters: pd.Series = None, add_const: bool = False):
    X_use = sm.add_constant(X, has_constant="add") if add_const else X
    model = sm.OLS(y.astype(float).values, X_use.astype(float).values)
    if clusters is not None:
        res = model.fit(cov_type="cluster", cov_kwds={"groups": clusters.values})
    else:
        res = model.fit()
    res.X_names = list(X_use.columns)
    return res


def short_print(title: str, res):
    print("\n" + "="*80)
    print(title)
    print("-"*80)
    params = pd.Series(res.params, index=res.X_names)
    print(params)
    print(f"\n nobs={int(res.nobs)}  R2={getattr(res,'rsquared',np.nan):.4f}")
    print("="*80)


def fisher_test_fe(df_levels: pd.DataFrame, y: str, X_cols: list):
    d = df_levels[[id_col, y] + X_cols].dropna().copy()
    for c in [y] + X_cols:
        d[c] = pd.to_numeric(d[c], errors="coerce")
    d = d.dropna(subset=[y] + X_cols)
    X_r = sm.add_constant(d[X_cols], has_constant="add").astype(float)
    yv = d[y].astype(float).values
    res_r = sm.OLS(yv, X_r.values).fit()
    dums = pd.get_dummies(d[id_col], drop_first=True, prefix="fe", dtype=float)
    X_u = pd.concat([X_r, dums], axis=1).astype(float)
    res_u = sm.OLS(yv, X_u.values).fit()
    q = dums.shape[1]
    n = len(d)
    k_u = X_u.shape[1]
    F = ((res_r.ssr - res_u.ssr) / q) / (res_u.ssr / (n - k_u))
    p = 1 - stats.f.cdf(F, q, n - k_u)
    return {"F": float(F), "df1": int(q), "df2": int(n - k_u), "p": float(p)}

def ancova_poolability_test(df_levels: pd.DataFrame, y: str, X_cols: list, add_time_fe: bool = False):
    d = df_levels[[id_col, time_col, y] + X_cols].dropna().copy()
    for c in [y] + X_cols:
        d[c] = pd.to_numeric(d[c], errors="coerce")
    d[time_col] = pd.to_numeric(d[time_col], errors="coerce")
    d = d.dropna(subset=[time_col, y] + X_cols)
    Xbase = sm.add_constant(d[X_cols], has_constant="add").astype(float)
    fe = pd.get_dummies(d[id_col], drop_first=True, prefix="fe", dtype=float)
    parts = [Xbase, fe]
    if add_time_fe:
        te = pd.get_dummies(d[time_col].astype(int), drop_first=True, prefix="t", dtype=float)
        parts.append(te)
    X_base = pd.concat(parts, axis=1).astype(float)
    inter_blocks = []
    for x in X_cols:
        inter_blocks.append(fe.mul(d[x].astype(float), axis=0))
    X_inter = pd.concat(inter_blocks, axis=1).astype(float)
    X_inter.columns = [f"{c}#c.{x}" for x in X_cols for c in fe.columns]

    X_unres = pd.concat([X_base, X_inter], axis=1).astype(float)

    yv = d[y].astype(float).values
    res_base = sm.OLS(yv, X_base.values).fit()
    res_unres = sm.OLS(yv, X_unres.values).fit()

    q = X_inter.shape[1]
    n = len(d)
    k_u = X_unres.shape[1]
    F = ((res_base.ssr - res_unres.ssr) / q) / (res_unres.ssr / (n - k_u))
    p = 1 - stats.f.cdf(F, q, n - k_u)
    return {"F": float(F), "df1": int(q), "df2": int(n - k_u), "p": float(p), "time_fe": add_time_fe}



def chow_group_test(df_levels: pd.DataFrame, y: str, X_cols: list, group_col: str):
    d = df_levels[[y, group_col] + X_cols].dropna().copy()
    g = d[group_col].astype(float)

    X0 = sm.add_constant(d[X_cols], has_constant="add")
    inter = pd.concat([(g * d[x]).rename(f"{group_col}*{x}") for x in X_cols], axis=1)
    X = pd.concat([X0, g.rename(group_col), inter], axis=1)

    res = sm.OLS(d[y].values, X.values).fit()

    k0 = X0.shape[1] + 1  # const + X + g
    R = np.zeros((len(X_cols), X.shape[1]))
    for i in range(len(X_cols)):
        R[i, k0 + i] = 1.0
    wald = res.wald_test(R)
    return {"chi2": float(wald.statistic), "df": len(X_cols), "p": float(wald.pvalue)}

def chow_time_break(df_levels: pd.DataFrame, y: str, X_cols: list, break_year: int):
    d = df_levels[[time_col, y] + X_cols].dropna().copy()
    X_full = sm.add_constant(d[X_cols], has_constant="add")
    res_full = sm.OLS(d[y].values, X_full.values).fit()

    d1 = d[d[time_col] < break_year]
    d2 = d[d[time_col] >= break_year]
    X1 = sm.add_constant(d1[X_cols], has_constant="add")
    X2 = sm.add_constant(d2[X_cols], has_constant="add")
    res1 = sm.OLS(d1[y].values, X1.values).fit()
    res2 = sm.OLS(d2[y].values, X2.values).fit()

    SSR_pooled = res_full.ssr
    SSR_split  = res1.ssr + res2.ssr
    k = X_full.shape[1]
    n1, n2 = len(d1), len(d2)

    F = ((SSR_pooled - SSR_split) / k) / (SSR_split / (n1 + n2 - 2*k))
    p = 1 - stats.f.cdf(F, k, (n1 + n2 - 2*k))
    return {"F": F, "df1": k, "df2": (n1 + n2 - 2*k), "p": p, "break_year": break_year}

if 'X_ALL' not in globals():
    try:
        X_ALL = Explanatory_Variables + Control_Variables
    except NameError:
        X_ALL = x_vars + control_vars

else:
    try:
        for c in Control_Variables:
            if c not in X_ALL:
                X_ALL.append(c)
    except NameError:
        for c in control_vars:
            if c not in X_ALL:
                X_ALL.append(c)

df_levels = df_Final[[id_col, time_col, dep_var] + X_ALL].dropna().copy()
ID_COL = id_col
T_COL = time_col
DEP = dep_var
VIF_CUT = VIF_cutoff
CLUSTER = cluster
if 'GROUP_COL' not in globals():
    GROUP_COL = None
if 'BREAK_YEAR' not in globals():
    BREAK_YEAR = None

print("\n>>> Fisher FE test (are country effects jointly significant?)")
print(fisher_test_fe(df_levels, DEP, X_ALL))

print("\n>>> ANCOVA / poolability test (common slopes) one-way base")
print(ancova_poolability_test(df_levels, DEP, X_ALL, add_time_fe=False))

print("\n>>> ANCOVA / poolability test (common slopes) two-way base")
print(ancova_poolability_test(df_levels, DEP, X_ALL, add_time_fe=True))

if GROUP_COL is not None:
    print("\n>>> Chow group test (0/1)")
    print(chow_group_test(df_levels, DEP, X_ALL, GROUP_COL))
if BREAK_YEAR is not None:
    print("\n>>> Chow time-break test")
    print(chow_time_break(df_levels, DEP, X_ALL, BREAK_YEAR))


print("\n>>> BETWEEN (BE) regression: OLS on country means (includes constant)")
between = df_levels.groupby(ID_COL)[[DEP] + X_ALL].mean().reset_index()

y_be = between[DEP]
X_be_raw = between[X_ALL]

X_be, vif_be = drop_high_vif(X_be_raw, VIF_CUT, verbose=True)
print("  BE regressors kept:", list(X_be.columns))
print("  VIF (BE):\n", vif_be)

res_be = fit_ols(y_be, X_be, clusters=None, add_const=True)
short_print("BETWEEN (BE)", res_be)

print("\n>>> WITHIN (one-way FE): within-transformed OLS (no constant)")
df_w = df_levels.sort_values([ID_COL, T_COL]).copy()
for c in [DEP] + X_ALL:
    df_w[c + "_within"] = df_w[c] - df_w.groupby(ID_COL)[c].transform("mean")

y_w = df_w[DEP + "_within"]
X_w_raw = df_w[[c + "_within" for c in X_ALL]]

X_w, vif_w = drop_high_vif(X_w_raw, VIF_CUT, verbose=True)
print("  FE(within) regressors kept:", list(X_w.columns))
print("  VIF (FE/within):\n", vif_w)

clusters = df_w[ID_COL] if CLUSTER else None
res_fe = fit_ols(y_w, X_w, clusters=clusters, add_const=False)
short_print("WITHIN / FE (one-way)", res_fe)



print("\n>>> MUNDLAK (RE + country means of ALL X): RandomEffects + constant")

df_m = df_levels.sort_values([ID_COL, T_COL]).copy()
for c in X_ALL:
    df_m[c + "_mean_i"] = df_m.groupby(ID_COL)[c].transform("mean")

mundlak_cols = X_ALL + [c + "_mean_i" for c in X_ALL]
X_m_raw = df_m[mundlak_cols]

X_m, vif_m = drop_high_vif(X_m_raw, VIF_CUT, verbose=True)
print("  Mundlak regressors kept:", list(X_m.columns))
print("  VIF (Mundlak):\n", vif_m)

panel_m = df_m.set_index([ID_COL, T_COL])
y_m = panel_m[DEP]
X_m_panel = sm.add_constant(panel_m[X_m.columns], has_constant="add")

re_mod = RandomEffects(y_m, X_m_panel)
re_res = re_mod.fit(cov_type="clustered", cluster_entity=True) if CLUSTER else re_mod.fit()
print(re_res.summary)



print("\n>>> TWO-WAY FE: OLS on your TWFE-transformed variables (no constant)")

needed = [DEP] + X_ALL
df_t = df_twfe[[ID_COL, T_COL] + needed].dropna().copy()


for col in needed:
    if col + "_twfe" not in df_twfe.columns:
        mi = df_t.groupby(ID_COL)[col].transform("mean")
        mt = df_t.groupby(T_COL)[col].transform("mean")
        mall = df_t[col].mean()
        df_t[col + "_twfe"] = df_t[col] - mi - mt + mall
    else:
        df_t[col + "_twfe"] = df_twfe.loc[df_t.index, col + "_twfe"]

y_twfe = df_t[DEP + "_twfe"]
X_twfe_raw = df_t[[c + "_twfe" for c in X_ALL]]

X_twfe, vif_twfe = drop_high_vif(X_twfe_raw, VIF_CUT, verbose=True)
print("  TWFE regressors kept:", list(X_twfe.columns))
print("  VIF (TWFE):\n", vif_twfe)

clusters = df_t[ID_COL] if CLUSTER else None
res_twfe = fit_ols(y_twfe, X_twfe, clusters=clusters, add_const=False)
short_print("TWO-WAY FE (TWFE transform)", res_twfe)

print("\n>>> FIRST DIFFERENCES: Δ-OLS (no constant)")

df_fd = df_levels.sort_values([ID_COL, T_COL]).copy()
for c in [DEP] + X_ALL:
    df_fd["d_" + c] = df_fd.groupby(ID_COL)[c].diff()

df_fd = df_fd.dropna(subset=["d_" + DEP] + ["d_" + c for c in X_ALL]).copy()
y_fd = df_fd["d_" + DEP]
X_fd_raw = df_fd[["d_" + c for c in X_ALL]]

X_fd, vif_fd = drop_high_vif(X_fd_raw, VIF_CUT, verbose=True)
print("  FD regressors kept:", list(X_fd.columns))
print("  VIF (FD):\n", vif_fd)

clusters = df_fd[ID_COL] if CLUSTER else None
res_fd = fit_ols(y_fd, X_fd, clusters=clusters, add_const=False)
short_print("FIRST DIFFERENCES (FD)", res_fd)

def coef_table():
    def pack_sm(res, name):
        return pd.Series(res.params, index=res.X_names, name=name)

    out = pd.concat([
        pack_sm(res_be, "BE"),
        pack_sm(res_fe, "FE_within"),
        pd.Series(re_res.params, name="Mundlak_RE"),
        pack_sm(res_twfe, "TWFE"),
        pack_sm(res_fd, "FD"),
    ], axis=1)

    return out

print("\n>>> Coefficient table:")
print(coef_table())
```

```
>>> Fisher FE test (are country effects jointly significant?)
{'F': 2.3613973272170266, 'df1': 8, 'df2': 1101, 'p': 0.016070775014142447}

>>> ANCOVA / poolability test (common slopes) one-way base
{'F': 8.212777334695192, 'df1': 48, 'df2': 1053, 'p': 1.1102230246251565e-16, 'time_fe': False}

>>> ANCOVA / poolability test (common slopes) two-way base
{'F': 25.757016923937602, 'df1': 48, 'df2': 927, 'p': 1.1102230246251565e-16, 'time_fe': True}

>>> BETWEEN (BE) regression: OLS on country means (includes constant)
    [VIF] dropping 'liquid_liabilities' (VIF=15.30 > 10.0)
  BE regressors kept: ['ln_gdppc_lag5', 'mcap_gdp', 'bank_loans_gdp', 'ln_cpi', 'ln_trade_openness']
  VIF (BE):
 ln_gdppc_lag5        2.400444
mcap_gdp             2.475139
bank_loans_gdp       1.189789
ln_cpi               1.961900
ln_trade_openness    1.568157
Name: VIF, dtype: float64

================================================================================
BETWEEN (BE)
--------------------------------------------------------------------------------
const                0.496256
ln_gdppc_lag5       -0.040350
mcap_gdp            -0.040641
bank_loans_gdp      -0.001434
ln_cpi               0.003367
ln_trade_openness    0.000969
dtype: float64

 nobs=9  R2=0.6695
================================================================================

>>> WITHIN (one-way FE): within-transformed OLS (no constant)
  FE(within) regressors kept: ['ln_gdppc_lag5_within', 'liquid_liabilities_within', 'mcap_gdp_within', 'bank_loans_gdp_within', 'ln_cpi_within', 'ln_trade_openness_within']
  VIF (FE/within):
 ln_gdppc_lag5_within         2.161711
liquid_liabilities_within    1.582851
mcap_gdp_within              1.655373
bank_loans_gdp_within        2.099611
ln_cpi_within                1.660694
ln_trade_openness_within     1.235519
Name: VIF, dtype: float64

================================================================================
WITHIN / FE (one-way)
--------------------------------------------------------------------------------
ln_gdppc_lag5_within        -0.000615
liquid_liabilities_within   -0.128656
mcap_gdp_within              0.031731
bank_loans_gdp_within       -0.009952
ln_cpi_within                0.001994
ln_trade_openness_within    -0.002723
dtype: float64

 nobs=1116  R2=0.0399
================================================================================

>>> MUNDLAK (RE + country means of ALL X): RandomEffects + constant
    [VIF] dropping 'ln_trade_openness_mean_i' (VIF=47.90 > 10.0)
    [VIF] dropping 'liquid_liabilities_mean_i' (VIF=13.02 > 10.0)
  Mundlak regressors kept: ['ln_gdppc_lag5', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp', 'ln_cpi', 'ln_trade_openness', 'ln_gdppc_lag5_mean_i', 'mcap_gdp_mean_i', 'bank_loans_gdp_mean_i', 'ln
_cpi_mean_i']
  VIF (Mundlak):
 ln_gdppc_lag5            2.184988
liquid_liabilities       2.445527
mcap_gdp                 2.286675
bank_loans_gdp           4.315489
ln_cpi                   1.704548
ln_trade_openness        2.324764
ln_gdppc_lag5_mean_i     2.639788
mcap_gdp_mean_i          3.428634
bank_loans_gdp_mean_i    3.055163
ln_cpi_mean_i            2.408766
Name: VIF, dtype: float64
                        RandomEffects Estimation Summary
================================================================================
Dep. Variable:       ln_gdp_growth_5y   R-squared:                        0.0518
Estimator:              RandomEffects   R-squared (Between):              0.6372
No. Observations:                1116   R-squared (Within):               0.0383
Date:                Sat, Dec 27 2025   R-squared (Overall):              0.0518
Time:                        17:40:44   Log-likelihood                    823.86
Cov. Estimator:             Clustered
                                        F-statistic:                      6.0386
Entities:                           9   P-value                           0.0000
Avg Obs:                       124.00   Distribution:                 F(10,1105)
Min Obs:                       118.00
Max Obs:                       127.00   F-statistic (robust):          5.437e+12
                                        P-value                           0.0000
Time periods:                     127   Distribution:                 F(10,1105)
Avg Obs:                       8.7874
Min Obs:                       6.0000
Max Obs:                       9.0000

                              Parameter Estimates
=====================================================================================
                       Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
-------------------------------------------------------------------------------------
const                     0.3404     0.2700     1.2607     0.2077     -0.1894      0.8703
ln_gdppc_lag5            -0.0023     0.0128    -0.1772     0.8594     -0.0273      0.0228
liquid_liabilities       -0.1133     0.0480    -2.3620     0.0183     -0.2074     -0.0192
mcap_gdp                  0.0299     0.0129     2.3204     0.0205      0.0046      0.0552
bank_loans_gdp           -0.0158     0.0504    -0.3127     0.7546     -0.1147      0.0832
ln_cpi                    0.0026     0.0005     4.8734     0.0000      0.0015      0.0036
ln_trade_openness         0.0047     0.0019     2.5080     0.0123      0.0010      0.0083
ln_gdppc_lag5_mean_i     -0.0164     0.0334    -0.4900     0.6242     -0.0820      0.0492
mcap_gdp_mean_i          -0.0242     0.0315    -0.7690     0.4420     -0.0860      0.0376
bank_loans_gdp_mean_i     0.0435     0.0415     1.0475     0.2951     -0.0380      0.1249
ln_cpi_mean_i            -0.0019     0.0018    -1.0187     0.3086     -0.0055      0.0017
=====================================================================================

>>> TWO-WAY FE: OLS on your TWFE-transformed variables (no constant)
  TWFE regressors kept: ['ln_gdppc_lag5_twfe', 'liquid_liabilities_twfe', 'mcap_gdp_twfe', 'bank_loans_gdp_twfe', 'ln_cpi_twfe', 'ln_trade_openness_twfe']
  VIF (TWFE):
 ln_gdppc_lag5_twfe         1.077105
liquid_liabilities_twfe    1.506131
mcap_gdp_twfe              1.082749
bank_loans_gdp_twfe        1.406294
ln_cpi_twfe                1.386821
ln_trade_openness_twfe     1.211558
Name: VIF, dtype: float64

================================================================================
TWO-WAY FE (TWFE transform)
--------------------------------------------------------------------------------
ln_gdppc_lag5_twfe        -0.175289
liquid_liabilities_twfe   -0.116228
mcap_gdp_twfe             -0.032740
bank_loans_gdp_twfe        0.041976
ln_cpi_twfe                0.000224
ln_trade_openness_twfe     0.008031
dtype: float64

 nobs=1116  R2=0.0839
================================================================================

>>> FIRST DIFFERENCES: Δ-OLS (no constant)
  FD regressors kept: ['d_ln_gdppc_lag5', 'd_liquid_liabilities', 'd_mcap_gdp', 'd_bank_loans_gdp', 'd_ln_cpi', 'd_ln_trade_openness']
  VIF (FD):
 d_ln_gdppc_lag5         1.005319
d_liquid_liabilities    1.513046
d_mcap_gdp              1.016562
d_bank_loans_gdp        1.500274
d_ln_cpi                1.052425
d_ln_trade_openness     1.050522
Name: VIF, dtype: float64

================================================================================
FIRST DIFFERENCES (FD)
--------------------------------------------------------------------------------
d_ln_gdppc_lag5        -0.924066
d_liquid_liabilities   -0.458900
d_mcap_gdp              0.005001
d_bank_loans_gdp        0.115599
d_ln_cpi               -0.006872
d_ln_trade_openness    -0.001844
dtype: float64

 nobs=1107  R2=0.5342
================================================================================

>>> Coefficient table:
                                 BE  FE_within  Mundlak_RE      TWFE        FD
const                      0.496256        NaN    0.340436       NaN       NaN
ln_gdppc_lag5             -0.040350        NaN   -0.002259       NaN       NaN
mcap_gdp                  -0.040641        NaN    0.029933       NaN       NaN
bank_loans_gdp            -0.001434        NaN   -0.015767       NaN       NaN
ln_cpi                     0.003367        NaN    0.002565       NaN       NaN
ln_trade_openness          0.000969        NaN    0.004680       NaN       NaN
ln_gdppc_lag5_within            NaN  -0.000615         NaN       NaN       NaN
liquid_liabilities_within       NaN  -0.128656         NaN       NaN       NaN
mcap_gdp_within                 NaN   0.031731         NaN       NaN       NaN
bank_loans_gdp_within           NaN  -0.009952         NaN       NaN       NaN
ln_cpi_within                   NaN   0.001994         NaN       NaN       NaN
ln_trade_openness_within        NaN  -0.002723         NaN       NaN       NaN
liquid_liabilities              NaN        NaN   -0.113279       NaN       NaN
ln_gdppc_lag5_mean_i            NaN        NaN   -0.016380       NaN       NaN
mcap_gdp_mean_i                 NaN        NaN   -0.024222       NaN       NaN
bank_loans_gdp_mean_i           NaN        NaN    0.043466       NaN       NaN
ln_cpi_mean_i                   NaN        NaN   -0.001874       NaN       NaN
ln_gdppc_lag5_twfe              NaN        NaN         NaN -0.175289       NaN
liquid_liabilities_twfe         NaN        NaN         NaN -0.116228       NaN
mcap_gdp_twfe                   NaN        NaN         NaN -0.032740       NaN
bank_loans_gdp_twfe             NaN        NaN         NaN  0.041976       NaN
ln_cpi_twfe                     NaN        NaN         NaN  0.000224       NaN
ln_trade_openness_twfe          NaN        NaN         NaN  0.008031       NaN
d_ln_gdppc_lag5                 NaN        NaN         NaN       NaN -0.924066
d_liquid_liabilities            NaN        NaN         NaN       NaN -0.458900
d_mcap_gdp                      NaN        NaN         NaN       NaN  0.005001
d_bank_loans_gdp                NaN        NaN         NaN       NaN  0.115599
d_ln_cpi                        NaN        NaN         NaN       NaN -0.006872
d_ln_trade_openness             NaN        NaN         NaN       NaN -0.001844
```

In [6]:

```python
def compute_vif(X: pd.DataFrame) -> pd.Series:
    Xc = X.copy()
    Xc = sm.add_constant(Xc, has_constant="add")
    vifs = {}
    for i, col in enumerate(Xc.columns):
        if col == "const":
            continue
        vifs[col] = variance_inflation_factor(Xc.values, i)
    return pd.Series(vifs).sort_values(ascending=False)

def mundlak_re(
    df: pd.DataFrame,
    y_col: str,
    X_cols: list[str],
    entity_col: str,
    time_col: str,
    vif_threshold: float = 10.0,
    robust: str = "clustered"
):
    d = df.copy()
    d = d.set_index([entity_col, time_col]).sort_index()

    for x in X_cols:
        d[f"{x}_mean"] = d.groupby(level=0)[x].transform("mean")

    Z_cols = X_cols + [f"{x}_mean" for x in X_cols]

    d_model = d[[y_col] + Z_cols].dropna()
    X_for_vif = d_model[Z_cols].astype(float)
    vif = compute_vif(X_for_vif)

    kept = list(X_for_vif.columns)
    dropped = []
    while True:
        vif = compute_vif(d_model[kept].astype(float))
        max_vif = vif.max()
        if max_vif <= vif_threshold:
            break
        worst = vif.idxmax()
        kept.remove(worst)
        dropped.append((worst, float(max_vif)))
    y = d_model[y_col]
    X = sm.add_constant(d_model[kept], has_constant="add")

    re_mod = RandomEffects(y, X)
    if robust == "clustered":
        re_res = re_mod.fit(cov_type="clustered", cluster_entity=True)
    elif robust == "robust":
        re_res = re_mod.fit(cov_type="robust")
    else:
        re_res = re_mod.fit()

    return {
        "result": re_res,
        "kept": kept,
        "dropped": dropped,
        "vif_final": compute_vif(d_model[kept].astype(float)),
        "nobs": int(re_res.nobs)
    }


y_col = "ln_gdp_growth_5y"
X_cols = ["ln_gdppc_lag5","liquid_liabilities","mcap_gdp","bank_loans_gdp","ln_cpi","ln_trade_openness"]
out = mundlak_re(df_twfe, y_col, X_cols, entity_col="country", time_col="year", vif_threshold=10.0)

print("Kept:", out["kept"])
print("Dropped (var, VIF at drop time):", out["dropped"])
print(out["vif_final"])
print(out["result"].summary)
```

```
Kept: ['ln_gdppc_lag5', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp', 'ln_cpi', 'ln_trade_openness', 'ln_gdppc_lag5_mean', 'mcap_gdp_mean', 'bank_loans_gdp_mean', 'ln_cpi_mean']
Dropped (var, VIF at drop time): [('ln_trade_openness_mean', 49.15483502336927), ('liquid_liabilities_mean', 13.69216475847366)]
bank_loans_gdp         4.356425
mcap_gdp_mean          3.403809
bank_loans_gdp_mean    3.183421
liquid_liabilities     2.492652
ln_gdppc_lag5_mean     2.349465
mcap_gdp               2.287390
ln_trade_openness      2.265604
ln_cpi_mean            2.251461
ln_gdppc_lag5          2.184780
ln_cpi                 1.700617
dtype: float64
                        RandomEffects Estimation Summary
================================================================================
Dep. Variable:       ln_gdp_growth_5y   R-squared:                        0.0516
Estimator:              RandomEffects   R-squared (Between):              0.6272
No. Observations:                1116   R-squared (Within):               0.0383
Date:                Sat, Dec 27 2025   R-squared (Overall):              0.0516
Time:                        17:41:38   Log-likelihood                    823.73
Cov. Estimator:             Clustered
                                        F-statistic:                      6.0097
Entities:                           9   P-value                           0.0000
Avg Obs:                       124.00   Distribution:                 F(10,1105)
Min Obs:                       118.00
Max Obs:                       127.00   F-statistic (robust):         -4.046e+11
                                        P-value                           1.0000
Time periods:                     127   Distribution:                 F(10,1105)
Avg Obs:                       8.7874
Min Obs:                       6.0000
Max Obs:                       9.0000

                              Parameter Estimates
=====================================================================================
                     Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
-------------------------------------------------------------------------------------
const                   0.3747     0.2813     1.3319     0.1832     -0.1773      0.9266
ln_gdppc_lag5          -0.0024     0.0127    -0.1874     0.8514     -0.0273      0.0226
liquid_liabilities     -0.1119     0.0486    -2.3012     0.0216     -0.2073     -0.0165
mcap_gdp                0.0293     0.0128     2.2824     0.0227      0.0041      0.0545
bank_loans_gdp         -0.0146     0.0508    -0.2871     0.7741     -0.1143      0.0851
ln_cpi                  0.0026     0.0005     5.0785     0.0000      0.0016      0.0036
ln_trade_openness       0.0046     0.0019     2.4121     0.0160      0.0009      0.0084
ln_gdppc_lag5_mean     -0.0203     0.0346    -0.5868     0.5575     -0.0881      0.0476
mcap_gdp_mean          -0.0222     0.0294    -0.7575     0.4489     -0.0799      0.0354
bank_loans_gdp_mean     0.0439     0.0442     0.9924     0.3212     -0.0429      0.1307
ln_cpi_mean            -0.0017     0.0015    -1.1802     0.2382     -0.0047      0.0012
=====================================================================================
```

In [7]:

```python
X_cols = ["ln_gdppc_lag5","liquid_liabilities","mcap_gdp","bank_loans_gdp","ln_cpi","ln_trade_openness"]
y_col = "ln_gdp_growth_5y"

# 1) Check coverage by country for each required variable
req = [y_col] + X_cols
coverage = (df_twfe.groupby("country")[req]
            .apply(lambda g: g.notna().mean())
           )

# Countries with low coverage in any required variable
worst = coverage.min(axis=1).sort_values()
print("Worst countries by minimum non-missing share across required vars:")
print(worst.head(20))

# 2) Countries that survive the complete-case requirement
complete_rows = df_twfe[req].notna().all(axis=1)
survivors = df_twfe.loc[complete_rows, "country"].unique()
print("\nCountries with at least one complete row:", sorted(survivors))
print("Count:", len(survivors))

# 3) For countries that dropped out, show which variable is the blocker
dropped = sorted(set(df_twfe["country"].unique()) - set(survivors))
print("\nDropped countries:", dropped)

for c in dropped:
    g = df_twfe[df_twfe["country"] == c]
    miss_rate = 1 - g[req].notna().mean()
    print(f"\n{c} missing-rate by variable:")
    print(miss_rate.sort_values(ascending=False))
```

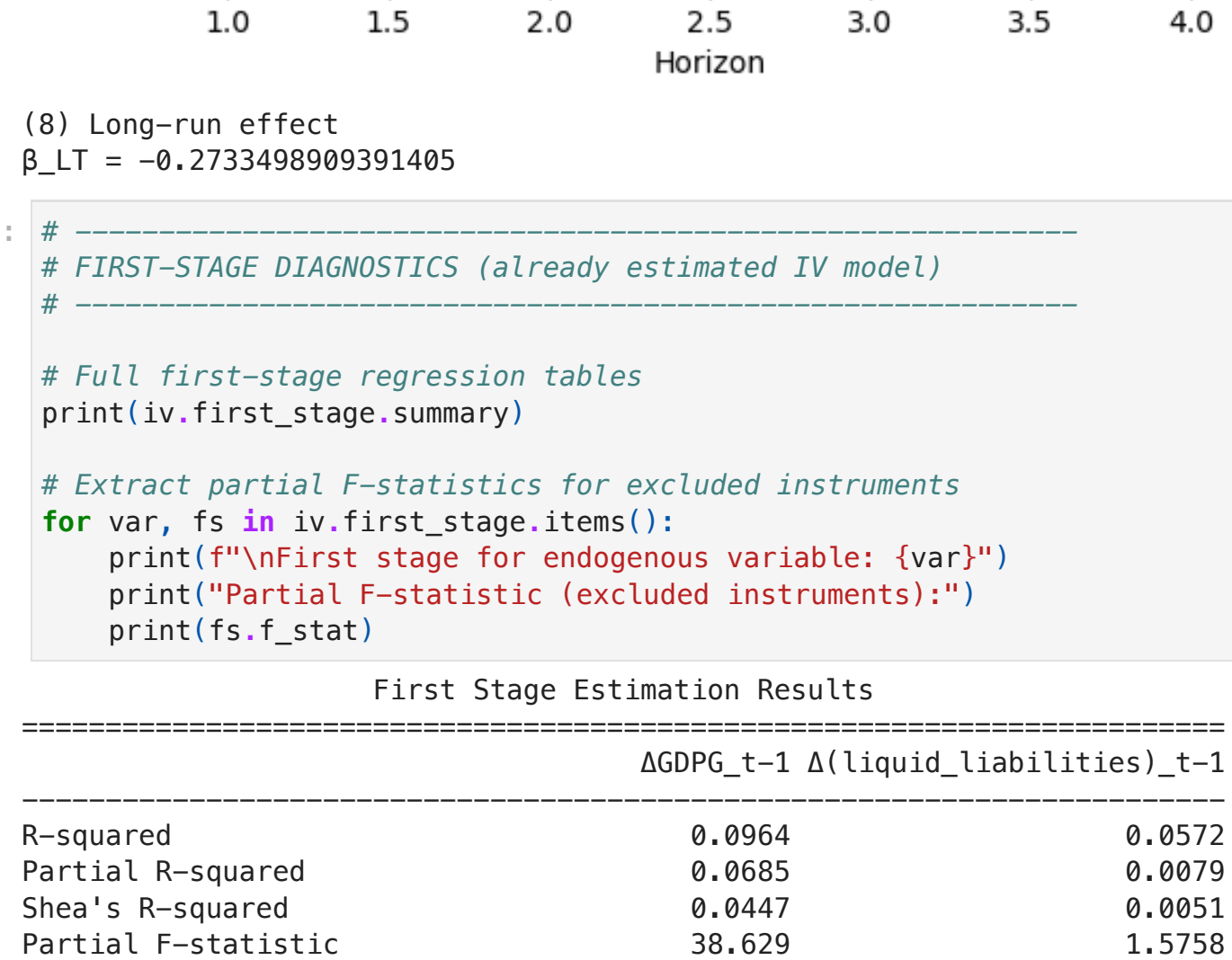
```
Worst countries by minimum non-missing share across required vars:
country
Finland           0.000000
France            0.000000
Italy             0.000000
Japan             0.000000
Netherlands       0.000000
Portugal          0.000000
Spain             0.000000
Australia         0.848921
Canada            0.848921
United States     0.848921
Denmark           0.913669
Germany           0.913669
Norway            0.913669
Sweden            0.913669
Switzerland       0.913669
United Kingdom    0.913669
dtype: float64

Countries with at least one complete row: ['Australia', 'Canada', 'Denmark', 'Germany', 'Norway', 'Sweden', 'Switzerland', 'United Kingdom', 'United States']
Count: 9

Dropped countries: ['Finland', 'France', 'Italy', 'Japan', 'Netherlands', 'Portugal', 'Spain']

Finland missing-rate by variable:
mcap_gdp              1.000000
ln_gdp_growth_5y      0.035971
ln_gdppc_lag5         0.035971
liquid_liabilities    0.000000
bank_loans_gdp        0.000000
ln_cpi                0.000000
ln_trade_openness     0.000000
dtype: float64

France missing-rate by variable:
bank_loans_gdp        1.000000
mcap_gdp              0.086331
ln_gdp_growth_5y      0.035971
ln_gdppc_lag5         0.035971
liquid_liabilities    0.000000
ln_cpi                0.000000
ln_trade_openness     0.000000
dtype: float64

Italy missing-rate by variable:
mcap_gdp              1.000000
ln_gdp_growth_5y      0.035971
ln_gdppc_lag5         0.035971
liquid_liabilities    0.000000
bank_loans_gdp        0.000000
ln_cpi                0.000000
ln_trade_openness     0.000000
dtype: float64

Japan missing-rate by variable:
mcap_gdp              1.000000
ln_gdp_growth_5y      0.035971
ln_gdppc_lag5         0.035971
liquid_liabilities    0.000000
bank_loans_gdp        0.000000
ln_cpi                0.000000
ln_trade_openness     0.000000
dtype: float64

Netherlands missing-rate by variable:
mcap_gdp              1.000000
bank_loans_gdp        1.000000
ln_gdp_growth_5y      0.035971
ln_gdppc_lag5         0.035971
liquid_liabilities    0.000000
ln_cpi                0.000000
ln_trade_openness     0.000000
dtype: float64

Portugal missing-rate by variable:
bank_loans_gdp        1.000000
mcap_gdp              0.086331
ln_gdp_growth_5y      0.035971
ln_gdppc_lag5         0.035971
liquid_liabilities    0.000000
```

```
ln_cpi                0.000000
ln_trade_openness     0.000000
dtype: float64

Spain missing-rate by variable:
bank_loans_gdp        1.000000
mcap_gdp              0.151079
ln_gdp_growth_5y      0.035971
ln_gdppc_lag5         0.035971
liquid_liabilities    0.000000
ln_cpi                0.000000
ln_trade_openness     0.000000
dtype: float64
```

In [10]:

```
df = df_twfe.copy()

ENTITY = "country"
TIME   = "year"


Y_raw = "ln_gdp_growth_5y"
X_raw = "liquid_liabilities"
CONTROLS_raw = ["ln_cpi", "ln_trade_openness"]


Y = "GDPG"
X = "liquid_liabilities"


rename_map = {Y_raw: Y, X_raw: X}
rename_map.update({c: c for c in CONTROLS_raw})
df = df.rename(columns=rename_map)

CONTROLS = CONTROLS_raw[:]

df = df.sort_values([ENTITY, TIME]).copy()

vars_needed = [Y, X] + CONTROLS
for v in vars_needed:
    df[v] = pd.to_numeric(df[v], errors="coerce")


df[f"{Y}_l1"] = df.groupby(ENTITY)[Y].shift(1)
df[f"{Y}_l2"] = df.groupby(ENTITY)[Y].shift(2)
df[f"{X}_l1"] = df.groupby(ENTITY)[X].shift(1)
df[f"{X}_l2"] = df.groupby(ENTITY)[X].shift(2)
df[f"d_{Y}"] = df[Y] - df[f"{Y}_l1"]
df[f"d_{X}"] = df[X] - df[f"{X}_l1"]
df[f"d_{Y}_l1"] = df.groupby(ENTITY)[f"d_{Y}"].shift(1)
df[f"d_{X}_l1"] = df.groupby(ENTITY)[f"d_{X}"].shift(1)
for c in CONTROLS:
    df[f"{c}_l1"] = df.groupby(ENTITY)[c].shift(1)
    df[f"d_{c}"] = df[c] - df[f"{c}_l1"]
needed = [
    f"d_{Y}", f"d_{Y}_l1",
    f"d_{X}", f"d_{X}_l1",
    f"{Y}_l2", f"{X}_l2"
] + [f"d_{c}" for c in CONTROLS]

df = df.dropna(subset=needed).copy()

print("Observations:", len(df))
print("Countries:", df[ENTITY].nunique())
print("\nAutocorrelation of Δy (lag 1):")
print(df.groupby(ENTITY)[f"d_{Y}"].apply(lambda s: s.autocorr(1)).describe())
def univar(s):
    return pd.Series({
        "mean": s.mean(),
        "median": s.median(),
        "std": s.std(),
        "stderr": s.std() / np.sqrt(len(s)),
        "skew": s.skew(),
        "kurt": s.kurt(),
        "min": s.min(),
        "max": s.max(),
        "n": len(s)
    })

vars_u = [f"d_{Y}", f"d_{Y}_l1", f"d_{X}", f"d_{X}_l1", f"{Y}_l2", f"{X}_l2"]
univar_table = pd.DataFrame({v: univar(df[v]) for v in vars_u}).T
print("\n(1) Univariate statistics")
print(univar_table)
corr_vars = [f"d_{Y}_l1", f"d_{X}", f"d_{X}_l1", f"{Y}_l2", f"{X}_l2"]
print("\n(2) Correlation matrix")
print(df[corr_vars].corr())
def fisher_adf(panel, var, entity_col, maxlag=1, regression="c", autolag=None, p_floor=1e-16):
    pvals = []
    used = 0
    skipped = 0

    for ent, g in panel.groupby(entity_col):
        s = g[var].dropna().values
        if len(s) <= maxlag + 2:
            skipped += 1
            continue
        try:
            res = adfuller(s, maxlag=maxlag, regression=regression, autolag=autolag)
            p = float(res[1])
            p = max(p, p_floor)  # avoid log(0)
            pvals.append(p)
            used += 1
        except Exception:
            skipped += 1
            continue

    if used == 0:
        return np.nan, np.nan, {"used": 0, "skipped": skipped}

    fisher_stat = -2.0 * np.sum(np.log(pvals))
    df_chi = 2 * used
    pval = chi2.sf(fisher_stat, df_chi)

    meta = {"used": used, "skipped": skipped, "df": df_chi, "min_p": min(pvals), "max_p": max(pvals)}
    return fisher_stat, pval, meta

print("(3) Fisher-ADF tests")
stat_y, p_y, meta_y = fisher_adf(df, f"d_{Y}", entity_col=ENTITY, maxlag=1, regression="c", autolag=None)
stat_x, p_x, meta_x = fisher_adf(df, f"d_{X}", entity_col=ENTITY, maxlag=1, regression="c", autolag=None)

print("Δy:", (stat_y, p_y), meta_y["used"], "entities used,", meta_y["skipped"], "skipped")
print("Δx:", (stat_x, p_x), meta_x["used"], "entities used,", meta_x["skipped"], "skipped")
y_ols = df[f"d_{Y}"]
X_ols = df[[f"d_{Y}_l1", f"d_{X}", f"d_{X}_l1"] + [f"d_{c}" for c in CONTROLS]].copy()
X_ols = X_ols.rename(columns={
    f"d_{Y}_l1": f"Δ{Y}_t-1",
    f"d_{X}":    f"Δ({X})_t",
    f"d_{X}_l1": f"Δ({X})_t-1",
    **{f"d_{c}": f"Δ{c}_t" for c in CONTROLS}
})

X_ols = sm.add_constant(X_ols)

ols = sm.OLS(y_ols, X_ols).fit(cov_type="HC1")

print("\n(4) OLS FD-ARDL")
print(ols.summary())
dep = df[f"d_{Y}"].copy()
exog = df[[f"d_{X}"] + [f"d_{c}" for c in CONTROLS]].copy()
exog = exog.rename(columns={
    f"d_{X}": f"Δ({X})_t",
    **{f"d_{c}": f"Δ{c}_t" for c in CONTROLS}
})
exog = sm.add_constant(exog)
endog = df[[f"d_{Y}_l1", f"d_{X}_l1"]].copy()
endog = endog.rename(columns={
    f"d_{Y}_l1": f"Δ{Y}_t-1",
    f"d_{X}_l1": f"Δ({X})_t-1"
})
instr = df[[f"{Y}_l2", f"{X}_l2"]].copy()
instr = instr.rename(columns={
    f"{Y}_l2": f"{Y}_t-2 (level)",
    f"{X}_l2": f"({X})_t-2 (level)"
})

iv = IV2SLS(dep, exog, endog, instr).fit(cov_type="robust")

print("\n(5) Anderson-Hsiao IV")
print(iv.summary)
print("\n(6) First-stage regressions")
print(iv.first_stage.summary)

# ---------------------------------------------------------------
# 9. (7) IMPULSE RESPONSE FUNCTIONS (safe: use names, not positions)
#    βy = coeff on ΔGDPG_{t-1}
#    β1 = coeff on ΔX_t
#    β2 = coeff on ΔX_{t-1}
# ---------------------------------------------------------------
beta_y_name = f"Δ{Y}_t-1"
beta_1_name = f"Δ({X})_t"
beta_2_name = f"Δ({X})_t-1"

βy = float(iv.params[beta_y_name])
β1 = float(iv.params[beta_1_name])
β2 = float(iv.params[beta_2_name])

IRF = [
    β1,
    βy * β1 + β2,
    (βy**2) * β1 + βy * β2,
    (βy**3) * β1 + (βy**2) * β2
]

irf_df = pd.DataFrame({"t": [1, 2, 3, 4], "IRF": IRF})
print("\n(7) IRFs")
print(irf_df)

plt.plot(irf_df["t"], irf_df["IRF"], marker="o")
plt.xlabel("Horizon")
plt.ylabel("Δ GDPG")
plt.title("Impulse Response: 1-unit increase in X")
plt.show()

β_LT = (β1 + β2) / (1 - βy)
print("\n(8) Long-run effect")
print("β_LT =", β_LT)
```

```
Observations: 2112
Countries: 16

Autocorrelation of Δy (lag 1):
count    16.000000
mean      0.207681
std       0.182037
min      -0.185585
25%       0.125790
50%       0.227598
75%       0.317863
max       0.466952
Name: d_GDPG, dtype: float64

(1) Univariate statistics
                          mean    median       std    stderr      skew  \
d_GDPG               -0.000266 -0.001509  0.074836  0.001628  0.667090
d_GDPG_l1             0.000106 -0.001302  0.074651  0.001624  0.666517
d_liquid_liabilities  0.005504  0.005124  0.068793  0.001497 -5.830477
d_liquid_liabilities_l1  0.004378  0.004927  0.067442  0.001468 -6.243464
GDPG_l2               0.096491  0.100122  0.132004  0.002872 -1.097063
liquid_liabilities_l2  0.595839  0.551024  0.249750  0.005434  1.318177

                           kurt       min       max       n
d_GDPG                19.581472 -0.766403  0.791372  2112.0
d_GDPG_l1             19.776906 -0.766403  0.791372  2112.0
d_liquid_liabilities  130.266391 -1.493865  0.609135  2112.0
d_liquid_liabilities_l1  140.558559 -1.493865  0.609135  2112.0
GDPG_l2                9.395162 -0.946174  0.803192  2112.0
liquid_liabilities_l2  3.603346  0.057906  2.259868  2112.0

(2) Correlation matrix
                        d_GDPG_l1  d_liquid_liabilities  \
d_GDPG_l1                1.000000             -0.136539
d_liquid_liabilities    -0.136539              1.000000
d_liquid_liabilities_l1 -0.304802              0.222596
GDPG_l2                 -0.284774              0.171308
liquid_liabilities_l2    0.027101             -0.126784

                        d_liquid_liabilities_l1   GDPG_l2  \
d_GDPG_l1                             -0.304802 -0.284774
d_liquid_liabilities                   0.222596  0.171308
d_liquid_liabilities_l1                1.000000  0.098385
GDPG_l2                                0.098385  1.000000
liquid_liabilities_l2                 -0.095948 -0.143237

                        liquid_liabilities_l2
d_GDPG_l1                            0.027101
d_liquid_liabilities                -0.126784
d_liquid_liabilities_l1             -0.095948
GDPG_l2                             -0.143237
liquid_liabilities_l2                1.000000
(3) Fisher-ADF tests
Δy: (np.float64(797.7323444661399), np.float64(4.865575826677976e-147)) 16 entities used, 0 skipped
Δx: (np.float64(793.0860928203259), np.float64(4.550937451556444e-146)) 16 entities used, 0 skipped

(4) OLS FD-ARDL
                            OLS Regression Results
==============================================================================
Dep. Variable:                 d_GDPG   R-squared:                       0.149
Model:                            OLS   Adj. R-squared:                  0.147
Method:                 Least Squares   F-statistic:                     10.10
Date:                Sat, 27 Dec 2025   Prob (F-statistic):           1.44e-09
Time:                        17:44:25   Log-Likelihood:                 2648.8
No. Observations:                2112   AIC:                            -5286.
Df Residuals:                    2106   BIC:                            -5252.
Df Model:                           5
Covariance Type:                  HC1
==============================================================================
                             coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                      0.0019      0.002      1.126      0.260      -0.001       0.005
ΔGDPG_t-1                  0.2104      0.054      3.866      0.000       0.104       0.317
Δ(liquid_liabilities)_t   -0.3311      0.076     -4.339      0.000      -0.481      -0.182
Δ(liquid_liabilities)_t-1  0.0764      0.051      1.501      0.133      -0.023       0.176
Δln_cpi_t                 -0.0130      0.004     -3.385      0.001      -0.020      -0.005
Δln_trade_openness_t       0.0085      0.024      0.357      0.721      -0.038       0.055
==============================================================================
Omnibus:                      501.570   Durbin-Watson:                   1.944
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            26271.819
Skew:                           0.094   Prob(JB):                         0.00
Kurtosis:                      20.277   Cond. No.                         17.6
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

(5) Anderson-Hsiao IV
                          IV-2SLS Estimation Summary
==============================================================================
Dep. Variable:                 d_GDPG   R-squared:                     -1.3326
Estimator:                    IV-2SLS   Adj. R-squared:                -1.3382
No. Observations:                2112   F-statistic:                    81.960
Date:                Sat, Dec 27 2025   P-value (F-stat)                0.0000
Time:                        17:44:25   Distribution:                  chi2(5)
Cov. Estimator:                robust

                                  Parameter Estimates
=========================================================================================
                           Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
-----------------------------------------------------------------------------------------
const                         0.0002     0.0035     0.0587     0.9532     -0.0066     0.0070
Δ(liquid_liabilities)_t      -0.2138     0.1498    -1.4272     0.1535     -0.5074     0.0798
Δln_cpi_t                    -0.0155     0.0028    -5.6042     0.0000     -0.0209    -0.0101
Δln_trade_openness_t         -0.0338     0.0254    -1.3304     0.1834     -0.0836     0.0160
ΔGDPG_t-1                     1.4965     0.2731     5.4796     0.0000      0.9613     2.0318
Δ(liquid_liabilities)_t-1     0.3495     0.7240     0.4828     0.6293     -1.0695     1.7685
=========================================================================================

Endogenous: ΔGDPG_t-1, Δ(liquid_liabilities)_t-1
Instruments: GDPG_t-2 (level), (liquid_liabilities)_t-2 (level)
Robust Covariance (Heteroskedastic)
Debiased: False

(6) First-stage regressions
                      First Stage Estimation Results
=================================================================
                                   ΔGDPG_t-1 Δ(liquid_liabilities)_t-1
-----------------------------------------------------------------
R-squared                             0.0964                0.0572
Partial R-squared                     0.0685                0.0079
Shea's R-squared                      0.0447                0.0051
Partial F-statistic                   38.629                1.5758
P-value (Partial F-stat)           4.092e-09                0.4548
Partial F-stat Distn                 chi2(2)               chi2(2)
================================ =========== ===========
const                                 0.0194                0.0107
                                    (3.1883)              (1.1897)
Δ(liquid_liabilities)_t              -0.0943                0.2009
                                   (-2.2878)              (2.5645)
Δln_cpi_t                             0.0011               -0.0016
                                    (0.2848)             (-0.4369)
Δln_trade_openness_t                  0.0258                0.0009
                                    (3.2732)              (0.1057)
GDPG_t-2 (level)                     -0.1497                0.0278
                                   (-6.1391)              (1.2446)
(liquid_liabilities)_t-2 (level)     -0.0075               -0.0169
                                   (-0.8619)             (-0.9056)
-----------------------------------------------------------------

T-stats reported in parentheses
T-stats use same covariance type as original model

(7) IRFs
   t       IRF
0  1 -0.213792
1  2  0.029573
2  3  0.044258
3  4  0.066233
```


Impulse Response: 1-unit increase in X

```
(8) Long-run effect
β_LT = -0.2733498909391405
```

In [11]:

```
# ---------------------------------------------------------------
# FIRST-STAGE DIAGNOSTICS (already estimated IV model)
# ---------------------------------------------------------------

# Full first-stage regression tables
print(iv.first_stage.summary)

# Extract partial F-statistics for excluded instruments
for var, fs in iv.first_stage.items():
    print(f"\nFirst stage for endogenous variable: {var}")
    print("Partial F-statistic (excluded instruments):")
    print(fs.f_stat)
```

```
                      First Stage Estimation Results
=================================================================
                                   ΔGDPG_t-1 Δ(liquid_liabilities)_t-1
-----------------------------------------------------------------
R-squared                             0.0964                0.0572
Partial R-squared                     0.0685                0.0079
Shea's R-squared                      0.0447                0.0051
Partial F-statistic                   38.629                1.5758
P-value (Partial F-stat)           4.092e-09                0.4548
Partial F-stat Distn                 chi2(2)               chi2(2)
================================ =========== ===========
const                                 0.0194                0.0107
                                    (3.1883)              (1.1897)
Δ(liquid_liabilities)_t              -0.0943                0.2009
                                   (-2.2878)              (2.5645)
Δln_cpi_t                             0.0011               -0.0016
                                    (0.2848)             (-0.4369)
Δln_trade_openness_t                  0.0258                0.0009
                                    (3.2732)              (0.1057)
GDPG_t-2 (level)                     -0.1497                0.0278
                                   (-6.1391)              (1.2446)
(liquid_liabilities)_t-2 (level)     -0.0075               -0.0169
                                   (-0.8619)             (-0.9056)
-----------------------------------------------------------------

T-stats reported in parentheses
T-stats use same covariance type as original model
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[11], line 9
      6 print(iv.first_stage.summary)
      8 # Extract partial F-statistics for excluded instruments
----> 9 for var, fs in iv.first_stage.items():
     10     print(f"\nFirst stage for endogenous variable: {var}")
     11     print("Partial F-statistic (excluded instruments):")

AttributeError: 'FirstStageResults' object has no attribute 'items'
```

In [12]:

```
df_fe = df_twfe.copy()
df_fe = df_fe.set_index([ENTITY, TIME])

# Build lags in LEVELS
df_fe["GDPG_l1"] = df_fe.groupby(level=0)["ln_gdp_growth_5y"].shift(1)
df_fe["X_l1"]    = df_fe.groupby(level=0)["liquid_liabilities"].shift(1)

fe_vars = [
    "ln_gdp_growth_5y",
    "GDPG_l1",
    "liquid_liabilities",
    "X_l1",
    "ln_cpi",
    "ln_trade_openness"
]

df_fe = df_fe[fe_vars].dropna()

fe_model = PanelOLS(
    df_fe["ln_gdp_growth_5y"],
    sm.add_constant(df_fe[[
        "GDPG_l1",
        "liquid_liabilities",
        "X_l1",
        "ln_cpi",
        "ln_trade_openness"
    ]]),
    entity_effects=True,
    time_effects=True
)

fe_res = fe_model.fit(cov_type="robust")
print(fe_res.summary)
```

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:       ln_gdp_growth_5y   R-squared:                        0.6850
Estimator:                   PanelOLS   R-squared (Between):             -1.0889
No. Observations:                2128   R-squared (Within):               0.7279
Date:                Sat, Dec 27 2025   R-squared (Overall):              0.7021
Time:                        17:44:41   Log-likelihood                    2919.8
Cov. Estimator:                Robust
                                        F-statistic:                      858.88
Entities:                          16   P-value                           0.0000
Avg Obs:                       133.00   Distribution:                  F(5,1975)
Min Obs:                       133.00
Max Obs:                       133.00   F-statistic (robust):             220.01
                                        P-value                           0.0000
Time periods:                     133   Distribution:                  F(5,1975)
Avg Obs:                       16.000
Min Obs:                       16.000
Max Obs:                       16.000

                              Parameter Estimates
======================================================================================
                    Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
--------------------------------------------------------------------------------------
const                  0.0448     0.0221     2.0278     0.0427      0.0015      0.0881
GDPG_l1                0.8202     0.0279     29.430     0.0000      0.7656      0.8749
liquid_liabilities    -0.2641     0.0701    -3.7676     0.0002     -0.4016     -0.1266
X_l1                   0.2538     0.0749     3.3870     0.0007      0.1069      0.4008
ln_cpi                 0.0005     0.0006     0.9101     0.3629     -0.0006      0.0016
ln_trade_openness      0.0050     0.0047     1.0749     0.2825     -0.0041      0.0142
======================================================================================

F-test for Poolability: 3.1379
P-value: 0.0000
Distribution: F(147,1975)

Included effects: Entity, Time
```

In [57]:

```python
import pandas as pd
from pathlib import Path
import os
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde, skew, kurtosis, norm
from scipy.stats import pearsonr
import statsmodels.api as sm
import seaborn as sns
from statsmodels.nonparametric.smoothers_lowess import lowess
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
from scipy import stats
from linearmodels.panel import RandomEffects
from linearmodels.panel import PanelOLS
import statsmodels.formula.api as smf
from statsmodels.tsa.stattools import adfuller
from statsmodels.stats.diagnostic import acorr_breusch_godfrey
from statsmodels.tsa.stattools import coint
from statsmodels.tsa.vector_ar.vecm import coint_johansen
```

In [21]:

```python
final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final = pd.read_excel(final_data_path / "final_panel_dataset.xlsx")

df_Final = df_Final.sort_values(["country", "year"])
df_Final = df_Final[['country', 'year', 'gdppc', 'labour_productivity', 'cpi', 'trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']]
df_Final = df_Final.sort_values(["country", "year"])

# Logging GDP per capita first
df_Final['ln_gdppc'] = np.log(df_Final['gdppc'])
df_Final["ln_gdppc_lag_2"] = df_Final.groupby("country")["ln_gdppc"].shift(2)
# Creating the 5 year lagged GDP per capita growth variable
df_Final["ln_gdp_growth_2y"] = df_Final["ln_gdppc"] - df_Final["ln_gdppc_lag_2"]

# Doing of rthe alternative dependent variable: labour productivity growth
df_Final['ln_labour_productivity'] = np.log(df_Final['labour_productivity'])
df_Final["ln_labour_productivity_lag2"] = df_Final.groupby("country")["ln_labour_productivity"].shift(2)
df_Final["ln_labour_productivity_growth_2y"] = df_Final["ln_labour_productivity"] - df_Final["ln_labour_productivity_lag2"]

# taking teh logarithm of the other variables as well
df_Final['ln_cpi'] = np.log(df_Final['cpi'])
df_Final['ln_trade_openness'] = np.log(df_Final['trade_openness'])

variables = [ 'ln_gdppc','ln_labour_productivity', 'ln_cpi', 'ln_trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp', "ln_gdp_growth_2y", "ln_labour_productivity_growth_2y" ]

y = 'ln_gdp_growth_2y'
X = ['liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']
Con = ['ln_cpi', 'ln_trade_openness']

# Splitting my data into the seperate regimes -> pre 1928, 1929–1971, post 1971

df_Regime = df_Final.copy()

conditions = [
        (df_Regime["year"] < 1928),
        (df_Regime["year"] >= 1928) & (df_Regime["year"] <= 1971),
        (df_Regime["year"] >= 1974)
]
regimes = ['Gold Standard', 'Bretton Woods','Finanzialisation']

regime_arr = np.select(conditions, regimes, default='')
df_Regime['regime'] = pd.Series(regime_arr, index=df_Regime.index).replace('', np.nan)

df_GS = df_Regime[df_Regime['regime'] == 'Gold Standard']
df_BW = df_Regime[df_Regime['regime'] == 'Bretton Woods']
df_Fin = df_Regime[df_Regime['regime'] == 'Finanzialisation']
```

In [25]:

```python
import numpy as np
from scipy import stats
import statsmodels.formula.api as smf

def poolability_test(df, y, x, controls, entity):

    rhs_vars = x + controls
    rhs_common = " + ".join(rhs_vars)

    # Restricted
    restricted_formula = f"{y} ~ {rhs_common} + C({entity})"

    # Unrestricted
    interactions = " + ".join([f"{var}:C({entity})" for var in rhs_vars])
    unrestricted_formula = (
        f"{y} ~ {rhs_common} + C({entity}) + {interactions}"
    )

    #  estimate models
    m_r = smf.ols(restricted_formula, data=df).fit()
    m_u = smf.ols(unrestricted_formula, data=df).fit()

    # SSR
    SSR_r = np.sum(m_r.resid ** 2)
    SSR_u = np.sum(m_u.resid ** 2)

    # dof
    q = int(m_u.df_model - m_r.df_model)   # number of restrictions
    df_u = int(m_u.df_resid)

    # F star
    F = ((SSR_r - SSR_u) / q) / (SSR_u / df_u)
    p_value = 1 - stats.f.cdf(F, q, df_u)

    return {
        "Restricted formula": restricted_formula,
        "Unrestricted formula": unrestricted_formula,
        "F statistic": F,
        "p-value": p_value,
        "q restrictions": q,
        "df residual (unrestricted)": df_u
    }

y = "ln_gdp_growth_2y"
X = ['liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']
Con = ['ln_cpi', 'ln_trade_openness']

results = {}

for name, df_sub in {
    "Gold Standard (GS)": df_GS,
    "Bretton Woods (BW)": df_BW,
    "Financialization (FIN)": df_Fin
}.items():

    res = poolability_test(
        df=df_sub,
        y=y,
        x=X,
        controls=Con,
        entity="country"
    )

    results[name] = res

    print(f"\n{name}")
    print("-" * 50)
    print(f"F-statistic : {res['F statistic']:.3f}")
    print(f"p-value     : {res['p-value']:.4f}")
    print(f"Restrictions: {res['q restrictions']}")
```

```

Gold Standard (GS)
--------------------------------------------------
F-statistic : 2.753
p-value     : 0.0000
Restrictions: 40

Bretton Woods (BW)
--------------------------------------------------
F-statistic : 5.474
p-value     : 0.0000
Restrictions: 40

Financialization (FIN)
--------------------------------------------------
F-statistic : 3.365
p-value     : 0.0000
Restrictions: 40
```

In [33]:

```python
# Adf Test for stationarity

def adf_test(series, maxlag=None, regression = 'c'):

    series = series.dropna()
    if len(series) < 10:
        return None
    result = adfuller(series, maxlag=maxlag, regression=regression, autolag="AIC")
    return {
        "ADF statistic": result[0],
        "p-value": result[1],
        "used lags": result[2],
        "nobs": result[3],
        "5% crit": result[4]["5%"],
        "stationary_5pct": result[1] < 0.05
    }

vars_to_test = [
    "ln_gdp_growth_2y",
    "liquid_liabilities",
    "bank_loans_gdp",
    "mcap_gdp",
    "ln_cpi",
    "ln_trade_openness"
]

def run_adf_by_country(df, variables, regime_name):
    results = []

    for country in df["country"].unique():
        df_c = df[df["country"] == country]

        for var in variables:
            res = adf_test(df_c[var], regression="c")
            if res is None:
                continue

            results.append({
                "regime": regime_name,
                "country": country,
                "variable": var,
                "ADF stat": res["ADF statistic"],
                "p-value": res["p-value"],
                "stationary (5%)": res["stationary_5pct"],
                "nobs": res["nobs"]
            })

    return pd.DataFrame(results)

adf_GS  = run_adf_by_country(df_GS,  vars_to_test, "GS")
adf_BW  = run_adf_by_country(df_BW,  vars_to_test, "BW")
adf_FIN = run_adf_by_country(df_Fin, vars_to_test, "FIN")

adf_results = pd.concat([adf_GS, adf_BW, adf_FIN], ignore_index=True)

adf_results.head()
```

Out[33]:

	regime	country	variable	ADF stat	p-value	stationary (5%)	nobs
0	GS	Australia	ln_gdp_growth_2y	-3.126785	0.024633	True	43
1	GS	Australia	liquid_liabilities	-2.948312	0.040011	True	41
2	GS	Australia	bank_loans_gdp	-0.527711	0.886494	False	45
3	GS	Australia	mcap_gdp	-2.288818	0.175600	False	28
4	GS	Australia	ln_cpi	-0.583640	0.874636	False	42

In [34]:

```python
stationarity_by_var = (
    adf_results
    .groupby("variable")["stationary (5%)"]
    .mean()
    .reset_index()
    .rename(columns={"stationary (5%)": "share_stationary"})
)

stationarity_by_var["share_nonstationary"] = 1 - stationarity_by_var["share_stationary"]

print(stationarity_by_var)
```

```
             variable  share_stationary  share_nonstationary
0      bank_loans_gdp          0.000000             1.000000
1  liquid_liabilities          0.062500             0.937500
2              ln_cpi          0.270833             0.729167
3    ln_gdp_growth_2y          0.500000             0.500000
4   ln_trade_openness          0.125000             0.875000
5            mcap_gdp          0.166667             0.833333
```

In [ ]:

```python
#serial Autocorrelation test

def bg_test_one_country(df_country, y, x, controls, nlags=2):
    rhs = " + ".join(x + controls)
    formula = f"{y} ~ {rhs}"


    model = smf.ols(formula, data=df_country).fit()

    # Breusch–Godfrey test
    lm_stat, lm_pvalue, f_stat, f_pvalue = acorr_breusch_godfrey(model, nlags=nlags)

    return {
        "LM stat": lm_stat,
        "LM p-value": lm_pvalue,
        "F stat": f_stat,
        "F p-value": f_pvalue,
        "nobs": int(model.nobs)
    }

def bg_test_by_regime(df_regime, regime_name, y, x, controls, nlags=2, min_obs=15):
    results = []

    for country in sorted(df_regime["country"].unique()):
        df_c = df_regime[df_regime["country"] == country].dropna(subset=[y] + x + controls)

        # skip if too few observations
        if len(df_c) < min_obs:
            continue

        out = bg_test_one_country(df_c, y, x, controls, nlags=nlags)
        results.append({
            "regime": regime_name,
            "country": country,
            "nlags": nlags,
            **out
        })

    return pd.DataFrame(results)

y = "ln_gdp_growth_2y"
X = ['liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']
Con = ['ln_cpi', 'ln_trade_openness']

# oicking the lag order
nlags = 2

bg_GS  = bg_test_by_regime(df_GS,  "GS",  y, X, Con, nlags=nlags)
bg_BW  = bg_test_by_regime(df_BW,  "BW",  y, X, Con, nlags=nlags)
bg_FIN = bg_test_by_regime(df_Fin, "FIN", y, X, Con, nlags=nlags)

bg_results = pd.concat([bg_GS, bg_BW, bg_FIN], ignore_index=True)

# Flag rejection of no autocorrelation
bg_results["reject_no_autocorr_5pct"] = bg_results["LM p-value"] < 0.05

# summary
summary_bg = (
    bg_results.groupby("regime")["reject_no_autocorr_5pct"]
    .mean()
    .reset_index(name="share_reject_BG_5pct")
)

print(summary_bg)

# see dem p value
print(bg_results.sort_values(["regime", "LM p-value"]).head())
print(bg_results)
```

```
  regime  share_reject_BG_5pct
0     BW              1.000000
1    FIN              1.000000
2     GS              0.888889
   regime         country  nlags    LM stat    LM p-value     F stat  \
9      BW       Australia      2  29.818532  3.349563e-07  37.847532
10     BW          Canada      2  26.829070  1.493281e-06  28.124467
16     BW  United Kingdom      2  25.064821  3.607806e-06  23.826909
17     BW   United States      2  20.264650  3.977289e-05  15.367951
12     BW         Germany      2  18.438536  9.911121e-05  12.984141

       F p-value  nobs  reject_no_autocorr_5pct
9   1.408748e-09    44                     True
10  4.406976e-08    44                     True
16  2.562720e-07    44                     True
17  1.496144e-05    44                     True
12  5.680608e-05    44                     True
   regime         country  nlags    LM stat    LM p-value     F stat  \
0      GS       Australia      2   9.482135  8.729325e-03   5.101091
1      GS          Canada      2  12.817247  1.647291e-03   8.316328
2      GS         Denmark      2   9.907001  7.058658e-03   5.289752
3      GS         Germany      2  10.070108  6.505846e-03   5.408242
4      GS          Norway      2   4.683625  9.615320e-02   2.108704
5      GS          Sweden      2   7.824262  1.999784e-02   3.889348
6      GS     Switzerland      2   6.355874  4.167153e-02   3.012822
7      GS  United Kingdom      2  16.338233  2.832681e-04  11.313643
8      GS   United States      2   8.730515  1.271138e-02   4.522582
9      BW       Australia      2  29.818532  3.349563e-07  37.847532
10     BW          Canada      2  26.829070  1.493281e-06  28.124467
11     BW         Denmark      2  15.899787  3.526996e-04  10.184840
12     BW         Germany      2  18.438536  9.911121e-05  12.984141
13     BW          Norway      2   9.512039  8.599772e-03   4.964536
14     BW          Sweden      2  17.734827  1.409066e-04  12.153999
15     BW     Switzerland      2  15.381977  4.569263e-04   9.674867
16     BW  United Kingdom      2  25.064821  3.607806e-06  23.826909
17     BW   United States      2  20.264650  3.977289e-05  15.367951
18    FIN       Australia      2  13.741450  1.037725e-03   8.218978
19    FIN          Canada      2  17.865961  1.319641e-04  12.439478
20    FIN         Denmark      2  14.642900  6.612026e-04   9.036564
21    FIN         Germany      2   9.005892  1.107632e-02   4.636189
22    FIN          Norway      2  24.570773  4.618750e-06  23.331881
23    FIN          Sweden      2  11.261491  3.585901e-03   6.209368
24    FIN     Switzerland      2  18.423299  9.986919e-05  13.118430
25    FIN  United Kingdom      2  17.089328  1.945806e-04  11.542087
26    FIN   United States      2  15.116786  5.217129e-04   9.487564

       F p-value  nobs  reject_no_autocorr_5pct
0   1.564445e-02    29                     True
1   2.187114e-03    29                     True
2   1.077005e-02    38                     True
3   9.869266e-03    38                     True
4   1.390312e-01    38                    False
5   3.148579e-02    38                     True
6   6.421591e-02    38                     True
7   2.180992e-04    38                     True
8   2.326320e-02    29                     True
9   1.408748e-09    44                     True
10  4.406976e-08    44                     True
11  3.123379e-04    44                     True
12  5.680608e-05    44                     True
13  1.247002e-02    44                     True
14  9.261826e-05    44                     True
15  4.338756e-04    44                     True
16  2.562720e-07    44                     True
17  1.496144e-05    44                     True
18  1.185021e-03    43                     True
19  8.296445e-05    43                     True
20  6.853047e-04    43                     True
21  1.636311e-02    43                     True
22  3.636442e-07    43                     True
23  4.921145e-03    43                     True
24  5.603571e-05    43                     True
25  1.413119e-04    43                     True
26  5.102717e-04    43                     True
```

In [43]:

```python
# testing to see if there are some structural breask int tyeh data set

formula = f"{y} ~ " + " + ".join(X + Con)


def chow_test_known_break(df_country, formula, break_year):

    df_country = df_country.dropna().copy()
    df_country = df_country.sort_values("year")

    df_pre  = df_country[df_country["year"] <= break_year]
    df_post = df_country[df_country["year"] > break_year]

    # Need enough obs on both sides
    if len(df_pre) < 10 or len(df_post) < 10:
        return None

    m_full = smf.ols(formula, data=df_country).fit()
    m_pre  = smf.ols(formula, data=df_pre).fit()
    m_post = smf.ols(formula, data=df_post).fit()

    SSR_full = float(np.sum(m_full.resid**2))
    SSR_pre  = float(np.sum(m_pre.resid**2))
    SSR_post = float(np.sum(m_post.resid**2))

    k = int(m_full.df_model + 1)  # number of parameters incl intercept
    n = int(m_full.nobs)

    # Chow F-stat
    num = (SSR_full - (SSR_pre + SSR_post)) / k
    den = (SSR_pre + SSR_post) / (n - 2*k)
    if den <= 0:
        return None

    F = num / den
    p = 1 - stats.f.cdf(F, k, n - 2*k)

    return {"break_year": break_year, "F": F, "p_value": p, "n": n, "k": k}

breaks_by_regime = {
    "GS":  [1893, 1907, 1914],
    "BW":  [1933, 1945, 1958],
    "FIN": [1987, 2001, 2008],
}

def chow_tests_regime(df_regime, regime_name, formula, break_years):
    out = []
    for country in sorted(df_regime["country"].unique()):
        df_c = df_regime[df_regime["country"] == country].copy()
        for by in break_years:
            res = chow_test_known_break(df_c, formula, by)
            if res is None:
                continue
            out.append({"regime": regime_name, "country": country, **res})
    return pd.DataFrame(out)

chow_GS  = chow_tests_regime(df_GS,  "GS",  formula, breaks_by_regime["GS"])
chow_BW  = chow_tests_regime(df_BW,  "BW",  formula, breaks_by_regime["BW"])
chow_FIN = chow_tests_regime(df_Fin, "FIN", formula, breaks_by_regime["FIN"])

chow_results = pd.concat([chow_GS, chow_BW, chow_FIN], ignore_index=True)

# Flag significant breaks
chow_results["reject_stability_5pct"] = chow_results["p_value"] < 0.05

# Summary table: share of (country × breakdate) tests rejecting stability
chow_summary = (
    chow_results.groupby("regime")["reject_stability_5pct"]
    .mean().reset_index(name="share_reject_chow_5pct")
)
print(chow_summary)

# (Optional) see the strongest breaks
print(chow_results.sort_values("p_value").head(15))
```

```
  regime  share_reject_chow_5pct
0     BW                0.611111
1    FIN                0.722222
2     GS                0.400000
   regime         country  break_year          F       p_value   n  k  \
17     BW          Canada        1945  14.082841  8.839473e-08  44  6
45    FIN     Switzerland        1987  10.841699  1.711632e-06  43  6
31     BW   United States        1945   8.018613  2.505816e-05  44  6
19     BW         Denmark        1945   7.861060  2.986636e-05  44  6
46    FIN     Switzerland        2001   7.083086  8.142059e-05  43  6
13     GS  United Kingdom        1914   7.705412  1.277131e-04  35  6
29     BW  United Kingdom        1945   6.509265  1.456766e-04  44  6
25     BW          Sweden        1945   6.063597  2.539212e-04  44  6
49    FIN   United States        1987   4.982212  1.124784e-03  43  6
33    FIN       Australia        1987   4.962267  1.155454e-03  43  6
32     BW   United States        1958   4.807013  1.342247e-03  44  6
50    FIN   United States        2001   4.693941  1.665748e-03  43  6
15     BW       Australia        1945   4.287258  2.795966e-03  44  6
42    FIN          Norway        2001   4.117362  3.747029e-03  43  6
11     GS     Switzerland        1914   4.444258  3.987821e-03  35  6

    reject_stability_5pct
17                   True
45                   True
31                   True
19                   True
46                   True
13                   True
29                   True
25                   True
49                   True
33                   True
32                   True
50                   True
15                   True
42                   True
11                   True
```

In [53]:

```python
y = "ln_gdp_growth_2y"

finance_vars = ["liquid_liabilities", "bank_loans_gdp", "mcap_gdp"]
controls = ["ln_cpi", "ln_trade_openness"]
base_rhs = finance_vars + controls

def add_break_dummies(df):
    df = df.copy()
    df["D_1914"] = (df["year"] >= 1914).astype(int)
    df["D_1945"] = (df["year"] >= 1945).astype(int)
    df["D_1987"] = (df["year"] >= 1987).astype(int)
    return df

df_GS  = add_break_dummies(df_GS)
df_BW  = add_break_dummies(df_BW)
df_FIN = add_break_dummies(df_Fin)

def country_hac_regressions(df, regime, breaks, maxlags=2, min_obs=15):
    rows = []
    rhs = base_rhs + breaks
    formula = f"{y} ~ " + " + ".join(rhs)

    # columns required for estimation (so we can dropna explicitly)
    required = [y] + rhs + ["country", "year"]

    for c in sorted(df["country"].unique()):
        df_c = df[df["country"] == c].copy()

        # Drop missing *before* calling statsmodels
        df_c = df_c.dropna(subset=required)

        # Need enough obs after dropping missing
        if len(df_c) < min_obs:
            continue

        # HAC maxlags must be < number of observations (rule of thumb)
        ml = min(maxlags, max(1, len(df_c)//4))

        try:
            model = smf.ols(formula, data=df_c).fit(
                cov_type="HAC",
                cov_kwds={"maxlags": ml}
            )
        except Exception as e:
            # Keep track of failures rather than crashing
            rows.append({
                "regime": regime,
                "country": c,
                "variable": "__MODEL_FAILED__",
                "coef": np.nan,
                "se": np.nan,
                "t": np.nan,
                "p": np.nan,
                "error": str(e),
                "nobs": len(df_c)
            })
            continue

        for v in finance_vars:
            rows.append({
                "regime": regime,
                "country": c,
                "variable": v,
                "coef": model.params.get(v, np.nan),
                "se": model.bse.get(v, np.nan),
                "t": model.tvalues.get(v, np.nan),
                "p": model.pvalues.get(v, np.nan),
                "nobs": int(model.nobs),
                "maxlags_used": ml
            })

    return pd.DataFrame(rows)

res_GS  = country_hac_regressions(df_GS,  "GS",  breaks=["D_1914"])
res_BW  = country_hac_regressions(df_BW,  "BW",  breaks=["D_1945"])
res_FIN = country_hac_regressions(df_Fin, "FIN", breaks=["D_1987"])

results = pd.concat([res_GS, res_BW, res_FIN], ignore_index=True)

# Optional: see if any country models failed
failed = results[results["variable"] == "__MODEL_FAILED__"]
print("Model failures:", len(failed))
if len(failed) > 0:
    print(failed[["regime","country","nobs","error"]].head(10))

# Keep only the finance coefficient rows for hypothesis testing
results_finance = results[results["variable"].isin(finance_vars)].copy()
print(results_finance)
```

```
Model failures: 0
   regime         country            variable      coef        se         t  \
0      GS       Australia  liquid_liabilities -0.371831  0.342576 -1.085399
1      GS       Australia      bank_loans_gdp -0.303282  0.295894 -1.024971
2      GS       Australia            mcap_gdp  0.212603  0.232890  0.912887
3      GS          Canada  liquid_liabilities -1.514826  0.474964 -3.189353
4      GS          Canada      bank_loans_gdp -0.788429  0.445138 -1.771201
..    ...             ...                 ...       ...       ...       ...
76    FIN  United Kingdom      bank_loans_gdp -0.015603  0.068070 -0.229223
77    FIN  United Kingdom            mcap_gdp  0.039865  0.015359  2.595539
78    FIN   United States  liquid_liabilities -0.148901  0.083706 -1.778848
79    FIN   United States      bank_loans_gdp  0.149359  0.127875  1.168008
80    FIN   United States            mcap_gdp  0.040465  0.017685  2.288171

           p  nobs  maxlags_used
0   0.277745    29             2
1   0.305377    29             2
2   0.361302    29             2
3   0.001426    29             2
4   0.076527    29             2
..       ...   ...           ...
76  0.818696    43             2
77  0.009444    43             2
78  0.075265    43             2
79  0.242803    43             2
80  0.022128    43             2

[81 rows x 9 columns]
```

In [48]:

```python
H1_test = (
    results
    .groupby(["regime", "variable"])["coef"]
    .mean()
    .unstack("regime")
)

print("H1 – Mean coefficients by regime")
print(H1_test)
```

```
H1 – Mean coefficients by regime
regime                    BW       FIN        GS
variable
bank_loans_gdp     -0.302238 -0.056103 -0.535979
liquid_liabilities -0.007194 -0.033775  0.061137
mcap_gdp            0.132139  0.047373  0.163516
```

In [49]:

```python
H2_test = (
    results[results["variable"].isin(finance_vars)]
    .groupby(["regime", "variable"])["coef"]
    .mean()
    .reset_index()
)

print("H2 – Average finance coefficients by regime")
print(H2_test)
```

```
H2 – Average finance coefficients by regime
  regime            variable      coef
0     BW      bank_loans_gdp -0.302238
1     BW  liquid_liabilities -0.007194
2     BW            mcap_gdp  0.132139
3    FIN      bank_loans_gdp -0.056103
4    FIN  liquid_liabilities -0.033775
5    FIN            mcap_gdp  0.047373
6     GS      bank_loans_gdp -0.535979
7     GS  liquid_liabilities  0.061137
8     GS            mcap_gdp  0.163516
```

In [54]:

```python
H3_test = (
    results[(results["regime"] == "BW") &
            (results["variable"].isin(["bank_loans_gdp", "mcap_gdp"]))]
    .groupby("variable")["coef"]
    .agg(["mean", "std", "count"])
)

print("H3 – Bretton Woods: credit vs markets")
print(H3_test)
```

```
H3 – Bretton Woods: credit vs markets
                    mean       std  count
variable
bank_loans_gdp -0.302238  0.343999      9
mcap_gdp        0.132139  1.019018      9
```

In [51]:

```python
H4_test = (
    results[(results["regime"] == "FIN") &
            (results["variable"] == "mcap_gdp")]
)

print("H4 – Financialization: market finance")
print(H4_test[["country", "coef", "p"]])

print("Share negative (FIN, mcap):",
      (H4_test["coef"] < 0).mean())
```

```
H4 – Financialization: market finance
           country      coef         p
56       Australia  0.052057  0.033426
59          Canada  0.012853  0.446123
62         Denmark  0.018977  0.398099
65         Germany  0.086768  0.030018
68          Norway  0.096951  0.038000
71          Sweden  0.051456  0.000557
74     Switzerland  0.026962  0.001230
77  United Kingdom  0.039865  0.009444
80   United States  0.040465  0.022128
Share negative (FIN, mcap): 0.0
```

In [ ]:

```python
def engle_granger_by_country(df, y_level, x_level, entity="country", trend="c", min_obs=25):
    rows = []
    for c in sorted(df[entity].unique()):
        d = df[df[entity] == c].dropna(subset=[y_level, x_level]).sort_values("year")
        if len(d) < min_obs:
            continue

        y = d[y_level].astype(float).to_numpy()
        x = d[x_level].astype(float).to_numpy()

        # coint returns: t_stat, p_value, critical_values
        t_stat, pval, crit = coint(y, x, trend=trend)
        rows.append({
            "country": c,
            "y": y_level,
            "x": x_level,
            "nobs": len(d),
            "EG_tstat": t_stat,
            "p_value": pval,
            "coint_5pct": pval < 0.05
        })

    return pd.DataFrame(rows)


pairs = ["mcap_gdp", "bank_loans_gdp", "liquid_liabilities"]

eg_results = []
for xvar in pairs:
    out = engle_granger_by_country(df_Final, y_level="ln_gdppc", x_level=xvar, trend="c", min_obs=25)
    eg_results.append(out)

eg_results = pd.concat(eg_results, ignore_index=True)

# summary
summary_eg = eg_results.groupby("x")["coint_5pct"].mean().reset_index(name="share_cointegrated_5pct")
print(eg_results)


print(eg_results.sort_values(["x","p_value"]).head(20))
```

```
           country         y                   x  nobs  EG_tstat   p_value  \
0        Australia  ln_gdppc            mcap_gdp   118 -2.949309  0.122752
1           Canada  ln_gdppc            mcap_gdp   118 -1.594901  0.723037
2          Denmark  ln_gdppc            mcap_gdp   127 -1.607282  0.717806
3           France  ln_gdppc            mcap_gdp   127 -0.400644  0.970699
4          Germany  ln_gdppc            mcap_gdp   127 -1.450367  0.779728
5           Norway  ln_gdppc            mcap_gdp   127 -0.947918  0.912288
6         Portugal  ln_gdppc            mcap_gdp   127 -1.913392  0.573200
7            Spain  ln_gdppc            mcap_gdp   118 -2.086631  0.483357
8           Sweden  ln_gdppc            mcap_gdp   127 -1.200631  0.857736
9      Switzerland  ln_gdppc            mcap_gdp   127 -1.669266  0.690790
10  United Kingdom  ln_gdppc            mcap_gdp   127 -1.531910  0.748753
11   United States  ln_gdppc            mcap_gdp   118 -2.005303  0.525820
12       Australia  ln_gdppc      bank_loans_gdp   139 -1.239708  0.847194
13          Canada  ln_gdppc      bank_loans_gdp   139 -1.722689  0.666459
14         Denmark  ln_gdppc      bank_loans_gdp   139 -1.044797  0.894016
15         Finland  ln_gdppc      bank_loans_gdp   139 -1.649428  0.699584
16         Germany  ln_gdppc      bank_loans_gdp   139 -0.902102  0.919898
17           Italy  ln_gdppc      bank_loans_gdp   139 -2.701507  0.199037
18           Japan  ln_gdppc      bank_loans_gdp   139 -2.022342  0.516947
19          Norway  ln_gdppc      bank_loans_gdp   139 -1.259289  0.841684
20          Sweden  ln_gdppc      bank_loans_gdp   139 -1.418188  0.791211
21     Switzerland  ln_gdppc      bank_loans_gdp   139 -0.992608  0.904240
22  United Kingdom  ln_gdppc      bank_loans_gdp   139 -1.759187  0.649326
23   United States  ln_gdppc      bank_loans_gdp   139 -2.128399  0.461534
24       Australia  ln_gdppc  liquid_liabilities   139 -1.455467  0.777870
25          Canada  ln_gdppc  liquid_liabilities   139 -1.418396  0.791138
26         Denmark  ln_gdppc  liquid_liabilities   139  0.582803  0.993130
27         Finland  ln_gdppc  liquid_liabilities   139 -0.444751  0.968040
28          France  ln_gdppc  liquid_liabilities   139 -0.126827  0.982478
29         Germany  ln_gdppc  liquid_liabilities   139 -1.109153  0.880114
30           Italy  ln_gdppc  liquid_liabilities   139 -1.965723  0.546343
31           Japan  ln_gdppc  liquid_liabilities   139 -1.597104  0.722111
32     Netherlands  ln_gdppc  liquid_liabilities   139 -1.910146  0.574852
33          Norway  ln_gdppc  liquid_liabilities   139  0.209773  0.989747
34        Portugal  ln_gdppc  liquid_liabilities   139 -2.878593  0.141954
35           Spain  ln_gdppc  liquid_liabilities   139 -2.270839  0.388190
36          Sweden  ln_gdppc  liquid_liabilities   139  0.157483  0.988960
37     Switzerland  ln_gdppc  liquid_liabilities   139 -1.506525  0.758682
38  United Kingdom  ln_gdppc  liquid_liabilities   139 -1.463221  0.775023
39   United States  ln_gdppc  liquid_liabilities   139 -1.650292  0.699203

    coint_5pct
0        False
1        False
2        False
3        False
4        False
5        False
6        False
7        False
8        False
9        False
10       False
11       False
12       False
13       False
14       False
15       False
16       False
17       False
18       False
19       False
20       False
21       False
22       False
23       False
24       False
25       False
26       False
27       False
28       False
29       False
30       False
31       False
32       False
33       False
34       False
35       False
36       False
37       False
38       False
39       False
           country         y                   x  nobs  EG_tstat   p_value  \
17           Italy  ln_gdppc      bank_loans_gdp   139 -2.701507  0.199037
23   United States  ln_gdppc      bank_loans_gdp   139 -2.128399  0.461534
18           Japan  ln_gdppc      bank_loans_gdp   139 -2.022342  0.516947
22  United Kingdom  ln_gdppc      bank_loans_gdp   139 -1.759187  0.649326
13          Canada  ln_gdppc      bank_loans_gdp   139 -1.722689  0.666459
15         Finland  ln_gdppc      bank_loans_gdp   139 -1.649428  0.699584
20          Sweden  ln_gdppc      bank_loans_gdp   139 -1.418188  0.791211
19          Norway  ln_gdppc      bank_loans_gdp   139 -1.259289  0.841684
12       Australia  ln_gdppc      bank_loans_gdp   139 -1.239708  0.847194
14         Denmark  ln_gdppc      bank_loans_gdp   139 -1.044797  0.894016
21     Switzerland  ln_gdppc      bank_loans_gdp   139 -0.992608  0.904240
16         Germany  ln_gdppc      bank_loans_gdp   139 -0.902102  0.919898
34        Portugal  ln_gdppc  liquid_liabilities   139 -2.878593  0.141954
35           Spain  ln_gdppc  liquid_liabilities   139 -2.270839  0.388190
30           Italy  ln_gdppc  liquid_liabilities   139 -1.965723  0.546343
32     Netherlands  ln_gdppc  liquid_liabilities   139 -1.910146  0.574852
39   United States  ln_gdppc  liquid_liabilities   139 -1.650292  0.699203
31           Japan  ln_gdppc  liquid_liabilities   139 -1.597104  0.722111
37     Switzerland  ln_gdppc  liquid_liabilities   139 -1.506525  0.758682
38  United Kingdom  ln_gdppc  liquid_liabilities   139 -1.463221  0.775023

    coint_5pct
17       False
23       False
```

```
18        False
22        False
13        False
15        False
20        False
19        False
12        False
14        False
21        False
16        False
34        False
35        False
30        False
32        False
39        False
31        False
37        False
38        False
```